On Douglas' conjecture for m-Kronecker quivers*

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Abstract

We prove Douglas' conjecture for m-Kronecker quivers for each $m \geq 3$ on the normalized logarithms of Euler characteristics of moduli spaces. Also, we state a log-concavity conjecture on Euler characteristics near slope one for the 3-Kronecker quiver.

1 Introduction and statement of the theorem

Let $m \in \mathbb{Z}_{\geq 3}$. Let $K(m)$ be the quiver with two vertices and m arrows from the first vertex to the second. All quiver representations are over $\mathbb{C}$. Let χ denote the ordinary topological Euler characteristic. For positive coprime integers a and b, let $\mathcal{M}^s_{(a,b)}(K(m))$ denote the moduli space of stable representations of $K(m)$ of dimension (a,b), where a is the dimension at the source and b at the target. Let

$$E_m(a,b) = \chi(\mathcal{M}^s_{(a,b)}(K(m))).$$

Our stability requires

$$av - bu > 0 \tag{1.1}$$

for every nonzero proper subrepresentation of dimension (u,v). For

$$\lambda_\pm = \frac{m \pm \sqrt{m^2-4}}{2}, \qquad I_m = [\lambda_-, \lambda_+],$$

let

$$\mathcal{A}_m = \{(a,b) \in \mathbb{Z}^2_{>0} : \gcd(a,b) = 1,\ \mathcal{M}^s_{(a,b)}(K(m)) \neq \emptyset\}.$$

We prove Douglas' conjecture in the following [19, Conjecture 6.1] (see also [11]). The study of black hole entropies motivates Douglas' conjecture [3, 4, 5, 7, 10, 19].

Theorem 1.1. *For each integer $m \geq 3$, there is a continuous function $F_m : I_m \to \mathbb{R}_{\geq 0}$ such that*

$$F_m(\lambda_-) = F_m(\lambda_+) = 0, \qquad 0 < F_m(r) < \infty \quad (\lambda_- < r < \lambda_+)$$

and

$$\lim_{\substack{a\to\infty,\ (a,b)\in\mathcal{A}_m \\ \frac{b}{a}\to r\in I_m}} \frac{\log E_m(a,b)}{a} = F_m(r). \tag{1.2}$$

By [18, Theorem 4.1] and [17, Theorem 6.1], we have

$$\mathcal{A}_m = \{(a,b) \in \mathbb{Z}^2_{>0} : \gcd(a,b) = 1,\ a^2 + b^2 - mab \leq 1\}. \tag{1.3}$$

For $a^2 + b^2 - mab \leq 0$, see also [15, Theorems 2.7(1) and 4.1(1)]. Furthermore, (1.2) is equivalent to the following: for every $r \in I_m$ and $\varepsilon > 0$, there are $\delta > 0$ and $A > 0$ such that

$$\left|\frac{\log E_m(a,b)}{a} - F_m(r)\right| < \varepsilon \quad \text{if } (a,b) \in \mathcal{A}_m,\ a > A, \left|\frac{b}{a} - r\right| < \delta.$$

*Some TEX and Lean formalization files are available at https://github.com/so-okada/kronecker-euler-sources.

1.1 The counting model and the comparison argument

We use a counting formula for $mE_m(a,b)$ [1] with quiver representations. The finite model counts colored gradings on paths. We construct maps between these sets of gradings and obtain a product estimate for their cardinalities. The product estimate gives an exponential growth rate under repeated concatenation.

We first prove the limit in (1.2) for rational r. We then obtain the limit for irrational r by comparison with rational values. Finally, we prove continuity of F_m and the endpoint values $F_m(\lambda_-) = F_m(\lambda_+) = 0$.

2 A finite model for Euler characteristics at coprime dimensions

2.1 Euler characteristics

By [8, Theorem 4.1 and Proposition 4.3] (see also [14, Proposition 3.5 and Section 3.5]), we identify the stable moduli $\mathcal{M}^s_{(a,b)}(K(m))$ with a quasi-projective complex algebraic variety in its analytic topology.

By [6, Corollary 6.10], an algebraic variety has the homotopy type of a finite CW complex. Consequently, ordinary rational singular homology is finite-dimensional in each degree and vanishes in all sufficiently high degrees. Thus,

$$\chi(X) = \sum_{i \geq 0} (-1)^i \dim_{\mathbb{Q}} H_i(X; \mathbb{Q})$$

is a finite sum for $X = \mathcal{M}^s_{(a,b)}(K(m))$.

2.2 Vertex exchange and reflection

Lemma 2.1. *Let a and b be positive coprime integers. Then, we have the following.*

1. *We have*
$$E_m(a,b) = E_m(b,a) = E_m(a, ma-b) \tag{2.1}$$
provided $ma - b > 0$.

2. *We have the following properties for the transformations from (a,b) to (b,a) and from (a,b) to $(a, ma-b)$.*

   (a) *They preserve coprimality.*

   (b) *They preserve the quadratic form $mab - a^2 - b^2$.*

Proof. Proof of Claim 1. Let us identify the dimension transformations that preserve the Euler characteristic. By vertex exchange and the reflection isomorphism of [19, Proposition 4.3], we obtain (2.1). On dimension vectors, the three steps are

$$(a,b) \xrightarrow{\text{exchange}} (b,a) \xrightarrow{\text{reflection}} (ma-b, a) \xrightarrow{\text{exchange}} (a, ma-b).$$

Transpose duality exchanges subrepresentations with quotients. Then, vertex exchange respects (1.1).

Proof of Claim 2(a). We have $\gcd(b,a) = \gcd(a,b)$ and $\gcd(a, ma-b) = \gcd(a,b)$.

Proof of Claim 2(b). Let us check preservation of the quadratic form. The form $mab - a^2 - b^2$ is symmetric in a and b. We have

$$ma(ma-b) - a^2 - (ma-b)^2 = mab - a^2 - b^2.$$

□

2.3 Connected colored gradings

For $a \in \mathbb{Z}$, $b \in \mathbb{Z}$, and $n \in \mathbb{Z}$, let

$$[a,b]_{\mathbb{Z}} = \{j \in \mathbb{Z} : a \le j \le b\}, \qquad [n]_{\mathbb{Z}} = [1,n]_{\mathbb{Z}}.$$

We recall the notion of NE paths from [1] and fix the necessary notation.

Definition 2.2. *An NE path is a finite lattice path P in $(\mathbb{Z}\times\mathbb{R})\cup(\mathbb{R}\times\mathbb{Z})$ such that P starts from $(0,0)$ and P consists of unit east steps E and north steps N. For an NE path P and $\mathbf{t} \in \mathbb{Z}^2$, we call $P+\mathbf{t}$ a translated NE path (tNE path).*

1. *For a tNE path U and $\zeta \in \mathbb{Z}^2$, define*

$$\mathrm{Trans}(U,\zeta) = U + \zeta.$$

   We call every path with no edges the empty path and denote such a path by $\emptyset$.

2. *Let P be a tNE path.*

   (a) *Let $|P|$, $|P|_E$, and $|P|_N$ denote its numbers of edges, horizontal edges, and vertical edges, respectively.*

   (b) *We label its horizontal edges $e_1,\dots,e_{|P|_E}$ and its vertical edges $n_1,\dots,n_{|P|_N}$ in traversal order.*

   (c) *For $k \in [|P|]_{\mathbb{Z}}$, let P_k denote its k-th edge.*

   (d) *For $P \neq \emptyset$, let $\mathrm{Start}_{\mathrm{pt}}(P)$ and $\mathrm{End}_{\mathrm{pt}}(P)$ denote its starting and ending points, respectively. We define its starting and ending edges by*

$$\mathrm{Start}_{\mathrm{NE}}(P) = P_1, \qquad \mathrm{End}_{\mathrm{NE}}(P) = P_{|P|}.$$

3. *For nonempty tNE paths U and V, their concatenation $U \dotplus V$ follows U and then*

$$\mathrm{Trans}(V, \mathrm{End}_{\mathrm{pt}}(U) - \mathrm{Start}_{\mathrm{pt}}(V)).$$

   We define $\emptyset \dotplus V = V$ and $U \dotplus \emptyset = U$. For $p \in \mathbb{Z}$, $q \in \mathbb{Z}$, and tNE paths U_i indexed by $i \in [p,q]_{\mathbb{Z}}$, define

$$\dot{\bigcup_{i=p}^{q}} U_i = \begin{cases} U_p \dotplus \cdots \dotplus U_q, & p \le q, \\ \emptyset, & p > q. \end{cases}$$

   For an integer $\nu \ge 1$, we define the ν-fold concatenation by

$$U^{\nu} = \underbrace{U \dotplus \cdots \dotplus U}_{\nu \text{ copies}}.$$

   We define $U^0 = \emptyset$.

4. *For an integer $j \ge 1$, let $R_j = E \dotplus N^j$ be an NE path. We call a translate of R_j a simple run of vertical length j. A run is a concatenation of one or more simple runs.*

5. *For a tNE path P, a subpath of P is a path consisting of the edges $P_{i+1},\dots,P_j$ for some $i \in [0,|P|]_{\mathbb{Z}}$ and $j \in [0,|P|]_{\mathbb{Z}}$ with $i \le j$. We denote this subpath by $P[i,j]$. We define $P[i,i] = \emptyset$.*

6. *For tNE paths A and B, let $A \subseteq B$ mean that A is a subpath of B. We call them disjoint if they have no common edge.*

Definition 2.3. *Let U and V be tNE paths.*

1. *For $i \in [0, |U|]_{\mathbb{Z}}$, we define the insertion*

$$\mathrm{Ins}(U, i, V) = U[0, i] + V + U[i, |U|].$$

2. *For a vertical edge $v = U_k$ of U, we define the deletion*

$$\mathrm{Del}(U, v) = U[0, k-1] + \mathrm{Trans}(U[k, |U|], (0, -1)).$$

3. *Suppose $U \neq \emptyset$. Let L and R be NE paths. Let A and B be simple runs such that $\mathrm{Start}_{\mathrm{pt}}(A) = \mathrm{Start}_{\mathrm{pt}}(B) = (0, 0)$. When $U = \mathrm{Trans}(L + A + B + R, \mathrm{Start}_{\mathrm{pt}}(U))$, we define the exchange*

$$\mathrm{Exc}(U; |L|, A, B) = \mathrm{Trans}(L + B + A + R, \mathrm{Start}_{\mathrm{pt}}(U)).$$

4. *Let $n \in \mathbb{Z}_{>0}$, let U be a run, and let $A_1, \ldots, A_n$ be simple runs. Assume*

$$U = \mathrm{Trans}(A_1 + \cdots + A_n, \mathrm{Start}_{\mathrm{pt}}(U)), \qquad \mathrm{Start}_{\mathrm{pt}}(A_i) = (0, 0) \quad (i \in [n]_{\mathbb{Z}}).$$

   For $j \in [0, n]_{\mathbb{Z}}$, we define the cyclic rotation

$$\mathrm{CRot}(U, j) = \mathrm{Trans}(A_{j+1} + \cdots + A_n + A_1 + \cdots + A_j, \mathrm{Start}_{\mathrm{pt}}(U)).$$

The grading convention comes from [1].

Definition 2.4. *Let U be a tNE path.*

1. *Let $E(U)$, $N(U)$, and $\mathrm{NE}(U)$ be the sets of horizontal, vertical, and all edges of U, respectively.*

2. *We refer to a function $g\colon \mathrm{NE}(U) \to \mathbb{Z}_{\geq 0}$ as a grading of U. For each $v \in \mathrm{NE}(U)$, we call $g(v)$ the grade of v. Let $G(U)$ denote the set of gradings of U. For each integer $h \geq 0$, we define*

$$G_h(U) = \{g \in G(U) : g(v) \leq h \text{ for every } v \in \mathrm{NE}(U)\}.$$

   For $g \in G(U)$, $h \in G(U)$, and $e \in \mathrm{NE}(U)$, we define

$$\mathrm{Diff}_U(g, h; e) = g(e) - h(e).$$

3. *Consider $g \in G(U)$.*

   (a) We define the boundary balance function of g

$$\mathrm{bBal}_{U,g}\colon [0, |U|]_{\mathbb{Z}} \to \mathbb{Z}$$

   such that

$$\mathrm{bBal}_{U,g}(i) = \#E(U[0, i]) - \sum_{v \in N(U[0,i])} g(v).$$

   (b) For $k \in [|U|]_{\mathbb{Z}}$, we define the edge balance at the k-th edge U_k of U by

$$\mathrm{eBal}_{U,g}(U_k) = \mathrm{bBal}_{U,g}(k).$$

4. *Consider $g \in G(U)$. We use the convention $\binom{m}{r} = 0$ for integers $r > m$. We call a function $c\colon \mathrm{NE}(U) \to \mathbb{Z}_{>0}$ a color function of g if, for each $v \in \mathrm{NE}(U)$, its color $c(v)$ satisfies*

$$c(v) \in \left[\binom{m}{g(v)}\right]_{\mathbb{Z}}.$$

Let $\mathrm{Col}(U, g, m)$ denote the set of color functions of g. We call (g, c) a colored grading of U.

5. *Let $g \in G(U)$ and $c \in \mathrm{Col}(U, g, m)$. Let $\mathbf{t} \in \mathbb{Z}^2$. We define the translated grading $g + \mathbf{t}$ and translated color function $c + \mathbf{t}$ on $\mathrm{Trans}(U, \mathbf{t})$ by*

$$(g + \mathbf{t})(v + \mathbf{t}) = g(v), \qquad (c + \mathbf{t})(v + \mathbf{t}) = c(v) \qquad (v \in \mathrm{NE}(U)).$$

6. *Let A be a subpath of U, $g \in G(U)$, and $c \in \mathrm{Col}(U, g, m)$. The restricted colored grading on A is*

$$(g|_A, c|_A) = (g|_{\mathrm{NE}(A)}, c|_{\mathrm{NE}(A)}).$$

Let g and c also denote these restrictions when the ambient path is specified.

7. *Let $j \in \mathbb{Z}_{\geq 0}$. For $s \in [j]_{\mathbb{Z}}$, let U_s be a tNE path and let (g_s, c_s) be a colored grading of U_s. Let $W = U_1 \dotplus \cdots \dotplus U_j$. For $s \in [j]_{\mathbb{Z}}$, define $\iota_s : \mathrm{NE}(U_s) \to \mathrm{NE}(W)$ by*

$$\iota_s((U_s)_h) = W_{|U_1|+\cdots+|U_{s-1}|+h} \qquad (h \in [|U_s|]_{\mathbb{Z}}).$$

The concatenated colored grading $(g_1 \dotplus \cdots \dotplus g_j, c_1 \dotplus \cdots \dotplus c_j)$ is the pair (g, c) such that

$$g(\iota_s(e)) = g_s(e), \qquad c(\iota_s(e)) = c_s(e) \qquad (s \in [j]_{\mathbb{Z}},\ e \in \mathrm{NE}(U_s)).$$

Example 2.5. Let $m = 3$. Let $U = E \dotplus E \dotplus N \dotplus E \dotplus N = e_1 \dotplus e_2 \dotplus n_1 \dotplus e_3 \dotplus n_2$. Consider $g \in G(U)$ and $c \in \mathrm{Col}(U, g, m)$ such that

x	e_1	e_2	n_1	e_3	n_2
$g(x)$	2	0	1	1	2
$c(x)$	3	1	2	1	3

Then, the boundary balances are

i	0	1	2	3	4	5
$\mathrm{bBal}_{U,g}(i)$	0	1	2	1	2	0

The edge balances are

x	e_1	e_2	n_1	e_3	n_2
$\mathrm{eBal}_{U,g}(x)$	1	2	1	2	0

Definition 2.6. *Let U be a tNE path. Consider $g \in G(U)$.*

1. *We define an h-precursor $P_{U,g}\colon N(U) \to E(U)$ of g by the following conditions for each $U_k \in N(U)$.*

   (a) $P_{U,g}(U_k) \in E(U[0, k-1])$.

   (b) We have

$$\#\{w \in N(U[0, k-1]) : P_{U,g}(w) = P_{U,g}(U_k)\} < g(P_{U,g}(U_k)).$$

(c) $P_{U,g}(U_k)$ is the last horizontal edge satisfying (a) and (b).

For each $v \in N(U)$, we say that $P_{U,g}(v)$ is an h-precursor of v.

2. *Suppose that an h-precursor $P_{U,g}$ exists. Consider $U_i \in E(U)$ with*

$$\#P_{U,g}^{-1}(\{U_i\}) = g(U_i) > 0.$$

Let $j \in [|U|]_{\mathbb{Z}}$ be maximal such that $U_j \in P_{U,g}^{-1}(\{U_i\})$.

(a) We define the node

$$\mathrm{Node}_{U,g}(U_i) = U[i-1, j].$$

(b) For each $w \in P_{U,g}^{-1}(\{U_i\})$, we call $\mathrm{Node}_{U,g}(U_i)$ the owner of w.

3. *Suppose that an h-precursor $P_{U,g}$ exists. Let $i \in [0, |U|]_{\mathbb{Z}}$. Let $x \in E(U[0,i])$. We define the debt of x at i by*

$$\mathrm{Debt}_{U,g}(x, i) = g(x) - \#\{v \in N(U[0,i]) : P_{U,g}(v) = x\}.$$

Lemma 2.7. *Let U be a tNE path. Consider $g \in G(U)$. Then, we have the following.*

1. *There is at most one h-precursor of g.*
2. *The h-precursor $P_{U,g}$ exists if and only if*

$$\sum_{x \in E(U[0,i])} g(x) \geq \#N(U[0,i]) \qquad (i \in [0, |U|]_{\mathbb{Z}}). \tag{2.2}$$

Proof. Proof of Claim 1. Let us prove uniqueness by induction on the edge index. Suppose that $P\colon N(U) \to E(U)$ and $Q\colon N(U) \to E(U)$ are h-precursors of g. We prove by induction on $k \in [0, |U|]_{\mathbb{Z}}$ that P and Q agree on $N(U[0,k])$. For $k = 0$, we have $N(U[0,0]) = \emptyset$.

Let us prove the induction step. Let $k \in [|U|]_{\mathbb{Z}}$. Suppose that P and Q agree on $N(U[0,k-1])$. If U_k is horizontal, then $N(U[0,k]) = N(U[0,k-1])$. Suppose that U_k is vertical. For each $x \in E(U[0,k-1])$, the induction hypothesis gives the common value

$$\begin{aligned}\nu(x) &= \#\{w \in N(U[0,k-1]) : P(w) = x\} \\ &= \#\{w \in N(U[0,k-1]) : Q(w) = x\}.\end{aligned}$$

Let

$$A_k = \{x \in E(U[0,k-1]) : \nu(x) < g(x)\}.$$

Conditions (a) and (b) of Definition 2.6(1) imply that both $P(U_k)$ and $Q(U_k)$ belong to A_k. Applying Definition 2.6(1)(c) to P with $x = Q(U_k)$ and to Q with $x = P(U_k)$ shows that neither edge follows the other. Hence, $P(U_k) = Q(U_k)$. Taking $k = |U|$ gives $P = Q$.

Proof of Claim 2. Let us obtain the prefix inequalities from an h-precursor. Suppose that $P_{U,g}$ exists. Definition 2.6(1)(a) gives $P_{U,g}(N(U[0,i])) \subseteq E(U[0,i])$. Definition 2.6(1)(b) gives $\#P_{U,g}^{-1}(\{x\}) \leq g(x)$ for each $x \in E(U)$. For $i \in [0, |U|]_{\mathbb{Z}}$, we therefore have

$$\#N(U[0,i]) = \sum_{x \in E(U[0,i])} \#\{w \in N(U[0,i]) : P_{U,g}(w) = x\} \leq \sum_{x \in E(U[0,i])} g(x).$$

Let us prove the converse by constructing $P_{U,g}$ in increasing edge order under (2.2). Suppose that we have defined its values on $N(U[0,k-1])$ and that U_k is vertical. For $x \in E(U[0,k-1])$, let

$$\nu_k(x) = \#\{w \in N(U[0,k-1]) : P_{U,g}(w) = x\}.$$

The induction hypothesis gives $\nu_k(x) \leq g(x)$. Inequality (2.2) for $i = k$ gives

$$\sum_{x \in E(U[0,k-1])} (g(x) - \nu_k(x)) = \sum_{x \in E(U[0,k-1])} g(x) - \#N(U[0,k-1]) \geq 1.$$

Thus, there is an x with $\nu_k(x) < g(x)$. Among these edges, let $P_{U,g}(U_k)$ be the last in edge order. Induction defines $P_{U,g}(v)$ for every $v \in N(U)$. □

Example 2.8. Let $U = E \dotplus E \dotplus N \dotplus N \dotplus N = e_1 \dotplus e_2 \dotplus n_1 \dotplus n_2 \dotplus n_3$. Consider $g \in G(U)$ such that

$$(g(e_1), g(e_2), g(n_1), g(n_2), g(n_3)) = (2, 1, 0, 0, 0).$$

Then, the h-precursor $P_{U,g}$ satisfies

$$P_{U,g}(n_1) = e_2, \qquad P_{U,g}(n_2) = P_{U,g}(n_3) = e_1.$$

Definition 2.9. *Let U be a tNE path.*

1. *Consider $g \in G(U)$. We call g connected if the following conditions hold.*

   (a) *The h-precursor $P_{U,g}$ exists.*

   (b) *For every node $\mathrm{Node}_{U,g}(x)$ and every vertical edge v owned by $\mathrm{Node}_{U,g}(x)$, we have*

   $$\mathrm{eBal}_{U,g}(v) \geq \mathrm{eBal}_{U,g}(x). \tag{2.3}$$

   We call this condition the node balance condition.

   (c) *$U = \mathrm{Node}_{U,g}(x)$ for some $x \in E(U)$.*

   Let $\mathrm{CtdG}(U)$ denote the set of connected gradings of U. For each integer $h \geq 0$, let

   $$\mathrm{CtdG}_h(U) = \mathrm{CtdG}(U) \cap G_h(U).$$

2. *For $g \in \mathrm{CtdG}(U)$, we define the surplus $\mathrm{Surp}(U, g)$ by*

   $$\mathrm{Surp}(U, g) = \#E(U) - 1 - \sum_{v \in N(U)} g(v). \tag{2.4}$$

3. *Consider $g \in G(U)$ and $c \in \mathrm{Col}(U, g, m)$. We call (g, c) connected if g is connected. For each integer $d \geq 0$, we define*

   $$\mathcal{C}_d(U) = \left\{ (g, c) \;\middle|\; \begin{array}{ll} g \in \mathrm{CtdG}(U), & c \in \mathrm{Col}(U, g, m), \\ \mathrm{Surp}(U, g) = d & \end{array} \right\}.$$

   We define

   $$C_d(U) = \#\mathcal{C}_d(U).$$

Lemma 2.10. *Let U be a tNE path. Let $\mathbf{t} \in \mathbb{Z}^2$ and $W = \mathrm{Trans}(U, \mathbf{t})$. For every integer $d \geq 0$, we have*

$$C_d(U) = C_d(W).$$

Proof. Let us define the map

$$\mathcal{C}_d(U) \longrightarrow \mathcal{C}_d(W), \qquad (g, c) \longmapsto (g + \mathbf{t}, c + \mathbf{t}).$$

Translation preserves edge order and each grade. Thus, for $v \in N(U)$,

$$P_{W,g+\mathbf{t}}(v + \mathbf{t}) = P_{U,g}(v) + \mathbf{t}.$$

Each node is translated by $\mathbf{t}$, and its edge balances are unchanged. The translated grading is therefore connected and has surplus d. The color bounds are unchanged because the grades are unchanged. Translation by $-\mathbf{t}$ defines the inverse map. Hence, the displayed map is a bijection. □

Lemma 2.11. *Let U be a tNE path. Consider $g \in \mathrm{CtdG}(U)$. Then,* $\mathrm{Surp}(U,g) \geq 0$.

Proof. Let us apply (2.3) at $\mathrm{End}_{\mathrm{NE}}(U)$. Let $x = \mathrm{Start}_{\mathrm{NE}}(U)$ and $v = \mathrm{End}_{\mathrm{NE}}(U)$. By Definition 2.9(1)(c), $U = \mathrm{Node}_{U,g}(x)$. Since $x = \mathrm{Start}_{\mathrm{NE}}(U) \in E(U)$, we have $\mathrm{eBal}_{U,g}(x) = 1$. Inequality (2.3) gives

$$\begin{aligned}\mathrm{Surp}(U,g) &= \#E(U) - 1 - \sum_{w \in N(U)} g(w) \\ &= \mathrm{eBal}_{U,g}(v) - \mathrm{eBal}_{U,g}(x) \geq 0.\end{aligned}$$

□

Lemma 2.12. *Let U be a tNE path and $d \in \mathbb{Z}_{\geq 0}$. Then,*

$$\begin{aligned}C_d(U) &= \sum_{\substack{g \in \mathrm{CtdG}_m(U) \\ \mathrm{Surp}(U,g)=d}} \#\,\mathrm{Col}(U,g,m) \\ &= \sum_{\substack{g \in \mathrm{CtdG}_m(U) \\ \mathrm{Surp}(U,g)=d}} \prod_{v \in \mathrm{NE}(U)} \binom{m}{g(v)}.\end{aligned} \tag{2.5}$$

Proof. Let us count the color functions for a fixed grading $g \in G(U)$. By Definition 2.4, choosing a color function is equivalent to choosing one element of $[\binom{m}{g(v)}]_{\mathbb{Z}}$ for each edge v. Hence,

$$\mathrm{Col}(U,g,m) \simeq \prod_{v \in \mathrm{NE}(U)} \left[\binom{m}{g(v)}\right]_{\mathbb{Z}}, \qquad \#\,\mathrm{Col}(U,g,m) = \prod_{v \in \mathrm{NE}(U)} \binom{m}{g(v)}.$$

If $g(v) > m$ for some edge v, the corresponding factor is empty. Thus, only gradings in $G_m(U)$ contribute to $C_d(U)$. Partitioning $\mathcal{C}_d(U)$ according to its grading gives (2.5). □

Lemma 2.13 (Prefix counts for h-precursors)**.** *Let $U = U_1 + \cdots + U_n$ be a tNE path. Let $g \in G(U)$. Suppose that $P = P_{U,g}$ exists. For $e \in E(U)$ and $k \in [0,n]_{\mathbb{Z}}$, define*

$$F_e = \{j \in [n]_{\mathbb{Z}} : U_j \in N(U),\ P(U_j) = e\}, \qquad \mathrm{Num}(e,k) = \#(F_e \cap [k]_{\mathbb{Z}}).$$

Then, we have the following.

1. *For $e \in E(U)$ and integers $0 \leq k \leq \ell \leq n$, we have*

$$0 = \mathrm{Num}(e,0) \leq \mathrm{Num}(e,k) \leq \mathrm{Num}(e,\ell) \leq g(e).$$

   If $\mathrm{Num}(e,k) = g(e)$, then $F_e \subseteq [k]_{\mathbb{Z}}$ and $\mathrm{Num}(e,\ell) = g(e)$ for $k \leq \ell \leq n$.

2. *Suppose that $U_s \in N(U)$ and $P(U_s) = U_r$. Then, $r < s$. For $k \in [r, s-1]_{\mathbb{Z}}$, we have*

$$\mathrm{Num}(U_r,k) \leq \mathrm{Num}(U_r, s-1) \leq g(U_r) - 1, \qquad \mathrm{Debt}_{U,g}(U_r,k) \geq 1.$$

   For each $U_k \in N(U)$ with $r < k \leq s$, let p_k be the unique index satisfying $P(U_k) = U_{p_k}$. Then, $r \leq p_k < k$.

3. *Under the hypotheses of Claim 2, let $r < i < s$ and $U_i \in E(U)$. Then,*

$$\mathrm{Num}(U_i, s-1) = \mathrm{Num}(U_i, n) = g(U_i), \qquad F_{U_i} \subseteq [i+1, s-1]_{\mathbb{Z}}.$$

   If $g(U_i) > 0$, the index $q_i = \max F_{U_i}$ satisfies

$$i < q_i < s, \qquad \mathrm{Node}_{U,g}(U_i) = U[i-1, q_i].$$

Proof. Proof of Claim 1. Let us prove $\#F_e \le g(e)$. The inclusions $F_e \cap [k]_{\mathbb{Z}} \subseteq F_e \cap [\ell]_{\mathbb{Z}} \subseteq F_e$ give monotonicity. If $F_e = \emptyset$, every count is zero. Otherwise, let $q = \max F_e$. Definition 2.6(1)(b) gives

$$\#F_e - 1 = \mathrm{Num}(e, q-1) < g(e).$$

Integrality gives $\#F_e \le g(e)$. If $\mathrm{Num}(e,k) = g(e)$, then

$$g(e) = \#(F_e \cap [k]_{\mathbb{Z}}) \le \#F_e \le g(e).$$

The two finite sets $F_e \cap [k]_{\mathbb{Z}} \subseteq F_e$ have equal cardinalities. Thus, they are equal, and $\mathrm{Num}(e,\ell) = g(e)$ for $k \le \ell \le n$.

Proof of Claim 2. Let us bound the debt before an assigned vertical edge. Conditions (a) and (b) of Definition 2.6(1) give $r < s$ and $\mathrm{Num}(U_r, s-1) < g(U_r)$. For $k \in [r, s-1]_{\mathbb{Z}}$, Claim 1 gives

$$\mathrm{Num}(U_r, k) \le \mathrm{Num}(U_r, s-1) \le g(U_r) - 1.$$

Hence, $\mathrm{Debt}_{U,g}(U_r, k) = g(U_r) - \mathrm{Num}(U_r, k) \ge 1$. For $r < k \le s$, the edge U_r precedes U_k and $\mathrm{Num}(U_r, k-1) < g(U_r)$. Definition 2.6(1)(c) gives $r \le p_k$; condition (a) gives $p_k < k$.

Proof of Claim 3. Let us prove $\mathrm{Num}(U_i, s-1) = g(U_i)$. Claim 1 gives $\mathrm{Num}(U_i, s-1) \le g(U_i)$. If $\mathrm{Num}(U_i, s-1) < g(U_i)$, Definition 2.6(1)(c) at U_s would give $i \le r$, contrary to $r < i$. Thus, $\mathrm{Num}(U_i, s-1) = g(U_i)$. Claim 1 gives $F_{U_i} \subseteq [s-1]_{\mathbb{Z}}$ and $\mathrm{Num}(U_i, n) = g(U_i)$. For each $t \in F_{U_i}$, we have $P(U_t) = U_i$. Definition 2.6(1)(a) gives $i < t$. Hence, $F_{U_i} \subseteq [i+1, s-1]_{\mathbb{Z}}$. If $g(U_i) > 0$, then $F_{U_i} \neq \emptyset$ and its maximum q_i satisfies $i < q_i < s$. The definition of a node in Definition 2.6 gives $\mathrm{Node}_{U,g}(U_i) = U[i-1, q_i]$. □

Lemma 2.14 (Restriction to an interval closed under h-precursors)**.** *Let $U = U_1 \dashv\!\vdash \cdots \dashv\!\vdash U_n$ be a tNE path. Let $g \in G(U)$. Suppose that $P = P_{U,g}$ exists. Let $a, b \in [0,n]_{\mathbb{Z}}$ satisfy $a < b$, and let $\ell = b - a$. Put*

$$X = U[a,b], \qquad X_j = U_{a+j}, \qquad g_X(X_j) = g(U_{a+j}) \quad (j \in [\ell]_{\mathbb{Z}}).$$

Suppose that $P(N(X)) \subseteq E(X)$. For $X_j \in N(X)$, let p_j be the unique index satisfying $P(U_{a+j}) = U_{p_j}$. Use $\mathrm{Num}(e,k) = \#(F_e \cap [k]_{\mathbb{Z}})$ from Lemma 2.13. Then, we have the following.

1. *For $X_j \in N(X)$, we have $a < p_j < a + j$. Define*

$$Q\colon N(X) \longrightarrow E(X), \qquad Q(X_j) = X_{p_j - a}.$$

For $X_i \in E(X)$ and $j \in [0,\ell]_{\mathbb{Z}}$,

$$\#\{k \in [j]_{\mathbb{Z}} : X_k \in N(X),\ Q(X_k) = X_i\} = \mathrm{Num}(U_{a+i}, a+j).$$

The map Q is the h-precursor P_{X,g_X}.

2. *Let $X_i \in E(X)$ and $i < j \le \ell$. Then,*

$$\mathrm{Node}_{X,g_X}(X_i) = X[i-1, j] \quad \Longleftrightarrow \quad \mathrm{Node}_{U,g}(U_{a+i}) = U[a+i-1, a+j]. \tag{2.6}$$

Each side of (2.6) asserts that its node exists and equals the specified subpath.

3. *For $j \in [0,\ell]_{\mathbb{Z}}$, we have*

$$\mathrm{bBal}_{X,g_X}(j) = \mathrm{bBal}_{U,g}(a+j) - \mathrm{bBal}_{U,g}(a).$$

For $j \in [\ell]_{\mathbb{Z}}$, we have

$$\mathrm{eBal}_{X,g_X}(X_j) = \mathrm{eBal}_{U,g}(U_{a+j}) - \mathrm{bBal}_{U,g}(a).$$

If g satisfies the node balance condition, then g_X satisfies that condition.

Proof. Proof of Claim 1. Let us verify $a < p_j < a + j$ and the formula for Q. For $X_j \in N(X)$, the hypothesis $P(N(X)) \subseteq E(X)$ gives $a < p_j \leq b$. Definition 2.6(1)(a) gives $p_j < a + j$. Hence, $X_{p_j - a} \in E(X)$ and $1 \leq p_j - a < j$.

Let us prove the equality of cardinalities in Claim 1. For $X_i \in E(X)$ and $j \in [0, \ell]_{\mathbb{Z}}$, the map

$$\{k \in [j]_{\mathbb{Z}} : X_k \in N(X),\ Q(X_k) = X_i\} \longrightarrow F_{U_{a+i}} \cap [a + j]_{\mathbb{Z}}, \qquad k \longmapsto a + k$$

is a bijection. Its inverse is $t \mapsto t - a$: every $t \in F_{U_{a+i}}$ satisfies $t > a + i > a$ by Definition 2.6(1)(a).

Let us verify Definition 2.6(1)(b) and (c) for Q. For $X_j \in N(X)$, apply the cardinality identity proved using $k \mapsto a + k$ with $i = p_j - a$ at boundary $j - 1$. This gives

$$\#\{k \in [j-1]_{\mathbb{Z}} : X_k \in N(X),\ Q(X_k) = X_{p_j - a}\} = \mathrm{Num}(U_{p_j}, a + j - 1) < g(U_{p_j}) = g_X(X_{p_j - a}),$$

where the strict inequality follows from Definition 2.6(1)(b). For $X_i \in E(X)$ with $i < j$, suppose

$$\#\{k \in [j-1]_{\mathbb{Z}} : X_k \in N(X),\ Q(X_k) = X_i\} < g_X(X_i).$$

Then, $\mathrm{Num}(U_{a+i}, a + j - 1) < g(U_{a+i})$. Definition 2.6(1)(c) gives $a + i \leq p_j$. Thus, $i \leq p_j - a$. Claim 1 of Lemma 2.7 gives $Q = P_{X, g_X}$.

Proof of Claim 2. First, suppose $\mathrm{Node}_{X, g_X}(X_i) = X[i-1, j]$. Then, $\#Q^{-1}(\{X_i\}) = g_X(X_i) = g(U_{a+i}) > 0$. Claim 1 at boundary ℓ gives $\mathrm{Num}(U_{a+i}, b) = g(U_{a+i})$. Claim 1 of Lemma 2.13 gives $F_{U_{a+i}} \subseteq [b]_{\mathbb{Z}}$. For every $t \in F_{U_{a+i}}$, Definition 2.6(1)(a) gives $t > a + i > a$. Hence,

$$F_{U_{a+i}} = \{a + k : k \in [\ell]_{\mathbb{Z}},\ X_k \in N(X),\ Q(X_k) = X_i\}.$$

The maximum of $\{k \in [\ell]_{\mathbb{Z}} : X_k \in N(X),\ Q(X_k) = X_i\}$ is j. Thus, $\max F_{U_{a+i}} = a + j$ and $\mathrm{Node}_{U, g}(U_{a+i}) = U[a + i - 1, a + j]$.

Second, suppose $\mathrm{Node}_{U, g}(U_{a+i}) = U[a + i - 1, a + j]$. Then, $F_{U_{a+i}} \subseteq [a + i + 1, a + j]_{\mathbb{Z}}$, $\#F_{U_{a+i}} = g(U_{a+i}) > 0$, and $\max F_{U_{a+i}} = a + j$. Claim 1 gives

$$\{k \in [\ell]_{\mathbb{Z}} : X_k \in N(X),\ Q(X_k) = X_i\} = \{t - a : t \in F_{U_{a+i}}\}.$$

This set has cardinality $g_X(X_i)$ and maximum j. Definition 2.6(2) gives $\mathrm{Node}_{X, g_X}(X_i) = X[i-1, j]$.

Proof of Claim 3. Let us prove the two balance identities in Claim 3. For $k \in [n]_{\mathbb{Z}}$, Definition 2.4(3)(a) gives

$$c_k := \mathrm{bBal}_{U, g}(k) - \mathrm{bBal}_{U, g}(k-1) = \begin{cases} 1, & U_k \in E(U), \\ -g(U_k), & U_k \in N(U). \end{cases}$$

Since $X_k = U_{a+k}$ and $g_X(X_k) = g(U_{a+k})$ for $k \in [\ell]_{\mathbb{Z}}$, for $j \in [0, \ell]_{\mathbb{Z}}$ we obtain

$$\mathrm{bBal}_{X, g_X}(j) = \sum_{k=a+1}^{a+j} c_k = \mathrm{bBal}_{U, g}(a + j) - \mathrm{bBal}_{U, g}(a).$$

For $j = 0$, the sum is empty and both sides are zero. For $j \in [\ell]_{\mathbb{Z}}$, Definition 2.4(3)(b) gives

$$\mathrm{eBal}_{X, g_X}(X_j) = \mathrm{bBal}_{X, g_X}(j), \qquad \mathrm{eBal}_{U, g}(U_{a+j}) = \mathrm{bBal}_{U, g}(a + j).$$

Substitution proves the edge-balance identity in Claim 3. Suppose that g satisfies the node balance condition. Let $Y = \mathrm{Node}_{X, g_X}(X_i) = X[i-1, j]$ and let X_k be owned by Y. Claims 1 and 2 give

$$\mathrm{Node}_{U, g}(U_{a+i}) = U[a + i - 1, a + j], \qquad P(U_{a+k}) = U_{a+i}.$$

Inequality (2.3) for g gives

$$\mathrm{eBal}_{X,g_X}(X_k) - \mathrm{eBal}_{X,g_X}(X_i) = \mathrm{eBal}_{U,g}(U_{a+k}) - \mathrm{eBal}_{U,g}(U_{a+i}) \geq 0.$$

This is the node balance condition for g_X. □

Lemma 2.15 (Structure of nodes)**.** *Let U be a tNE path. Consider $g \in G(U)$. Suppose that the h-precursor $P_{U,g}$ exists. Let $x = U_r \in E(U)$. Consider a node $X = \mathrm{Node}_{U,g}(x)$. Let $v = \mathrm{End}_{\mathrm{NE}}(X) = U_s$. Then, we have the following.*

1. *The following statements hold.*
   (a) *For every $w \in N(X)$, we have $P_{U,g}(w) \in E(X)$.*
   (b) *For each $y \in E(X)$ with $g(y) > 0$, the node $\mathrm{Node}_{U,g}(y)$ exists and $\mathrm{Node}_{U,g}(y) \subseteq X$.*
   (c) *For every $y \in E(X) \setminus \{x\}$ with $g(y) > 0$, $\mathrm{End}_{\mathrm{NE}}(\mathrm{Node}_{U,g}(y))$ precedes v.*
2. *For each pair of distinct edges $y \in E(U)$ and $z \in E(U)$ such that $\mathrm{Node}_{U,g}(y)$ and $\mathrm{Node}_{U,g}(z)$ exist, these nodes are disjoint or*
$$\mathrm{Node}_{U,g}(y) \subsetneq \mathrm{Node}_{U,g}(z) \quad \text{or} \quad \mathrm{Node}_{U,g}(z) \subsetneq \mathrm{Node}_{U,g}(y).$$
3. *Suppose that g satisfies the node balance condition. The following statements hold.*
   (a) *For every $i \in [r,s]_{\mathbb{Z}}$, we have*
$$\mathrm{bBal}_{U,g}(i) \geq \mathrm{eBal}_{U,g}(x).$$
   (b) *Let $y \in E(U)$ be such that $\mathrm{Node}_{U,g}(y) \subsetneq X$. Then, $\mathrm{eBal}_{U,g}(y) > \mathrm{eBal}_{U,g}(x)$.*
4. *Suppose that g satisfies the node balance condition. The following statements hold.*
   (a) *The grading $g|_{\mathrm{NE}(X)}$ is connected.*
   (b) *We have $\mathrm{Surp}(X, g|_{\mathrm{NE}(X)}) \geq 0$.*
   (c) *We have*
$$\mathrm{eBal}_{U,g}(v) - \mathrm{bBal}_{U,g}(r-1) = \mathrm{Surp}(X, g|_{\mathrm{NE}(X)}) + 1. \tag{2.7}$$

Proof. Proof of Claim 1(a). Let us locate the h-precursors inside the node. Definition 2.6(2) gives $P_{U,g}(U_s) = U_r$. For $U_k \in N(X)$, Claim 2 of Lemma 2.13 gives $r \leq p_k < k \leq s$, where $P_{U,g}(U_k) = U_{p_k}$. Thus, $P_{U,g}(U_k) \in E(X)$.

Proof of Claim 1(b). Let us prove containment of the smaller nodes. If $y = x$, then $\mathrm{Node}_{U,g}(y) = X$. For $y = U_i \in E(X) \setminus \{x\}$ with $g(y) > 0$, we have $r < i < s$. Claim 3 of Lemma 2.13 gives $q_i = \max F_{U_i} < s$ and $\mathrm{Node}_{U,g}(y) = U[i-1, q_i] \subseteq X$.

Proof of Claim 1(c). Let us compare the node endpoints. For $y = U_i \in E(X) \setminus \{x\}$ with $g(y) > 0$, Claim 3 of Lemma 2.13 gives $q_i < s$. Thus, the endpoint U_{q_i} precedes U_s.

Proof of Claim 2. Let us compare $\mathrm{Node}_{U,g}(y)$ and $\mathrm{Node}_{U,g}(z)$. Interchange y and z if necessary so that y precedes z. Let t be the endpoint of $\mathrm{Node}_{U,g}(y)$. If t precedes z, then $\mathrm{Node}_{U,g}(y)$ and $\mathrm{Node}_{U,g}(z)$ are disjoint. Otherwise, z lies in that node and is not its start. Claims 1(b) and 1(c) give $\mathrm{Node}_{U,g}(z) \subsetneq \mathrm{Node}_{U,g}(y)$.

Proof of Claim 3(a). Let us bound every boundary balance by the initial node balance. Let $B_j = \mathrm{bBal}_{U,g}(j)$. We prove $B_i \geq B_r$ by strong induction on $i \in [r,s]_{\mathbb{Z}}$. The case $i = r$ is equality. For $i > r$, assume $B_j \geq B_r$ for every $r \leq j < i$. If U_i is horizontal, then $B_i = B_{i-1} + 1 \geq B_r + 1$. If U_i is vertical, Claim 1(a) gives $P_{U,g}(U_i) = U_{p_i}$ for $r \leq p_i < i$.

Definition 2.6(1)(b) implies $g(U_{p_i}) > 0$. Thus, Claim 1(b) supplies its node. The node balance condition and the induction hypothesis at p_i give

$$B_i = \mathrm{eBal}_{U,g}(U_i) \geq \mathrm{eBal}_{U,g}(U_{p_i}) = B_{p_i} \geq B_r.$$

Since $B_r = \mathrm{eBal}_{U,g}(x)$, this proves Claim 3(a).

Proof of Claim 3(b). Let us prove the strict balance inequality for descendants. Write $y = U_i$. Proper containment gives $r < i \leq s$. Claim 3(a) at $i-1$ gives

$$\mathrm{eBal}_{U,g}(y) = \mathrm{bBal}_{U,g}(i-1) + 1 \geq \mathrm{eBal}_{U,g}(x) + 1 > \mathrm{eBal}_{U,g}(x).$$

Proof of Claim 4(a). Let us prove connectedness of the restricted grading. Let $g_X = g|_{\mathrm{NE}(X)}$. Claim 1(a) gives $P_{U,g}(N(X)) \subseteq E(X)$. Apply Lemma 2.14 with $a = r-1$ and $b = s$. Claim 1 of Lemma 2.14 gives the h-precursor P_{X,g_X}. Claim 2 of Lemma 2.14 gives $\mathrm{Node}_{X,g_X}(x) = X$. Claim 3 of Lemma 2.14 gives the node balance condition for g_X. Thus, g_X is connected by Definition 2.9(1).

Proof of Claim 4(b). Claim 4(a) and Lemma 2.11 give $\mathrm{Surp}(X, g|_{\mathrm{NE}(X)}) \geq 0$.

Proof of Claim 4(c). Let us compute the surplus from the endpoint balances. Claim 3 of Lemma 2.14 and (2.4) give

$$\mathrm{eBal}_{U,g}(v) - \mathrm{bBal}_{U,g}(r-1) = \#E(X) - \sum_{w \in N(X)} g(w) = \mathrm{Surp}(X, g|_{\mathrm{NE}(X)}) + 1.$$

This proves (2.7). □

Definition 2.16. *Let U be a tNE path and $g \in G(U)$. Let X and Y be nodes of (U, g).*

1. *We call X the parent of Y and Y a child of X if $Y \subsetneq X$ and there is no node Z such that $Y \subsetneq Z \subsetneq X$.*
2. *We call X an ancestor of Y and Y a descendant of X if $Y \subsetneq X$.*

Lemma 2.17. *Let U be a tNE path and $g \in G(U)$. Each node of (U, g) has at most one parent.*

Proof. Let us prove that two parents of the same node are equal. Suppose X and Z are parents of a node Y. Both contain Y, so they are not disjoint. Claim 2 of Lemma 2.15 gives $X \subseteq Z$ or $Z \subseteq X$. If $X \neq Z$, interchanging their names if necessary gives

$$Y \subsetneq X \subsetneq Z.$$

This contradicts the condition that Z is a parent of Y in Definition 2.16(1). Hence, $X = Z$. □

2.4 Compatibility and counting colored gradings

Definition 2.18. *Let U be a nonempty tNE path. Let u and v be edges of U.*

1. *Let $U\langle u, v\rangle$ denote the cyclic subpath of U read from u in traversal order, returning from $\mathrm{End}_{\mathrm{NE}}(U)$ to $\mathrm{Start}_{\mathrm{NE}}(U)$, and ending at the first occurrence of v. We include both end edges.*
2. *For $P = U\langle u, v\rangle$, let $\mathrm{NE}(P) = \mathrm{NE}(U, u, v)$, $E(P) = E(U, u, v)$, and $N(P) = N(U, u, v)$ denote the sets of all, horizontal, and vertical edges of U traversed by P, respectively.*

3. *We define the NE-path realization* $\mathrm{NEReal}(U,u,v)$ *of* $U\langle u,v\rangle$ *to be the NE path from* $(0,0)$ *with the same ordered sequence of east and north steps as* $U\langle u,v\rangle$*. Each edge of* $\mathrm{NEReal}(U,u,v)$ *keeps its name and color, if any. Let* $X = \mathrm{NEReal}(U,u,v)$*. For* $e \in \mathrm{NE}(U,u,v)$*, let* $\widetilde{e} \in \mathrm{NE}(X)$ *be its realized edge. For* $g \in G(U)$*, we define the transported grading* $g_{U,u,v}$ *of* X *by*

$$g_{U,u,v}(\widetilde{e}) = g(e) \qquad (e \in \mathrm{NE}(U,u,v)).$$

4. *Let* x *and* y *be edges of* U*. We call* $U\langle u,v\rangle$ *a subpath of* $U\langle x,y\rangle$ *and let*

$$U\langle u,v\rangle \subseteq U\langle x,y\rangle$$

if the edge sequence of $U\langle u,v\rangle$ *occurs consecutively in the edge sequence of* $U\langle x,y\rangle$*.*

5. *Let* x *and* y *be edges of* U*. We call* $U\langle u,v\rangle$ *and* $U\langle x,y\rangle$ *disjoint if*

$$\mathrm{NE}(U,u,v) \cap \mathrm{NE}(U,x,y) = \varnothing.$$

We define the notions of compatibility and shadows [1] for tNE paths and fix some notation for later use.

Definition 2.19. *Let* U *be a tNE path and* $g \in G(U)$*.*

1. *For edges* u *and* t *of* U*, we define*

$$\mathrm{NGrDiff}_{U,g}(u,t) = \#N(U,u,t) - \sum_{x \in E(U,u,t)} g(x),$$
$$\mathrm{EGrDiff}_{U,g}(u,t) = \#E(U,u,t) - \sum_{x \in N(U,u,t)} g(x).$$

We call $\mathrm{NGrDiff}_{U,g}(u,t) = 0$ *the NGrDiff-zero condition. We call* $\mathrm{EGrDiff}_{U,g}(u,t) = 0$ *the EGrDiff-zero condition.*

2. *We call* g *compatible if for every horizontal edge* u *and vertical edge* v *of* U *with* $g(u)g(v) > 0$*, there is an edge* $t \in \mathrm{NE}(U,u,v)$ *satisfying one of the following conditions.*

   (a) t *is vertical,* $t \neq v$*, and* $\mathrm{NGrDiff}_{U,g}(u,t) = 0$*.*

   (b) t *is horizontal,* $t \neq u$*, and* $\mathrm{EGrDiff}_{U,g}(t,v) = 0$*.*

   We call such an edge t *a witness for* (u,v)*.*

3. *Let* u *be a horizontal edge of* U*. If there is a vertical edge* t *of* U *such that*

$$\mathrm{NGrDiff}_{U,g}(u,t) = 0,$$

let $t_{U,g}(u)$ *be the first such edge in cyclic order starting at* u*. Define the horizontal shadow by*

$$\mathrm{Sh}_{U,g}(u) = N(U,u,t_{U,g}(u)).$$

If no vertical edge satisfies the NGrDiff-zero condition, let $\mathrm{Sh}_{U,g}(u) = N(U)$*.*

4. *Let* v *be a vertical edge of* U*. If there is a horizontal edge* t *of* U *such that*

$$\mathrm{EGrDiff}_{U,g}(t,v) = 0,$$

let $s_{U,g}(v)$ *be the first such edge in reverse cyclic order starting at* v*. We define the vertical shadow* $\mathrm{Sh}_{U,g}(v)$ *of* v *by*

$$\mathrm{Sh}_{U,g}(v) = E(U, s_{U,g}(v), v).$$

If no horizontal edge satisfies the EGrDiff-zero condition, let $\mathrm{Sh}_{U,g}(v) = E(U)$. *For* $C \subseteq \mathrm{NE}(U)$, *we define*

$$\mathrm{Sh}_{U,g}(C) = \bigcup_{e \in C} \mathrm{Sh}_{U,g}(e).$$

5. *We define the total vertical grade and total horizontal grade of* g *by*

$$\mathrm{TNG}_{U,g} = \sum_{v \in N(U)} g(v), \qquad \mathrm{TEG}_{U,g} = \sum_{u \in E(U)} g(u),$$

respectively.

6. *We define the sets of positively graded horizontal and vertical edges by*

$$E_g(U) = \{u \in E(U) : g(u) > 0\}, \qquad N_g(U) = \{v \in N(U) : g(v) > 0\}.$$

7. *Let* $\mathcal{P}_{U,g}$ *be the set of cyclic subpaths* $U\langle u, t_{U,g}(u)\rangle$ *for all* $u \in E_g(U)$ *with* $t_{U,g}(u)$ *existing. Let* $P \in \mathcal{P}_{U,g}$. *We call* P *a maximal h-shadow subpath of* g *if there is no* $Q \in \mathcal{P}_{U,g}$ *such that* $P \subsetneq Q$.

We use the definitions in [1] for maximal Dyck paths, the following function π_{d_1,d_2}, compatible gradings with fixed totals, and tightness.

Definition 2.20. *Let* d_1 *and* d_2 *be positive integers.*

1. *The maximal Dyck path* $\mathcal{D} = \mathcal{D}_{(d_1,d_2)}$ *is the highest NE path from* $(0,0)$ *to* (d_1,d_2) *that stays on or below the line segment from* $(0,0)$ *to* (d_1,d_2).

2. *For* $e \in \mathrm{NE}(\mathcal{D})$, *let* (x,y) *be its left endpoint if* $e \in E(\mathcal{D})$ *and its lower endpoint if* $e \in N(\mathcal{D})$. *We define* $\pi_{d_1,d_2} \colon \mathrm{NE}(\mathcal{D}) \to [0, d_1+d_2-1]_{\mathbb{Z}}$ *by*

$$\pi_{d_1,d_2}(e) = xd_2 - yd_1.$$

3. *Let* p *and* q *be nonnegative integers. Let* $\mathrm{CG}(d_1,d_2,p,q)$ *denote the set of compatible gradings* $g \in G(\mathcal{D})$ *satisfying*

$$\mathrm{TNG}_{\mathcal{D},g} = p, \qquad \mathrm{TEG}_{\mathcal{D},g} = q.$$

4. *Let* $g \in \mathrm{CG}(d_1,d_2,p,q)$. *We call* g *tight if there exists* $\varepsilon \in \{-1,1\}$ *satisfying*

$$pd_2 - qd_1 = \varepsilon \gcd(p,q)$$

and the corresponding support condition below [1, Definition 4.8 and Remark 4.9].

   (a) *If* $\varepsilon = 1$, *then* $E_g(\mathcal{D}) \subseteq \mathrm{Sh}_{\mathcal{D},g}(N_g(\mathcal{D}))$.

   (b) *If* $\varepsilon = -1$, *then* $N_g(\mathcal{D}) \subseteq \mathrm{Sh}_{\mathcal{D},g}(E_g(\mathcal{D}))$.

   We define

$$\mathrm{TCG}(d_1,d_2,p,q) = \{g \in \mathrm{CG}(d_1,d_2,p,q) : g \text{ is tight}\}.$$

Lemma 2.21 (Horizontal defect as a sum of debts)**.** *Let* $U = U_1 + \cdots + U_n$ *be a tNE path. Let* $h \in G(U)$. *Suppose that* $P = P_{U,h}$ *exists. Let* $X = \mathrm{Node}_{U,h}(U_r) = U[r-1,s]$. *Use the counts* $\mathrm{Num}(y,k)$ *from Lemma 2.13 for the grading* h. *For* $k \in [r,s]_{\mathbb{Z}}$, *put*

$$H_{r,k} = E(U[r-1,k]), \qquad \Delta_r(k) = \mathrm{NGrDiff}_{U,h}(U_r, U_k).$$

Then, we have the following.

1. *For $k \in [r,s]_{\mathbb{Z}}$, we have*

$$\Delta_r(k) = -\sum_{y \in H_{r,k}} (h(y) - \mathrm{Num}(y,k)) = -\sum_{y \in H_{r,k}} \mathrm{Debt}_{U,h}(y,k).$$

2. *We have*

$$\Delta_r(k) \le -1 \quad (r \le k < s), \qquad \Delta_r(s) = 0.$$

In particular, $t_{U,h}(U_r) = U_s$.

Proof. Proof of Claim 1. Let us count the vertical edges by their h-precursors. For $U_j \in N(U[r-1,k])$, Claim 1(a) of Lemma 2.15 places $P(U_j)$ in $E(X)$. Since $P(U_j)$ precedes U_j, it belongs to $H_{r,k}$. For $y = U_i \in H_{r,k}$ and $t \in F_y$, Definition 2.6(1)(a) gives $t > i \ge r$. Consequently,

$$\#N(U[r-1,k]) = \sum_{y \in H_{r,k}} \mathrm{Num}(y,k).$$

The subpath $U[r-1,k]$ does not cross from U_n to U_1. Subtracting $\sum_{y \in H_{r,k}} h(y)$ proves Claim 1.

Proof of Claim 2. First, let us prove $\Delta_r(k) < 0$ for $k \in [r, s-1]_{\mathbb{Z}}$. Fix such an index k. Claim 1 of Lemma 2.13 gives

$$h(y) - \mathrm{Num}(y,k) \ge 0 \qquad (y \in H_{r,k}).$$

Claim 2 of Lemma 2.13, applied to $P(U_s) = U_r$, gives $h(U_r) - \mathrm{Num}(U_r,k) \ge 1$. Since $U_r \in H_{r,k}$, Claim 1 gives

$$\Delta_r(k) = -\sum_{y \in H_{r,k}} (h(y) - \mathrm{Num}(y,k)) \le -1.$$

Second, let us prove $\Delta_r(s) = 0$. For $y \in H_{r,s}$ with $h(y) > 0$, Claim 1(b) of Lemma 2.15 gives

$$\mathrm{Node}_{U,h}(y) \subseteq X, \qquad P^{-1}(\{y\}) \subseteq N(X).$$

The node definition gives $\#P^{-1}(\{y\}) = h(y)$, so $\mathrm{Num}(y,s) = h(y)$. For $y \in H_{r,s}$ with $h(y) = 0$, Claim 1 of Lemma 2.13 gives $\mathrm{Num}(y,s) = 0$. Thus, Claim 1 gives

$$\Delta_r(s) = -\sum_{y \in H_{r,s}} (h(y) - \mathrm{Num}(y,s)) = 0.$$

Let us identify the first vertical zero of NGrDiff from U_r. The indices $r, \dots, s$ occur in this order before the cyclic path returns to U_1. For every $j \in [r,s]_{\mathbb{Z}}$ with $U_j \in N(U)$, Claim 1 gives

$$\mathrm{NGrDiff}_{U,h}(U_r, U_j) = \Delta_r(j).$$

The strict inequalities for $j < s$ and equality at $j = s$ prove the claim. □

Lemma 2.22 (An explicit witness and a first failing pair)**.** *Let $U = U_1 + \cdots + U_n$ be a tNE path. Let $h \in G(U)$. Suppose that $P = P_{U,h}$ exists. Suppose that every horizontal edge of positive grade has a node. For each $U_k \in N(U)$, let p_k be the unique index satisfying $P(U_k) = U_{p_k}$. Put $B_i = \mathrm{bBal}_{U,h}(i)$ for $i \in [0,n]_{\mathbb{Z}}$. The following statements hold.*

1. *For $i \in [n]_{\mathbb{Z}}$, we have*

$$B_i - B_{i-1} = \begin{cases} 1, & U_i \in E(U), \\ -h(U_i), & U_i \in N(U). \end{cases}$$

2. *Let $U_r \in E(U)$, $U_s \in N(U)$, and $r < s$. Suppose that $h(U_r)h(U_s) > 0$ and $B_s \geq B_r$. The set*

$$J_{r,s} = \{i \in [r+1, s-1]_{\mathbb{Z}} : B_i > B_s\}$$

is nonempty. Its minimum $j_* = \min J_{r,s}$ satisfies

$$U_{j_*} \in E(U), \qquad B_{j_*-1} = B_s, \qquad B_{j_*} = B_s + 1, \qquad \mathrm{EGrDiff}_{U,h}(U_{j_*}, U_s) = 0.$$

Thus, U_{j_*} is a compatibility witness for (U_r, U_s).

3. *Define*

$$S = \{k \in [n]_{\mathbb{Z}} : U_k \in N(U),\ B_k < B_{p_k}\}.$$

If $S \neq \emptyset$, put $s = \min S$, $r = p_s$, and $z = B_s$. Then,

$$h(U_r) > 0, \qquad h(U_s) > 0, \qquad r < s, \qquad B_i \geq B_r > z \quad (r \leq i < s).$$

For $r < j < s$, we have

$$\begin{cases} \mathrm{EGrDiff}_{U,h}(U_j, U_s) = z - B_{j-1} < 0, & U_j \in E(U), \\ \mathrm{NGrDiff}_{U,h}(U_r, U_j) < 0, & U_j \in N(U). \end{cases}$$

The pair (U_r, U_s) has positive grades and no compatibility witness. Its cyclic interval is the ordinary interval from U_r to U_s.

Proof. Proof of Claim 1. Let us compute the balance change at each edge. By Definition 2.4(3)(a), a horizontal edge increases $\mathrm{bBal}_{U,h}$ by 1. A vertical edge U_i decreases it by $h(U_i)$.

Proof of Claim 2. Let us determine $j_* = \min J_{r,s}$. Since $h(U_s) > 0$, we have $B_{s-1} = B_s + h(U_s) > B_s \geq B_r$. Thus, $s - 1 > r$ and $s - 1 \in J_{r,s}$. Minimality and $B_r \leq B_s$ give

$$B_{j_*-1} \leq B_s < B_{j_*}.$$

For $U_i \in N(U)$, Claim 1 gives $B_i - B_{i-1} = -h(U_i) \leq 0$. Hence, U_{j_*} is horizontal and $B_{j_*} = B_{j_*-1} + 1$. Integrality gives $B_{j_*-1} = B_s$. By Definition 2.19(1) and Claim 1, summing $B_i - B_{i-1}$ for $i \in [j_*, s]_{\mathbb{Z}}$ gives

$$\mathrm{EGrDiff}_{U,h}(U_{j_*}, U_s) = B_s - B_{j_*-1} = 0.$$

The bounds $r < j_* < s$ verify the witness location and $U_{j_*} \neq U_r$.

Proof of Claim 3. Let us identify the node of U_r. Definition 2.6(1)(a) and (b) gives

$$r < s, \qquad 0 \leq \mathrm{Num}(U_r, s-1) < h(U_r).$$

Here $\mathrm{Num}(U_r, s-1)$ is the count in Lemma 2.13. Thus, $h(U_r) > 0$. Let $t \in [r, n]_{\mathbb{Z}}$ satisfy

$$\mathrm{Node}_{U,h}(U_r) = U[r-1, t].$$

Since $P(U_s) = U_r$, we have $s \leq t$.

Let us prove $B_i \geq B_r$ for $i \in [r, s-1]_{\mathbb{Z}}$ by strong induction on i. For $i = r$, the inequality is equality. Fix $i \in [r+1, s-1]_{\mathbb{Z}}$ and assume

$$B_j \geq B_r \qquad (j \in [r, i-1]_{\mathbb{Z}}).$$

First, let us prove the induction step when $U_i \in E(U)$. Claim 1 and the induction hypothesis give

$$B_i = B_{i-1} + 1 \geq B_r + 1 > B_r.$$

Second, let us prove the induction step when $U_i \in N(U)$. Recall that $P(U_i) = U_{p_i}$. Claim 1(a) of Lemma 2.15 gives $r \le p_i < i$. Since $i < s = \min S$, we have $i \notin S$. Therefore,

$$B_i \ge B_{p_i} \ge B_r.$$

The inequality $B_{p_i} \ge B_r$ follows from the induction hypothesis, since $r \le p_i < i$.

Let us prove $h(U_s) > 0$. The definition of S gives $B_r > B_s = z$. Claim 1 and the bound at $i = s - 1$ give

$$h(U_s) = B_{s-1} - B_s \ge B_r - z > 0.$$

First, let us exclude a horizontal interior witness. Let $j \in [r+1, s-1]_{\mathbb{Z}}$ satisfy $U_j \in E(U)$. Definition 2.19(1) and Claim 1 give

$$\mathrm{EGrDiff}_{U,h}(U_j, U_s) = B_s - B_{j-1} = z - B_{j-1} \le z - B_r < 0.$$

Second, let us exclude a vertical interior witness. Let $j \in [r+1, s-1]_{\mathbb{Z}}$ satisfy $U_j \in N(U)$. Since $r < j < s \le t$, Claim 2 of Lemma 2.21, applied to the node $U[r-1, t]$, gives

$$\mathrm{NGrDiff}_{U,h}(U_r, U_j) < 0.$$

Let us exclude the endpoint witnesses. Definition 2.19(2) requires an EGrDiff witness different from U_r and an NGrDiff witness different from U_s. Thus, (U_r, U_s) has no compatibility witness. □

Lemma 2.23 (Witnesses under cyclic realization and zero extension)**.** *Let $D = D_1 \mathbin{+\!\!+} \cdots \mathbin{+\!\!+} D_N$ be a nonempty tNE path and let $q \in [N]_{\mathbb{Z}}$ and $\ell \in [N]_{\mathbb{Z}}$. For an integer a, let $a \bmod N$ denote its remainder in $[0, N-1]_{\mathbb{Z}}$. Define*

$$\eta(j) = 1 + ((q + j - 2) \bmod N), \qquad \theta(k) = 1 + ((k - q) \bmod N) \quad (j, k \in [N]_{\mathbb{Z}}).$$

Put

$$U = \mathrm{NEReal}(D, D_q, D_{\eta(\ell)}), \qquad \iota(U_j) = D_{\eta(j)} \quad (j \in [\ell]_{\mathbb{Z}}).$$

Let $h \in G(U)$. Suppose that $P_{U,h}$ exists and that every horizontal edge of positive grade has a node. Let $g \in G(D)$ satisfy $g(D_{\eta(j)}) = h(U_j)$ for $j \in [\ell]_{\mathbb{Z}}$. Then, we have the following.

1. *The following statements hold.*

   (a) *The maps η and θ are inverse permutations of $[N]_{\mathbb{Z}}$. For $j \in [N]_{\mathbb{Z}}$ and $k \in [N]_{\mathbb{Z}}$, we have*

   $$\theta(\eta(j)) = j, \qquad \eta(\theta(k)) = k.$$

   (b) *We have $|U| = \ell$. For $1 \le i \le j \le \ell$, the map $\iota(U_k) = D_{\eta(k)}$ for $k \in [i, j]_{\mathbb{Z}}$ identifies $U[i-1, j]$ with $D\langle D_{\eta(i)}, D_{\eta(j)}\rangle$ in edge order. For $k \in [\ell]_{\mathbb{Z}}$, we have*

   $$U_k \in E(U) \iff D_{\eta(k)} \in E(D), \qquad U_k \in N(U) \iff D_{\eta(k)} \in N(D),$$
   $$h(U_k) = g(D_{\eta(k)}).$$

   (c) *For $1 \le i \le j \le \ell$, we have*

   $$\mathrm{NGrDiff}_{D,g}(D_{\eta(i)}, D_{\eta(j)}) = \mathrm{NGrDiff}_{U,h}(U_i, U_j),$$
   $$\mathrm{EGrDiff}_{D,g}(D_{\eta(i)}, D_{\eta(j)}) = \mathrm{EGrDiff}_{U,h}(U_i, U_j).$$

   (d) *Suppose that $U_i \in E(U)$, $U_j \in N(U)$, $i < j$, and $h(U_i)h(U_j) > 0$. For every $k \in [i, j]_{\mathbb{Z}}$, the edge U_k is a witness for (U_i, U_j) if and only if $D_{\eta(k)}$ is a witness for $(D_{\eta(i)}, D_{\eta(j)})$. Thus, ι restricts to a bijection between these two witness sets.*

2. *Suppose that $U_i \in E(U)$, $U_j \in N(U)$, $j < i$, and $h(U_i)h(U_j) > 0$. Define*

$$F_i = \{k \in [\ell]_{\mathbb{Z}} : U_k \in N(U),\ P_{U,h}(U_k) = U_i\}, \qquad t_i = \max F_i.$$

Then, U_{t_i} is a witness for (U_i, U_j), and $D_{\eta(t_i)}$ is a witness for $(D_{\eta(i)}, D_{\eta(j)})$.

3. *If g is compatible, then h is compatible. Define*

$$g_0(D_k) = \begin{cases} h(U_{\theta(k)}), & \theta(k) \le \ell, \\ 0, & \theta(k) > \ell. \end{cases}$$

Then, g_0 is compatible if and only if h is compatible.

Proof. Proof of Claim 1(a). Let us check that the two index maps are inverses. Let $j \in [N]_{\mathbb{Z}}$ and $k \in [N]_{\mathbb{Z}}$. The definitions of η and θ give

$$\theta(\eta(j)) = 1 + ((j-1) \bmod N) = j,$$
$$\eta(\theta(k)) = 1 + ((k-1) \bmod N) = k,$$

because $0 \le j - 1 < N$ and $0 \le k - 1 < N$. Hence, $\theta = \eta^{-1}$ on $[N]_{\mathbb{Z}}$.

Proof of Claim 1(b). Let us identify the edges of U. We have $\eta(1) = q$ and

$$\eta(k+1) = \begin{cases} \eta(k) + 1, & \eta(k) < N, \\ 1, & \eta(k) = N \end{cases} \qquad (k \in [N-1]_{\mathbb{Z}}).$$

Claim 1(a) makes $\eta(1), \dots, \eta(\ell)$ pairwise distinct. Thus, the cyclic traversal from D_q first reaches $D_{\eta(\ell)}$ after exactly ℓ edges, in the order $D_{\eta(1)}, \dots, D_{\eta(\ell)}$. Definition 2.18 gives $|U| = \ell$ and preserves the direction of every edge under ι. For $1 \le i \le j \le \ell$, the sequence $D_{\eta(i)}, \dots, D_{\eta(j)}$ has no repeated edge and ends at its first occurrence of $D_{\eta(j)}$. Hence, it is precisely $D\langle D_{\eta(i)}, D_{\eta(j)}\rangle$. The hypothesis $g(D_{\eta(k)}) = h(U_k)$ supplies grade preservation for each $k \in [\ell]_{\mathbb{Z}}$.

Proof of Claim 1(c). Let us compare the grading differences under the cyclic index map. Let $1 \le i \le j \le \ell$. By Claim 1(b) and Definition 2.19(1), we have

$$\begin{aligned} \mathrm{NGrDiff}_{D,g}(D_{\eta(i)}, D_{\eta(j)}) &= \#\{k \in [i,j]_{\mathbb{Z}} : U_k \in N(U)\} - \sum_{\substack{k \in [i,j]_{\mathbb{Z}} \\ U_k \in E(U)}} h(U_k) \\ &= \mathrm{NGrDiff}_{U,h}(U_i, U_j), \\ \mathrm{EGrDiff}_{D,g}(D_{\eta(i)}, D_{\eta(j)}) &= \#\{k \in [i,j]_{\mathbb{Z}} : U_k \in E(U)\} - \sum_{\substack{k \in [i,j]_{\mathbb{Z}} \\ U_k \in N(U)}} h(U_k) \\ &= \mathrm{EGrDiff}_{U,h}(U_i, U_j). \end{aligned}$$

Proof of Claim 1(d). Let us transport the compatibility witnesses in both directions. Let $k \in [i,j]_{\mathbb{Z}}$. By Definition 2.19(2), the edge U_k is a witness for (U_i, U_j) precisely when one of the following holds:

$$U_k \in N(U), \quad k \ne j, \quad \mathrm{NGrDiff}_{U,h}(U_i, U_k) = 0;$$
$$U_k \in E(U), \quad k \ne i, \quad \mathrm{EGrDiff}_{U,h}(U_k, U_j) = 0.$$

Claim 1(a) gives $k \ne j \iff \eta(k) \ne \eta(j)$ and $k \ne i \iff \eta(k) \ne \eta(i)$. Claims 1(b) and 1(c) therefore identify these two conditions with the two witness conditions for $D_{\eta(k)}$ and $(D_{\eta(i)}, D_{\eta(j)})$. Claim 1(b) includes every edge of $D\langle D_{\eta(i)}, D_{\eta(j)}\rangle$. The inverse on witnesses is $D_{\eta(k)} \mapsto U_{\theta(\eta(k))} = U_k$.

Proof of Claim 2. Let us find a witness when the edge order is reversed. The node at U_i exists. Thus, F_i is nonempty and its maximum t_i satisfies $i < t_i \le \ell$. Claim 2 of Lemma 2.21 gives $\mathrm{NGrDiff}_{U,h}(U_i, U_{t_i}) = 0$. Claim 1(c) gives $\mathrm{NGrDiff}_{D,g}(D_{\eta(i)}, D_{\eta(t_i)}) = 0$.

Let us verify that the two vertical edges lie strictly between the two edges of each pair. In cyclic order on U, the distance from index i to $j < i$ is $\ell - i + j$. We have

$$0 < t_i - i \le \ell - i < \ell - i + j.$$

In cyclic order on D, the distance from $\eta(i)$ to $\eta(j)$ is $N - i + j$. We have

$$t_i - i \le \ell - i \le N - i < N - i + j.$$

The edges U_{t_i} and $D_{\eta(t_i)}$ are vertical. The inequalities for $t_i - i$ place them strictly before U_j and $D_{\eta(j)}$ in the respective cyclic intervals. The two NGrDiff identities and Definition 2.19(2) therefore make them witnesses for (U_i, U_j) and $(D_{\eta(i)}, D_{\eta(j)})$, respectively.

Proof of Claim 3. Let us prove compatibility in both directions. Suppose first that g is compatible. Let $U_i \in E(U)$ and $U_j \in N(U)$ satisfy $h(U_i)h(U_j) > 0$. Their images have positive g-grades. If $i < j$, Claim 1(d) transports a witness from D to U. If $j < i$, Claim 2 supplies a witness in U. The directions exclude $i = j$. Thus, h is compatible. Claim 1(a) gives $g_0(D_{\eta(j)}) = h(U_j)$ for $j \le \ell$. The implication g compatible $\Longrightarrow$ h compatible therefore applies to $g = g_0$.

Let us prove compatibility of the zero extension. Conversely, let h be compatible and let (D_a, D_b) be a positive horizontal–vertical pair for g_0. Positivity gives the explicit indices $i = \theta(a) \le \ell$ and $j = \theta(b) \le \ell$, and $h(U_i)h(U_j) > 0$. If $i < j$, Claim 1(d) transports a witness from U to D. If $j < i$, Claim 2 supplies an ambient witness. Thus, g_0 is compatible. The argument includes $\ell = N$. □

Lemma 2.24 (Nodes and horizontal shadows)**.** *Let $\mathcal{D}$ be a maximal Dyck path and $g \in G(\mathcal{D})$. Consider a maximal h-shadow subpath $\mathcal{D}\langle u_0, v_0\rangle$. Let $U = \mathrm{NEReal}(\mathcal{D}, u_0, v_0)$ and $h = g_{\mathcal{D},u_0,v_0}$. Then, we have the following.*

1. *The following statements hold.*
   (a) *The map $P_{U,h}$ exists.*
   (b) *For each $x \in E(U)$ with $h(x) > 0$, the node $\mathrm{Node}_{U,h}(x)$ exists. The edge $\mathrm{End}_{\mathrm{NE}}(\mathrm{Node}_{U,h}(x))$ corresponds to $t_{\mathcal{D},g}(x)$.*
   (c) *We have $\mathrm{Node}_{U,h}(u_0) = U$.*
2. *The grading h of U is compatible if and only if h satisfies the node balance condition.*
3. *The following statements hold.*
   (a) *Define $g_0 \in G(\mathcal{D})$ by*

$$g_0(e) = \begin{cases} g(e), & e \in \mathrm{NE}(\mathcal{D}, u_0, v_0), \\ 0, & e \in \mathrm{NE}(\mathcal{D}) \setminus \mathrm{NE}(\mathcal{D}, u_0, v_0). \end{cases}$$

   Then, g_0 is compatible if and only if h is connected.
   (b) *If g is compatible, then h is connected.*

Proof. Proof of Claim 1(a). Let us prove that $P_{U,h}$ exists. Write $n = |U|$, $U_1 = u_0$, $U_n = v_0$, and

$$A_k = \#N(U[0,k]) - \sum_{y \in E(U[0,k])} h(y) \quad (k \in [0,n]_{\mathbb{Z}}).$$

The realization preserves the ordered steps and grades from u_0 through v_0. Definition 2.19(3) gives $A_n = 0$ and $A_1 = -h(u_0) < 0$. A horizontal step decreases A_k by a nonnegative integer; a vertical step increases it by one. If $A_k \geq 0$ for some $k \in [n-1]_{\mathbb{Z}}$, let k be the least such index. Since $A_1 < 0$, we have $k \geq 2$. The edge U_k is vertical and $A_k = A_{k-1} + 1 = 0$, contrary to the definition of v_0. Thus,

$$A_k < 0 \quad (1 \leq k < n), \qquad A_n = 0. \tag{2.8}$$

Since $A_0 = 0$, Claim 2 of Lemma 2.7 gives $P_{U,h}$.

Proof of Claim 1(b). Let us prove saturation of every fiber. Use the counts $\mathrm{Num}(y,k)$ from Lemma 2.13. Claim 1 of Lemma 2.13 gives $\mathrm{Num}(y,n) \leq h(y)$. Each vertical edge has one h-precursor, so (2.8) gives

$$\sum_{y \in E(U)} (h(y) - \mathrm{Num}(y,n)) = \sum_{y \in E(U)} h(y) - \#N(U) = 0.$$

Every summand is nonnegative. Hence, $\mathrm{Num}(y,n) = h(y)$ for every $y \in E(U)$. For $y = U_r \in E(U)$ with $h(y) > 0$, the fiber $P_{U,h}^{-1}(\{y\})$ is nonempty and saturated. Thus, its node exists; write $\mathrm{Node}_{U,h}(y) = U[r-1,s]$. Claim 2 of Lemma 2.21 gives

$$\mathrm{NGrDiff}_{U,h}(U_r, U_j) < 0 \quad (r \leq j < s), \qquad \mathrm{NGrDiff}_{U,h}(U_r, U_s) = 0.$$

Claims 1(b) and 1(c) of Lemma 2.23 transport $\mathrm{NGrDiff}_{U,h}(U_r, U_j) < 0$ for $r \leq j < s$ and $\mathrm{NGrDiff}_{U,h}(U_r, U_s) = 0$ to $\mathcal{D}$. Thus, U_s corresponds to $t_{\mathcal{D},g}(y)$.

Proof of Claim 1(c). Let us identify the node at the first edge. Apply Claim 1(b) to u_0. Its first zero is v_0. Thus, $\mathrm{Node}_{U,h}(u_0) = U$.

Proof of Claim 2. Let us prove that node balance implies compatibility. Suppose that h satisfies the node balance condition. Let $U_r \in E(U)$ and $U_s \in N(U)$ satisfy $h(U_r)h(U_s) > 0$. Let t be the ending index of the node

$$\mathrm{Node}_{U,h}(U_r) = U[r-1,t].$$

First, let us find a witness when $U_s \notin \mathrm{NE}(U[r-1,t])$. Then, $s < r$ or $s > t$. In either case, U_t occurs before U_s in cyclic order from U_r, and $U_t \neq U_s$. Claim 2 of Lemma 2.21 gives

$$\mathrm{NGrDiff}_{U,h}(U_r, U_t) = 0.$$

Thus, U_t is a compatibility witness.

Second, let us find a witness when $U_s \in \mathrm{NE}(U[r-1,t])$. Since U_r is horizontal and U_s is vertical, we have $r < s \leq t$. Claim 3(a) of Lemma 2.15 gives

$$\mathrm{bBal}_{U,h}(s) \geq \mathrm{bBal}_{U,h}(r).$$

Claim 2 of Lemma 2.22 therefore gives an EGrDiff witness for (U_r, U_s).

Let us prove that compatibility implies node balance. Suppose that the node balance condition fails. For $U_k \in N(U)$, let p_k satisfy $P_{U,h}(U_k) = U_{p_k}$. Then, the set

$$S = \{k \in [|U|]_{\mathbb{Z}} : U_k \in N(U),\ \mathrm{bBal}_{U,h}(k) < \mathrm{bBal}_{U,h}(p_k)\}$$

is nonempty. Claim 1(b) gives a node for every horizontal edge of positive grade. Claim 3 of Lemma 2.22, applied to $s = \min S$ and $r = p_s$, gives

$$h(U_r)h(U_s) > 0$$

and no compatibility witness for (U_r, U_s). Thus, h is not compatible.

Proof of Claim 3(a). Let us check compatibility of the zero extension. Claim 3 of Lemma 2.23 gives compatibility of g_0 if and only if compatibility of h. Claim 2 identifies compatibility of h with the node balance condition for h. Claims 1(a) and 1(c) supply the h-precursor $P_{U,h}$ and the identity $\mathrm{Node}_{U,h}(U_1) = U$.

Proof of Claim 3(b). Let us prove connectedness of the restriction. Claim 3 of Lemma 2.23 sends compatibility of g to compatibility of h. Claim 2 gives the node balance condition. Claims 1(a) and 1(c) give connectedness. □

Definition 2.25. *Let $b > a \geq 2$ be coprime integers. Let $t \in [a-1]_{\mathbb{Z}}$ be the unique integer satisfying $bt \equiv 1 \pmod{a}$. We define*

$$\begin{gathered}\mu_1(a,b) = a + t, \qquad \mu_2(a,b) = \frac{b\,\mu_1(a,b) - 1}{a}, \\ \Gamma(a,b) = \frac{\mu_2(a,b)}{\mu_1(a,b)} = \frac{b}{a} - \frac{1}{a\,\mu_1(a,b)}.\end{gathered} \tag{2.9}$$

For $i \in [a]_{\mathbb{Z}}$, we define $\Lambda_i(a,b) = \lfloor i\,\Gamma(a,b) \rfloor - \lfloor (i-1)\Gamma(a,b) \rfloor$. We define the run

$$\mathcal{U}(a,b) = R_{\Lambda_1(a,b)} \mathbin{+\!\!+} \cdots \mathbin{+\!\!+} R_{\Lambda_a(a,b)} \mathbin{+\!\!+} R_1. \tag{2.10}$$

Lemma 2.26 (Decomposition into maximal horizontal shadows)**.** *Let $b > a \geq 2$ be coprime integers, and let $\mu_j = \mu_j(a,b)$ for $j \in [2]_{\mathbb{Z}}$. Let $\mathcal{D} = \mathcal{D}_{(\mu_1,\mu_2)}$ and $g \in \mathrm{TCG}(\mu_1,\mu_2,a,b)$. The maximal h-shadow subpaths of g are pairwise disjoint paths $\mathcal{D}\langle u_i, v_i\rangle$, indexed by $i \in [M]_{\mathbb{Z}}$ for some $M \geq 1$. Every edge outside these subpaths has grade zero.*

For $i \in [M]_{\mathbb{Z}}$, let

$$U_i = \mathrm{NEReal}(\mathcal{D}, u_i, v_i), \qquad g_i = g_{\mathcal{D},u_i,v_i}, \qquad q_i = \#N(U_i), \qquad p_i = \#E(U_i) - 1.$$

Then, g_i is connected, and

$$\mathrm{TEG}_{U_i,g_i} = q_i, \qquad \mathrm{TNG}_{U_i,g_i} \leq p_i, \qquad \sum_{i=1}^{M} q_i = b, \qquad \sum_{i=1}^{M} \mathrm{TNG}_{U_i,g_i} = a.$$

Proof. Let us fix the ambient dimensions and the total grades. Let $n = \mu_1 + \mu_2$ and write $\mathcal{D} = D_1 \mathbin{+\!\!+} \cdots \mathbin{+\!\!+} D_n$. Equation (2.9) and Definition 2.20(3) give

$$\mu_1 \geq a+1, \qquad \mu_2 \geq b+1, \qquad b\mu_1 - a\mu_2 = 1, \qquad \sum_{u \in E(\mathcal{D})} g(u) = b, \qquad \sum_{v \in N(\mathcal{D})} g(v) = a.$$

Let us define the sums used to choose a cyclic cut. For $i \in \mathbb{Z}_{>0}$, define

$$\bar{i} = 1 + ((i-1) \bmod n), \qquad \delta_i = \begin{cases} g(D_{\bar{i}}), & D_{\bar{i}} \in E(\mathcal{D}), \\ -1, & D_{\bar{i}} \in N(\mathcal{D}). \end{cases}$$

For $t \in \mathbb{Z}_{\geq 0}$, define $A(t) = \sum_{i=1}^{t} \delta_i$. Then,

$$A(0) = 0, \qquad A(n) = b - \mu_2 < 0, \qquad A(t+n) = A(t) + A(n) \quad (t \in \mathbb{Z}_{\geq 0}).$$

Let us choose the cut. Choose $c \in [0, n-1]_{\mathbb{Z}}$ satisfying $A(c) = \min\{A(j) : j \in [0,n-1]_{\mathbb{Z}}\}$. For $i \in [n]_{\mathbb{Z}}$, define $\eta(i) = 1 + ((c+i-1) \bmod n)$, and let

$$W = \mathrm{NEReal}(\mathcal{D}, D_{\eta(1)}, D_{\eta(n)}), \qquad h = g_{\mathcal{D},D_{\eta(1)},D_{\eta(n)}}.$$

The edge W_i corresponds to $D_{\eta(i)}$, and $h(W_i) = g(D_{\eta(i)})$.

Let us compute the sums after the cut. For $t \in [0,n]_{\mathbb{Z}}$, define

$$B(t) = A(c+t) - A(c) = \sum_{u \in E(W[0,t])} h(u) - \#N(W[0,t]).$$

Thus, $B(0) = 0$ and $B(n) = A(n)$. If $c + t < n$, minimality of $A(c)$ gives $B(t) \geq 0 > B(n)$. If $c + t \geq n$, it gives

$$B(t) = A(c+t-n) - A(c) + A(n) \geq A(n) = B(n).$$

Therefore, $B(n) \leq B(t)$ for every $t \in [0,n]_{\mathbb{Z}}$.

Let us check the effect of the cut on compatibility and shadows. The map $W_i \mapsto D_{\eta(i)}$ preserves directions, grades, and cyclic edge order. For $r, s \in [n]_{\mathbb{Z}}$, the sums in Definition 2.19(1) therefore give

$$\mathrm{NGrDiff}_{W,h}(W_r, W_s) = \mathrm{NGrDiff}_{\mathcal{D},g}(D_{\eta(r)}, D_{\eta(s)}),$$
$$\mathrm{EGrDiff}_{W,h}(W_r, W_s) = \mathrm{EGrDiff}_{\mathcal{D},g}(D_{\eta(r)}, D_{\eta(s)}).$$

Hence, h is compatible. Since $a\mu_2 - b\mu_1 = -1$, tightness gives $N_h(W) \subseteq \mathrm{Sh}_{W,h}(E_h(W))$.

Let us show that every positive horizontal grade closes before the end of W. Let $r \in [n]_{\mathbb{Z}}$ satisfy $W_r \in E(W)$ and $h(W_r) > 0$. We have

$$B(r) > B(r-1), \qquad B(n) \leq B(r-1).$$

Let us find the first return to $B(r-1)$. A horizontal edge increases B by its nonnegative grade, and a vertical edge decreases B by 1. Thus, the first index $t \in [r,n]_{\mathbb{Z}}$ with $B(t) \leq B(r-1)$ satisfies

$$r < t, \qquad W_t \in N(W), \qquad B(t) = B(r-1), \qquad B(j) > B(r-1) \quad (r \leq j < t).$$

Write $t(r) = t$. Since $\mathrm{NGrDiff}_{W,h}(W_r, W_j) = B(r-1) - B(j)$ for $r \leq j \leq t(r)$, we have $t_{W,h}(W_r) = W_{t(r)}$.

Let us determine the grade of a vertical edge after a minimum. Let $s \in [0, n-1]_{\mathbb{Z}}$ satisfy $B(s) \leq B(j)$ for every $j \in [0,s]_{\mathbb{Z}}$. Suppose that W_{s+1} is vertical and $h(W_{s+1}) > 0$. The shadow support condition gives an $r \in [s]_{\mathbb{Z}}$ such that

$$W_r \in E_h(W), \qquad s+1 \leq t(r).$$

The strict inequality at $j = s$ then gives $B(s) > B(r-1)$, contrary to the choice of s. Therefore, $h(W_{s+1}) = 0$.

Let us divide W into zero-grade edges and complete horizontal shadows. Start at the character boundary $s = 0$. At each boundary $s < n$, maintain $B(s) \leq B(j)$ for $j \in [0,s]_{\mathbb{Z}}$. If $h(W_{s+1}) = 0$, pass to $s+1$; the value of B either stays fixed or decreases by 1. Thus, $B(s+1) \leq B(j)$ for every $j \in [0, s+1]_{\mathbb{Z}}$.

Let us give the step for a positive grade. If $h(W_{s+1}) > 0$, then $W_{s+1} \in E(W)$: under $B(s) \leq B(j)$ for $j \in [0,s]_{\mathbb{Z}}$, we proved $W_{s+1} \in N(W) \Longrightarrow h(W_{s+1}) = 0$. Record the subpath $W[s, t(s+1)]$ and pass to $t(s+1)$. On $W[s, t(s+1)]$, we have

$$B(t(s+1)) = B(s), \qquad B(j) > B(s) \quad (s < j < t(s+1)).$$

Thus, $B(t(s+1)) \leq B(j)$ for every $j \in [0, t(s+1)]_{\mathbb{Z}}$ in the positive-grade case.

Let us record the resulting intervals. Each step increases s, so the procedure ends at $s = n$. Let the recorded subpaths be $W[r_i - 1, t_i]$ for $i \in [M]_{\mathbb{Z}}$, in their order in W. They satisfy

$$1 \leq r_i < t_i \leq n, \qquad t_i = t(r_i), \qquad t_i < r_{i+1} \quad (1 \leq i < M).$$

Every edge outside these subpaths has grade zero. Since $\sum_{u\in E(W)} h(u) = b > 0$, we have $M \geq 1$.

Let us prove that these are exactly the maximal h-shadow subpaths. Let $W_r \in E_h(W)$ belong to $W[r_i - 1, t_i]$. The first-return inequalities for r_i give $B(t_i) = B(r_i - 1) \leq B(r - 1)$. Since $B(r) > B(r - 1)$ and each decrease of B is 1, the first return from r satisfies

$$r < t(r) \leq t_i.$$

Thus, every h-shadow subpath starting at a positive horizontal grade is contained in a recorded subpath. The recorded subpaths are disjoint, and each is itself an h-shadow subpath. They are therefore exactly the maximal h-shadow subpaths.

Let us transport these subpaths to $\mathcal{D}$. For $i \in [M]_{\mathbb{Z}}$, let

$$u_i = D_{\eta(r_i)}, \qquad v_i = D_{\eta(t_i)}, \qquad U_i = \mathrm{NEReal}(\mathcal{D}, u_i, v_i), \qquad g_i = g_{\mathcal{D},u_i,v_i}.$$

The cyclic index map identifies U_i with $W[r_i - 1, t_i]$ in edge order and preserves all grades. Consequently, $\mathcal{D}\langle u_i, v_i\rangle$ for $i \in [M]_{\mathbb{Z}}$ are the maximal h-shadow subpaths of g. Every edge of $\mathcal{D}$ outside them has grade zero.

Let us compute the totals on these subpaths. For $i \in [M]_{\mathbb{Z}}$, define $q_i = \#N(U_i)$ and $p_i = \#E(U_i) - 1$. The equality $B(t_i) = B(r_i - 1)$ gives $\mathrm{TEG}_{U_i,g_i} = q_i$. Therefore,

$$\sum_{i=1}^{M} q_i = b, \qquad \sum_{i=1}^{M} \mathrm{TNG}_{U_i,g_i} = a.$$

Let us bound the total vertical grade of each subpath. Claim 3(b) of Lemma 2.24 makes each g_i connected. Lemma 2.11 gives

$$p_i - \mathrm{TNG}_{U_i,g_i} = \mathrm{Surp}(U_i, g_i) \geq 0 \quad (i \in [M]_{\mathbb{Z}}).$$

Summing over $i \in [M]_{\mathbb{Z}}$ gives $\sum_{i=1}^{M} p_i \geq a$. □

Lemma 2.27 (The initial path and its horizontal shadow)**.** *Let $b > a \geq 2$ be coprime integers, and let $\mu_j = \mu_j(a, b)$ for $j \in [2]_{\mathbb{Z}}$. Let $\mathcal{D} = \mathcal{D}_{(\mu_1,\mu_2)}$.*

1. *The path $\mathcal{U}(a, b)$ is an initial subpath of $\mathcal{D}$, and*

$$\sum_{i=1}^{a} \Lambda_i(a, b) = b - 1, \qquad \#E(\mathcal{U}(a, b)) = a + 1, \qquad \#N(\mathcal{U}(a, b)) = b.$$

2. *Let $g \in \mathrm{TCG}(\mu_1, \mu_2, a, b)$. The unique maximal h-shadow subpath of g is the initial subpath $\mathcal{U}(a, b)$ of $\mathcal{D}$. Every edge outside $\mathcal{U}(a, b)$ has grade zero. The restriction $g|_{\mathcal{U}(a,b)}$ is connected and has surplus zero.*

Proof. Proof of Claim 1. Let us identify the initial subpath. Equation (2.9) gives $\mu_1 \geq a + 1$ and $\Gamma(a, b) > 1$. The first a simple runs of $\mathcal{D}$ have vertical lengths $\Lambda_1(a, b), \ldots, \Lambda_a(a, b)$, and

$$\sum_{i=1}^{a} \Lambda_i(a, b) = \lfloor a\Gamma(a, b)\rfloor = \left\lfloor b - \frac{1}{\mu_1}\right\rfloor = b - 1.$$

Since $\Gamma(a, b) > 1$, the next simple run starts with a horizontal edge followed by a vertical edge. Appending these two edges gives $\mathcal{U}(a, b)$ by (2.10). Its horizontal and vertical edge counts are $a + 1$ and b.

Proof of Claim 2. This shadow localization is the case $p = a$, $q = b$, and $(d_1, d_2) = (\mu_1, \mu_2)$ of [1, Lemmas 4.12 and 4.13]. We give a direct proof. Let us fix the maximal h-shadow subpaths. Use M, u_i, v_i, U_i, g_i, p_i, and q_i from Lemma 2.26. Let $n = \mu_1 + \mu_2$ and write $\mathcal{D} = D_1 + \cdots + D_n$. Equation (2.9) gives $b\mu_1 - a\mu_2 = 1$ and $\gcd(\mu_1, \mu_2) = 1$.

Let us prove that the edge coordinates are distinct. For $j \in [n]_{\mathbb{Z}}$, let (x_j, y_j) be the starting point of D_j. Then, $x_j + y_j = j - 1$, so

$$\pi_{\mu_1,\mu_2}(D_j) = \mu_2(j-1) - ny_j.$$

If $\pi_{\mu_1,\mu_2}(D_i) = \pi_{\mu_1,\mu_2}(D_j)$ for $i, j \in [n]_{\mathbb{Z}}$, then n divides $\mu_2(i-j)$. Since $\gcd(\mu_2, n) = \gcd(\mu_1, \mu_2) = 1$ and $|i-j| < n$, we obtain $i = j$. In particular, D_1 is the unique edge with coordinate 0.

Let us bound the number of subpaths. Lemma 2.26 gives $\sum_{i=1}^{M} q_i = b$ and $\sum_{i=1}^{M} p_i \geq a$. Moving past a horizontal edge changes π_{μ_1,μ_2} by μ_2, and moving past a vertical edge changes it by $-\mu_1$. The coordinate increments $+\mu_2$ on horizontal edges and $-\mu_1$ on vertical edges also apply from D_n to D_1, since their total over $\mathcal{D}$ is $\mu_1\mu_2 - \mu_2\mu_1 = 0$. Since u_i is horizontal and v_i is vertical, Definition 2.20(2) gives

$$\begin{aligned} q_i\mu_1 - p_i\mu_2 &= \pi_{\mu_1,\mu_2}(u_i) + \mu_1 + \mu_2 - \pi_{\mu_1,\mu_2}(v_i) \\ &\geq \pi_{\mu_1,\mu_2}(u_i) + 1 \geq 1 \qquad (i \in [M]_{\mathbb{Z}}). \end{aligned} \tag{2.11}$$

Summing (2.11) gives

$$1 \leq M \leq b\mu_1 - \mu_2 \sum_{i=1}^{M} p_i \leq b\mu_1 - a\mu_2 = 1. \tag{2.12}$$

Hence, $M = 1$.

Let us determine the endpoints of this subpath. We have $q_1 = b$ and $p_1 \geq a$. Equations (2.11) and (2.12) give

$$1 \leq b\mu_1 - p_1\mu_2 = 1 - (p_1 - a)\mu_2 \leq 1.$$

Thus, $p_1 = a$, and

$$\begin{gathered} 1 = \pi_{\mu_1,\mu_2}(u_1) + \mu_1 + \mu_2 - \pi_{\mu_1,\mu_2}(v_1), \\ \pi_{\mu_1,\mu_2}(u_1) = 0, \qquad \pi_{\mu_1,\mu_2}(v_1) = \mu_1 + \mu_2 - 1. \end{gathered}$$

Let us identify the subpath with $\mathcal{U}(a, b)$. Coordinate injectivity gives $u_1 = D_1$. The subpath has $a + 1$ horizontal edges and b vertical edges. Claim 1 identifies the first $a + b + 1$ edges of $\mathcal{D}$ with $\mathcal{U}(a, b)$. Hence,

$$\mathcal{D}\langle u_1, v_1\rangle = \mathcal{U}(a, b), \qquad \operatorname{Surp}(\mathcal{U}(a, b), g_1) = p_1 - a = 0.$$

□

Lemma 2.28 (Restriction and zero extension)**.** *Let $b > a \geq 2$ be coprime integers, and let $\mu_j = \mu_j(a,b)$ for $j \in [2]_{\mathbb{Z}}$. Let $\mathcal{D} = \mathcal{D}_{(\mu_1,\mu_2)}$ and $V = \mathcal{U}(a,b)$. Restriction defines a bijection*

$$\mathrm{TCG}(\mu_1,\mu_2,a,b) \longrightarrow \{f \in \mathrm{CtdG}(V) : \mathrm{Surp}(V,f) = 0\}, \qquad g \longmapsto g|_V.$$

Its inverse sends f to the grading $g \in G(\mathcal{D})$ defined by

$$g(e) = \begin{cases} f(e), & e \in \mathrm{NE}(V), \\ 0, & e \in \mathrm{NE}(\mathcal{D}) \setminus \mathrm{NE}(V). \end{cases}$$

Proof. Let us check that restriction has the stated values. Claim 2 of Lemma 2.27 identifies V with the unique maximal h-shadow subpath of each tight grading. It makes the restricted grading connected with surplus zero.

Let us fix a grading to extend by zero. Let $f \in \mathrm{CtdG}(V)$ satisfy $\mathrm{Surp}(V,f) = 0$, and let g be its zero extension. Claim 1 of Lemma 2.27 gives $\#E(V) = a+1$ and $\#N(V) = b$. Let $h_0 = V_1$ and $v_* = V_{a+b+1}$.

Let us compute the two total grades of f. The definition of connectedness gives $\mathrm{Node}_{V,f}(h_0) = V$. Claim 1(b) of Lemma 2.15 gives a node for each $x \in E(V)$ with $f(x) > 0$. Thus, $\#P_{V,f}^{-1}(\{x\}) = f(x)$ for each such x. For $x \in E(V)$ with $f(x) = 0$, the equality $\#P_{V,f}^{-1}(\{x\}) = 0$ follows from Claim 1 of Lemma 2.13.

Let us sum these fiber sizes. Each vertical edge has exactly one h-precursor, so

$$\mathrm{TEG}_{V,f} = \sum_{x \in E(V)} \#P_{V,f}^{-1}(\{x\}) = \#N(V) = b.$$

Since $\mathrm{Surp}(V,f) = 0$ and $\#E(V) = a+1$, we also have $\mathrm{TNG}_{V,f} = a$.

Let us identify the horizontal shadow of h_0. Claim 2 of Lemma 2.21 gives

$$\mathrm{NGrDiff}_{V,f}(h_0, V_j) < 0 \quad (1 \leq j < a+b+1), \qquad \mathrm{NGrDiff}_{V,f}(h_0, v_*) = 0.$$

For $j \in [a+b+1]_{\mathbb{Z}}$, we have $g(V_j) = f(V_j)$. Thus, $t_{\mathcal{D},g}(h_0) = v_*$.

Let us prove that this h-shadow subpath is maximal. Let $x \in E(V)$ satisfy $f(x) > 0$. Claim 1(b) of Lemma 2.15 gives a node at x contained in V. Claim 2 of Lemma 2.21 identifies the endpoint of this node with $t_{\mathcal{D},g}(x)$. Thus, every h-shadow subpath of g is contained in $V = \mathcal{D}\langle h_0, v_*\rangle$.

Let us verify compatibility and tightness. Claim 3(a) of Lemma 2.24, applied to this maximal h-shadow subpath, makes g compatible. Its total grades are a and b, and

$$N_g(\mathcal{D}) \subseteq N(V) = \mathrm{Sh}_{\mathcal{D},g}(h_0), \qquad h_0 \in E_g(\mathcal{D}).$$

Definition 2.20(4)(b) gives $g \in \mathrm{TCG}(\mu_1,\mu_2,a,b)$.

Let us verify that the two maps are inverse. A tight grading is zero outside V by Claim 2 of Lemma 2.27. Thus, extending its restriction by zero recovers that grading. Conversely, restricting the zero extension of f to V gives f. □

Proposition 2.29 (Counting colored gradings on a run)**.** *Let $b > a \geq 2$ be coprime integers. Then, we have the following.*

1. *The total vertical length of the first a simple runs of $\mathcal{U}(a,b)$ is*

$$\sum_{i=1}^{a} \Lambda_i(a,b) = b - 1.$$

2. *We have*

$$C_0(\mathcal{U}(a,b)) = mE_m(a,b). \tag{2.13}$$

Proof. Proof of Claim 1. Apply Claim 1 of Lemma 2.27.

Proof of Claim 2. Let us apply the finite-enumeration formula. Let $\mu_j = \mu_j(a,b)$ for $j \in [2]_{\mathbb{Z}}$, $\mathcal{D} = \mathcal{D}_{(\mu_1,\mu_2)}$, and $V = \mathcal{U}(a,b)$. For incoming polynomials $(1+x)^m$, the finite enumeration in [1, Theorems 3.2 and 4.10] and [16, Theorem 12.1] gives

$$\sum_{g \in \mathrm{TCG}(\mu_1,\mu_2,a,b)} \prod_{w \in \mathrm{NE}(\mathcal{D})} \binom{m}{g(w)} = mE_m(a,b).$$

Let us count the color functions after restriction. Lemma 2.28 identifies the tight gradings with the connected gradings of V having surplus zero. For $g \in \mathrm{TCG}(\mu_1,\mu_2,a,b)$, Lemma 2.28 gives $g(w) = 0$ for $w \in \mathrm{NE}(\mathcal{D}) \setminus \mathrm{NE}(V)$. Since $\binom{m}{0} = 1$,

$$\prod_{w \in \mathrm{NE}(\mathcal{D})} \binom{m}{g(w)} = \prod_{w \in \mathrm{NE}(V)} \binom{m}{(g|_V)(w)}.$$

Equation (2.5) therefore gives (2.13). □

Example 2.30. We have $\mu_1(2,5) = 3$, $\mu_2(2,5) = 7$, and

$$\mathcal{U}(2,5) = R_2^2 + R_1 = E + N^2 + E + N^2 + E + N.$$

The run $\mathcal{U}(2,5)$ passes through $(2,4)$ and ends at $(3,5)$. Since $4 > \frac{5}{3} \cdot 2$, $\mathcal{U}(2,5)$ does not stay below the line segment joining its endpoints.

For a connected grading g of surplus zero, we have $g(n_1) = g(n_2) = 0$. The other grades have the following possibilities:

$(g(e_1), g(e_2), g(e_3))$	$(g(n_3), g(n_4), g(n_5))$
$(3,2,0)$	$(0,0,2)$
$(4,1,0)$	$(0,0,2),\ (0,1,1)$
$(5,0,0)$	$(0,0,2),\ (1,0,1),\ (0,1,1).$

Therefore,

$$C_0(R_2^2 + R_1) = \binom{m}{3}\binom{m}{2}^2 + m\binom{m}{4}\left(\binom{m}{2} + m^2\right) + \binom{m}{5}\left(\binom{m}{2} + 2m^2\right).$$

For $m = 3$, only the first row contributes. Thus, $C_0(R_2^2 + R_1) = 9$ and $E_3(2,5) = 3$.

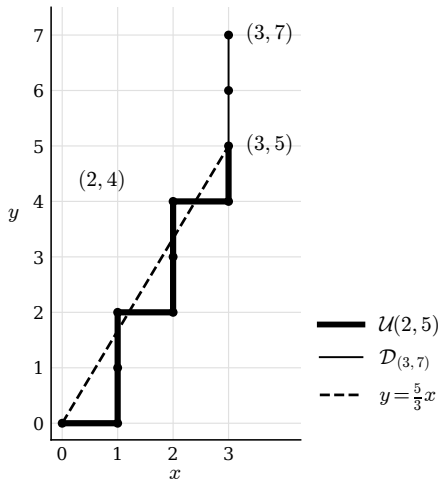


Figure 1: The thick line shows $\mathcal{U}(2,5)$. The thin line continues $\mathcal{U}(2,5)$ to $\mathcal{D}_{(3,7)}$. The dashed segment joins $(0,0)$ to $(3,5)$.

3 Shell paths and gradings

3.1 Shell constructions and periodic growth

Definition 3.1. *Let $j \in \mathbb{Z}_{\geq 0}$ and $\epsilon \in \mathbb{Z}_{\geq 0}$ satisfy $j+1+\epsilon \leq m$. For $s \in [j]_{\mathbb{Z}}$, let X_s be a tNE path. Let $u \in [m-1]_{\mathbb{Z}}$ and $v \in [m-1]_{\mathbb{Z}}$. Let $X = X_1 \mathbin{+\!\!+} \cdots \mathbin{+\!\!+} X_j$. For $j = 0$, let $X = \emptyset$.*

1. *We define the shell path*

$$Y = \mathrm{Shell}(X_1, \ldots, X_j; u, v, \epsilon) = R_u \mathbin{+\!\!+} X \mathbin{+\!\!+} R_v.$$

   For $j = 0$, use the notation $\mathrm{Shell}(; u, v, \epsilon)$.

2. *For $s \in [j]_{\mathbb{Z}}$, let $g_s \in \mathrm{CtdG}_m(X_s)$. Let f be the concatenated grading on X. For $j = 0$, let $f = \emptyset$. We define the shell grading $g = \mathrm{GShell}_f(X_1, \ldots, X_j; u, v, \epsilon)$ on Y by*

$$g(Y_k) = \begin{cases} u+1, & k = 1, \\ f(X_{k-u-1}), & k \in [u+2, u+1+|X|]_{\mathbb{Z}}, \\ v-1, & k = u+|X|+2, \\ j+1+\epsilon, & k = |Y|, \\ 0, & \textit{otherwise} \end{cases} \qquad (k \in [|Y|]_{\mathbb{Z}}). \tag{3.1}$$

   For $j = 0$, use the notation $\mathrm{GShell}_{\emptyset}(; u, v, \epsilon)$.

3. *We call a shell path* $\mathrm{Shell}(X_1, \ldots, X_j; u, v, \epsilon)$ *unary if $j = 1$.*

4. *We call a shell path* $\mathrm{Shell}(X_1, \ldots, X_j; u, v, \epsilon)$ *binary if $j = 2$.*

5. *We call a shell grading* $\mathrm{GShell}_f(X_1, \ldots, X_j; u, v, \epsilon)$ *preserving if $\epsilon = 0$.*

6. *We call a shell grading* $\mathrm{GShell}_f(X_1, \ldots, X_j; u, v, \epsilon)$ *draining if $j = \epsilon = 1$.*

Definition 3.2. *Let U and V be tNE paths. Let $S \subseteq \mathrm{NE}(U)$ and $\iota \colon S \hookrightarrow \mathrm{NE}(V)$. Let $g \in G_m(U)$, $\gamma \in \mathrm{Col}(U, g, m)$, and $h \in G_m(V)$. We define the recoloring operator*

$$\mathrm{Recol}(U, V, \iota, h; g, \gamma) = \gamma' \in \mathrm{Col}(V, h, m)$$

such that γ' satisfies

$$\gamma'(e) = \begin{cases} \gamma(x), & e = \iota(x),\ x \in S,\ h(e) = g(x), \\ 1, & \text{otherwise} \end{cases} \qquad (e \in \mathrm{NE}(V)). \tag{3.2}$$

Lemma 3.3. *Let $j \in \mathbb{Z}_{\geq 0}$ and $\epsilon \in \mathbb{Z}_{\geq 0}$ satisfy $j+1+\epsilon \leq m$. Let $u \in [m-1]_{\mathbb{Z}}$ and $v \in [m-1]_{\mathbb{Z}}$. For $s \in [j]_{\mathbb{Z}}$, let X_s be a tNE path and let $g_s \in \mathrm{CtdG}_m(X_s)$. Let $X = X_1 +\!\!+ \cdots +\!\!+ X_j$ and let f be the concatenated grading on X. For $j = 0$, let $X = \emptyset$ and $f = \emptyset$. Suppose that one of the following conditions holds.*

(a) $j \in [0,2]_{\mathbb{Z}}$ and $\epsilon = 0$.

(b) $j = \epsilon = 1$ and $\mathrm{Surp}(X_1, g_1) \geq 1$.

Let $Y = \mathrm{Shell}(X_1, \ldots, X_j; u, v, \epsilon)$ and $g = \mathrm{GShell}_f(X_1, \ldots, X_j; u, v, \epsilon)$. Put

$$r_i = u + 1 + \sum_{s=1}^{i} |X_s| \qquad (i \in [0, j]_{\mathbb{Z}}).$$

The empty sum is 0. Then, we have the following.

1. *$g \in G_m(Y)$.*
2. *$g \in \mathrm{CtdG}_m(Y)$.*
3. *We have*
$$\mathrm{Surp}(Y, g) = \sum_{i=1}^{j} \mathrm{Surp}(X_i, g_i) - \epsilon.$$
4. *Fix $X_1, \ldots, X_j$, u, v, and ϵ. Let $c_s \in \mathbb{Z}_{\geq 0}$ for $s \in [j]_{\mathbb{Z}}$. Assume $\sum_{s=1}^{j} c_s \geq \epsilon$. For $(g_s, \gamma_s) \in \mathcal{C}_{c_s}(X_s)$, let η be the concatenated coloring. If $j = 0$, let $\eta = \emptyset$. We define*
$$\iota\colon \mathrm{NE}(X) \hookrightarrow \mathrm{NE}(Y), \qquad \iota(X_k) = Y_{u+1+k} \quad (k \in [|X|]_{\mathbb{Z}}).$$
Let
$$\gamma = \mathrm{Recol}(X, Y, \iota, g; f, \eta). \tag{3.3}$$
Then, the map
$$\Phi_{\mathrm{sh}}\colon \prod_{s \in [j]_{\mathbb{Z}}} \mathcal{C}_{c_s}(X_s) \longrightarrow \mathcal{C}_{\sum_{s=1}^{j} c_s - \epsilon}(Y), \qquad \Phi_{\mathrm{sh}}((g_s, \gamma_s)_{s \in [j]_{\mathbb{Z}}}) = (g, \gamma)$$
is injective. Also, we have
$$(g(\iota(X_k)), \gamma(\iota(X_k))) = (f(X_k), \eta(X_k)) \qquad (k \in [|X|]_{\mathbb{Z}}).$$
5. *For $i \in [j]_{\mathbb{Z}}$, $k \in [|X_i|]_{\mathbb{Z}}$, and $e = (X_i)_k$, define the copy of e in Y by*
$$\widetilde{e} = Y_{r_{i-1}+k}. \tag{3.4}$$
Then, we have
$$P_{Y,g}(\widetilde{w}) = \widetilde{P_{X_i,g_i}(w)} \qquad (i \in [j]_{\mathbb{Z}},\ w \in N(X_i)). \tag{3.5}$$
6. *For $i \in [0, j]_{\mathbb{Z}}$ and $e \in E(Y[0, r_i])$, we have*
$$\mathrm{Debt}_{Y,g}(e, r_i) = \begin{cases} 1, & e = Y_1, \\ 0, & e \neq Y_1. \end{cases}$$

7. *We have*

$$\mathrm{bBal}_{Y,g}(r_i) = 1 + i + \sum_{s=1}^{i} \mathrm{Surp}(X_s, g_s) \qquad (i \in [0, j]_{\mathbb{Z}}).$$

Proof. Proof of Claim 1. Let us verify that every shell grade lies in the prescribed range. The bounds $u \in [m-1]_{\mathbb{Z}}$, $v \in [m-1]_{\mathbb{Z}}$, $j+1+\epsilon \le m$, and $g_i \in \mathrm{CtdG}_m(X_i)$ for $i \in [j]_{\mathbb{Z}}$ in Definition 3.1 give

$$g_i(e) \le m \qquad (i \in [j]_{\mathbb{Z}},\ e \in \mathrm{NE}(X_i)),$$
$$\max\{u+1, v-1, j+1+\epsilon\} \le m.$$

Every edge of Y inherits a grade from some g_i or has grade 0, $u+1$, $v-1$, or $j+1+\epsilon$. Hence, $g \in G_m(Y)$.

Proof of Claim 2. Let us verify the three conditions for connectedness in Definition 2.9(1). Let $y = \mathrm{Start}_{\mathrm{NE}}(Y)$ and $z = \mathrm{End}_{\mathrm{NE}}(Y)$. We construct $P_{Y,g}$ in edge order, using Definition 2.6(1). The grades $g(Y_1) = u+1$ and $g(Y_{k+1}) = 0$ for $k \in [u]_{\mathbb{Z}}$ give

$$P_{Y,g}(Y_{k+1}) = y \qquad (k \in [u]_{\mathbb{Z}}),$$
$$\mathrm{Debt}_{Y,g}(y, r_0) = g(y) - u = (u+1) - u = 1.$$

Let us determine the h-precursors in each copy of X_i. The subpath $Y[r_{i-1}, r_i]$ is the translated copy of X_i with the grading translated from g_i. At each vertical edge $\widetilde{w}$ of this copy, the translated h-precursor from X_i has positive debt and follows every horizontal edge before the copy. Definition 2.6(1) therefore selects $\widetilde{P_{X_i,g_i}(w)}$ as the h-precursor of $\widetilde{w}$.

Let us compute the debt of y after each copy of X_i. Since g_i is connected, Claim 1(b) of Lemma 2.15 shows that every horizontal edge of X_i has debt zero at $|X_i|$. Thus,

$$\mathrm{Debt}_{Y,g}(y, r_i) = 1 \qquad (i \in [j]_{\mathbb{Z}}).$$

Let us compute the change in boundary balance across the copies. The boundary balances satisfy

$$\mathrm{bBal}_{Y,g}(r_i) - \mathrm{bBal}_{Y,g}(r_{i-1}) = |X_i|_E - \mathrm{TNG}_{X_i,g_i} = \mathrm{Surp}(X_i, g_i) + 1 \qquad (i \in [j]_{\mathbb{Z}}),$$

and hence

$$\mathrm{bBal}_{Y,g}(r_j) - \mathrm{bBal}_{Y,g}(r_0) = \sum_{i=1}^{j} \mathrm{Surp}(X_i, g_i) + j.$$

Let us prove $\mathrm{Node}_{Y,g}(y) = Y$. The grades $g(Y_{r_j+1}) = v-1$ and $g(Y_{r_j+1+k}) = 0$ for $k \in [v-1]_{\mathbb{Z}}$ give

$$P_{Y,g}(Y_{r_j+1+k}) = Y_{r_j+1} \qquad (k \in [v-1]_{\mathbb{Z}}),$$
$$\mathrm{bBal}_{Y,g}(r_j + v) - \mathrm{bBal}_{Y,g}(r_j) = 1 - \sum_{k=1}^{v-1} g(Y_{r_j+1+k}) = 1,$$
$$\mathrm{Debt}_{Y,g}(Y_{r_j+1}, r_j + v) = (v-1) - (v-1) = 0.$$

Immediately before z, the edge y has debt one and every other horizontal edge has debt zero. Hence,

$$P_{Y,g}(z) = y, \qquad \mathrm{Debt}_{Y,g}(y, |Y|) = 0.$$

This constructs $P_{Y,g}$ and gives $\mathrm{Node}_{Y,g}(y) = Y$.

Let us verify the node balance condition. For $i \in [j]_{\mathbb{Z}}$, $w \in N(X_i)$, and $a = P_{X_i,g_i}(w)$, the identity $P_{Y,g}(\widetilde{w}) = \widetilde{a}$ and the connectedness of g_i give

$$\begin{aligned}&\mathrm{eBal}_{Y,g}(\widetilde{w}) - \mathrm{eBal}_{Y,g}(\widetilde{a})\\ &\quad = \mathrm{eBal}_{X_i,g_i}(w) - \mathrm{eBal}_{X_i,g_i}(a) \geq 0.\end{aligned}$$

For Y_{k+1} with $k \in [u]_{\mathbb{Z}}$ and Y_{r_j+1+k} with $k \in [v-1]_{\mathbb{Z}}$, we have

$$\begin{aligned}\mathrm{eBal}_{Y,g}(Y_{k+1}) = 1 = \mathrm{eBal}_{Y,g}(y) \quad (k \in [u]_{\mathbb{Z}}),\\ \mathrm{eBal}_{Y,g}(Y_{r_j+1+k}) = \mathrm{eBal}_{Y,g}(Y_{r_j+1}) \quad (k \in [v-1]_{\mathbb{Z}}).\end{aligned}$$

By Lemma 2.11, each $\mathrm{Surp}(X_i, g_i)$ is nonnegative. Under condition (a) or (b), the edge balance at z satisfies

$$\mathrm{eBal}_{Y,g}(z) = \mathrm{EGrDiff}_{Y,g}(y,z) = 1 + \sum_{i=1}^{j} \mathrm{Surp}(X_i, g_i) - \epsilon \geq 1 = \mathrm{eBal}_{Y,g}(y).$$

Therefore, g is connected. By Claim 1, $g \in \mathrm{CtdG}_m(Y)$.

Proof of Claim 3. By (2.4),

$$\mathrm{Surp}(Y,g) = \mathrm{EGrDiff}_{Y,g}(y,z) - 1 = \sum_{i=1}^{j} \mathrm{Surp}(X_i, g_i) - \epsilon.$$

Proof of Claim 4. Let us define the shell map and prove injectivity. Claims 2 and 3 give $(g,\gamma) \in \mathcal{C}_{\sum_{s=1}^{j} c_s - \epsilon}(Y)$. Equation (3.1) gives $g(\iota(X_k)) = f(X_k)$. Definition 3.2 then gives $\gamma(\iota(X_k)) = \eta(X_k)$. Restriction to each copy of X_s recovers (g_s, γ_s), proving injectivity.

Proof of Claim 5. Let us compare the h-precursors in X_i and its copy in Y. Fix $i \in [j]_{\mathbb{Z}}$. The copy map (3.4) preserves edge order and grades. For $w = (X_i)_k \in N(X_i)$, Definition 2.6(1) gives $\mathrm{Debt}_{X_i,g_i}(P_{X_i,g_i}(w), k-1) > 0$. Induction on k preserves the debts of the horizontal edges in $X_i[0,k-1]$ under the copy map. Every horizontal edge before the copy of X_i precedes $\widetilde{P_{X_i,g_i}(w)}$. Thus, Definition 2.6(1) selects $\widetilde{P_{X_i,g_i}(w)}$ as the h-precursor of $\widetilde{w}$. Hence, (3.5) holds.

Proof of Claim 6. Let us compute the debts at r_i. The first u vertical edges have h-precursor Y_1. Hence,

$$\mathrm{Debt}_{Y,g}(Y_1, r_0) = g(Y_1) - u = 1.$$

By Claim 5, every vertical edge in a copy of X_s has its h-precursor in that copy. Claim 1(b) of Lemma 2.15 gives debt zero at $|X_s|$ for every horizontal edge of X_s. Thus, at r_i, the edge Y_1 has debt 1 and every horizontal edge in the copies of $X_1, \dots, X_i$ has debt 0.

Proof of Claim 7. The definition of bBal and (2.4) give

$$\mathrm{bBal}_{Y,g}(r_i) = 1 + \sum_{s=1}^{i} (|X_s|_E - \mathrm{TNG}_{X_s,g_s}) = 1 + i + \sum_{s=1}^{i} \mathrm{Surp}(X_s, g_s) \qquad (i \in [0,j]_{\mathbb{Z}}).$$

□

Definition 3.4 (Fixed simple-run grading)**.** *Let $j \in [m]_{\mathbb{Z}}$. Define $g_j^{\mathrm{run}} \in G_m(R_j)$ and $\gamma_j^{\mathrm{run}} \in \mathrm{Col}(R_j, g_j^{\mathrm{run}}, m)$ by*

$$g_j^{\mathrm{run}}(\mathrm{Start}_{\mathrm{NE}}(R_j)) = j, \qquad g_j^{\mathrm{run}}(v) = 0 \quad (v \in N(R_j)), \qquad \gamma_j^{\mathrm{run}}(e) = 1 \quad (e \in \mathrm{NE}(R_j)).$$

Lemma 3.5. *Let U be a run with ν simple runs. Let L be an integer. Assume the following.*

(a) $L \in [m-1]_{\mathbb{Z}}$.

(b) Each simple run of U has vertical length at most L.

Then, we have

$$m^{\lceil \frac{\nu}{2} \rceil} \le C_0(U) \le 2^{m(L+1)\nu}.$$

Proof. Let us prove the lower bound first for one and two simple runs. By Lemma 2.10, we translate U so that $\mathrm{Start}_{\mathrm{pt}}(U) = (0,0)$.

First, let us prove the bound for $\nu = 1$. Let $U = R_j$ and let $g = g_j^{\mathrm{run}}$. This grading is connected and satisfies $\mathrm{Surp}(U,g) = 0$. By assumptions (a) and (b), $1 \le j \le L \le m-1$. Thus, the horizontal edge has $\binom{m}{j} \ge m$ colors.

Second, let us prove the bound for $\nu = 2$. Let $U = R_u +\!\!+ R_v$ and use $g = \mathrm{GShell}_{\emptyset}(;u,v,0)$. Claims 2 and 3 of Lemma 3.3 give a connected grading of surplus zero. The edge $\mathrm{End}_{\mathrm{NE}}(U)$ has grade one and m colors.

Let us extend the lower bound by induction on ν. For $\nu \ge 3$, let $U = R_u +\!\!+ X +\!\!+ R_v$ with $|X|_E = \nu - 2$. For each $(g,\gamma) \in \mathcal{C}_0(X)$, Claims 2–4 of Lemma 3.3, followed by Lemma 2.10, give a colored grading $(h,\delta) \in \mathcal{C}_0(U)$ such that

$$h(U_{u+1+k}) = g(X_k), \qquad \delta(U_{u+1+k}) = \gamma(X_k) \quad (k \in [|X|]_{\mathbb{Z}}). \tag{3.6}$$

The edge $\mathrm{End}_{\mathrm{NE}}(U)$ has grade two and $\binom{m}{2} \ge m$ colors. Equation (3.6) recovers (g,γ) from (h,δ). Induction gives

$$C_0(U) \ge \binom{m}{2} C_0(X) \ge m\, m^{\lceil \frac{\nu-2}{2} \rceil} = m^{\lceil \frac{\nu}{2} \rceil}.$$

Let us prove the upper bound by summing the choices of grade and color on each edge. By assumption (b), there are at most $(L+1)\nu$ edges. Hence,

$$C_0(U) \le \prod_{e \in \mathrm{NE}(U)} \sum_{r=0}^{m} \binom{m}{r} = 2^{m|U|} \le 2^{m(L+1)\nu}.$$

□

Lemma 3.6. *Let X be a tNE path. Let A and B be runs with $P \ge 1$ and $S \ge 1$ simple runs, respectively. Assume the following.*

(a) Each simple run of A and B has vertical length at most $m-1$.

(b) $\max(P,S) \le 2\min(P,S)$.

Then, for every integer $d \ge 0$, there is an injective map

$$i_d \colon \mathcal{C}_d(X) \longrightarrow \mathcal{C}_d(A +\!\!+ X +\!\!+ B)$$

with the following property. For each $(g,\gamma) \in \mathcal{C}_d(X)$ and $(h,\delta) = i_d(g,\gamma)$, we have

$$h((A +\!\!+ X +\!\!+ B)_{|A|+k}) = g(X_k), \qquad \delta((A +\!\!+ X +\!\!+ B)_{|A|+k}) = \gamma(X_k) \quad (k \in [|X|]_{\mathbb{Z}}). \tag{3.7}$$

Proof. Let us construct shell gradings that preserve the surplus d. Let $(g,\gamma) \in \mathcal{C}_d(X)$. Let $u \in [m-1]_{\mathbb{Z}}$, $v \in [m-1]_{\mathbb{Z}}$, and $t \in [m-1]_{\mathbb{Z}}$. Give R_t the colored grading $(g_t^{\mathrm{run}}, \gamma_t^{\mathrm{run}})$. The shell paths

$$\begin{gathered} \mathrm{Shell}(X;u,v,0), \\ \mathrm{Shell}(R_t,X;u,v,0), \\ \mathrm{Shell}(X,R_t;u,v,0) \end{gathered} \tag{3.8}$$

add $(1,1)$, $(2,1)$, and $(1,2)$ simple runs before and after X, respectively. For each shell path Y, let h be its shell grading from (3.1). Claims 2 and 3 of Lemma 3.3 give $h \in \mathrm{CtdG}_m(Y)$ and

$$\mathrm{Surp}(Y,h) = \mathrm{Surp}(X,g) = d.$$

Let us choose the numbers of shell paths to add exactly P simple runs before X and S after X. By assumption (b), the following decomposition has nonnegative integer coefficients:

$$(P,S) = \begin{cases} (P-S)(2,1) + (2S-P)(1,1), & S \le P \le 2S, \\ (S-P)(1,2) + (2P-S)(1,1), & P \le S \le 2P. \end{cases} \tag{3.9}$$

Let us match the simple runs of A and B in order. Let $a_1, \dots, a_P$ and $b_1, \dots, b_S$ be the ordered vertical lengths of the simple runs of A and B. By assumption (a),

$$a_i \in [m-1]_{\mathbb{Z}} \quad (i \in [P]_{\mathbb{Z}}), \qquad b_i \in [m-1]_{\mathbb{Z}} \quad (i \in [S]_{\mathbb{Z}}).$$

Let us specify one step of the construction. Let $p \in [0,P]_{\mathbb{Z}}$ and $q \in [0,S]_{\mathbb{Z}}$. Suppose that Y is a translated copy of $R_{a_{p+1}} \dotplus \cdots \dotplus R_{a_P} \dotplus X \dotplus R_{b_1} \dotplus \cdots \dotplus R_{b_q}$. For the three shell operators in (3.8) applied to Y, use

$$(u,v) = (a_p, b_{q+1}), \qquad (u,t,v) = (a_{p-1}, a_p, b_{q+1}), \qquad (u,t,v) = (a_p, b_{q+1}, b_{q+2}),$$

respectively. The first choice requires $p \ge 1$ and $q < S$. The second requires $p \ge 2$ and $q < S$. The third requires $p \ge 1$ and $q+2 \le S$.

Let us complete the construction. Construct the binary shell paths before the unary shell paths. Starting at $(p,q) = (P,0)$, the numbers of shell paths in (3.9) give $(p,q) = (0,S)$. At each step, define the shell grading by (3.1) and its coloring by (3.3). At $(p,q) = (0,S)$, Y is a translated copy of $A \dotplus X \dotplus B$. Lemma 2.10 transports the colored grading on Y to $A \dotplus X \dotplus B$.

Let us verify (3.7) and prove that i_d is injective. By Claim 4 of Lemma 3.3, the map i_d copies the grade and color of X_k to $(A \dotplus X \dotplus B)_{|A|+k}$ for every $k \in [|X|]_{\mathbb{Z}}$. This proves (3.7). For $(g_1,\gamma_1) \in \mathcal{C}_d(X)$ and $(g_2,\gamma_2) \in \mathcal{C}_d(X)$, equation (3.7) gives

$$i_d(g_1,\gamma_1) = i_d(g_2,\gamma_2) \implies \begin{cases} g_1(X_k) = g_2(X_k), \\ \gamma_1(X_k) = \gamma_2(X_k) \end{cases} \qquad (k \in [|X|]_{\mathbb{Z}}).$$

Hence, $(g_1,\gamma_1) = (g_2,\gamma_2)$ and i_d is injective. □

Lemma 3.7 (A shifted supermultiplicative limit)**.** *Let $(A_n)_{n \in \mathbb{Z}_{>0}}$ be a sequence of real numbers and let $C \ge 0$. Assume*

$$A_n \ge 1, \qquad \log A_n \le Cn \quad (n \in \mathbb{Z}_{>0}), \qquad A_{i+j+2} \ge A_i A_j \quad (i,j \in \mathbb{Z}_{>0}).$$

Then, the following statements hold.

1. *The limit exists and satisfies*

$$L := \lim_{n\to\infty} \frac{\log A_n}{n} = \sup_{n \ge 1} \frac{\log A_n}{n+2} \in [0,C].$$

2. *For $n \in \mathbb{Z}_{>0}$, we have $A_n \le e^{(n+2)L}$.*

Proof. Proof of Claim 1. Let us write $\nu = n+2$. Let $B_\nu = \log A_{\nu-2}$ for $\nu \in \mathbb{Z}_{\geq 3}$ and let

$$L_0 = \sup_{\nu \geq 3} \frac{B_\nu}{\nu}.$$

We have $0 \leq B_\nu \leq C\nu$ for $\nu \in \mathbb{Z}_{\geq 3}$ and $B_{u+v} \geq B_u + B_v$ for $u \in \mathbb{Z}_{\geq 3}$ and $v \in \mathbb{Z}_{\geq 3}$. Thus, $L_0 \in [0, C]$. Fix $t \in \mathbb{Z}_{\geq 3}$. For $\nu \geq 3$, let $\nu = kt + r$ for $k \in \mathbb{Z}_{\geq 0}$ and $r \in [3, t+2]_{\mathbb{Z}}$. Induction on k gives

$$B_\nu \geq kB_t + B_r \geq kB_t.$$

Since $\frac{k}{\nu} \to \frac{1}{t}$, we obtain

$$\liminf_{\nu \to \infty} \frac{B_\nu}{\nu} \geq \frac{B_t}{t}.$$

Taking the supremum over t gives a lower bound of L_0. The definition of L_0 gives $\frac{B_\nu}{\nu} \leq L_0$. Hence, $\frac{B_\nu}{\nu} \to L_0$ and

$$\frac{\log A_n}{n} = \frac{n+2}{n} \frac{B_{n+2}}{n+2} \longrightarrow L_0.$$

Replacing ν by $n+2$ in the definition of L_0 proves Claim 1.

Proof of Claim 2. Let us bound the error in the finite growth estimate. Claim 1 gives $\log A_n \leq (n+2)L$. Exponentiation gives the assertion. □

Proposition 3.8 (Periodic growth)**.** *Let W be a run with q simple runs. Assume the following.*

(a) Every simple run of W has vertical length at most $m-1$.

(b) Let $A_j = C_0(W^j)$ for $j \geq 1$.

Then, we have the following.

1. *For all integers $i \geq 1$ and $j \geq 1$, we have*

$$A_{i+j+2} \geq A_i A_j.$$

2. *We have*

$$\lim_{j \to \infty} \frac{\log A_j}{j} = \sup_{j \geq 1} \frac{\log A_j}{j+2} \in (0, \infty). \tag{3.10}$$

3. *For every $j \geq 1$, we have*

$$A_j \leq \exp\left((j+2) \lim_{k \to \infty} \frac{\log A_k}{k}\right).$$

Proof. Proof of Claim 1. Let us construct a colored grading on W^{i+j+2} from colored gradings on W^i and W^j. Let $i \geq 1$ and $j \geq 1$. Let $a_1, \ldots, a_q$ be the vertical lengths of the simple runs of W. Let $X = W^i$ and $Y = W^j$. For $(g_1, \gamma_1) \in \mathcal{C}_0(X)$ and $(g_2, \gamma_2) \in \mathcal{C}_0(Y)$, form the shell paths

$$Z_1 = \mathrm{Shell}(X, Y; a_q, a_1, 0),$$
$$Z_{s+1} = \mathrm{Shell}(Z_s; a_{q-s}, a_{s+1}, 0) \quad (s \in [q-1]_{\mathbb{Z}}).$$

Give each Z_s its shell grading (3.1) and coloring (3.3). Assumption (a) and Claims 2 and 3 of Lemma 3.3 give a connected colored grading of surplus zero on Z_q. By Lemma 2.10, translate Z_q and its colored grading to $Z = W \dotplus X \dotplus Y \dotplus W = W^{i+j+2}$. Denote the resulting colored grading by (h, δ). This defines the map

$$\Phi_{i,j} \colon \mathcal{C}_0(W^i) \times \mathcal{C}_0(W^j) \longrightarrow \mathcal{C}_0(W^{i+j+2}), \qquad \Phi_{i,j}((g_1, \gamma_1), (g_2, \gamma_2)) = (h, \delta).$$

Let us recover (g_1, γ_1) and (g_2, γ_2) to prove that $\Phi_{i,j}$ is injective. Claim 4 of Lemma 3.3 gives

$$\begin{aligned} h(Z_{|W|+k}) &= g_1(X_k), \qquad \delta(Z_{|W|+k}) = \gamma_1(X_k) \quad (k \in [|X|]_{\mathbb{Z}}), \\ h(Z_{|W|+|X|+k}) &= g_2(Y_k), \quad \delta(Z_{|W|+|X|+k}) = \gamma_2(Y_k) \quad (k \in [|Y|]_{\mathbb{Z}}). \end{aligned} \tag{3.11}$$

Equation (3.11) determines both (g_1, γ_1) and (g_2, γ_2) from (h, δ). Hence, $\Phi_{i,j}$ is injective and $A_{i+j+2} \geq A_i A_j$.

Proof of Claim 2. Let us prove (3.10). Lemma 3.5 gives $A_j \geq 1$ and $\log A_j \leq m^2 qj \log 2$ for $j \geq 1$. Claim 1 gives $A_{i+j+2} \geq A_i A_j$ for $i \in \mathbb{Z}_{>0}$ and $j \in \mathbb{Z}_{>0}$. Claim 1 of Lemma 3.7 therefore gives (3.10). The lower bound in Lemma 3.5 also gives

$$\lim_{j\to\infty} \frac{\log A_j}{j} \geq \frac{q}{2} \log m > 0.$$

Proof of Claim 3. Claim 2 of Lemma 3.7 proves Claim 3. □

Definition 3.9. *Let W be a run whose simple runs have vertical lengths at most $m-1$. We define*

$$\ell(W) = \lim_{j\to\infty} \frac{\log C_0(W^j)}{j}.$$

3.2 A uniform upper bound for finite dimension pairs

Lemma 3.10 (Reduction of dimension pairs)**.** *Let $m \in \mathbb{Z}_{\geq 3}$ and let $a \in \mathbb{Z}_{>0}$ and $b \in \mathbb{Z}_{>0}$ be coprime. Let $Q = mab - a^2 - b^2 > 0$. Then, there exist coprime $u \in \mathbb{Z}_{>0}$ and $v \in \mathbb{Z}_{>0}$ such that*

$$u \leq v \leq \frac{mu}{2}, \qquad muv - u^2 - v^2 = Q, \qquad E_m(u, v) = E_m(a, b),$$

$$u + v \leq \sqrt{\frac{m+2}{m-2}}\sqrt{Q}.$$

Proof. Let us obtain (u, v) by decreasing $x + y$. Let

$$(x, y) = (\min(a, b), \max(a, b)).$$

If $2y > mx$, replace (x, y) by

$$(\min(x, mx - y), \max(x, mx - y)).$$

Claims 1, 2(a), and 2(b) of Lemma 2.1 show that each replacement preserves the Euler characteristic, coprimality, and Q. The inequality $mxy - x^2 - y^2 = Q > 0$ implies $y < mx$. Thus, both new entries are positive. Their sum changes by $mx - 2y < 0$. A strictly decreasing sequence of positive integers is finite. Thus, the replacements end at a pair $u \leq v \leq \frac{mu}{2}$.

Let us bound $u + v$. We have $mv - 2u \geq 0$ and $mu - 2v \geq 0$. Therefore,

$$(m+2)Q - (m-2)(u+v)^2 = (mv - 2u)(mu - 2v) \geq 0.$$

Dividing by $m - 2 > 0$ and taking square roots gives $u + v \leq \sqrt{\frac{m+2}{m-2}}\sqrt{Q}$. □

Proposition 3.11. *For every $(a, b) \in \mathcal{A}_m$, let $\rho(a, b) = \frac{b}{a}$. Then, we have*

$$\begin{aligned} 0 \leq \frac{\log E_m(a, b)}{a} \leq m \log 2 \sqrt{\frac{m+2}{m-2}} \sqrt{\max(m\rho(a, b) - 1 - \rho(a, b)^2, 0)} \\ + \frac{\log\left(\frac{2^m}{m}\right)}{a}. \end{aligned} \tag{3.12}$$

Proof. Let us separate the cases according to the sign of $Q = mab - a^2 - b^2$. First, suppose $Q > 0$. Lemma 3.10 gives a coprime pair (u, v) satisfying

$$u \le v \le \frac{mu}{2}, \qquad E_m(u, v) = E_m(a, b), \qquad u + v \le \sqrt{\frac{m+2}{m-2}}\sqrt{Q}.$$

Let us bound the Euler characteristic at the reduced pair (u, v). If $u \ge 2$, we have $v > u$ because u and v are coprime. Then, Claim 2 of Proposition 2.29 applies. Equation (2.9) gives $\Gamma(u, v) < \frac{v}{u} \le \frac{m}{2}$. The floor-difference formula in Definition 2.25 gives

$$\Lambda_i(u, v) \le \lceil \Gamma(u, v) \rceil \le \left\lceil \frac{m}{2} \right\rceil \le m - 1 \qquad (i \in [u]_{\mathbb{Z}}).$$

The final simple run is R_1 by (2.10). The lower bound in Lemma 3.5 with $L = m - 1$ and $\nu = u + 1$ gives $C_0(\mathcal{U}(u, v)) \ge m$.

Let us obtain the upper bound when $u \ge 2$. The run $\mathcal{U}(u, v)$ has exactly $u + v + 1$ edges. At each edge, the number of choices of a grade and its color is

$$\sum_{k=0}^{m} \binom{m}{k} = 2^m.$$

Therefore,

$$C_0(\mathcal{U}(u, v)) \le (2^m)^{u+v+1} = 2^{m(u+v+1)}.$$

Dividing by m in (2.13) gives

$$1 \le E_m(u, v) \le \frac{2^{m(u+v+1)}}{m}. \tag{3.13}$$

If $u = 1$, the stability condition (1.1) means that the images of the m arrow maps span $\mathbb{C}^v$. The moduli space $\mathcal{M}^s_{(1,v)}(K(m))$ is the Grassmannian of v-dimensional quotients of $\mathbb{C}^m$. Then, $E_m(1, v) = \binom{m}{v}$. The bounds in (3.13) also hold for $u = 1$.

Let us apply $u + v \le \sqrt{\frac{m+2}{m-2}}\sqrt{Q}$ from Lemma 3.10. Taking logarithms in (3.13) therefore yields

$$0 \le \log E_m(a, b) \le m \log 2 \sqrt{\frac{m+2}{m-2}Q} + \log\left(\frac{2^m}{m}\right). \tag{3.14}$$

Second, suppose $Q = 0$. No positive integral pair has this value, because $m \ge 3$ gives

$$(m-1)^2 < m^2 - 4 < m^2.$$

Third, suppose $Q < 0$. Let us prove $Q = -1$ and $E_m(a, b) = 1$. The nonempty stable moduli space $\mathcal{M}^s_{(a,b)}(K(m))$ has dimension $1 + Q$ [19, part (2) of Proposition 4.3]. Since Q is a negative integer, we obtain $Q = -1$.

Let us prove $E_m(a, b) = 1$ when $Q = -1$. The stable locus is a nonempty open subset of $\mathrm{Hom}_{\mathbb{C}}(\mathbb{C}^a, \mathbb{C}^b)^m$ and is therefore irreducible. The smooth geometric quotient $\mathcal{M}^s_{(a,b)}(K(m))$ is irreducible and zero-dimensional. Hence, $\mathcal{M}^s_{(a,b)}(K(m))$ consists of one point. Replacing Q by $\max(Q, 0)$ in (3.14) gives a bound on all of $\mathcal{A}_m$. Dividing by a gives (3.12). □

4 Run-edits

4.1 Insertion and deletion

Definition 4.1. *Let* () *denote the empty tuple.*

1. *Let U be a run. Let $i \in [0, |U|]_{\mathbb{Z}}$. Let $v \in N(U)$. Let $j \geq 1$ be an integer. We refer to any one of the operations*

$$U \mapsto \mathrm{Ins}(U, i, N), \qquad U \mapsto \mathrm{Del}(U, v), \qquad U \mapsto \mathrm{Ins}(U, i, R_j),$$

or an exchange of two adjacent simple runs as a run-edit of U.

2. *We refer to a run-edit that inserts or deletes a vertical edge as a vertical run-edit.*

3. *Let U_0 be a run and $\ell \in \mathbb{Z}_{\geq 0}$. For each $j \in [\ell]_{\mathbb{Z}}$, let ω_j be a run-edit of U_{j-1} and define*

$$U_j = \omega_j(U_{j-1}).$$

We call $\Omega = (\omega_j)_{j \in [\ell]_{\mathbb{Z}}}$ a run-edit list from U_0 to U_ℓ. We define

$$\mathrm{RunEResult}(\Omega, U_0) = U_\ell, \qquad \mathrm{RunEResult}((), U_0) = U_0.$$

Lemma 4.2 (H-precursors after insertion)**.** *Let $Y = Y_1 +\!\!+ \cdots +\!\!+ Y_n$ be a run, let $g \in \mathrm{CtdG}(Y)$. Let $q \in [1, n-1]_{\mathbb{Z}}$. Put*

$$h_0 = Y_1, \qquad W = \mathrm{Ins}(Y, q, N), \qquad w_* = W_{q+1}.$$

Define the map $\tau : \mathrm{NE}(Y) \to \mathrm{NE}(W)$ by

$$\tau(Y_k) = \begin{cases} W_k, & k \in [q]_{\mathbb{Z}}, \\ W_{k+1}, & k \in [q+1, n]_{\mathbb{Z}}. \end{cases}$$

Define $g' \in G(W)$ by

$$g'(w_*) = 0, \qquad g'(\tau(h_0)) = g(h_0) + 1, \qquad g'(\tau(e)) = g(e) \quad (e \in \mathrm{NE}(Y) \setminus \{h_0\}). \tag{4.1}$$

Let

$$H_q = \{r \in [q]_{\mathbb{Z}} : Y_r \in E(Y),\ \mathrm{Debt}_{Y,g}(Y_r, q) > 0\}.$$

The following statements hold.

1. *The set H_q contains 1. Its decreasing enumeration is uniquely determined:*

$$H_q = \{r_0, \ldots, r_d\}, \qquad r_0 > \cdots > r_d = 1.$$

For $j \in [0, d]_{\mathbb{Z}}$, put

$$a_j = Y_{r_j}, \qquad A_j = \mathrm{Node}_{Y,g}(a_j), \qquad t_j = \mathrm{End}_{\mathrm{NE}}(A_j) = Y_{s_j}.$$

Then,

$$A_0 \subsetneq A_1 \subsetneq \cdots \subsetneq A_d = Y, \qquad q < s_0 < \cdots < s_d = n, \tag{4.2}$$

and A_{j+1} is the parent of A_j for $j \in [0, d-1]_{\mathbb{Z}}$.

2. *Use the indices from Claim 1. The h-precursor $P_{W,g'}$ exists and satisfies*

$$\begin{aligned} P_{W,g'}(w_*) &= \tau(a_0), \\ P_{W,g'}(\tau(t_j)) &= \tau(a_{j+1}) \qquad (j \in [0, d-1]_{\mathbb{Z}}), \\ P_{W,g'}(\tau(v)) &= \tau(P_{Y,g}(v)) \quad \big(v \in N(Y) \setminus \{t_j : j \in [0, d-1]_{\mathbb{Z}}\}\big). \end{aligned} \tag{4.3}$$

For $d = 0$, the middle line has no instances.

3. *For $e \in E(Y)$, put $F_e = P_{Y,g}^{-1}(\{e\})$. For a set of edges F, write $\tau(F) = \{\tau(v) : v \in F\}$. Define*

$$b_0 = w_*, \qquad b_j = \tau(t_{j-1}) \quad (j \in [d]_{\mathbb{Z}}).$$

Then,

$$\begin{aligned} P_{W,g'}^{-1}(\{\tau(a_j)\}) &= \tau(F_{a_j} \setminus \{t_j\}) \sqcup \{b_j\} \quad (j \in [0, d-1]_{\mathbb{Z}}), \\ P_{W,g'}^{-1}(\{\tau(a_d)\}) &= \tau(F_{a_d}) \sqcup \{b_d\}, \\ P_{W,g'}^{-1}(\{\tau(e)\}) &= \tau(F_e) \qquad\qquad (e \in E(Y) \setminus \{a_0, \dots, a_d\}). \end{aligned} \tag{4.4}$$

Consequently, every $e \in E(Y)$ satisfies

$$\#P_{W,g'}^{-1}(\{\tau(e)\}) = g'(\tau(e)),$$

and

$$\mathrm{Node}_{W,g'}(\tau(h_0)) = W.$$

Proof. Proof of Claim 1. Let us enumerate H_q and the nodes $\mathrm{Node}_{Y,g}(Y_r)$ for $r \in H_q$. Connectedness gives $Y = \mathrm{Node}_{Y,g}(h_0)$. Every positive-grade horizontal edge has a node by Claim 1(b) of Lemma 2.15. If $e = Y_r \in E(Y[0,q])$ and its node ends at Y_s, then

$$\mathrm{Debt}_{Y,g}(e, q) > 0 \quad \Longleftrightarrow \quad q < s. \tag{4.5}$$

Let us order these nodes by inclusion. We have $P_{Y,g}(Y_n) = h_0$ and $q < n$. Equation (4.5) gives $1 \in H_q$. Every node indexed by H_q contains the boundary at q. Claim 2 of Lemma 2.15 therefore orders them by inclusion. The order $r_0 > \cdots > r_d = 1$ gives

$$A_0 \subsetneq A_1 \subsetneq \cdots \subsetneq A_d = Y.$$

Claim 1(c) of Lemma 2.15 gives

$$q < s_0 < \cdots < s_d = n.$$

Let us prove that A_{j+1} is the parent of A_j for $j \in [0, d-1]_{\mathbb{Z}}$. If a node $\mathrm{Node}_{Y,g}(Y_r)$ satisfied

$$A_j \subsetneq \mathrm{Node}_{Y,g}(Y_r) \subsetneq A_{j+1},$$

then it would contain the boundary at q. Hence,

$$r \in H_q, \qquad r_{j+1} < r < r_j.$$

This contradicts the consecutive enumeration of H_q. Definition 2.16(1) proves the parent assertion.

Proof of Claim 2. Let us verify Definition 2.6(1)(a)–(c) for Q. Define $Q : N(W) \to E(W)$ by (4.3). Define its candidate debt, before knowing that Q is an h-precursor, by

$$D_Q(x, \ell) = g'(x) - \#\{v \in N(W[0,\ell]) : Q(v) = x\}, \qquad x \in E(W[0,\ell]). \tag{4.6}$$

Write $\mathbf{1}_{\mathcal{P}} = 1$ when $\mathcal{P}$ holds and 0 otherwise. For $k \in [0,q]_{\mathbb{Z}}$ and $e \in E(Y[0,k])$, equations (4.1), (4.3), and (4.6) give

$$D_Q(\tau(e), k) = \mathrm{Debt}_{Y,g}(e, k) + \mathbf{1}_{e=h_0}. \tag{4.7}$$

For $k \in [q,n]_{\mathbb{Z}}$ and $e \in E(Y[0,k])$, equations (4.1), (4.3), and (4.6) give

$$\begin{aligned} D_Q(\tau(e), k+1) = \mathrm{Debt}_{Y,g}(e, k) &+ \mathbf{1}_{e=h_0} - \mathbf{1}_{e=a_0} \\ &+ \sum_{\substack{j \in [0,d-1]_{\mathbb{Z}} \\ s_j \le k}} (\mathbf{1}_{e=a_j} - \mathbf{1}_{e=a_{j+1}}). \end{aligned} \tag{4.8}$$

w_* consumes one unit at a_0. Changing the owner of t_j returns one unit at a_j and consumes one unit at a_{j+1}.

Let us verify $Q(\tau(Y_k)) = \tau(P_{Y,g}(Y_k))$ for $Y_k \in N(Y)$ with $k \le q$, and $Q(w_*) = \tau(a_0)$. The debt of h_0 is positive at every boundary before n. By (4.7), adding one unit to this first horizontal edge does not change the last horizontal edge of positive debt. Thus, Q agrees with the h-precursor rule before w_*. At boundary q, the last positive-debt edge is a_0. Thus, $Q(w_*) = \tau(a_0)$.

Let us verify $Q(\tau(Y_k))$ for $Y_k \in N(Y)$ with $k > q$. Put $s_{-1} = q$. For $j \in [0,d]_{\mathbb{Z}}$, $s_{j-1} < k \le s_j$, and $e \in E(Y[0,k-1])$, the sum in (4.8) telescopes to

$$D_Q(\tau(e),k) = \mathrm{Debt}_{Y,g}(e,k-1) + \mathbf{1}_{e=h_0} - \mathbf{1}_{e=a_j}. \tag{4.9}$$

Here boundary k of W is immediately before $\tau(Y_k) = W_{k+1}$.

First, suppose $j < d$ and Y_k is vertical with $k < s_j$. If $P_{Y,g}(Y_k) = a_j$, then both Y_k and t_j have owner A_j. Thus,

$$\mathrm{Debt}_{Y,g}(a_j,k-1) \ge 2, \qquad D_Q(\tau(a_j),k) \ge 1. \tag{4.10}$$

If $P_{Y,g}(Y_k) \ne a_j$, its h-precursor follows a_j, because a_j has positive debt until s_j. Equation (4.9) leaves its debt unchanged. By (4.9) and (4.10), in both cases $D_Q(\tau(P_{Y,g}(Y_k)),k) > 0$ and $D_Q(\tau(Y_r),k) = 0$ for every horizontal edge Y_r strictly between $P_{Y,g}(Y_k)$ and Y_k. Hence, $Q(\tau(Y_k)) = \tau(P_{Y,g}(Y_k))$.

Second, suppose $j < d$ and $k = s_j$. We have $t_j = Y_{s_j}$, $P_{Y,g}(t_j) = a_j$, and $s_j = \max\{k : Y_k \in N(Y),\ P_{Y,g}(Y_k) = a_j\}$. Its parent node ends at $s_{j+1} > s_j$ by (4.2). Hence, (4.9) gives

$$\begin{aligned} \mathrm{Debt}_{Y,g}(a_j,s_j-1) &= 1, \qquad D_Q(\tau(a_j),s_j) = 0, \\ \mathrm{Debt}_{Y,g}(a_{j+1},s_j-1) &\ge 1, \quad D_Q(\tau(a_{j+1}),s_j) \ge 1. \end{aligned} \tag{4.11}$$

Let us exclude a later horizontal edge of positive candidate debt when $j < d$ and $k = s_j$. Let $r \in [r_{j+1}+1, s_j-1]_{\mathbb{Z}} \setminus \{r_j\}$ satisfy $Y_r \in E(Y)$. If $g(Y_r) > 0$, its node is contained in A_j, or is disjoint from A_j and ends before A_j starts. Claim 1 excludes a node strictly between A_j and A_{j+1}. Claims 1(c) and 2 of Lemma 2.15 give

$$\mathrm{Debt}_{Y,g}(Y_r, s_j - 1) = 0.$$

If $g(Y_r) = 0$, then $\mathrm{Debt}_{Y,g}(Y_r, s_j - 1) = 0$ by Claim 1 of Lemma 2.13 and Definition 2.6(3). Equation (4.9) therefore gives

$$D_Q(\tau(Y_r), s_j) = 0.$$

The equalities $D_Q(\tau(Y_r), s_j) = 0$ for $r \in [r_{j+1}+1, s_j-1]_{\mathbb{Z}} \setminus \{r_j\}$ satisfying $Y_r \in E(Y)$, together with (4.11), show that $\tau(a_{j+1})$ is the last horizontal edge of positive candidate debt. Thus, $Q(\tau(t_j)) = \tau(a_{j+1})$.

Third, suppose $j = d$. The two indicator terms in (4.9) cancel because $a_d = h_0$. Thus, $Q(\tau(Y_k)) = \tau(P_{Y,g}(Y_k))$ for every $Y_k \in N(Y)$ with $s_{d-1} < k \le s_d$. Induction through the edges verifies all three conditions of Definition 2.6(1). Uniqueness in Claim 1 of Lemma 2.7 gives $Q = P_{W,g'}$.

Proof of Claim 3. Let us count the sets $P^{-1}_{W,g'}(\{\tau(e)\})$ for $e \in E(Y)$. For $j < d$, the fiber at a_j loses t_j and gains exactly b_j. The fiber at a_d gains b_d and loses no edge. Every other fiber is copied by τ. These unions are disjoint: w_* is new. For $j \in [d]_{\mathbb{Z}}$, we have $P_{Y,g}(t_{j-1}) = a_{j-1} \ne a_j$. This proves (4.4). Connectedness of g gives $\#F_e = g(e)$, including $F_e = \varnothing$ when $g(e) = 0$. Therefore, $\#P^{-1}_{W,g'}(\{\tau(e)\}) = g'(\tau(e))$ for every $e \in E(Y)$. The last edge of $P^{-1}_{W,g'}(\{\tau(h_0)\})$ is $\tau(Y_n) = W_{n+1}$. Thus, $\mathrm{Node}_{W,g'}(\tau(h_0)) = W$. For $d = 0$, this argument reads $P^{-1}_{W,g'}(\{\tau(h_0)\}) = \tau(F_{h_0}) \sqcup \{w_*\}$. □

Lemma 4.3 (Balances after insertion)**.** *Use the hypotheses and construction of Lemma 4.2. Then, the following statements hold.*

1. *For $\ell \in [0, n+1]_{\mathbb{Z}}$,*

$$\mathrm{bBal}_{W,g'}(\ell) = \begin{cases} \mathrm{bBal}_{Y,g}(\ell), & 0 \le \ell \le q, \\ \mathrm{bBal}_{Y,g}(\ell - 1), & q+1 \le \ell \le n+1. \end{cases}$$

   Consequently,

$$\mathrm{eBal}_{W,g'}(\tau(e)) = \mathrm{eBal}_{Y,g}(e) \quad (e \in \mathrm{NE}(Y)), \qquad \mathrm{eBal}_{W,g'}(w_*) = \mathrm{bBal}_{Y,g}(q).$$

2. *We have*

$$g' \in \mathrm{CtdG}(W), \qquad \mathrm{Surp}(W, g') = \mathrm{Surp}(Y, g).$$

   If $m \in \mathbb{Z}_{\ge 0}$, $g \in G_m(Y)$, and $g(h_0) + 1 \le m$, then $g' \in \mathrm{CtdG}_m(W)$.

Proof. Proof of Claim 1. Let us compute the two terms in the boundary balance. Definition 2.4(3)(a) gives

$$\mathrm{bBal}_{W,g'}(\ell) = \#E(W[0,\ell]) - \sum_{v \in N(W[0,\ell])} g'(v) \qquad (\ell \in [0, n+1]_{\mathbb{Z}}).$$

First, let us compute this expression for $\ell \in [0, q]_{\mathbb{Z}}$. The insertion map τ gives

$$E(W[0,\ell]) = \tau(E(Y[0,\ell])), \qquad N(W[0,\ell]) = \tau(N(Y[0,\ell])).$$

Equation (4.1) gives $g'(\tau(v)) = g(v)$ for every $v \in N(Y)$. Hence,

$$\mathrm{bBal}_{W,g'}(\ell) = \#E(Y[0,\ell]) - \sum_{v \in N(Y[0,\ell])} g(v) = \mathrm{bBal}_{Y,g}(\ell).$$

Second, let us compute the expression for $\ell \in [q+1, n+1]_{\mathbb{Z}}$. The prefix contains the new vertical edge w_*, and

$$E(W[0,\ell]) = \tau(E(Y[0,\ell-1])),$$
$$N(W[0,\ell]) = \tau(N(Y[0,\ell-1])) \sqcup \{w_*\}.$$

Since $g'(w_*) = 0$, we obtain

$$\mathrm{bBal}_{W,g'}(\ell) = \#E(Y[0,\ell-1]) - \sum_{v \in N(Y[0,\ell-1])} g(v) - 0 = \mathrm{bBal}_{Y,g}(\ell - 1).$$

Let us evaluate the boundary balance after each edge. For $k \in [q]_{\mathbb{Z}}$, we have $\tau(Y_k) = W_k$. For $k \in [q+1, n]_{\mathbb{Z}}$, we have $\tau(Y_k) = W_{k+1}$. Substituting $\mathrm{bBal}_{W,g'}(k) = \mathrm{bBal}_{Y,g}(k)$ for $k \le q$ and $\mathrm{bBal}_{W,g'}(k+1) = \mathrm{bBal}_{Y,g}(k)$ for $k > q$ gives

$$\mathrm{eBal}_{W,g'}(\tau(Y_k)) = \mathrm{bBal}_{Y,g}(k) = \mathrm{eBal}_{Y,g}(Y_k) \qquad (k \in [n]_{\mathbb{Z}}).$$

Since $w_* = W_{q+1}$, they also give

$$\mathrm{eBal}_{W,g'}(w_*) = \mathrm{bBal}_{W,g'}(q+1) = \mathrm{bBal}_{Y,g}(q).$$

Proof of Claim 2. Let us verify the node balance condition. Lemma 4.2 gives the h-precursor, every positive-grade horizontal node, and the root node W. Let $v \in N(Y)$, write $a = P_{Y,g}(v)$. Let b satisfy $P_{W,g'}(\tau(v)) = \tau(b)$. Claim 2 of Lemma 4.2 gives either $b = a$ or

$$\mathrm{Node}_{Y,g}(a) \subsetneq \mathrm{Node}_{Y,g}(b).$$

The node balance condition for g, Claim 3(b) of Lemma 2.15, and Claim 1 give

$$\mathrm{eBal}_{W,g'}(\tau(v)) = \mathrm{eBal}_{Y,g}(v) \geq \mathrm{eBal}_{Y,g}(a) \geq \mathrm{eBal}_{Y,g}(b) = \mathrm{eBal}_{W,g'}(\tau(b)).$$

The inequality $\mathrm{eBal}_{Y,g}(a) \geq \mathrm{eBal}_{Y,g}(b)$ is equality when $a = b$.

Let us check the balance at the inserted edge. For the inserted edge, Claim 1 of Lemma 4.2 gives $r_0 \leq q < s_0$. Claim 3(a) of Lemma 2.15 applied to $A_0 = Y[r_0 - 1, s_0]$ yields

$$\mathrm{eBal}_{W,g'}(w_*) = \mathrm{bBal}_{Y,g}(q) \geq \mathrm{eBal}_{Y,g}(a_0) = \mathrm{eBal}_{W,g'}(P_{W,g'}(w_*)).$$

Definition 2.9(1) now gives connectedness.

Let us compute the surplus. We have

$$\#E(W) = \#E(Y), \qquad \sum_{v \in N(W)} g'(v) = \sum_{v \in N(Y)} g(v) + 0.$$

Substitution in Definition 2.9(2) gives $\mathrm{Surp}(W, g') = \mathrm{Surp}(Y, g)$. If $g \in G_m(Y)$ and $g(h_0) + 1 \leq m$, equation (4.1) gives $g'(\tau(h_0)) \leq m$ and $g'(e) \leq m$ for every $e \in \mathrm{NE}(W) \setminus \{\tau(h_0)\}$. Thus, $g' \in G_m(W)$. □

Lemma 4.4 (Insertion)**.** *Let $M \in [m-2]_{\mathbb{Z}}$, $u \in [M]_{\mathbb{Z}}$, and $v \in [M]_{\mathbb{Z}}$. Let X be a run such that $\#N(A) \leq M$ for every simple run $A \subseteq X$. Let $c \in \mathbb{Z}_{\geq 0}$. Consider $(g, \gamma) \in \mathcal{C}_c(X)$. Use the following construction.*

(a) Let $Y = \mathrm{Shell}(X; u, v, 0)$ with the shell grading $\widetilde{g} = \mathrm{GShell}_g(X; u, v, 0)$.

(b) Let $i \in [|X|]_{\mathbb{Z}}$. Let

$$W = \mathrm{Ins}(Y, u + 1 + i, N) = R_u +\!\!+ \mathrm{Ins}(X, i, N) +\!\!+ R_v.$$

Let $q = u + 1 + i$. Define $g' \in G(W)$ by

$$g'(W_k) = \begin{cases} u + 2, & k = 1, \\ \widetilde{g}(Y_k), & k \in [2, q]_{\mathbb{Z}}, \\ 0, & k = q + 1, \\ \widetilde{g}(Y_{k-1}), & k \in [q + 2, |Y| + 1]_{\mathbb{Z}}. \end{cases}$$

(c) Define $\kappa\colon \mathrm{NE}(X) \hookrightarrow \mathrm{NE}(W)$ by

$$\kappa(X_k) = \begin{cases} W_{u+1+k}, & k \in [i]_{\mathbb{Z}}, \\ W_{u+2+k}, & k \in [i+1, |X|]_{\mathbb{Z}}. \end{cases} \tag{4.12}$$

Let

$$\gamma' = \mathrm{Recol}(X, W, \kappa, g'; g, \gamma).$$

Then, we have the following.

1. *The grading g' has the following properties.*
   (a) $g' \in \mathrm{CtdG}(W)$.
   (b) We have $\mathrm{Surp}(W, g') = \mathrm{Surp}(X, g)$.
2. *We have*

$$g'(\kappa(X_k)) = g(X_k), \qquad \gamma'(\kappa(X_k)) = \gamma(X_k) \quad (k \in [|X|]_{\mathbb{Z}}). \tag{4.13}$$

3. Fix X, i, u, and v. The construction (a)–(c) gives an injective map

$$\Phi_c\colon \mathcal{C}_c(X) \longrightarrow \mathcal{C}_c(W)$$

such that $\Phi_c(g,\gamma) = (g',\gamma')$ satisfies (4.13) for every $(g,\gamma) \in \mathcal{C}_c(X)$.

Proof. Proof of Claim 1(a). Let us apply the insertion lemmas to the shell grading. Put $q = u+1+i$ and $h_0 = \mathrm{Start}_{\mathrm{NE}}(Y)$. The hypotheses give

$$1 \le q < |Y|, \qquad \widetilde{g}(h_0) + 1 = u + 2 \le M + 2 \le m.$$

Claim 2 of Lemma 3.3 gives $\widetilde{g} \in \mathrm{CtdG}_m(Y)$. Apply Lemma 4.2 to $(Y, \widetilde{g}, q)$. Claim 2 of Lemma 4.3 gives $g' \in \mathrm{CtdG}_m(W)$.

Proof of Claim 1(b). Claim 2 of Lemma 4.3 and Claim 3 of Lemma 3.3 give

$$\mathrm{Surp}(W, g') = \mathrm{Surp}(Y, \widetilde{g}) = \mathrm{Surp}(X, g).$$

Proof of Claim 2. Let us verify (4.13). By (4.12), the copy of X_k in W has index $u+1+k$ for $k \in [i]_{\mathbb{Z}}$ and $u+2+k$ for $k \in [i+1, |X|]_{\mathbb{Z}}$. Assumptions (b) and (c) change no grade or color on these edges. Thus, the map (4.12) satisfies (4.13).

Proof of Claim 3. Let us recover the input from the inserted grading. Fix X, i, u, and v. For $(g_1,\gamma_1) \in \mathcal{C}_c(X)$ and $(g_2,\gamma_2) \in \mathcal{C}_c(X)$, equation (4.13) gives

$$\Phi_c(g_1,\gamma_1) = \Phi_c(g_2,\gamma_2) \implies (g_1(X_k), \gamma_1(X_k)) = (g_2(X_k), \gamma_2(X_k)) \quad (k \in [|X|]_{\mathbb{Z}}).$$

Hence, $(g_1,\gamma_1) = (g_2,\gamma_2)$. □

Lemma 4.5 (H-precursors after deletion)**.** *Let Z be a tNE path. Let $g \in G(Z)$. Suppose that the h-precursor $P_{Z,g}$ exists. Let $v = Z_r \in N(Z)$ and put $x = P_{Z,g}(v)$. Let $D = \mathrm{Del}(Z,v)$. Define*

$$\tau_v\colon \mathrm{NE}(Z) \setminus \{v\} \longrightarrow \mathrm{NE}(D), \qquad \tau_v(Z_k) = \begin{cases} D_k, & k < r, \\ D_{k-1}, & k > r. \end{cases}$$

Define $g_v^- \in G(D)$ by

$$g_v^-(\tau_v(e)) = \begin{cases} g(x) - 1, & e = x, \\ g(e), & e \ne x \end{cases} \qquad (e \in \mathrm{NE}(Z) \setminus \{v\}).$$

Then, the h-precursor P_{D,g_v^-} exists and satisfies

$$P_{D,g_v^-}(\tau_v(w)) = \tau_v(P_{Z,g}(w)) \qquad (w \in N(Z) \setminus \{v\}). \tag{4.14}$$

Proof. Let us define $Q\colon N(D) \to E(D)$ by

$$Q(\tau_v(w)) = \tau_v(P_{Z,g}(w)) \qquad (w \in N(Z) \setminus \{v\}).$$

The map τ_v preserves edge order and direction. Condition (b) of Definition 2.6(1) gives $g(x) > 0$. Thus, g_v^- has nonnegative grades. For $Z_s \in N(Z)$ and $f \in E(Z[0, s-1])$, define

$$n_f(s) = \#\{w \in N(Z[0,s-1]) : P_{Z,g}(w) = f\}.$$

For $s \ne r$, define

$$n'_f(s) = \#\{w \in N(Z[0,s-1]) \setminus \{v\} : Q(\tau_v(w)) = \tau_v(f)\}.$$

The definition of Q gives

$$n'_f(s) = \begin{cases} n_f(s) - 1, & s > r \text{ and } f = x, \\ n_f(s), & s < r \text{ or } f \neq x. \end{cases}$$

Let us verify that Q satisfies the h-precursor conditions of Definition 2.6(1) at $\tau_v(Z_s)$ when $s < r$. If $f = P_{Z,g}(Z_s) \neq x$, then $g_v^-(\tau_v(f)) = g(f)$ and $n'_f(s) = n_f(s)$. If $P_{Z,g}(Z_s) = x$, then

$$n_x(s) + 1 \leq n_x(r) < g(x), \qquad n'_x(s) < g(x) - 1.$$

Thus, $Q(\tau_v(Z_s))$ satisfies conditions (a) and (b) of Definition 2.6(1). For every $f \in E(Z[0, s-1])$, we have $n'_f(s) = n_f(s)$ and $g_v^-(\tau_v(f)) \leq g(f)$. Hence, every horizontal edge satisfying these two conditions after deletion also satisfies them before deletion. Condition (c) follows.

Let us verify Definition 2.6(1)(a)–(c) for Q at $\tau_v(Z_s)$ when $s > r$. For $f \neq x$, we have $g_v^-(\tau_v(f)) = g(f)$ and $n'_f(s) = n_f(s)$. For $f = x$ and $s > r$, we have $v = Z_r \in N(Z[0, s-1])$ and $P_{Z,g}(v) = x$. Thus,

$$n_x(s) \geq 1, \qquad n'_x(s) < g_v^-(\tau_v(x)) \iff n_x(s) - 1 < g(x) - 1 \iff n_x(s) < g(x).$$

Thus, the horizontal edges satisfying conditions (a) and (b) correspond under τ_v. The map τ_v preserves their order. Thus, condition (c) also holds. Therefore, Q is an h-precursor of g_v^-. Claim 1 of Lemma 2.7 gives (4.14). □

Lemma 4.6 (Deletion)**.** *Let $M \in [m-1]_{\mathbb{Z}}$. Let X be a run such that $\#N(A) \leq M$ for every simple run $A \subseteq X$. Let $v = X_r \in N(X)$. Let $c \in \mathbb{Z}_{\geq 0}$. Consider $(g, \gamma) \in \mathcal{C}_c(X)$. Let $h = g(v)$. Let $u_0 \in [M]_{\mathbb{Z}}$ and $v_0 \in [M]_{\mathbb{Z}}$.*

(a) Let $Y = \mathrm{Shell}(X; u_0, v_0, 0)$ with the shell grading $\widetilde{g} = \mathrm{GShell}_g(X; u_0, v_0, 0)$.

(b) Let

$$W = \mathrm{Del}(Y, Y_{u_0+1+r}) = R_{u_0} +\!\!+ \mathrm{Del}(X, v) +\!\!+ R_{v_0}.$$

Let $q = u_0 + 1 + r$. For each $x = X_j \in E(X[0, r-1])$ satisfying $g(x) > 0$, define $g'_x \in G(W)$ by

$$g'_x(W_k) = \begin{cases} g(x) - 1, & k = u_0 + 1 + j, \\ \widetilde{g}(Y_k), & k \in [1, q-1]_{\mathbb{Z}} \setminus \{u_0 + 1 + j\}, \\ \widetilde{g}(Y_{k+1}), & k \in [q, |Y| - 1]_{\mathbb{Z}}. \end{cases}$$

(c) Define $\kappa\colon \mathrm{NE}(X) \setminus \{v\} \hookrightarrow \mathrm{NE}(W)$ by

$$\kappa(X_k) = \begin{cases} W_{u_0+1+k}, & k \in [r-1]_{\mathbb{Z}}, \\ W_{u_0+k}, & k \in [r+1, |X|]_{\mathbb{Z}}. \end{cases}$$

For each $x \in E(X[0, r-1])$ satisfying $g(x) > 0$, define

$$\gamma'_x = \mathrm{Recol}(X, W, \kappa, g'_x; g, \gamma).$$

Then, we have the following.

1. *There exists $y \in E(X[0, r-1])$ with $g(y) > 0$ such that $g'_y \in \mathrm{CtdG}(W)$.*
2. *For every $x \in E(X[0, r-1])$ with $g(x) > 0$, we have*

$$\#E(W) - 1 - \sum_{w \in N(W)} g'_x(w) = \mathrm{Surp}(X, g) + h.$$

3. *For every $x \in E(X[0,r-1])$ with $g(x) > 0$, we have*

$$\begin{aligned} &g'_x(\kappa(x)) = g(x) - 1, \\ &g'_x(\kappa(e)) = g(e), \qquad \gamma'_x(\kappa(e)) = \gamma(e) \quad (e \in \mathrm{NE}(X) \setminus \{x, v\}). \end{aligned} \tag{4.15}$$

4. *Fix X, v, u_0, and v_0. Also, for each $(g,\gamma) \in \mathcal{C}_c(X)$, fix an edge $y(g,\gamma)$ satisfying Claim 1. The construction (a)–(c) gives a map*

$$\Phi_c \colon \mathcal{C}_c(X) \longrightarrow \bigcup_{s \in [c, c+m]_{\mathbb{Z}}} \mathcal{C}_s(W)$$

such that $\Phi_c(g,\gamma) = (g'_{y(g,\gamma)}, \gamma'_{y(g,\gamma)})$. The identities (4.15) hold at $x = y(g,\gamma)$. Also, we have

$$\mathrm{Surp}(W, g'_{y(g,\gamma)}) = c + g(v).$$

Proof. Proof of Claim 1. Let us choose $y \in E(X[0,r-1])$. Let $y = P_{X,g}(v)$. Definition 2.6(1) gives $y \in E(X[0,r-1])$ and $g(y) > 0$. Let $q = u_0 + 1 + r$ and let $\widetilde{y}$ be the copy of y in Y. Claim 2 of Lemma 3.3 gives $\widetilde{g} \in \mathrm{CtdG}_m(Y)$. Claim 5 of Lemma 3.3 gives $P_{Y,\widetilde{g}}(Y_q) = \widetilde{y}$.

Let us apply the deletion formula to the copied edge Y_q. Let $\tau \colon \mathrm{NE}(Y) \setminus \{Y_q\} \to \mathrm{NE}(W)$ be the map τ_{Y_q} of Lemma 4.5. That lemma applies to Y, $\widetilde{g}$, and Y_q and gives

$$P_{W,g'_y}(\tau(w)) = \tau(P_{Y,\widetilde{g}}(w)) \qquad (w \in N(Y) \setminus \{Y_q\}). \tag{4.16}$$

Let us verify the node balance condition. For $k \in [|Y|]_{\mathbb{Z}} \setminus \{q\}$, deletion of the vertical grade h gives

$$\mathrm{eBal}_{W,g'_y}(\tau(Y_k)) = \mathrm{eBal}_{Y,\widetilde{g}}(Y_k) + \begin{cases} 0, & k < q, \\ h, & k > q. \end{cases} \tag{4.17}$$

Let $w = Y_k \in N(Y) \setminus \{Y_q\}$ and let $P_{Y,\widetilde{g}}(w) = Y_j$. We have $j < k$. Thus, (4.17) gives

$$\begin{aligned} &\mathrm{eBal}_{W,g'_y}(\tau(w)) - \mathrm{eBal}_{W,g'_y}(\tau(Y_j)) \\ &\quad \geq \mathrm{eBal}_{Y,\widetilde{g}}(w) - \mathrm{eBal}_{Y,\widetilde{g}}(Y_j) \geq 0. \end{aligned}$$

The inequality $\mathrm{eBal}_{Y,\widetilde{g}}(w) - \mathrm{eBal}_{Y,\widetilde{g}}(Y_j) \geq 0$ follows from connectedness of $\widetilde{g}$ and Claim 1(b) of Lemma 2.15. Together with (4.16), this proves the node balance condition for g'_y.

Let us verify $\mathrm{Node}_{W,g'_y}(W_1) = W$. The edges Y_1 and $Y_{|Y|}$ belong to the copies of R_{u_0} and R_{v_0}, respectively. Deleting Y_q keeps Y_1 and $Y_{|Y|}$ and gives $g'_y(W_1) = \widetilde{g}(Y_1)$. We have $P_{Y,\widetilde{g}}(Y_q) = \widetilde{y} \neq Y_1$. Equation (4.16) therefore gives $P_{W,g'_y}(\tau(w)) = W_1$ for every $w \in P^{-1}_{Y,\widetilde{g}}(\{Y_1\})$. Since $\mathrm{Node}_{Y,\widetilde{g}}(Y_1) = Y$, we obtain

$$\mathrm{Node}_{W,g'_y}(W_1) = W.$$

Definition 2.9(1) gives $g'_y \in \mathrm{CtdG}(W)$. The grading g'_y decreases only the positive grade $g(y)$. Since $\widetilde{g} \in G_m(Y)$, we also have $g'_y \in G_m(W)$.

Proof of Claim 2. Let us sum the vertical grades. Let $x \in E(X[0,r-1])$ satisfy $g(x) > 0$. Deletion preserves the number of horizontal edges and removes vertical grade h. By (2.4) and Claim 3 of Lemma 3.3,

$$\#E(W) - 1 - \sum_{w \in N(W)} g'_x(w) = \#E(Y) - 1 - \left(\sum_{w \in N(Y)} \widetilde{g}(w) - h \right) = \mathrm{Surp}(X,g) + h.$$

Proof of Claim 3. Let us identify the unchanged grades and colors. Let $x \in E(X[0,r-1])$ with $g(x) > 0$. Assumption (b) gives the grade identities in (4.15). For $e \in \mathrm{NE}(X) \setminus \{x, v\}$, we have $g'_x(\kappa(e)) = g(e)$. Definition 3.2 and assumption (c) therefore give $\gamma'_x(\kappa(e)) = \gamma(e)$.

Proof of Claim 4. Let us prove $\Phi_c(g,\gamma) \in \bigcup_{s\in[c,c+m]_{\mathbb{Z}}} \mathcal{C}_s(W)$. Let $y = y(g,\gamma)$. For the chosen y, Claim 1 gives $g'_y \in \mathrm{CtdG}(W)$. Definition 3.2 gives $\gamma'_y \in \mathrm{Col}(W, g'_y, m)$. Definition 2.9(2), Claim 2, and $g(v) \in [0,m]_{\mathbb{Z}}$ give

$$c \le \mathrm{Surp}(W, g'_y) = c + g(v) \le c + m.$$

Claim 3 gives (4.15) at $x = y$. □

Lemma 4.7 (Insertion of a simple run)**.** *Let Q be a tNE path, $c \in \mathbb{Z}_{\ge 0}$, $q \in [1, |Q|-1]_{\mathbb{Z}}$, and $j \in [m]_{\mathbb{Z}}$. Let $Q' = \mathrm{Ins}(Q, q, R_j)$. Define $\tau\colon \mathrm{NE}(Q) \hookrightarrow \mathrm{NE}(Q')$ by*

$$\tau(Q_k) = \begin{cases} Q'_k, & k \in [q]_{\mathbb{Z}}, \\ Q'_{k+j+1}, & k \in [q+1, |Q|]_{\mathbb{Z}}. \end{cases} \tag{4.18}$$

Let $a_* = Q'_{q+1}$ and $v_t = Q'_{q+1+t}$ for $t \in [j]_{\mathbb{Z}}$. For $(h,\eta) \in \mathcal{C}_c(Q)$, define $h' \in G_m(Q')$ by

$$h' = h|_{Q[0,q]} +\!\!+ g_j^{\mathrm{run}} +\!\!+ h|_{Q[q,|Q|]}. \tag{4.19}$$

Let

$$\eta' = \mathrm{Recol}(Q, Q', \tau, h'; h, \eta). \tag{4.20}$$

Then, the map

$$\Phi_{q,j,c}\colon \mathcal{C}_c(Q) \longrightarrow \mathcal{C}_{c+1}(Q'), \qquad \Phi_{q,j,c}(h,\eta) = (h',\eta')$$

is well-defined and injective.

Proof. Let us determine the h-precursor after insertion of the simple run. The new horizontal edge a_* is the h-precursor of the j new vertical edges. The debt at a_* decreases from j to zero before the next original edge. Definition 2.6(1) gives

$$\begin{aligned} P_{Q',h'}(v_t) &= a_* && (t \in [j]_{\mathbb{Z}}), \\ P_{Q',h'}(\tau(w)) &= \tau(P_{Q,h}(w)) && (w \in N(Q)), \\ \mathrm{Node}_{Q',h'}(a_*) &= Q'[q, q+j+1]. \end{aligned} \tag{4.21}$$

For each $e \in E(Q)$, equation (4.21) gives

$$\#P^{-1}_{Q',h'}(\{\tau(e)\}) = \#P^{-1}_{Q,h}(\{e\}) = h(e) = h'(\tau(e)).$$

For $a_0 = \mathrm{Start}_{\mathrm{NE}}(Q)$, we therefore have

$$\mathrm{Node}_{Q',h'}(\tau(a_0)) = Q'.$$

Let us verify the node balance condition. For $w = Q_k$, let $\varepsilon(w) = 0$ when $k \le q$ and $\varepsilon(w) = 1$ when $k > q$. Equation (4.19) gives

$$\begin{aligned} \mathrm{eBal}_{Q',h'}(\tau(w)) &= \mathrm{eBal}_{Q,h}(w) + \varepsilon(w), \\ \mathrm{eBal}_{Q',h'}(a_*) &= \mathrm{eBal}_{Q',h'}(v_t) = \mathrm{bBal}_{Q,h}(q) + 1 \quad (t \in [j]_{\mathbb{Z}}). \end{aligned} \tag{4.22}$$

For $w \in N(Q)$ and $a = P_{Q,h}(w)$, the edge a precedes w. Hence, $\varepsilon(a) \le \varepsilon(w)$ and

$$\begin{aligned} &\mathrm{eBal}_{Q',h'}(\tau(w)) - \mathrm{eBal}_{Q',h'}(\tau(a)) \\ &\quad = \mathrm{eBal}_{Q,h}(w) - \mathrm{eBal}_{Q,h}(a) + \varepsilon(w) - \varepsilon(a) \ge 0. \end{aligned}$$

Together with (4.21) and (4.22), this proves that h' is connected. Finally,

$$|Q'|_E = |Q|_E + 1, \qquad \mathrm{TNG}_{Q',h'} = \mathrm{TNG}_{Q,h}, \qquad \mathrm{Surp}(Q',h') = \mathrm{Surp}(Q,h) + 1. \tag{4.23}$$

Let us recover (h,η) from $\Phi_{q,j,c}(h,\eta)$. Definition 3.2 and (4.19) give

$$(h'(\tau(e)), \eta'(\tau(e))) = (h(e), \eta(e)) \qquad (e \in \mathrm{NE}(Q)).$$

Restriction to the image of τ therefore recovers (h,η). Hence, $\Phi_{q,j,c}$ is injective. □

4.2 Composition of run-edits

Lemma 4.8 (Surplus reduction)**.** *Let $m \in \mathbb{Z}_{\geq 3}$ and $n \in \mathbb{Z}_{\geq 0}$. Let Q be a tNE path. Fix $\alpha = (a_j)_{j\in[n]_\mathbb{Z}}$ and $\beta = (b_j)_{j\in[n]_\mathbb{Z}}$ with $a_j \in [m-1]_\mathbb{Z}$ and $b_j \in [m-1]_\mathbb{Z}$. Define*

$$Q_0 = Q, \qquad Q_j = R_{a_j} \dashv Q_{j-1} \dashv R_{b_j} \quad (j \in [n]_\mathbb{Z}), \qquad \rho = \sum_{j=1}^{n} (a_j + 1).$$

Then, there is an injective map

$$\Delta_{Q,\alpha,\beta} \colon \bigcup_{c\in[0,n]_\mathbb{Z}} \mathcal{C}_c(Q) \longrightarrow \mathcal{C}_0(Q_n) \tag{4.24}$$

such that for each $(g,\gamma) \in \bigcup_{c\in[0,n]_\mathbb{Z}} \mathcal{C}_c(Q)$, $(g',\gamma') = \Delta_{Q,\alpha,\beta}(g,\gamma)$ satisfies

$$(g'((Q_n)_{\rho+k}), \gamma'((Q_n)_{\rho+k})) = (g(Q_k), \gamma(Q_k)) \qquad (k \in [|Q|]_\mathbb{Z}). \tag{4.25}$$

Proof. Let us construct the shell gradings and colorings. Let $(g,\gamma) \in \bigcup_{c\in[0,n]_\mathbb{Z}} \mathcal{C}_c(Q)$. Let $c = \mathrm{Surp}(Q,g)$ and $(g_0,\gamma_0) = (g,\gamma)$. For $j \in [n]_\mathbb{Z}$, define

$$\epsilon_j(c) = \begin{cases} 1, & j \leq c, \\ 0, & j > c, \end{cases} \qquad g_j = \mathrm{GShell}_{g_{j-1}}(Q_{j-1}; a_j, b_j, \epsilon_j(c)).$$

Define $\iota_j \colon \mathrm{NE}(Q_{j-1}) \hookrightarrow \mathrm{NE}(Q_j)$ by

$$\iota_j((Q_{j-1})_k) = (Q_j)_{a_j+1+k} \qquad (k \in [|Q_{j-1}|]_\mathbb{Z}).$$

Let

$$\gamma_j = \mathrm{Recol}(Q_{j-1}, Q_j, \iota_j, g_j; g_{j-1}, \gamma_{j-1}).$$

Let us prove that the final surplus is zero. Claims 2–4 of Lemma 3.3 give, by induction,

$$(g_j,\gamma_j) \in \mathcal{C}_{\max(c-j,0)}(Q_j) \qquad (j \in [0,n]_\mathbb{Z}).$$

For $j \leq c$, we have $\epsilon_j(c) = 1$ and $\mathrm{Surp}(Q_{j-1}, g_{j-1}) = c - j + 1 \geq 1$. Thus, condition (b) of Lemma 3.3 applies. For $j > c$, we have $\epsilon_j(c) = 0$ and use condition (a) of Lemma 3.3. Claim 3 of Lemma 3.3 gives $\mathrm{Surp}(Q_j, g_j) = \mathrm{Surp}(Q_{j-1}, g_{j-1}) - \epsilon_j(c)$. Since $c \leq n$, define

$$\Delta_{Q,\alpha,\beta}(g,\gamma) = (g_n,\gamma_n) \in \mathcal{C}_0(Q_n).$$

Let us recover (g,γ) from (g_n,γ_n). Claim 4 of Lemma 3.3 gives

$$(g_j(\iota_j(e)), \gamma_j(\iota_j(e))) = (g_{j-1}(e), \gamma_{j-1}(e)) \qquad (e \in \mathrm{NE}(Q_{j-1})). \tag{4.26}$$

Composing (4.26) for $j \in [n]_\mathbb{Z}$ gives (4.25). Restriction to the copy of Q therefore recovers (g,γ). Thus, $\Delta_{Q,\alpha,\beta}$ is injective on its whole domain. □

Definition 4.9. *For $\nu \in \mathbb{Z}_{\geq 0}$ and $D \in \mathbb{Z}_{\geq 0}$, define*

$$\mathrm{Fib}_m(\nu, D) = \binom{\nu + D}{D} 2^{2mD}.$$

Proposition 4.10. *Let $M \in [m-1]_\mathbb{Z}$. Let U be a run with ν simple runs. Let $K \in \mathbb{Z}_{\geq 0}$, $A \in \mathbb{Z}_{\geq 0}$, $D \in \mathbb{Z}_{\geq 0}$, and $\ell = K + A + D$. Assume the following.*

(a) *Let* $\Omega = (\omega_s)_{s\in[\ell]_\mathbb{Z}}$ *be a run-edit list starting at* U*. Let*

$$U_s = \mathrm{RunEResult}((\omega_1,\dots,\omega_s),U) \quad (s\in[0,\ell]_\mathbb{Z}), \qquad V = \mathrm{RunEResult}(\Omega,U).$$

Let I_R*,* I_+*, and* I_- *satisfy*

$$[\ell]_\mathbb{Z} = I_R \sqcup I_+ \sqcup I_-, \qquad \#I_R = K, \qquad \#I_+ = A, \qquad \#I_- = D.$$

For $s\in I_R\cup I_+$*, fix* $i_s\in[0,|U_{s-1}|]_\mathbb{Z}$*. For* $s\in I_R$*, fix* $j_s\in\mathbb{Z}_{>0}$*. For* $s\in I_-$*, fix* $v_s\in N(U_{s-1})$*. The run-edits satisfy*

$$U_s = \begin{cases} \mathrm{Ins}(U_{s-1},i_s,R_{j_s}), & s\in I_R,\\ \mathrm{Ins}(U_{s-1},i_s,N), & s\in I_+,\\ \mathrm{Del}(U_{s-1},v_s), & s\in I_-. \end{cases}$$

(b) *For* $s\in[0,\ell]_\mathbb{Z}$*, the path* U_s *is a run and every simple run* $B\subseteq U_s$ *satisfies* $\#N(B)\in[M]_\mathbb{Z}$*.*

(c) *If* $K+A>0$*, then* $m\ge 4$ *and* $M\le m-2$*.*

(d) *Let* $u\in[M]_\mathbb{Z}$ *and* $v\in[M]_\mathbb{Z}$*. Let* P *be an integer satisfying*

$$P \ge K + A + (m+1)D. \tag{4.27}$$

(e) *Fix the tNE paths* W_L *and* W_R*. Each path consists of* P *simple runs. Every simple run of these paths has vertical length in* $[M]_\mathbb{Z}$*. Let*

$$V_* = W_\mathrm{L} \mathbin{+\!\!+} R_u \mathbin{+\!\!+} V \mathbin{+\!\!+} R_v \mathbin{+\!\!+} W_\mathrm{R}. \tag{4.28}$$

Then, we have the following.

1. *There is a map*
$$\Phi\colon \mathcal{C}_0(U) \longrightarrow \mathcal{C}_0(V_*)$$
such that
$$\#\Phi^{-1}(\{\xi\}) \le \mathrm{Fib}_m(\nu,D) \qquad (\xi\in\mathcal{C}_0(V_*)).$$

2. *We have*
$$C_0(V_*) \ge \frac{C_0(U)}{\mathrm{Fib}_m(\nu,D)}. \tag{4.29}$$

Proof. Proof of Claim 1. Let us construct Φ. Let $(g,\gamma)\in\mathcal{C}_0(U)$. For u and v in assumption (d), let

$$Y = \mathrm{Shell}(U;u,v,0) = R_u \mathbin{+\!\!+} U \mathbin{+\!\!+} R_v.$$

Let $h = \mathrm{GShell}_g(U;u,v,0)$. Define $\iota\colon \mathrm{NE}(U)\hookrightarrow\mathrm{NE}(Y)$ by $\iota(U_k) = Y_{u+1+k}$ for $k\in[|U|]_\mathbb{Z}$ and let

$$\eta = \mathrm{Recol}(U,Y,\iota,h;g,\gamma).$$

Claims 2 and 3 of Lemma 3.3 give $(h,\eta)\in\mathcal{C}_0(Y)$. Suppose $P=0$. Inequality (4.27) gives $K=A=D=\ell=0$. Thus, $V=U$ and $W_\mathrm{L}=W_\mathrm{R}=\emptyset$. The paths Y and V_* are nonempty translated copies. Let

$$\mathbf{t} = \mathrm{Start}_\mathrm{pt}(V_*) - \mathrm{Start}_\mathrm{pt}(Y), \qquad \Phi(g,\gamma) = (h+\mathbf{t},\eta+\mathbf{t}).$$

Claim 4 of Lemma 3.3 and Lemma 2.10 show that $\Phi:\mathcal{C}_0(U)\to\mathcal{C}_0(V_*)$ is injective. Its fibers therefore have size at most $1=\mathrm{Fib}_m(\nu,0)$.

Second, suppose $P \geq 1$. Let us define (Q_s, h_s, η_s) for $s \in [0, \ell]_{\mathbb{Z}}$. Let $(Q_0, h_0, \eta_0) = (Y, h, \eta)$. For $s \in [\ell]_{\mathbb{Z}}$, let us construct (Q_s, h_s, η_s) recursively. The run-edit ω_s acts on the copy of U_{s-1} in Q_{s-1}. Here $h_s = h_s(g, \gamma)$ and $\eta_s = \eta_s(g, \gamma)$. Define $a_t \in [M]_{\mathbb{Z}}$ and $b_t \in [M]_{\mathbb{Z}}$ for $t \in [P]_{\mathbb{Z}}$ by

$$W_{\mathrm{L}} = \mathrm{Trans}(R_{a_P} \mathbin{+\!\!+} \cdots \mathbin{+\!\!+} R_{a_1}, \mathrm{Start}_{\mathrm{pt}}(W_{\mathrm{L}})),$$
$$W_{\mathrm{R}} = \mathrm{Trans}(R_{b_1} \mathbin{+\!\!+} \cdots \mathbin{+\!\!+} R_{b_P}, \mathrm{Start}_{\mathrm{pt}}(W_{\mathrm{R}})).$$

Let us insert a simple run at step $s \in I_R$. Let $Q_{s-1} = B_s^{\mathrm{L}} \mathbin{+\!\!+} U_{s-1} \mathbin{+\!\!+} B_s^{\mathrm{R}}$ and put $q_s = |B_s^{\mathrm{L}}| + i_s$. The paths B_s^{L} and B_s^{R} contain the outer simple runs. Thus, $q_s \in [1, |Q_{s-1}| - 1]_{\mathbb{Z}}$. Assumption (b) gives $j_s \in [M]_{\mathbb{Z}} \subseteq [m]_{\mathbb{Z}}$. Apply Lemma 4.7 with $Q = Q_{s-1}$, $q = q_s$, $j = j_s$, and $c = \mathrm{Surp}(Q_{s-1}, h_{s-1})$. Let (h_s, η_s) be the image of (h_{s-1}, η_{s-1}) under its map $\Phi_{q,j,c}$. The resulting path is

$$Q_s = \mathrm{Ins}(Q_{s-1}, q_s, R_{j_s}) = B_s^{\mathrm{L}} \mathbin{+\!\!+} U_s \mathbin{+\!\!+} B_s^{\mathrm{R}}.$$

Let us verify the hypotheses for the vertical run-edits. At step $s \in I_+ \cup I_-$, let

$$Q_{s-1} = B_s^{\mathrm{L}} \mathbin{+\!\!+} U_{s-1} \mathbin{+\!\!+} B_s^{\mathrm{R}}, \qquad t_s = \#((I_+ \cup I_-) \cap [s]_{\mathbb{Z}}).$$

The paths B_s^{L} and B_s^{R} consist of the simple runs added around U_{s-1}. Every simple run of Q_{s-1} has vertical length in $[M]_{\mathbb{Z}}$ by assumptions (b), (d), and (e).

First, let us insert a vertical edge when $s \in I_+$. Then, $A > 0$, and assumption (c) gives

$$m \geq 4, \qquad M \leq m - 2.$$

Claims 1–3 of Lemma 4.4 therefore apply after the first $|B_s^{\mathrm{L}}| + i_s$ edges.

Second, let us delete a vertical edge when $s \in I_-$. Let $v_s = (U_{s-1})_{r_s}$ and let

$$\widehat{v}_s = (Q_{s-1})_{|B_s^{\mathrm{L}}| + r_s}.$$

Claims 1–4 of Lemma 4.6 apply at $\widehat{v}_s$ because $M \leq m - 1$. For each connected colored grading (h, η) on Q_{s-1}, fix an edge $y_s(h, \eta)$ satisfying Claim 1 of Lemma 4.6 for $X = Q_{s-1}$, $v = \widehat{v}_s$, $u_0 = a_{t_s}$, and $v_0 = b_{t_s}$. At step s, use $y_s(h_{s-1}, \eta_{s-1})$.

Let us specify the shell path used at step $s \in I_+ \cup I_-$. Use the parameters a_{t_s} and b_{t_s} in the construction (a)–(c) of Lemma 4.4 or the construction (a)–(c) of Lemma 4.6, respectively. The resulting path is

$$Q_s = R_{a_{t_s}} \mathbin{+\!\!+} B_s^{\mathrm{L}} \mathbin{+\!\!+} U_s \mathbin{+\!\!+} B_s^{\mathrm{R}} \mathbin{+\!\!+} R_{b_{t_s}}.$$

Claim 3 of Lemma 4.4 and Claim 4 of Lemma 4.6 supply the colored grading (h_s, η_s).

Let us compute $\mathrm{Surp}(Q_\ell, h_\ell)$. For $s \in [0, \ell]_{\mathbb{Z}}$, let $c_s = \mathrm{Surp}(Q_s, h_s)$. The initial shell gives $c_0 = 0$. Equation (4.23), Claim 1(b) of Lemma 4.4, and Claim 2 of Lemma 4.6 give

$$c_s - c_{s-1} = \begin{cases} 1, & s \in I_R, \\ 0, & s \in I_+, \\ h_{s-1}(\widehat{v}_s), & s \in I_-. \end{cases}$$

Since $h_{s-1}(\widehat{v}_s) \in [0, m]_{\mathbb{Z}}$, summing gives

$$c_\ell = K + \sum_{s \in I_-} h_{s-1}(\widehat{v}_s) \leq K + mD.$$

Let us reduce the surplus to zero. Let $n = P - A - D$. Assumption (d) gives

$$n \geq K + mD \geq c_\ell.$$

Apply Lemma 4.8 with $Q = Q_\ell$ and

$$\alpha' = (a_{A+D+j})_{j\in[n]_\mathbb{Z}}, \qquad \beta' = (b_{A+D+j})_{j\in[n]_\mathbb{Z}}.$$

Let Q' denote the path Q_n in that application and let

$$(f, \delta) = \Delta_{Q_\ell,\alpha',\beta'}(h_\ell, \eta_\ell) \in \mathcal{C}_0(Q').$$

Let us identify Q' and define Φ. The paths Q_ℓ and Q' are translated copies of

$$R_{a_{A+D}} + \cdots + R_{a_1} + R_u + V + R_v + R_{b_1} + \cdots + R_{b_{A+D}}$$

and

$$R_{a_P} + \cdots + R_{a_1} + R_u + V + R_v + R_{b_1} + \cdots + R_{b_P},$$

respectively. When $A + D = 0$, the blocks indexed by $[A + D]_\mathbb{Z}$ are empty. Let

$$\mathbf{t} = \mathrm{Start}_{\mathrm{pt}}(V_*) - \mathrm{Start}_{\mathrm{pt}}(Q').$$

Then, $\mathrm{Trans}(Q', \mathbf{t}) = V_*$. Using the translated grading and color function from Definition 2.4(5), define

$$\Phi(g, \gamma) = (f + \mathbf{t}, \delta + \mathbf{t}).$$

Lemma 2.10 gives $\Phi(g, \gamma) \in \mathcal{C}_0(V_*)$.

Let us bound the fibers of Φ by recovering the colored gradings on U. Let $D_U \subseteq N(U)$ be the set of edges of U deleted by Ω in assumption (a). For $e \in \mathrm{NE}(U) \setminus D_U$, let $\widetilde{e} \in \mathrm{NE}(V_*)$ be its image under the inclusion $U \to Q_0$, the edge maps $Q_{s-1} \to Q_s$ at steps $s = 1, \ldots, \ell$, the inclusion $Q_\ell \to Q'$, and the translation by $\mathbf{t}$, in that order. Fix $(h, \eta) \in \mathcal{C}_0(V_*)$. Define $\widehat{h} \in G(U)$ by

$$\widehat{h}(e) = \begin{cases} h(\widetilde{e}), & e \notin D_U, \\ 0, & e \in D_U. \end{cases}$$

Let us recover an input colored grading from its differences and deleted edges. Let $e_1, \ldots, e_\nu$ be the horizontal edges of U in path order. For $(g, \gamma) \in \Phi^{-1}(\{(h, \eta)\})$, let

$$\delta_i = \mathrm{Diff}_U(g, \widehat{h}; e_i) = g(e_i) - h(\widetilde{e}_i) \qquad (i \in [\nu]_\mathbb{Z}).$$

Claim 4 of Lemma 3.3, Claim 2 of Lemma 4.4, and Lemma 4.8 preserve the grades and colors of the traced input edges. Equation (4.19) and Definition 3.2 give grade and color preservation for simple-run insertion. Claim 3 of Lemma 4.6 decreases one horizontal grade by 1 and preserves all other surviving grades and colors. Assumption (a) specifies D deletions. Hence, $\delta_i \geq 0$ for $i \in [\nu]_\mathbb{Z}$ and $\sum_{i=1}^{\nu} \delta_i \leq D$. If $\delta_i = 0$, no deletion decreases the grade of the traced edge, so its color is preserved. We recover (g, γ) from:

(i) the differences $\delta_1, \ldots, \delta_\nu$;

(ii) the colors $\gamma(e_i)$ for $i \in [\nu]_\mathbb{Z}$ such that $\delta_i > 0$;

(iii) the pairs $(g(v), \gamma(v))$ for $v \in D_U$.

We have

$$\begin{aligned} g(e_i) &= h(\widetilde{e}_i) + \delta_i && (i \in [\nu]_\mathbb{Z}), \\ \gamma(e_i) &= \eta(\widetilde{e}_i) && (\delta_i = 0), \\ g(v) &= h(\widetilde{v}), \qquad \gamma(v) = \eta(\widetilde{v}) && (v \in N(U) \setminus D_U). \end{aligned}$$

Let us count the difference vectors, horizontal colors, and grade–color pairs in (i)–(iii). The vector $(\delta_1, \ldots, \delta_\nu)$ has $\binom{\nu+D}{D}$ possibilities. At most D original horizontal colors change. Each

has at most 2^m choices. We delete at most D original vertical edges. Each has $\sum_{r=0}^{m}\binom{m}{r} = 2^m$ possible grade–color pairs. For each $\xi \in \mathcal{C}_0(V_*)$, we obtain

$$\#\Phi^{-1}(\{\xi\}) \le \binom{\nu+D}{D}(2^m)^D(2^m)^D = \mathrm{Fib}_m(\nu, D).$$

Proof of Claim 2. By Claim 1,

$$C_0(U) = \sum_{\xi\in\mathcal{C}_0(V_*)} \#\Phi^{-1}(\{\xi\}) \le \mathrm{Fib}_m(\nu, D)C_0(V_*).$$

□

Definition 4.11. *For $x > 0$, define*

$$\psi_m(x) = (1+x)\log(1+x) - x\log x + 2mx\log 2.$$

Let

$$\psi_m(0) = 0.$$

Lemma 4.12. *Let $\nu \in \mathbb{Z}_{\ge 0}$, $D \in \mathbb{Z}_{\ge 0}$, $N \in \mathbb{R}_{>0}$, and $x \in \mathbb{R}_{\ge 0}$ satisfy $\nu \le N$ and $D \le xN$. Then,*

$$\log \mathrm{Fib}_m(\nu, D) \le N\psi_m(x). \tag{4.30}$$

Proof. First, suppose $x = 0$. Then, $D = 0$ and $\log \mathrm{Fib}_m(\nu, 0) = 0 = N\psi_m(0)$.

Second, suppose $x > 0$. Let us prove (4.30). Let $t = \frac{x}{1+x} \in (0, 1)$. The binomial expansion gives

$$\binom{\nu+D}{D}t^D(1-t)^\nu \le \big(t + (1-t)\big)^{\nu+D} = 1.$$

Since $-\log t > 0$ and $-\log(1-t) > 0$, the bounds $\nu \le N$ and $D \le xN$ give

$$\begin{aligned}\log \mathrm{Fib}_m(\nu, D) &\le -D\log t - \nu\log(1-t) + 2mD\log 2\\ &\le -xN\log t - N\log(1-t) + 2mxN\log 2\\ &= N\psi_m(x).\end{aligned}$$

□

5 Rational slopes

5.1 Integer slopes for $m \ge 4$

Proposition 5.1. *Let m and k be integers and let $M = m - 2$. Assume the following.*

(a) $m \ge 4$.

(b) $k \in [M]_{\mathbb{Z}}$.

Let

$$\mathcal{S}_m = \left\{(a,b) \in \mathbb{Z}^2 : \begin{array}{c} b > a \ge 2, \quad \gcd(a,b) = 1,\\ |R|_N \in [M]_{\mathbb{Z}} \text{ for every simple run } R \subseteq \mathcal{U}(a,b)\end{array}\right\}.$$

Then, we have the following.

1. *We have*

$$\lim_{a\to\infty,\ \frac{b}{a}\to k,\ (a,b)\in\mathcal{A}_m\cap\mathcal{S}_m} \frac{\log E_m(a,b)}{a} = \ell(R_k).$$

2. *The following one-sided limits hold.*

   (a) *We have*

$$\lim_{\substack{a\to\infty,\ (a,b)\in\mathcal{A}_m\\ \frac{b}{a}\downarrow 1}} \frac{\log E_m(a,b)}{a} = \ell(R_1).$$

   (b) *We have*

$$\lim_{\substack{a\to\infty,\ (a,b)\in\mathcal{A}_m\\ \frac{b}{a}\uparrow M}} \frac{\log E_m(a,b)}{a} = \ell(R_M).$$

Proof. Proof of Claim 1. Let us estimate the number of vertical run-edits needed to obtain a constant run. Consider $(a,b)\in\mathcal{A}_m\cap\mathcal{S}_m$ as $a\to\infty$ and $\frac{b}{a}\to k$. For all sufficiently large a, the first a simple runs of $\mathcal{U}(a,b)$ have vertical lengths $k-1$ and k or k and $k+1$. The $(a+1)$-st simple run of $\mathcal{U}(a,b)$ is R_1. The total vertical run-edit distance to R_k^{a+1} is

$$\Delta = |b-1-ka| + (k-1) = o(a).$$

Let us prove the upper bound. Let $\nu = a+1$ and $P = (m+2)\Delta+1$. We change the run $\mathcal{U}(a,b)$ to the constant run R_k^ν by vertical run-edits. Every added outer simple run has vertical length k. Hence, the target is $R_k^{\nu+2P+2}$. Assumptions (a)–(b) and $(a,b)\in\mathcal{S}_m$ give the length and insertion hypotheses for Claim 2 of Proposition 4.10. Equations (4.29) and (2.13) give

$$mE_m(a,b) \le \mathrm{Fib}_m(\nu,D)\, C_0(R_k^{\nu+2P+2}), \qquad D\le\Delta.$$

Let us check the error term as $a\to\infty$ and $\frac{b}{a}\to k$ through $\mathcal{A}_m\cap\mathcal{S}_m$. Set $x=\frac{\Delta}{\nu}$. We have $x\to 0$ and $\frac{\nu}{a}\to 1$. Since $x\log x\to 0$ as $x\downarrow 0$, Definition 4.11 gives $\psi_m(x)\to 0$. Equation (4.30), applied with $N=\nu$, gives

$$0 \le \frac{\log \mathrm{Fib}_m(\nu,D)}{a} \le \frac{\nu}{a}\psi_m\left(\frac{\Delta}{\nu}\right) \longrightarrow 0.$$

Also, $P=o(\nu)$ gives $\frac{\nu+2P+2}{a}\to 1$ and $\nu+2P+2\to\infty$. Claim 2 of Proposition 3.8 therefore gives

$$\frac{\log E_m(a,b)}{a} \le \frac{\log \mathrm{Fib}_m(\nu,D)}{a} + \frac{\nu+2P+2}{a}\frac{\log C_0(R_k^{\nu+2P+2})}{\nu+2P+2} - \frac{\log m}{a},$$

$$\limsup_{\substack{a\to\infty,\ (a,b)\in\mathcal{A}_m\cap\mathcal{S}_m\\ \frac{b}{a}\to k}} \frac{\log E_m(a,b)}{a} \le \ell(R_k).$$

Let us prove the lower bound. Let $A_1,\dots,A_\nu$ be simple runs such that $\mathcal{U}(a,b) = A_1 +\!\!+ \cdots +\!\!+ A_\nu$. Let $l=\nu-2P-2$. Since $\frac{P}{\nu}\to 0$, we have $l\ge 1$ for all sufficiently large a. We start from R_k^l. For the current path Z, we use the run-edits $Z\mapsto \mathrm{Ins}(Z,i,N)$ at $i\in[|Z|]_{\mathbb{Z}}$ or $Z\mapsto\mathrm{Del}(Z,v)$ at a vertical edge v. At most Δ such run-edits give a translated copy of $A_{P+2} +\!\!+ \cdots +\!\!+ A_{\nu-P-1}$. We use $A_1,\dots,A_{P+1}$ and $A_{\nu-P},\dots,A_\nu$ for the outer simple runs. Equations (4.29) and (2.13) give

$$mE_m(a,b) \ge \frac{C_0(R_k^l)}{\mathrm{Fib}_m(l,D)}, \qquad D\le\Delta.$$

Since $\frac{l}{a}\to 1$ and $D=o(l)$, Claim 2 of Proposition 3.8 and (4.30) give

$$\frac{\log E_m(a,b)}{a} \ge \frac{l}{a}\frac{\log C_0(R_k^l)}{l} - \frac{\log \mathrm{Fib}_m(l,D)}{a} - \frac{\log m}{a},$$

$$\liminf_{\substack{a\to\infty,\ (a,b)\in\mathcal{A}_m\cap\mathcal{S}_m\\ \frac{b}{a}\to k}} \frac{\log E_m(a,b)}{a} \ge \ell(R_k).$$

Proof of Claim 2(a). Let us prove the one-sided limit as $\frac{b}{a} \downarrow 1$. We have $1 < \frac{b}{a} < M$ for all sufficiently large a. Since $b - a \geq 1$, equation (2.9) gives $1 < \Gamma(a,b) < M$. The floor differences in (2.10) lie in $[M]_{\mathbb{Z}}$. The $(a+1)$-st simple run of $\mathcal{U}(a,b)$ is R_1. Hence, $(a,b) \in \mathcal{S}_m$. Thus, Claim 1 applies with $k = 1$.

Proof of Claim 2(b). Let us prove the one-sided limit as $\frac{b}{a} \uparrow M$. We have $1 < \Gamma(a,b) < \frac{b}{a} < M$ for all sufficiently large a by (2.9). The floor differences in (2.10) and the appended R_1 have vertical lengths in $[M]_{\mathbb{Z}}$. Hence, $(a,b) \in \mathcal{S}_m$. Claim 1 applies with $k = M$. □

5.2 Mechanical run sequences near a noninteger rational slope

Definition 5.2. *Let $m \in \mathbb{Z}_{\geq 3}$, $p \in \mathbb{Z}_{>0}$, and $q \in \mathbb{Z}_{>0}$ satisfy $p \geq q$ and $\gcd(p,q) = 1$.*

1. *Define*

$$M_{\mathrm{run}}(m) = \begin{cases} 2, & m = 3, \\ m - 2, & m \geq 4. \end{cases}$$

For $i \in \mathbb{Z}_{>0}$, define

$$w_i(p,q) = \left\lfloor \frac{ip}{q} \right\rfloor - \left\lfloor \frac{(i-1)p}{q} \right\rfloor.$$

We define the rational run and the rational run sequence by

$$W_{p,q} = R_{w_1(p,q)} \mathbin{+\!\!+} \cdots \mathbin{+\!\!+} R_{w_q(p,q)}, \qquad \mathcal{R}_{p,q} = (R_{w_i(p,q)})_{i \in \mathbb{Z}_{>0}}.$$

For $j \in \mathbb{Z}_{>0}$ and $\alpha = \frac{p}{q} \in (1, M_{\mathrm{run}}(m))$, define

$$A_j(p,q) = C_0(W_{p,q}^j), \qquad \Upsilon_m(\alpha) = \frac{\ell(W_{p,q})}{q}. \tag{5.1}$$

2. *Let $s \geq 1$ and $u \in [0,1)$ be real numbers. Define*

$$\mathrm{MechRun}(s,u) = (R_{j_i})_{i \in \mathbb{Z}_{>0}}, \qquad j_i = \lfloor u + is \rfloor - \lfloor u + (i-1)s \rfloor.$$

We call $\mathrm{MechRun}(s,u)$ the mechanical run sequence of slope s and intercept u.

3. *Let $\alpha = \frac{p}{q}$. For $h \in \mathbb{Z}_{\geq 0}$, define*

$$\mathrm{RunPhase}_q(h) = h - q \left\lfloor \frac{h}{q} \right\rfloor.$$

We call $\mathrm{RunPhase}_q(h)$ the run phase at the start of the $(h+1)$-st simple run of $\mathcal{R}_{p,q}$.

4. *Let $n \in \mathbb{Z}_{>0}$ and let $A_1, \ldots, A_n$ be simple runs. Define $W = A_1 \mathbin{+\!\!+} \cdots \mathbin{+\!\!+} A_n$. Let $r \in \mathbb{Z}_{>0}$ and $\ell \in \mathbb{Z}_{>0}$ satisfy $r + \ell - 1 \leq n$. Define the (r,ℓ)-run-factor of W by*

$$\mathrm{RunFactor}_{r,\ell}(W) = W \left[\sum_{i=1}^{r-1} |A_i|, \sum_{i=1}^{r+\ell-1} |A_i| \right].$$

Let $\mathcal{A} = (A_i)_{i \in \mathbb{Z}_{>0}}$ be a sequence of simple runs. For $r \in \mathbb{Z}_{>0}$ and $\ell \in \mathbb{Z}_{>0}$, define

$$\mathrm{RunFactor}_{r,\ell}(\mathcal{A}) = \mathrm{RunFactor}_{r,\ell}(A_1 \mathbin{+\!\!+} \cdots \mathbin{+\!\!+} A_{r+\ell-1}).$$

Lemma 5.3. *Let $m \in \mathbb{Z}_{\geq 3}$. Let $p \in \mathbb{Z}_{>0}$ and $q \in \mathbb{Z}_{>0}$ satisfy $\gcd(p,q) = 1$, $q \geq 2$, and $1 < \frac{p}{q} < M_{\mathrm{run}}(m)$. Let $\alpha = \frac{p}{q}$. Let $s \geq 1$ and $u \in [0,1)$. Assume the following.*

(a) $|s-\alpha| < q^{-2}$.

Then, we have the following.

1. *For every* $r \in \mathbb{Z}_{>0}$ *and* $l \in [q-1]_{\mathbb{Z}}$, *there exists* $t \in \mathbb{Z}_{>0}$ *such that the path*

$$\mathrm{RunFactor}_{r,l}(\mathrm{MechRun}(s,u))$$

   is a translated copy of $\mathrm{RunFactor}_{t,l}(\mathcal{R}_{p,q})$.

2. *For* $r \in \mathbb{Z}_{>0}$, *let* $Z_r = \mathrm{RunFactor}_{r,q}(\mathrm{MechRun}(s,u))$. *If* $|Z_r|_N = p$, *then* Z_r *is a translated copy of a cyclic rotation of* $W_{p,q}$ *by simple runs.*

3. *Let* $k = \lfloor \alpha \rfloor$. *Then,* $w_i(p,q) \in [k, k+1]_{\mathbb{Z}}$ *for* $i \in [q]_{\mathbb{Z}}$ *and* $w_1(p,q) = k$.

4. *For* $i \in [0, q-1]_{\mathbb{Z}}$, *we have*

$$\lfloor is \rfloor = \lfloor i\alpha \rfloor .$$

Proof. Proof of Claim 1. Let us locate the floor thresholds within one period. For $i \in [q-1]_{\mathbb{Z}}$, define $T_i(t) = \left\lceil \frac{ip}{q} \right\rceil - it$ for $t \in \mathbb{R}$. Let us prove

$$T_i(s) < T_j(s) \iff T_i(\alpha) < T_j(\alpha) \qquad (i,j \in [q-1]_{\mathbb{Z}}). \tag{5.2}$$

Since $\alpha = \frac{p}{q}$, the numbers $T_i(\alpha)$ are distinct elements of $\{\frac{1}{q}, \ldots, \frac{q-1}{q}\}$. By assumption (a),

$$|T_i(s) - T_i(\alpha)| = i|s-\alpha| < \frac{1}{q}, \qquad 0 < T_i(s) < 1.$$

If $i \in [q-1]_{\mathbb{Z}}$ and $j \in [q-1]_{\mathbb{Z}}$ are distinct, then we also have

$$|(T_i(s) - T_j(s)) - (T_i(\alpha) - T_j(\alpha))| = |i-j|\,|s-\alpha| < \frac{1}{q} \le |T_i(\alpha) - T_j(\alpha)|.$$

Hence, equation (5.2) holds. For $t \in \{s, \alpha\}$ and $v \in [0,1)$,

$$\lfloor v + it \rfloor = \begin{cases} \lceil i\alpha \rceil - 1, & v < T_i(t), \\ \lceil i\alpha \rceil, & v \ge T_i(t). \end{cases}$$

Thus, equation (5.2) gives, for every $l \in [q-1]_{\mathbb{Z}}$,

$$\left\{ (\lfloor v + is \rfloor)_{i \in [l]_{\mathbb{Z}}} : v \in [0,1) \right\} = \left\{ (\lfloor v + i\alpha \rfloor)_{i \in [l]_{\mathbb{Z}}} : v \in [0,1) \right\}. \tag{5.3}$$

Let us apply (5.3) to $\mathrm{RunFactor}_{r,l}(\mathrm{MechRun}(s,u))$. Let $v_r = u + (r-1)s - \lfloor u + (r-1)s \rfloor$. For $i \in [l]_{\mathbb{Z}}$,

$$\begin{aligned} |\mathrm{RunFactor}_{r,i}(\mathrm{MechRun}(s,u))|_N &= \lfloor u + (r+i-1)s \rfloor - \lfloor u + (r-1)s \rfloor \\ &= \lfloor v_r + is \rfloor . \end{aligned} \tag{5.4}$$

Every vector in the right-hand side of (5.3) occurs at an intercept in $\{0, \frac{1}{q}, \ldots, \frac{q-1}{q}\}$. Since $\gcd(p,q) = 1$,

$$\{(t-1)\alpha - \lfloor (t-1)\alpha \rfloor : t \in [q]_{\mathbb{Z}}\} = \left\{ 0, \frac{1}{q}, \ldots, \frac{q-1}{q} \right\}.$$

Equations (5.3) and (5.4) therefore give $t \in [q]_{\mathbb{Z}}$ such that

$$|\mathrm{RunFactor}_{r,i}(\mathrm{MechRun}(s,u))|_N = |\mathrm{RunFactor}_{t,i}(\mathcal{R}_{p,q})|_N \qquad (i \in [l]_{\mathbb{Z}}).$$

Taking consecutive differences shows that $\mathrm{RunFactor}_{r,l}(\mathrm{MechRun}(s,u))$ is a translated copy of $\mathrm{RunFactor}_{t,l}(\mathcal{R}_{p,q})$.

Proof of Claim 2. Let us determine Z_r from its first $q-1$ simple runs and its total vertical length. Claim 1 gives $t \in \mathbb{Z}_{>0}$ such that the first $q-1$ simple runs of Z_r are a translated copy of $\mathrm{RunFactor}_{t,q-1}(\mathcal{R}_{p,q})$. Since $q\alpha = p \in \mathbb{Z}$, we have

$$|\mathrm{RunFactor}_{t,q}(\mathcal{R}_{p,q})|_N = \lfloor (t+q-1)\alpha \rfloor - \lfloor (t-1)\alpha \rfloor = p.$$

The assumption $|Z_r|_N = p$ therefore determines the last simple-run length. Thus, Z_r is a translated copy of $\mathrm{RunFactor}_{t,q}(\mathcal{R}_{p,q})$. The identity $\lfloor z+p \rfloor = \lfloor z \rfloor + p$ for $z \in \mathbb{R}$ shows that the simple-run lengths of $\mathcal{R}_{p,q}$ are q-periodic. Hence, $\mathrm{RunFactor}_{t,q}(\mathcal{R}_{p,q})$ is a cyclic rotation of $W_{p,q}$ up to translation.

Proof of Claim 3. Let us compute the periodic run lengths. Let $\beta = \alpha - k \in [0,1)$. For $i \in [q]_{\mathbb{Z}}$, we have

$$w_i(p,q) = k + \lfloor i\beta \rfloor - \lfloor (i-1)\beta \rfloor \in [k, k+1]_{\mathbb{Z}}.$$

At $i = 1$, we have $w_1(p,q) = k + \lfloor \beta \rfloor - \lfloor 0 \rfloor = k$.

Proof of Claim 4. Let us bound the distance of $i\alpha$ from the integers. For $i = 0$, both floors are zero. For $i \in [q-1]_{\mathbb{Z}}$, coprimality of p and q gives

$$\lfloor i\alpha \rfloor + \frac{1}{q} \le i\alpha \le \lfloor i\alpha \rfloor + 1 - \frac{1}{q}.$$

Assumption (a) gives

$$|is - i\alpha| < \frac{i}{q^2} < \frac{1}{q}.$$

Hence, $\lfloor i\alpha \rfloor < is < \lfloor i\alpha \rfloor + 1$, so $\lfloor is \rfloor = \lfloor i\alpha \rfloor$. □

Definition 5.4. *Let $m \in \mathbb{Z}_{\ge 3}$, $p \in \mathbb{Z}_{>0}$, and $q \in \mathbb{Z}_{>0}$ satisfy $\gcd(p,q) = 1$ and $q \ge 2$. Let $\alpha = \frac{p}{q} \in (1, M_{\mathrm{run}}(m))$. Let $n \in \mathbb{Z}_{>0}$ and let $A_1, \dots, A_n$ be simple runs. Define $W = A_1 +\!\!+ \cdots +\!\!+ A_n$. For $r \in [n-q+1]_{\mathbb{Z}}$, define*

$$Z_r = \mathrm{RunFactor}_{r,q}(W).$$

We call Z_r α-bad if $|Z_r|_N \neq p$. Define

$$D_{\mathrm{bad}}(W,p,q) = \#\{r \in [n-q+1]_{\mathbb{Z}} : |Z_r|_N \neq p\}.$$

Lemma 5.5 (Counting exceptional floor differences)**.** *Let $q \in \mathbb{Z}_{>0}$, $N \in \mathbb{Z}_{>0}$, $p \in \mathbb{Z}$, $s \in \mathbb{R}$, and $u \in \mathbb{R}$. Assume $|qs - p| < 1$. Then,*

$$\#\{i \in [0, N-1]_{\mathbb{Z}} : \lfloor u + (i+q)s \rfloor - \lfloor u + is \rfloor \neq p\} \le N|qs-p| + q.$$

Proof. Let us express the exceptional floors as jumps of an integer sequence. Let $\epsilon = qs - p$ and define

$$\delta_i = \lfloor u + (i+q)s \rfloor - \lfloor u + is \rfloor - p \qquad (i \in [0, N-1]_{\mathbb{Z}}).$$

Since $p \in \mathbb{Z}$ and $|\epsilon| < 1$,

$$\delta_i = \lfloor u + is + \epsilon \rfloor - \lfloor u + is \rfloor \in \begin{cases} [0,1]_{\mathbb{Z}}, & \epsilon \ge 0, \\ [-1,0]_{\mathbb{Z}}, & \epsilon < 0. \end{cases}$$

Thus, $\delta_i \neq 0$ implies $\delta_i = 1$ when $\epsilon \ge 0$ and $\delta_i = -1$ when $\epsilon < 0$.

Let us sum the floor differences in each residue class. For $r \in [0, q-1]_{\mathbb{Z}}$, let $n_r = \#\{i \in [0, N-1]_{\mathbb{Z}} : i \equiv r \pmod q\}$. Let us compute $\sum_{t=0}^{n_r-1} \delta_{r+tq}$. Telescoping gives

$$\sum_{t=0}^{n_r-1} \delta_{r+tq} = \lfloor u + (r+n_r q)s \rfloor - \lfloor u + rs \rfloor - n_r p,$$
$$\left| \sum_{t=0}^{n_r-1} \delta_{r+tq} - n_r \epsilon \right| < 1.$$

Since $\delta_i \in \{0, 1\}$ for $\epsilon \geq 0$ and $\delta_i \in \{-1, 0\}$ for $\epsilon < 0$, we obtain

$$\begin{aligned} \#\{i \in [0, N-1]_{\mathbb{Z}} : \delta_i \neq 0\} &= \sum_{r=0}^{q-1} \left| \sum_{t=0}^{n_r-1} \delta_{r+tq} \right| \\ &\leq \sum_{r=0}^{q-1} (n_r|\epsilon| + 1) = N|\epsilon| + q. \end{aligned}$$

□

Lemma 5.6. *Let $m \in \mathbb{Z}_{\geq 3}$, $p \in \mathbb{Z}_{>0}$, and $q \in \mathbb{Z}_{>0}$ satisfy $\gcd(p, q) = 1$ and $q \geq 2$. Let $\alpha = \frac{p}{q} \in (1, M_{\mathrm{run}}(m))$. Let $b > a \geq 2$ be coprime integers. Then, we have the following.*

1. *Assume the following.*
   - *(a) $a \geq q$.*
   - *(b) $|\Gamma(a, b) - \alpha| < q^{-2}$.*

   Then,
   $$D_{\mathrm{bad}}(\mathcal{U}(a, b), p, q) \leq |qb - pa| + q + 2. \tag{5.5}$$
2. *We have*
   $$D_{\mathrm{bad}}(\mathcal{U}(a, b), p, q) = o(a) \qquad \left(a \to \infty,\ \frac{b}{a} \to \alpha,\ b > a,\ \gcd(a, b) = 1\right).$$

Proof. Proof of Claim 1. Let us bound the number of α-bad run-factors contained in the first a simple runs of $\mathcal{U}(a, b)$. Assumptions (a) and (b) give $a - q + 1 \geq 1$ and $|q\Gamma(a, b) - p| < 1$. Lemma 5.5 with $N = a - q + 1$, $s = \Gamma(a, b)$, and $u = 0$ gives

$$\begin{aligned} &\#\left\{i \in [0, a-q]_{\mathbb{Z}} : \lfloor (i+q)\Gamma(a, b) \rfloor - \lfloor i\Gamma(a, b) \rfloor \neq p\right\} \\ &\quad \leq (a - q + 1)|q\Gamma(a, b) - p| + q. \end{aligned} \tag{5.6}$$

For $i \in [0, a-q]_{\mathbb{Z}}$, Definition 2.25 gives

$$|\mathrm{RunFactor}_{i+1,q}(\mathcal{U}(a, b))|_N = \lfloor (i+q)\Gamma(a, b) \rfloor - \lfloor i\Gamma(a, b) \rfloor.$$

By (2.10), the $(a+1)$-st simple run of $\mathcal{U}(a, b)$ is R_1. For $r \in [a - q + 2]_{\mathbb{Z}}$, the path $\mathrm{RunFactor}_{r,q}(\mathcal{U}(a, b))$ contains the $(a+1)$-st simple run if and only if $r = a - q + 2$. Hence, equation (5.6) gives

$$\begin{aligned} D_{\mathrm{bad}}(\mathcal{U}(a, b), p, q) &\leq 1 + (a - q + 1)|q\Gamma(a, b) - p| + q \\ &\leq aq|\Gamma(a, b) - \alpha| + q + 1. \end{aligned}$$

Assumption (a) and (2.9) give $\mu_1(a, b) \geq a + 1 > q$ and

$$aq|\Gamma(a, b) - \alpha| = \left| qb - pa - \frac{q}{\mu_1(a, b)} \right| \leq |qb - pa| + \frac{q}{\mu_1(a, b)}.$$

Since $\frac{q}{\mu_1(a,b)} < 1$, inequality (5.5) follows.

Proof of Claim 2. Let us verify the range of the ambient slope. Equation (2.9) gives

$$|\Gamma(a,b) - \alpha| \le \left|\frac{b}{a} - \alpha\right| + \frac{1}{a(a+1)} \longrightarrow 0.$$

For all sufficiently large a, Claim 1 applies and gives

$$0 \le \frac{D_{\mathrm{bad}}(\mathcal{U}(a,b),p,q)}{a} \le \left|q\frac{b}{a} - p\right| + \frac{q+2}{a} \longrightarrow 0.$$

□

Lemma 5.7. *Let $m \in \mathbb{Z}_{\ge 3}$, $p \in \mathbb{Z}_{>0}$, and $q \in \mathbb{Z}_{>0}$ satisfy $\gcd(p,q) = 1$ and $q \ge 2$. Let $\alpha = \frac{p}{q} \in (1, M_{\mathrm{run}}(m))$.*

1. *For every integer $J \ge 4$, each cyclic rotation V of $W_{p,q}^J$ by simple runs satisfies $C_0(V) \ge A_{J-3}(p,q)$.*
2. *Let $h \ge 1$ be an integer. Let $X_1, \dots, X_h$ be runs. Let $B_0, \dots, B_h$ each consist of three simple runs. Assume the following.*

   (a) Every simple run of $B_0, \dots, B_h$ has vertical length in $[M_{\mathrm{run}}(m)]_{\mathbb{Z}}$.

   Then,

$$C_0(B_0 +\!\!+ X_1 +\!\!+ B_1 +\!\!+ \cdots +\!\!+ X_h +\!\!+ B_h) \ge \prod_{i=1}^{h} C_0(X_i). \tag{5.7}$$

Proof. Proof of Claim 1. Let us apply Lemma 3.6 to a rotated periodic path. Let $t = J - 3$ and let $j \in [0, q-1]_{\mathbb{Z}}$. Let $U_j = \mathrm{CRot}(W_{p,q}, j)$. Let $P = 2q - j$ and $S = q + j$. Let $X = \mathrm{RunFactor}_{P+1,qt}(U_j^{t+3})$. Then,

$$X = \mathrm{Trans}(W_{p,q}^t, \mathrm{Start}_{\mathrm{pt}}(X)), \qquad P + qt + S = q(t+3).$$

Since $P + j = 2q$, we have $\mathrm{RunPhase}_q(P+j) = 0$. Also, $\max(P,S) \le 2\min(P,S)$. Claim 3 of Lemma 5.3 gives $|R|_N \in [M_{\mathrm{run}}(m)]_{\mathbb{Z}}$ for every simple run R of $W_{p,q}$. Lemma 3.6 gives

$$C_0(U_j^{t+3}) \ge C_0(W_{p,q}^t) = A_{J-3}(p,q).$$

Proof of Claim 2. Let us specify the shell paths. Let $B_i = R_{b_{i,1}} +\!\!+ R_{b_{i,2}} +\!\!+ R_{b_{i,3}}$ for $i \in [0,h]_{\mathbb{Z}}$. For $r \in [h]_{\mathbb{Z}}$ and $s \in [h]_{\mathbb{Z}}$ satisfying $r \le s$ and $(u,v) \in \{(2,2),(2,1),(1,3),(3,2)\}$, define

$$T_{uv}(r,s) = \left(\mathop{+\!\!+}_{j=4-u}^{3} R_{b_{r-1,j}}\right) +\!\!+ X_r +\!\!+ \left(\mathop{+\!\!+}_{i=r+1}^{s} (B_{i-1} +\!\!+ X_i)\right) +\!\!+ \left(\mathop{+\!\!+}_{j=1}^{v} R_{b_{s,j}}\right).$$

The iterated concatenations use increasing index order. We compare paths up to translation using Lemma 2.10. For $r = s$, we have

$$\begin{aligned}
T_{22}(r,r) &= \mathrm{Shell}(\mathrm{Shell}(X_r; b_{r-1,3}, b_{r,1}, 0); b_{r-1,2}, b_{r,2}, 0),\\
T_{21}(r,r) &= \mathrm{Shell}(R_{b_{r-1,3}}, X_r; b_{r-1,2}, b_{r,1}, 0),\\
T_{13}(r,r) &= \mathrm{Shell}(X_r, R_{b_{r,1}} +\!\!+ R_{b_{r,2}}; b_{r-1,3}, b_{r,3}, 0),\\
T_{32}(r,r) &= \mathrm{Shell}(T_{21}(r,r); b_{r-1,1}, b_{r,2}, 0).
\end{aligned}$$

For $r < s$, we have

$$\begin{aligned}
T_{22}(r,s) &= \mathrm{Shell}(\mathrm{Shell}(X_r; b_{r-1,3}, b_{r,1}, 0), T_{21}(r+1,s); b_{r-1,2}, b_{s,2}, 0),\\
T_{21}(r,s) &= \mathrm{Shell}(T_{13}(r,s-1), X_s; b_{r-1,2}, b_{s,1}, 0),\\
T_{13}(r,s) &= \mathrm{Shell}(X_r, T_{32}(r+1,s); b_{r-1,3}, b_{s,3}, 0),\\
T_{32}(r,s) &= \mathrm{Shell}(T_{22}(r,s-1), \mathrm{Shell}(X_s; b_{s-1,3}, b_{s,1}, 0); b_{r-1,1}, b_{s,2}, 0).
\end{aligned}$$

Let us count the resulting colored gradings. Assumption (a) gives $b_{i,j} \in [M_{\mathrm{run}}(m)]_{\mathbb{Z}} \subseteq [m-1]_{\mathbb{Z}}$. Lemma 3.5 gives $C_0(R_{b_{i,j}}) \geq 1$ and $C_0(R_{b_{i,1}} + R_{b_{i,2}}) \geq 1$. Claim 4 of Lemma 3.3 gives an injective map into colored gradings of surplus zero at each shell step. Induction on $s-r$ therefore gives

$$C_0(T_{uv}(r,s)) \geq \prod_{i=r}^{s} C_0(X_i).$$

Finally,

$$\mathrm{Shell}(T_{22}(1,h); b_{0,1}, b_{h,3}, 0) = B_0 + X_1 + B_1 + \cdots + X_h + B_h$$

up to translation. Claim 4 of Lemma 3.3 and Lemma 2.10 give (5.7). □

Theorem 5.8 (Noninteger rational all-sequence limit)**.** *Let $m \in \mathbb{Z}_{\geq 3}$, $p \in \mathbb{Z}_{>0}$, and $q \in \mathbb{Z}_{>0}$ satisfy $\gcd(p,q) = 1$ and $q \geq 2$. Let $\alpha = \frac{p}{q} \in (1, M_{\mathrm{run}}(m))$. Then,*

$$\lim_{\substack{a\to\infty,\ (a,b)\in\mathcal{A}_m \\ \frac{b}{a}\to\alpha}} \frac{\log E_m(a,b)}{a} = \frac{\ell(W_{p,q})}{q}. \tag{5.8}$$

Proof. Let us construct the initial factor and the separated periodic factors. Let $k = \lfloor \alpha \rfloor$. In the limit in (5.8), we have $a \geq q$ and $|\Gamma(a,b) - \alpha| < q^{-2}$ for all sufficiently large a.

Proof of the upper bound. First, we divide $\mathcal{U}(a,b)$ into periodic pieces. Equation (2.10) and Claims 3 and 4 of Lemma 5.3 give

$$\mathrm{RunFactor}_{1,1}(\mathcal{U}(a,b)) = \mathrm{RunFactor}_{1,1}(W_{p,q}) = R_k, \qquad \mathrm{RunPhase}_q(0) = 0.$$

Let $0 = c_0 < c_1 < \cdots < c_d = a$ satisfy

$$\{c_1, \ldots, c_{d-1}\} = \{r \in [a-q+1]_{\mathbb{Z}} : |\,\mathrm{RunFactor}_{r,q}(\mathcal{U}(a,b))|_N \neq p\}.$$

For $i \in [d]_{\mathbb{Z}}$, let

$$X_i = \mathrm{RunFactor}_{c_{i-1}+1, c_i - c_{i-1}}(\mathcal{U}(a,b)).$$

Each X_i contains no α-bad run-factor. We have $d \leq D_{\mathrm{bad}}(\mathcal{U}(a,b), p, q) + 1$. By Claim 1 of Lemma 5.3, a piece with $l < q$ simple runs is a translated copy of $\mathrm{RunFactor}_{t,l}(\mathcal{R}_{p,q})$ for some $t \in \mathbb{Z}_{>0}$. In a piece with at least q simple runs, every run-factor consisting of q simple runs has vertical length p. Subtracting the sums for consecutive run-factors gives

$$0 = \sum_{i=j+1}^{j+q} \Lambda_i(a,b) - \sum_{i=j}^{j+q-1} \Lambda_i(a,b) = \Lambda_{j+q}(a,b) - \Lambda_j(a,b).$$

Thus, $\Lambda_{j+q}(a,b) = \Lambda_j(a,b)$ for $i \in [d]_{\mathbb{Z}}$ and $c_{i-1} < j < j+q \leq c_i$. Claim 2 of Lemma 5.3 identifies the first q simple runs of each such piece with a cyclic rotation of $W_{p,q}$.

Second, let us align each piece with a run-factor of $\mathcal{R}_{p,q}$. Let us prove that no insertion is needed before the first piece. Claim 4 of Lemma 5.3, applied with $s = \Gamma(a,b)$, gives

$$\lfloor i\Gamma(a,b) \rfloor = \lfloor i\alpha \rfloor \qquad (i \in [q-1]_{\mathbb{Z}}).$$

If the first piece has at least q simple runs, its first q lengths sum to p. Thus, its q-th length also agrees with $w_q(p,q)$. The identities $\Lambda_{j+q}(a,b) = \Lambda_j(a,b)$ for $1 \le j < j+q \le c_1$ identify X_1 with $\mathrm{RunFactor}_{1,c_1}(\mathcal{R}_{p,q})$. Consequently, no insertion is needed before X_1.

Let us proceed by induction through $X_2, \dots, X_d$. For the current piece X, choose the least $t \in [q]_{\mathbb{Z}}$ such that X is a translated copy of $\mathrm{RunFactor}_{t,|X|_E}(\mathcal{R}_{p,q})$. Let $P +\!\!+ X +\!\!+ Z$ be the current path and let $h = |P|_E$. The induction hypothesis is that the simple-run lengths of P are $w_1(p,q), \dots, w_h(p,q)$. The identification of X_1 gives the base case. Define

$$j = \mathrm{RunPhase}_q(q + t - 1 - \mathrm{RunPhase}_q(h)) \in [0, q-1]_{\mathbb{Z}}.$$

Then, $\mathrm{RunPhase}_q(h+j) = t-1$. If $j > 0$, insert the simple runs of $\mathrm{RunFactor}_{h+1,j}(\mathcal{R}_{p,q})$ after P, producing

$$\mathrm{Ins}(P +\!\!+ X +\!\!+ Z, |P|, \mathrm{RunFactor}_{h+1,j}(\mathcal{R}_{p,q})).$$

If $j = 0$, the path is $P +\!\!+ X +\!\!+ Z$.

By q-periodicity, the simple-run lengths of X now agree with those at positions $h+j+1, \dots, h+j+|X|_E$ in $\mathcal{R}_{p,q}$. Thus, the prefix through X has simple-run lengths $w_1(p,q), \dots, w_{h+j+|X|_E}(p,q)$, which proves the induction step.

Let us complete the final period before R_1. Let h be the number of simple runs before the final R_1 of $\mathcal{U}(a,b)$ after these insertions. Let $j = \mathrm{RunPhase}_q(q - \mathrm{RunPhase}_q(h))$. If $j > 0$, insert $\mathrm{RunFactor}_{h+1,j}(\mathcal{R}_{p,q})$ before R_1. Then, $\mathrm{RunPhase}_q(h+j) = 0$. Let K be the number of inserted simple runs. Let $n_0 = \frac{a+K}{q}$. The result is

$$W_{p,q}^{n_0} +\!\!+ R_1, \qquad K \le (q-1)(D_{\mathrm{bad}}(\mathcal{U}(a,b), p, q) + 2).$$

For each insertion, use (4.18), (4.19), and (4.20). The resulting grading g satisfies $\mathrm{Surp}(W_{p,q}^{n_0} +\!\!+ R_1, g) = K$.

Third, let us add the outer simple runs. Apply Lemma 3.6 with $(P,S) = (q, q-1)$ to add $W_{p,q}$ before $W_{p,q}^{n_0} +\!\!+ R_1$ and $R_{w_2(p,q)} +\!\!+ \cdots +\!\!+ R_{w_q(p,q)}$ after $W_{p,q}^{n_0} +\!\!+ R_1$. The target is

$$Q_0 = W_{p,q}^{n_0+1} +\!\!+ R_1 +\!\!+ R_{w_2(p,q)} +\!\!+ \cdots +\!\!+ R_{w_q(p,q)}. \tag{5.9}$$

If $k = 1$, we have $Q_0 = W_{p,q}^{n_0+2}$. For $m = 3$, the condition $1 < \alpha < 2$ gives $k = 1$.

Let us obtain the initial run of length k. For $k > 1$, let us replace the factor R_1 in (5.9) by R_k. We have $m \ge 4$. Apply Claims 1–3 of Lemma 4.4 $k-1$ times. For each application, use q unary shell paths to add one copy of $W_{p,q}$ on each side. After the jth vertical-edge insertion, the path is

$$Q_j = W_{p,q}^{n_0+j+1} +\!\!+ R_{j+1} +\!\!+ R_{w_2(p,q)} +\!\!+ \cdots +\!\!+ R_{w_q(p,q)} +\!\!+ W_{p,q}^{j} \qquad (j \in [k-1]_{\mathbb{Z}}).$$

The transported shell grading h_j satisfies $\mathrm{Surp}(Q_j, h_j) = K$.

Let us reduce the surplus to zero. We have $Q_{k-1} = W_{p,q}^{n_0+2k}$. Let $H = \left\lceil \frac{K}{q} \right\rceil$. Apply Lemma 4.8 with $Q = W_{p,q}^{n_0+2k}$, $n = qH$, and

$$a_j = w_{q-\mathrm{RunPhase}_q(j-1)}(p,q), \qquad b_j = w_{1+\mathrm{RunPhase}_q(j-1)}(p,q) \qquad (j \in [qH]_{\mathbb{Z}}).$$

The final path is a translated copy of $W_{p,q}^{n_0+2k+2H}$.

Let us prove injectivity of the composition. At each simple-run insertion, equations (4.19) and (4.20) recover the input grading and coloring. At each vertical-edge insertion, Claim 3 of Lemma 4.4 gives injectivity. Lemma 3.6 gives injectivity of each outer extension. Lemma 2.10 and the injection (4.24) give

$$mE_m(a,b) \le A_N(p,q), \qquad N = \frac{a+K}{q} + 2k + 2\left\lceil \frac{K}{q} \right\rceil. \tag{5.10}$$

Let us obtain the upper limit from (5.10). By Claim 2 of Lemma 5.6, we have $K = o(a)$ with q fixed. Hence, $\frac{N}{a} \to \frac{1}{q}$. Claim 2 of Proposition 3.8 gives

$$\frac{\log E_m(a,b)}{a} \le \frac{N}{a}\frac{\log A_N(p,q)}{N} - \frac{\log m}{a},$$
$$\limsup_{\substack{a\to\infty,\ (a,b)\in\mathcal{A}_m \\ \frac{b}{a}\to\alpha}} \frac{\log E_m(a,b)}{a} \le \frac{\ell(W_{p,q})}{q}.$$

Proof of the lower bound. Let us select separated subpaths on which to count gradings. Let $J \ge 4$ be an integer and let

$$h = \left\lfloor \frac{a-2}{Jq+3} \right\rfloor$$

if a is sufficiently large and $h \ge 1$. Define

$$\begin{aligned}
X_i &= \mathrm{RunFactor}_{4+(i-1)(Jq+3),Jq}(\mathcal{U}(a,b)) && (i \in [h-1]_{\mathbb{Z}}),\\
X_h &= \mathrm{RunFactor}_{4+(h-1)(Jq+3),\,a-5-(h-1)(Jq+3)}(\mathcal{U}(a,b)),\\
B_i &= \mathrm{RunFactor}_{i(Jq+3)+1,3}(\mathcal{U}(a,b)) && (i \in [0,h-1]_{\mathbb{Z}}),\\
B_h &= \mathrm{RunFactor}_{a-1,3}(\mathcal{U}(a,b)).
\end{aligned}$$

Then, $\mathcal{U}(a,b) = B_0 \dotplus X_1 \dotplus B_1 \dotplus \cdots \dotplus X_h \dotplus B_h$ and $|X_h|_E \ge Jq$.

Let us count colored gradings on these subpaths. Let

$$\mathcal{I} = \{h\} \cup \{i \in [h-1]_{\mathbb{Z}} : D_{\mathrm{bad}}(X_i,p,q) > 0\}.$$

Then, $\#\mathcal{I} \le D_{\mathrm{bad}}(\mathcal{U}(a,b),p,q) + 1$. We call X_i exceptional for $i \in \mathcal{I}$. For $i \in [h]_{\mathbb{Z}} \setminus \mathcal{I}$, the path X_i is a translated copy of $\mathrm{RunFactor}_{t,Jq}(\mathcal{R}_{p,q})$ for some $t \in \mathbb{Z}_{>0}$. By Claim 1 of Lemma 5.7, such a subpath has at least $A_{J-3}(p,q)$ connected colored gradings of surplus zero. Each exceptional subpath X has a connected colored grading (g,γ) with $\mathrm{Surp}(X,g) = 0$ by Lemma 3.5. Claim 2 of Lemma 5.7 gives

$$mE_m(a,b) \ge A_{J-3}(p,q)^{\max(0,h-D_{\mathrm{bad}}(\mathcal{U}(a,b),p,q)-1)}.$$

Let us take the lower limit and then let J tend to infinity. If J and q are fixed, then (5.5) gives

$$\frac{h}{a} \longrightarrow \frac{1}{Jq+3}, \qquad \frac{D_{\mathrm{bad}}(\mathcal{U}(a,b),p,q)+1}{a} \longrightarrow 0 \quad \left(a\to\infty,\ \frac{b}{a}\to\alpha,\ (a,b)\in\mathcal{A}_m\right).$$

We then let $J \to \infty$ with q fixed. By Claim 2 of Proposition 3.8, the lower bound becomes

$$\liminf_{\substack{a\to\infty,\ (a,b)\in\mathcal{A}_m \\ \frac{b}{a}\to\alpha}} \frac{\log E_m(a,b)}{a} \ge \lim_{J\to\infty} \frac{\log A_{J-3}(p,q)}{Jq+3} = \frac{\ell(W_{p,q})}{q}.$$

□

6 Irrational slopes for $m \ge 4$

6.1 Existence of the limit

Lemma 6.1 (Vertical length comparison). *Let $M \in \mathbb{Z}_{>0}$ and $L \in \mathbb{Z}_{>0}$. Let $X_1,\dots,X_L$ and $Y_1,\dots,Y_L$ be simple runs. Put $X = X_1 \dotplus \cdots \dotplus X_L$, $Y = Y_1 \dotplus \cdots \dotplus Y_L$, $\alpha_i = |X_i|_N$, and $\beta_i = |Y_i|_N$. Assume $\alpha_i \in [M]_{\mathbb{Z}}$ and $\beta_i \in [M]_{\mathbb{Z}}$ for $i \in [L]_{\mathbb{Z}}$. Then, there are $n \in \mathbb{Z}_{\ge 0}$ and a run-edit list $\Omega = (\omega_1,\dots,\omega_n)$ satisfying the following.*

1. *The list Ω starts at X and satisfies*

$$\mathrm{RunEResult}(\Omega, X) = \mathrm{Trans}(Y, \mathrm{Start}_{\mathrm{pt}}(X) - \mathrm{Start}_{\mathrm{pt}}(Y)).$$

2. *The list Ω consists of exactly*

$$n = \sum_{i=1}^{L} |\alpha_i - \beta_i|$$

vertical run-edits.

3. *After each run-edit of Ω, every simple-run vertical length belongs to $[M]_{\mathbb{Z}}$.*

Proof. Proof of Claim 1. Let us construct Ω. Process the simple runs in the order $i = 1, \dots, L$. At the ith simple run, let W be the current path and write this simple run as $W[s, s+t+1]$. Here $s \in [0, |W| - 1]_{\mathbb{Z}}$ and $t \in \mathbb{Z}_{>0}$.

First, let us increase the vertical length when $\alpha_i < \beta_i$. For each $t \in [\alpha_i, \beta_i - 1]_{\mathbb{Z}}$ in increasing order, replace W by

$$\mathrm{Ins}(W, s+t+1, N).$$

This gives $\beta_i - \alpha_i$ insertions and final vertical length β_i.

Second, let us decrease the vertical length when $\beta_i < \alpha_i$. For each $t = \alpha_i, \alpha_i - 1, \dots, \beta_i + 1$, replace W by

$$\mathrm{Del}(W, W_{s+t+1}).$$

Since $t > \beta_i \geq 1$, every deletion has $t \geq 2$. This gives $\alpha_i - \beta_i$ deletions and final vertical length β_i.

Let us identify the resulting path. When $\alpha_i = \beta_i$, make no change at the ith simple run. The resulting vertical lengths are $\beta_1, \dots, \beta_L$. Thus, the resulting path is $\mathrm{Trans}(Y, \mathrm{Start}_{\mathrm{pt}}(X) - \mathrm{Start}_{\mathrm{pt}}(Y))$.

Proof of Claim 2. Let us count the vertical run-edits. For each $i \in [L]_{\mathbb{Z}}$, the construction in Claim 1 uses exactly $|\alpha_i - \beta_i|$ vertical run-edits. Summing over $i \in [L]_{\mathbb{Z}}$ gives $n = \sum_{i=1}^{L} |\alpha_i - \beta_i|$.

Proof of Claim 3. Let us bound the intermediate vertical lengths. During the run-edits of Ω on the ith simple run, its vertical length t satisfies

$$1 \leq \min(\alpha_i, \beta_i) \leq t \leq \max(\alpha_i, \beta_i) \leq M.$$

For $j \in [L]_{\mathbb{Z}} \setminus \{i\}$, these run-edits do not change the vertical length of the jth simple run. □

Definition 6.2. *Let $M \in \mathbb{Z}_{>0}$, $r \in \mathbb{R}$, $\eta > 0$, $\delta > 0$, and $B > 0$. Let $l \in \mathbb{Z}_{>0}$, $T \in \mathbb{Z}_{>0}$, $\sigma_1 \in [1, \infty)$, $\sigma_2 \in [1, \infty)$, $u_1 \in [0, 1)$, and $u_2 \in [0, 1)$. Let $A_1, \dots, A_l$ be simple runs. We call $(l, T, \sigma_1, \sigma_2, u_1, u_2, (A_1, \dots, A_l))$ (M, r, η, δ, B)-admissible when the following conditions hold.*

(a) We have

$$|A_i|_N = \lfloor u_1 + i\sigma_1 \rfloor - \lfloor u_1 + (i-1)\sigma_1 \rfloor \quad (i \in [l-1]_{\mathbb{Z}}), \qquad |A_l|_N \in [M]_{\mathbb{Z}}.$$

(b) We have

$$|\sigma_i - r| < \delta \quad (i \in [2]_{\mathbb{Z}}), \qquad T \geq (1+\eta)l + B.$$

Lemma 6.3 (Fractional parts near a rational slope)**.** *Let $p \in \mathbb{Z}_{>0}$ and $q \in \mathbb{Z}_{>0}$ satisfy $\gcd(p, q) = 1$. Let $\sigma \in \mathbb{R}$, $w \in \mathbb{R}$, and $0 < \eta < \frac{1}{4}$. Assume*

$$|\sigma - \frac{p}{q}| < \frac{2}{q^2}.$$

For $x \in \mathbb{R}$, let $\{x\} = x - \lfloor x \rfloor$. Let $0 \leq x_1 \leq \cdots \leq x_q < 1$ be the nondecreasing rearrangement of $(\{w + j\sigma\})_{j \in [0, q-1]_{\mathbb{Z}}}$. Let $x_{q+1} = x_1 + 1$. Then, the following hold.

1. *We have*
$$x_{i+1} - x_i \le \frac{5}{q} \qquad (i \in [q]_{\mathbb{Z}}).$$

2. *We have*
$$\#\{j \in [0, q-1]_{\mathbb{Z}} : \{w + j\sigma\} \in [0, 2\eta] \cup [1 - 2\eta, 1)\} \le 4\eta q + 10.$$

Proof. Proof of Claim 1. Let us compare the fractional parts with a rational grid. For $j \in [0, q-1]_{\mathbb{Z}}$, put $t_j = jp - q\lfloor w + j\sigma \rfloor$. Then,
$$\left| \{w + j\sigma\} - \left(w + \frac{t_j}{q} \right) \right| = j \left| \sigma - \frac{p}{q} \right| < \frac{2}{q}.$$
The integers t_j represent distinct residue classes modulo q because $t_j \equiv jp \pmod q$ and $\gcd(p, q) = 1$. Thus, each $t \in \mathbb{Z}$ has the form $t = t_j + nq$ for some $j \in [0, q-1]_{\mathbb{Z}}$ and $n \in \mathbb{Z}$. Suppose $x_{i+1} - x_i > \frac{5}{q}$. The interval $(x_i + \frac{2}{q}, x_{i+1} - \frac{2}{q})$ has length greater than $\frac{1}{q}$. The interval contains $w + \frac{t}{q}$ for some $t \in \mathbb{Z}$. Let $t = t_j + nq$. The number $\{w + j\sigma\} + n$ belongs to (x_i, x_{i+1}), contradicting the choice of x_i and x_{i+1}.

Proof of Claim 2. Let us count the grid points near the two endpoints. If $\{w + j\sigma\} \in [0, 2\eta] \cup [1 - 2\eta, 1)$, then
$$w + \frac{t_j}{q} \in \left[-\frac{2}{q}, 2\eta + \frac{2}{q} \right] \cup \left[1 - 2\eta - \frac{2}{q}, 1 + \frac{2}{q} \right].$$
Each interval has length $2\eta + \frac{4}{q}$ and contains at most $2\eta q + 5$ numbers of the form $w + \frac{t}{q}$, $t \in \mathbb{Z}$. Hence, at most $4\eta q + 10$ indices satisfy the condition in Claim 2. □

Lemma 6.4 (Phase alignment and prefix discrepancies)**.** *Let $p_0 \in \mathbb{Z}_{>0}$, $q_0 \in \mathbb{Z}_{>0}$, $L \in \mathbb{Z}_{>0}$, $0 < \eta < \frac{1}{4}$, $\sigma_1 \in [1, \infty)$, and $\sigma_2 \in [1, \infty)$. Assume*
$$\gcd(p_0, q_0) = 1, \qquad q_0 > \frac{20}{\eta}, \qquad q_0 \le \eta L,$$
$$\left| \sigma_a - \frac{p_0}{q_0} \right| < \frac{2}{q_0^2} \quad (a \in [2]_{\mathbb{Z}}), \qquad |\sigma_1 - \sigma_2| < \frac{\eta}{2L}.$$
Let $u_1 \in [0, 1)$, $u_2 \in [0, 1)$, and $b \in \mathbb{Z}_{\ge 0}$. For $j \in [0, q_0 - 1]_{\mathbb{Z}}$, define
$$v_j = \{u_2 + (b + j)\sigma_2\}, \qquad J = \{j \in [0, q_0 - 1]_{\mathbb{Z}} : |u_1 - v_j| < \eta\}.$$
The following statements hold.

1. *The set J is nonempty. Define the uniquely specified index and phase*
$$j_* = \min J, \qquad v = v_{j_*}.$$
Then, $0 \le j_* < q_0$ and $|u_1 - v| < \eta$.

2. *Use v from Claim 1. Define*
$$d_i = \lfloor u_1 + i\sigma_1 \rfloor - \lfloor v + i\sigma_2 \rfloor \quad (i \in [0, L]_{\mathbb{Z}}).$$
We have $d_0 = 0$. For $i \in [L]_{\mathbb{Z}}$,
$$d_i \in \{-1, 0, 1\}, \qquad d_i \ne 0 \Longrightarrow \{u_1 + i\sigma_1\} \in [0, 2\eta] \cup [1 - 2\eta, 1).$$
Moreover,
$$\sum_{i=1}^{L} |d_i| \le \left(4\eta + \frac{10}{q_0} \right) L + q_0 \le \frac{11}{2} \eta L.$$

3. *Define, for* $i \in [L]_{\mathbb{Z}}$,

$$\alpha_i = \lfloor u_1 + i\sigma_1 \rfloor - \lfloor u_1 + (i-1)\sigma_1 \rfloor, \qquad \beta_i = \lfloor v + i\sigma_2 \rfloor - \lfloor v + (i-1)\sigma_2 \rfloor.$$

Then,

$$\alpha_i - \beta_i = d_i - d_{i-1}, \qquad \sum_{i=1}^{L} |\alpha_i - \beta_i| \le 2 \sum_{i=1}^{L} |d_i| - |d_L| \le 11\eta L.$$

Proof. Proof of Claim 1. Let us prove that J is nonempty. Let $x_1 \le \cdots \le x_{q_0}$ be the ordered values of $v_0, \ldots, v_{q_0-1}$. Let $x_{q_0+1} = x_1 + 1$. Claim 1 of Lemma 6.3 with phase $u_2 + b\sigma_2$ gives

$$x_{a+1} - x_a \le \frac{5}{q_0} < \eta \quad (a \in [q_0]_{\mathbb{Z}}).$$

If $x_a \le u_1 \le x_{a+1}$ for some $a < q_0$, then $|u_1 - x_a| \le \frac{5}{q_0} < \eta$. If $u_1 \le x_1$, use

$$x_1 - u_1 \le x_1 \le x_1 + 1 - x_{q_0} \le \frac{5}{q_0}.$$

If $u_1 \ge x_{q_0}$, use

$$u_1 - x_{q_0} < 1 - x_{q_0} \le x_1 + 1 - x_{q_0} \le \frac{5}{q_0}.$$

Every x_i equals v_j for some $j \in [0, q_0 - 1]_{\mathbb{Z}}$. Hence, $J \neq \varnothing$. Since $j_* = \min J \in J$, we have $0 \le j_* < q_0$ and $|u_1 - v_{j_*}| < \eta$.

Proof of Claim 2. Let us locate every nonzero discrepancy. Since $u_1 \in [0, 1)$ and $v \in [0, 1)$, we have $d_0 = 0$. For $i \in [L]_{\mathbb{Z}}$, put

$$a_i = u_1 + i\sigma_1, \qquad b_i = v + i\sigma_2.$$

Claim 1 gives $|u_1 - v| < \eta$. Together with $|\sigma_1 - \sigma_2| < \frac{\eta}{2L}$ and $i \le L$, this gives

$$|a_i - b_i| \le |u_1 - v| + i|\sigma_1 - \sigma_2| < \eta + \frac{\eta}{2} = \frac{3\eta}{2} < 1.$$

Hence, the integer $d_i = \lfloor a_i \rfloor - \lfloor b_i \rfloor$ belongs to $\{-1, 0, 1\}$. If $d_i = 1$, then $b_i < \lfloor a_i \rfloor$. Thus,

$$\{a_i\} = a_i - \lfloor a_i \rfloor < a_i - b_i < \frac{3\eta}{2} < 2\eta.$$

If $d_i = -1$, then $b_i \ge \lfloor a_i \rfloor + 1$. Thus,

$$\{a_i\} \ge 1 - (b_i - a_i) > 1 - \frac{3\eta}{2} > 1 - 2\eta.$$

Thus, $d_i \neq 0$ implies $\{u_1 + i\sigma_1\} \in [0, 2\eta] \cup [1 - 2\eta, 1)$.

Let us count these indices. Write

$$H = \left\lfloor \frac{L}{q_0} \right\rfloor, \qquad s = L - Hq_0 \in [0, q_0 - 1]_{\mathbb{Z}}.$$

Partition $[L]_{\mathbb{Z}}$ into

$$B_z = [zq_0 + 1, (z+1)q_0]_{\mathbb{Z}} \quad (z \in [0, H-1]_{\mathbb{Z}})$$

and the final interval $[Hq_0 + 1, L]_{\mathbb{Z}}$, which has s elements. For $i = zq_0 + 1 + j$, $j \in [0, q_0 - 1]_{\mathbb{Z}}$, let

$$w_z = u_1 + (zq_0 + 1)\sigma_1.$$

Then, $\{u_1 + i\sigma_1\} = \{w_z + j\sigma_1\}$. Claim 2 of Lemma 6.3 bounds the number of nonzero d_i in B_z by $4\eta q_0 + 10$. Since $|d_i|$ equals 1 precisely at the nonzero discrepancies,

$$\sum_{i=1}^{L} |d_i| \le H(4\eta q_0 + 10) + s \le \left(4\eta + \frac{10}{q_0}\right) L + q_0.$$

The inequalities $\frac{10}{q_0} < \frac{\eta}{2}$ and $q_0 \le \eta L$ give the bound $\frac{11}{2}\eta L$. The estimate $\sum_{i=1}^{L} |d_i| \le H(4\eta q_0 + 10) + s$ also holds when $s = 0$ and $[Hq_0 + 1, L]_{\mathbb{Z}} = \varnothing$.

Proof of Claim 3. Let us express $\alpha_i - \beta_i$ as $d_i - d_{i-1}$ for $i \in [L]_{\mathbb{Z}}$. Expanding the definitions gives $\alpha_i - \beta_i = d_i - d_{i-1}$. Since $d_0 = 0$, the triangle inequality gives

$$\begin{aligned}\sum_{i=1}^{L} |\alpha_i - \beta_i| &\le \sum_{i=1}^{L} |d_i| + \sum_{i=1}^{L} |d_{i-1}| \\ &= 2\sum_{i=1}^{L} |d_i| - |d_L| \\ &\le 11\eta L.\end{aligned}$$

The inequality $2\sum_{i=1}^{L} |d_i| - |d_L| \le 11\eta L$ follows from Claim 2. □

Lemma 6.5 (Mechanical run-edits)**.** *Let $m \in \mathbb{Z}_{\ge 4}$ and $M = m-2$. Let $r \in (1, M)$ be irrational and let $0 < \eta < \frac{1}{4}$. Then, there are constants $\delta_m(r,\eta) > 0$ and $B_m(r,\eta) > 0$ depending only on m, r, and η with the following property. For each $(M, r, \eta, \delta_m(r,\eta), B_m(r,\eta))$-admissible tuple $(l, T, \sigma_1, \sigma_2, u_1, u_2, (A_1, \dots, A_l))$ and $U_0 = A_1 + \cdots + A_l$, there are $n \in \mathbb{Z}_{\ge 0}$, $\zeta \in \mathbb{Z}^2$, and a run-edit list $\Omega = (\omega_1, \dots, \omega_n)$ with the following properties.*

1. *The list Ω starts at U_0 and satisfies*

$$\mathrm{RunEResult}(\Omega, U_0) = \mathrm{Trans}(\mathrm{RunFactor}_{1,T}(\mathrm{MechRun}(\sigma_2, u_2)), \zeta).$$

2. *Ω consists of $K = T - l$ insertions of simple runs, I vertical-edge insertions, and D vertical-edge deletions. Their numbers satisfy*

$$n = K + I + D, \qquad I, D \in \mathbb{Z}_{\ge 0}, \qquad I + D \le 12\eta l + B_m(r,\eta). \tag{6.1}$$

3. *For $U_j = \mathrm{RunEResult}((\omega_1, \dots, \omega_j), U_0)$ $(j \in [0,n]_{\mathbb{Z}})$, every simple run $R \subseteq U_j$ satisfies $|R|_N \in [M]_{\mathbb{Z}}$.*

Proof. Proof of Claim 1. Let us choose p_0, q_0, L, $\delta_m(r,\eta)$, and $B_m(r,\eta)$. Put $k = \lfloor r \rfloor$. Choose relatively prime $p_0 \in \mathbb{Z}_{>0}$ and $q_0 \in \mathbb{Z}_{>0}$ such that

$$q_0 > \frac{20}{\eta}, \qquad |r - \frac{p_0}{q_0}| < \frac{1}{q_0^2}. \tag{6.2}$$

To obtain such a fraction, partition $[0,1)$ into Q intervals of length $\frac{1}{Q}$. Two of the $Q+1$ numbers $\{jr\}$, $j \in [0,Q]_{\mathbb{Z}}$, lie in the same interval. Subtracting their indices gives integers p and q such that

$$1 \le q \le Q, \qquad |qr - p| < \frac{1}{Q}, \qquad \left|r - \frac{p}{q}\right| < \frac{1}{qQ} \le \frac{1}{q^2}.$$

Let $\frac{p'}{q'}$ be $\frac{p}{q}$ in lowest terms. Then, $1 \le q' \le q$ and $\left|r - \frac{p'}{q'}\right| < \frac{1}{q^2} \le \frac{1}{q'^2}$. These reduced denominators are unbounded as $Q \to \infty$: otherwise, only finitely many reduced fractions in $[r-1, r+1]$ could occur, and their positive distances from the irrational number r could not

tend to zero. Thus, choose a reduced fraction with $q_0 > \frac{20}{\eta}$ satisfying (6.2). Choose $L \in \mathbb{Z}_{>0}$ such that $q_0 \le \eta L$ and let

$$\delta_m(r,\eta) = \frac{1}{2}\min\left\{\frac{1}{q_0^2}, \frac{\eta}{4L}, r-k, k+1-r\right\}, \qquad B_m(r,\eta) = (M-1)L+1. \tag{6.3}$$

These constants depend only on r, η, and M. Condition (b) of Definition 6.2 gives $k < \sigma_i < k+1$ for $i \in [2]_{\mathbb{Z}}$. Thus, every simple run of either mechanical sequence has vertical length k or $k+1$. Both lengths belong to $[M]_{\mathbb{Z}}$.

Let us compare the two slopes to the same rational approximation. For $\sigma \in \{\sigma_1, \sigma_2\}$, equations (6.2)–(6.3) give

$$|\sigma - \frac{p_0}{q_0}| < \frac{2}{q_0^2}.$$

Thus, Lemma 6.3 applies with $p = p_0$ and $q = q_0$.

Let us partition the run into factors of length L and one final factor. Let $H = \left\lfloor \frac{l-1}{L} \right\rfloor$. Let us construct the part of Ω acting on the first HL simple runs of U_0. Put $c_0 = 0$. For $z \in [0, H-1]_{\mathbb{Z}}$, suppose the run-edits on the first zL simple runs have produced the first c_z simple runs of $\mathrm{MechRun}(\sigma_2, u_2)$.

Let us choose the target intercept for the factor indexed by $z \in [0, H-1]_{\mathbb{Z}}$. Let $u = \{u_1 + zL\sigma_1\}$. Apply Claim 1 of Lemma 6.4 with source phase u and offset $b = c_z$. Let $j_z = \min J$ and $v = \{u_2 + (c_z + j_z)\sigma_2\}$ as in Claim 1 of Lemma 6.4. Then, $0 \le j_z < q_0$ and $|u - v| < \eta$. If $j_z > 0$, insert $\mathrm{RunFactor}_{c_z+1, j_z}(\mathrm{MechRun}(\sigma_2, u_2))$ immediately before the copy of $\mathrm{RunFactor}_{zL+1,L}(U_0)$.

Let us bound the number of vertical run-edits on the copy of $\mathrm{RunFactor}_{zL+1,L}(U_0)$. Define

$$X = \mathrm{RunFactor}_{1,L}(\mathrm{MechRun}(\sigma_1, u)), \qquad Y = \mathrm{RunFactor}_{1,L}(\mathrm{MechRun}(\sigma_2, v)).$$

Claim 3 of Lemma 6.4 gives

$$\sum_{i=1}^{L} |\,|\,\mathrm{RunFactor}_{i,1}(X)|_N - |\,\mathrm{RunFactor}_{i,1}(Y)|_N\,| \le 11\eta L.$$

By Lemma 6.1, there is a run-edit list $\Omega_z = (\omega_{z,1}, \ldots, \omega_{z,n_z})$ satisfying $\mathrm{RunEResult}(\Omega_z, X) = Y$ and consisting of at most $11\eta L$ vertical run-edits. Every simple-run vertical length produced by Ω_z belongs to $[M]_{\mathbb{Z}}$. Let $c_{z+1} = c_z + j_z + L$. For some $\zeta_z \in \mathbb{Z}^2$, the resulting first c_{z+1} simple runs form

$$\mathrm{Trans}(\mathrm{RunFactor}_{1,c_{z+1}}(\mathrm{MechRun}(\sigma_2, u_2)), \zeta_z).$$

Let us construct the run-edits on $A_{HL+1}, \ldots, A_l$. We have $l - HL \in [L]_{\mathbb{Z}}$. For $i \in [l-HL]_{\mathbb{Z}}$, change the length of the copy of A_{HL+i} to $|\,\mathrm{RunFactor}_{c_H+i,1}(\mathrm{MechRun}(\sigma_2, u_2))|_N$. Both lengths belong to $[M]_{\mathbb{Z}}$, including the length of A_l by Condition (a) of Definition 6.2. Lemma 6.1 gives at most $(M-1)L$ vertical run-edits and keeps every length in $[M]_{\mathbb{Z}}$. Now,

$$c_H + l - HL = l + \sum_{z=0}^{H-1} j_z \le l + Hq_0 \le (1+\eta)l < T.$$

Append

$$\mathrm{RunFactor}_{c_H+l-HL+1,\, T-c_H-l+HL}(\mathrm{MechRun}(\sigma_2, u_2)).$$

Let $Q = \mathrm{RunEResult}(\Omega, U_0)$ and $\zeta = \mathrm{Start}_{\mathrm{pt}}(Q)$. Then,

$$Q = \mathrm{Trans}(\mathrm{RunFactor}_{1,T}(\mathrm{MechRun}(\sigma_2, u_2)), \zeta).$$

Proof of Claim 2. Let us count the simple-run insertions and the vertical run-edits. The construction in Claim 1 uses exactly $K = T - l$ simple-run insertions. If I and D count its vertical-edge insertions and deletions, respectively, then $n = K + I + D$. The bound in (6.1) follows from

$$I + D \le 11\eta HL + (M-1)L \le 12\eta l + B_m(r,\eta).$$

Proof of Claim 3. Every inserted simple run has length k or $k+1$. Both belong to $[M]_{\mathbb{Z}}$. Lemma 6.1 keeps every intermediate vertical length in $[M]_{\mathbb{Z}}$ at each application in the construction of Ω. □

Lemma 6.6. *Let $m \in \mathbb{Z}_{\ge 3}$. For every $r \in I_m$, there is a sequence $((a_\nu, b_\nu))_{\nu\ge 1}$ in $\mathcal{A}_m$ satisfying*

$$\lim_{\nu\to\infty} a_\nu = \infty, \qquad \lim_{\nu\to\infty} \frac{b_\nu}{a_\nu} = r. \tag{6.4}$$

Proof. Let us choose pairs in $\mathcal{A}_m$ whose slopes tend to r. Choose distinct reduced fractions $\frac{b_\nu}{a_\nu} \in I_m$ tending to r. For each $A > 0$, the bounded interval I_m contains only finitely many reduced fractions with denominator at most A. Thus, $a_\nu \to \infty$. Since $\frac{b_\nu}{a_\nu} \in I_m$, we have $a_\nu^2 + b_\nu^2 - m a_\nu b_\nu \le 0$. Equation (1.3) gives $(a_\nu, b_\nu) \in \mathcal{A}_m$. □

Theorem 6.7. *Let $m \in \mathbb{Z}_{\ge 4}$ and $M = m - 2$. For every irrational $r \in (1, M)$, there is $\Upsilon_m(r) \in \mathbb{R}$ such that*

$$\lim_{\substack{a\to\infty,\ (a,b)\in\mathcal{A}_m \\ \frac{b}{a}\to r}} \frac{\log E_m(a,b)}{a} = \Upsilon_m(r).$$

Proof. Let us choose the comparison parameters for Lemma 6.5 and Claim 2 of Proposition 4.10. Let

$$C = 32(m+1), \qquad H = 2C + 16, \qquad 0 < \eta < \frac{1}{2H}, \qquad d = 1 - H\eta.$$

Use $\delta_m(r,\eta)$ and $B_m(r,\eta)$ from Lemma 6.5. Choose relatively prime $p \in \mathbb{Z}_{>0}$ and $q \in \mathbb{Z}_{>0}$ such that $q \ge 2$, $\alpha = \frac{p}{q} \in (1, M)$, and $|\alpha - r| < \delta_m(r,\eta)$. We fix α, p, and q while $a \to \infty$, $(a,b) \in \mathcal{A}_m$, and $\frac{b}{a} \to r$.

Let us verify (4.27) for Claim 2 of Proposition 4.10. For a run Z of ν simple runs and sufficiently large ν, let

$$P = \lfloor C\eta\nu \rfloor, \qquad T = \nu - 2P - 2, \qquad V = \mathrm{RunFactor}_{P+2,T}(Z).$$

We have $P > 0$ and $T > 0$. Define

$$W_{\mathrm{L}} = \mathrm{RunFactor}_{1,P}(Z), \qquad W_{\mathrm{R}} = \mathrm{RunFactor}_{\nu-P+1,P}(Z),$$
$$u = |\,\mathrm{RunFactor}_{P+1,1}(Z)|_N, \qquad v = |\,\mathrm{RunFactor}_{\nu-P,1}(Z)|_N.$$

Equation (4.28) gives $Z = W_{\mathrm{L}} \mathbin{+\!\!+} R_u \mathbin{+\!\!+} V \mathbin{+\!\!+} R_v \mathbin{+\!\!+} W_{\mathrm{R}}$. Let $l = d\nu + O(1)$. Then,

$$K = T - l, \qquad \lim_{\nu\to\infty} \frac{K}{\nu} = 16\eta, \qquad T \ge (1+\eta)l + B_m(r,\eta)$$

when ν is sufficiently large. If $I \in \mathbb{Z}_{\ge 0}$ and $D \in \mathbb{Z}_{\ge 0}$ satisfy $I + D \le 13\eta\nu$, then eventually

$$K + I + (m+1)D \le (17 + 13(m+1))\eta\nu \le P.$$

The inequality $(17 + 13(m+1))\eta\nu \le P$ uses $P = \lfloor C\eta\nu \rfloor$ and $(C - 17 - 13(m+1))\eta\nu \ge 1$ for sufficiently large ν, since $C = 32(m+1) > 17 + 13(m+1)$. Equation (4.30) gives

$$\log \mathrm{Fib}_m(l, D) \le \nu\psi_m(13\eta).$$

Let us verify admissibility in the two applications below. Equation (2.9) gives $\Gamma(a,b) \to r$. Thus, $1 < \Gamma(a,b) < M$ and $|\Gamma(a,b) - r| < \delta_m(r,\eta)$ for all sufficiently large a. The choice of α and the bound $T \geq (1+\eta)l + B_m(r,\eta)$ verify condition (b) of Definition 6.2 in both applications.

First, let us obtain a lower bound for $\frac{1}{a}\log E_m(a,b)$. Let $(a,b) \in \mathcal{A}_m$, $\nu = a+1$, and $j = \left\lfloor \frac{d\nu}{q} \right\rfloor$. Apply Lemma 6.5 with

$$\begin{gathered} U = W_{p,q}^j, \quad l = jq, \quad Z = \mathcal{U}(a,b), \quad V = \mathrm{RunFactor}_{P+2,T}(\mathcal{U}(a,b)), \\ (\sigma_1, \sigma_2, u_1, u_2) = (\alpha, \Gamma(a,b), 0, \{(P+1)\Gamma(a,b)\}), \\ A_i = R_{w_i(p,q)} \qquad (i \in [l]_{\mathbb{Z}}). \end{gathered}$$

Definition 5.2 verifies condition (a) of Definition 6.2 for these source runs. Lemma 6.5 gives $I + D \leq 12\eta l + B_m(r,\eta) \leq 13\eta\nu$ for all sufficiently large a. The last run $\mathrm{RunFactor}_{a+1,1}(\mathcal{U}(a,b))$ belongs to W_{R} and does not occur in V. Claim 2 of Proposition 4.10 and Lemma 2.10 apply with $V_* = \mathcal{U}(a,b)$. Equations (2.13), (4.29), and (4.30) give

$$\frac{\log E_m(a,b)}{a} \geq \frac{j}{a}\frac{\log C_0(W_{p,q}^j)}{j} - \frac{\nu}{a}\psi_m(13\eta) - \frac{\log m}{a}.$$

Since $\frac{j}{a} \to \frac{d}{q}$ and $\frac{\nu}{a} \to 1$, Claim 2 of Proposition 3.8 and (5.1) give

$$\liminf_{\substack{a\to\infty,\ (a,b)\in\mathcal{A}_m \\ \frac{b}{a}\to r}} \frac{\log E_m(a,b)}{a} \geq d\Upsilon_m(\alpha) - \psi_m(13\eta). \tag{6.5}$$

Second, let us obtain an upper bound for $\frac{1}{a}\log E_m(a,b)$. Let $(a,b) \in \mathcal{A}_m$, $l = a+1$, $N = \left\lceil \frac{l}{dq} \right\rceil$, and $\nu = Nq$. Apply Lemma 6.5 with

$$\begin{gathered} U = \mathcal{U}(a,b), \quad Z = W_{p,q}^N, \quad V = \mathrm{RunFactor}_{P+2,T}(W_{p,q}^N), \\ (\sigma_1, \sigma_2, u_1, u_2) = (\Gamma(a,b), \alpha, 0, \{(P+1)\alpha\}), \\ A_i = R_{\Lambda_i(a,b)} \quad (i \in [a]_{\mathbb{Z}}), \qquad A_l = R_1. \end{gathered}$$

Definition 2.25 verifies condition (a) of Definition 6.2, including $|A_l|_N = 1 \in [M]_{\mathbb{Z}}$. Lemma 6.5 gives $I + D \leq 12\eta l + B_m(r,\eta) \leq 13\eta\nu$ for all sufficiently large a. Claim 2 of Proposition 4.10 and Lemma 2.10 apply with $V_* = W_{p,q}^N$. Equations (4.29), (2.13), and (4.30) give

$$\frac{\log E_m(a,b)}{a} \leq \frac{N}{a}\frac{\log C_0(W_{p,q}^N)}{N} + \frac{\nu}{a}\psi_m(13\eta) - \frac{\log m}{a}.$$

Here, $\frac{N}{a} \to \frac{1}{dq}$ and $\frac{\nu}{a} \to \frac{1}{d}$. By Claim 2 of Proposition 3.8 and (5.1),

$$\limsup_{\substack{a\to\infty,\ (a,b)\in\mathcal{A}_m \\ \frac{b}{a}\to r}} \frac{\log E_m(a,b)}{a} \leq \frac{\Upsilon_m(\alpha) + \psi_m(13\eta)}{d}. \tag{6.6}$$

Third, let us compare (6.5) and (6.6). Lemma 3.5 gives $\Upsilon_m(\alpha) \leq m(M+1)\log 2$. The interval in (6.5)–(6.6) has width at most

$$m(M+1)\log 2\left(\frac{1}{d} - d\right) + \psi_m(13\eta)\left(1 + \frac{1}{d}\right).$$

As $\eta \downarrow 0$, we have $d = 1 - H\eta \to 1$ and

$$\frac{1}{d} - d \longrightarrow 0, \qquad \psi_m(13\eta) \longrightarrow 0.$$

The limit $\psi_m(13\eta) \to 0$ as $\eta \downarrow 0$ follows from $x \log x \to 0$ as $x \downarrow 0$ and Definition 4.11.

Let us prove that every sequence has the same limit. For each $i \in [2]_{\mathbb{Z}}$, let $((a_{i,n}, b_{i,n}))_{n\geq 1}$ be a sequence in $\mathcal{A}_m$ satisfying

$$\lim_{n\to\infty} a_{i,n} = \infty, \qquad \lim_{n\to\infty} \frac{b_{i,n}}{a_{i,n}} = r.$$

Define

$$L_i = \liminf_{n\to\infty} \frac{\log E_m(a_{i,n}, b_{i,n})}{a_{i,n}}, \qquad U_i = \limsup_{n\to\infty} \frac{\log E_m(a_{i,n}, b_{i,n})}{a_{i,n}}.$$

Lemma 3.5 and (2.13) make these four numbers finite.

Let us prove $U_i \leq L_j$ for $i, j \in [2]_{\mathbb{Z}}$. For each fixed η, choose α, $\delta_m(r,\eta)$, and $B_m(r,\eta)$ before taking either sequence limit. Equations (6.5) and (6.6) give, for $i, j \in [2]_{\mathbb{Z}}$,

$$U_i - L_j \leq m(M+1)\log 2 \left(\frac{1}{d} - d\right) + \psi_m(13\eta)\left(1 + \frac{1}{d}\right).$$

Letting $\eta \downarrow 0$ gives $U_i \leq L_j$. Taking $i = j$ and using $L_i \leq U_i$ gives $L_i = U_i$. Taking $(i,j) = (1,2)$ and $(2,1)$ gives $L_1 = U_1 = L_2 = U_2$. Lemma 6.6 supplies at least one such sequence. Define $\Upsilon_m(r)$ to be its limit. The equality $L_1 = U_1 = L_2 = U_2$ makes this definition independent of the chosen sequence. □

6.2 Continuity of the limit

Lemma 6.8. *Let $m \in \mathbb{Z}_{\geq 3}$. Let $h_1 \in \mathbb{R}$ and $h_2 \in \mathbb{R}$ satisfy $h_1 < h_2$. Let $J = [h_1, h_2]$. Let $F : J \to \mathbb{R}$. Assume the following.*

(a) For each $r \in J$ and every sequence $((a_n, b_n))_{n\geq 1}$ in $\mathcal{A}_m$ satisfying

$$\lim_{n\to\infty} a_n = \infty, \qquad \lim_{n\to\infty} \frac{b_n}{a_n} = r,$$

we have

$$\lim_{n\to\infty} \frac{\log E_m(a_n, b_n)}{a_n} = F(r).$$

(b) For each $r \in J$, there exists a sequence $((a_n, b_n))_{n\geq 1}$ in $\mathcal{A}_m$ such that

$$\lim_{n\to\infty} a_n = \infty, \qquad \lim_{n\to\infty} \frac{b_n}{a_n} = r.$$

Then, we have the following.

1. *For every $r \in J$ and $\varepsilon > 0$, there exist $A > 0$ and $\delta > 0$ such that*

$$\left|\frac{\log E_m(a,b)}{a} - F(r)\right| < \varepsilon \quad \left((a,b) \in \mathcal{A}_m,\ a > A,\ \left|\frac{b}{a} - r\right| < \delta\right). \tag{6.7}$$

2. *The function F is continuous on J.*

3. *At the endpoints of J, we have the following.*

   (a) For $r \in \{h_1, h_2\}$, the estimate in Claim 1 holds when additionally $\frac{b}{a} \in J$.

   (b) We have

$$\lim_{s\downarrow h_1} F(s) = F(h_1), \qquad \lim_{s\uparrow h_2} F(s) = F(h_2) \qquad (s \in J).$$

Proof. Proof of Claim 1. Let us prove (6.7) by contradiction. If (6.7) fails at r, there is $\varepsilon_0 > 0$ such that, for every positive integer ν, we find $(a_\nu, b_\nu) \in \mathcal{A}_m$ with

$$a_\nu > \nu, \qquad \left|\frac{b_\nu}{a_\nu} - r\right| < \frac{1}{\nu}, \qquad \left|\frac{\log E_m(a_\nu, b_\nu)}{a_\nu} - F(r)\right| \geq \varepsilon_0.$$

We have $a_\nu \to \infty$ and $\frac{b_\nu}{a_\nu} \to r$. Assumption (a) contradicts $|\frac{\log E_m(a_\nu,b_\nu)}{a_\nu} - F(r)| \geq \varepsilon_0$.

Proof of Claim 2. Let us obtain continuity from (6.7). Let $r \in J$ and $\varepsilon > 0$. Claim 1 gives $\delta > 0$ and $A > 0$ such that

$$\left|\frac{\log E_m(a, b)}{a} - F(r)\right| < \frac{\varepsilon}{2} \quad \left((a, b) \in \mathcal{A}_m,\ a > A,\ \left|\frac{b}{a} - r\right| < \delta\right).$$

Let us prove $|F(s)-F(r)| < \varepsilon$ when $s \in J$ and $|s-r| < \frac{\delta}{2}$. Fix such an s. By assumption (b), choose a sequence $((a_\nu, b_\nu))_{\nu\geq 1}$ in $\mathcal{A}_m$ such that

$$a_\nu \longrightarrow \infty, \qquad \frac{b_\nu}{a_\nu} \longrightarrow s \qquad (\nu \longrightarrow \infty).$$

Choose $\nu_0 \in \mathbb{Z}_{>0}$ such that

$$a_\nu > A, \qquad \left|\frac{b_\nu}{a_\nu} - s\right| < \frac{\delta}{2} \qquad (\nu \geq \nu_0).$$

For $\nu \geq \nu_0$, the triangle inequality gives

$$\left|\frac{b_\nu}{a_\nu} - r\right| \leq \left|\frac{b_\nu}{a_\nu} - s\right| + |s - r| < \delta.$$

Hence, (6.7) applies with ε replaced by $\frac{\varepsilon}{2}$ for every $\nu \geq \nu_0$.

Let us take the limit as $\nu \to \infty$. Assumption (a) and continuity of the absolute value give

$$|F(s) - F(r)| = \lim_{\nu\to\infty} \left|\frac{\log E_m(a_\nu, b_\nu)}{a_\nu} - F(r)\right| \leq \frac{\varepsilon}{2} < \varepsilon.$$

Proof of Claim 3(a). Claim 1 applies to $r \in \{h_1, h_2\}$ and pairs satisfying $\frac{b}{a} \in J$.

Proof of Claim 3(b). Let us identify the one-sided endpoint limits. The hypothesis $h_1 < h_2$ makes both one-sided domains nonempty. By Claim 2, for every $\varepsilon > 0$ there are $\delta_1 > 0$ and $\delta_2 > 0$ such that

$$|F(s) - F(h_1)| < \varepsilon \quad (s \in J,\ 0 < s - h_1 < \delta_1),$$
$$|F(s) - F(h_2)| < \varepsilon \quad (s \in J,\ 0 < h_2 - s < \delta_2).$$

These are the right limit at h_1 and the left limit at h_2, respectively. □

7 A uniform lower bound near slope one for the three-arrow quiver

Definition 7.1. *For positive coprime integers a and r, let*

$$E_r(a) = E_3(a, a + r).$$

From here to the end of the manuscript, let

$$\beta = \frac{256}{27}.$$

7.1 Graphs and quivers

Definition 7.2 (Graphs)**.** *For a set V, define*

$$\mathrm{Sym}^2(V) = (V \times V)/\!\sim, \qquad (u,v) \sim (u',v') \iff (u',v') = (u,v) \text{ or } (u',v') = (v,u).$$

Let $[\![u,v]\!]$ denote the class of (u,v). For $U \subseteq V$, we identify $\mathrm{Sym}^2(U)$ with its image in $\mathrm{Sym}^2(V)$. A graph G consists of a vertex set $\mathrm{Vert}(G)$, an edge set $\mathrm{Edge}(G)$, and an endpoint map

$$\partial_G : \mathrm{Edge}(G) \longrightarrow \mathrm{Sym}^2(\mathrm{Vert}(G)).$$

We call G finite if $\mathrm{Vert}(G)$ and $\mathrm{Edge}(G)$ are finite.

1. *An edge e is a loop at v if $\partial_G(e) = [\![v,v]\!]$.*
2. *Edges e and e' are parallel if $\partial_G(e) = \partial_G(e')$.*
3. *For $n \in \mathbb{Z}_{\geq 0}$, consider a tuple*

   $$\gamma = (v_0, e_1, v_1, \ldots, e_n, v_n)$$

   such that $v_i \in \mathrm{Vert}(G)$ for $i \in [0,n]_{\mathbb{Z}}$ and

   $$e_i \in \mathrm{Edge}(G), \qquad \partial_G(e_i) = [\![v_{i-1}, v_i]\!] \qquad (i \in [n]_{\mathbb{Z}}).$$

   The tuple γ starts at v_0 and ends at v_n. We call γ a simple path of length n if $v_0, \ldots, v_n$ are pairwise distinct.
4. *A cycle of length $n \in \mathbb{Z}_{\geq 1}$ is a tuple $(v_0, e_1, v_1, \ldots, e_n, v_n)$ such that $v_n = v_0$, the vertices $v_0, \ldots, v_{n-1} \in \mathrm{Vert}(G)$ are pairwise distinct, the edges $e_1, \ldots, e_n \in \mathrm{Edge}(G)$ are pairwise distinct, and*

   $$\partial_G(e_i) = [\![v_{i-1}, v_i]\!] \qquad (i \in [n]_{\mathbb{Z}}).$$

Definition 7.3 (Graph notions)**.** *Let G be a graph.*

1. *We call G connected if, for every $\varnothing \neq U \subsetneq \mathrm{Vert}(G)$, there exist $u \in U$, $v \in \mathrm{Vert}(G) \setminus U$, and $e \in \mathrm{Edge}(G)$ such that*

   $$\partial_G(e) = [\![u,v]\!].$$

2. *We call G simple if it has no loops and ∂_G is injective. For a simple graph, we identify each edge with its image under ∂_G and let*

   $$G = (\mathrm{Vert}(G), \mathrm{Edge}(G)), \qquad \mathrm{Edge}(G) \subseteq \{[\![u,v]\!] : u, v \in \mathrm{Vert}(G),\ u \neq v\}.$$

   We call a simple graph without cycles a forest and a connected forest with at least one vertex a tree.
3. *Let G' be a graph. A homomorphism $\psi : G \to G'$ is a pair of maps*

   $$\psi_0 : \mathrm{Vert}(G) \longrightarrow \mathrm{Vert}(G'), \qquad \psi_1 : \mathrm{Edge}(G) \longrightarrow \mathrm{Edge}(G')$$

   such that, for every $e \in \mathrm{Edge}(G)$, $u \in \mathrm{Vert}(G)$, and $v \in \mathrm{Vert}(G)$,

   $$\partial_G(e) = [\![u,v]\!] \quad \Longrightarrow \quad \partial_{G'}(\psi_1(e)) = [\![\psi_0(u), \psi_0(v)]\!].$$

   We call ψ an isomorphism if ψ_0 and ψ_1 are bijective. We compose homomorphisms componentwise and let $1_G = (1_{\mathrm{Vert}(G)}, 1_{\mathrm{Edge}(G)})$. If G and G' are simple, we denote a homomorphism $\psi : G \to G'$ by its vertex map ψ_0.

4. *Let* $v \in \mathrm{Vert}(G)$. *We define*

$$\mathrm{AdjVert}_G(v) = \{u \in \mathrm{Vert}(G) : \partial_G(e) = [\![v, u]\!] \text{ for some } e \in \mathrm{Edge}(G)\}.$$

We call u *and* v *adjacent if* $u \in \mathrm{AdjVert}_G(v)$. *We define the set of edges incident to* v *by*

$$\mathrm{IncEdge}_G(v) = \{e \in \mathrm{Edge}(G) : \partial_G(e) = [\![v, u]\!] \text{ for some } u \in \mathrm{Vert}(G)\}.$$

We define the degree of v *by*

$$\begin{aligned}\deg_G(v) = {} & \#\{e \in \mathrm{Edge}(G) : \partial_G(e) = [\![v, u]\!] \text{ for some } u \neq v\} \\ & + 2\#\{e \in \mathrm{Edge}(G) : \partial_G(e) = [\![v, v]\!]\}.\end{aligned}$$

We define the set of leaves by

$$\mathrm{Leaf}(G) = \{v \in \mathrm{Vert}(G) : \deg_G(v) = 1\}.$$

5. *A graph* H *is a subgraph of* G *if*

$$\mathrm{Vert}(H) \subseteq \mathrm{Vert}(G), \qquad \mathrm{Edge}(H) \subseteq \mathrm{Edge}(G)$$

and

$$\partial_H(e) = \partial_G(e) \qquad (e \in \mathrm{Edge}(H)).$$

6. *For* $U \subseteq \mathrm{Vert}(G)$, *the induced subgraph* $\mathrm{Subgraph}(G, U)$ *has vertex set* U *and edge set*

$$\mathrm{Edge}(\mathrm{Subgraph}(G, U)) = \{e \in \mathrm{Edge}(G) : \partial_G(e) \in \mathrm{Sym}^2(U)\}.$$

7. *Let* $\mathrm{CtdComp}(G)$ *denote the set of induced subgraphs* $\mathrm{Subgraph}(G, U)$ *for nonempty* $U \subseteq \mathrm{Vert}(G)$ *that are maximal under inclusion among connected induced subgraphs of* G. *We call an element of* $\mathrm{CtdComp}(G)$ *a connected component of* G.

8. *Let* C *be a set. An edge coloring is a map* $\kappa : \mathrm{Edge}(G) \to C$. *We call* (G, κ) *a colored graph.*

9. *A color-preserving homomorphism* $(G, \kappa) \to (G', \kappa')$ *is a graph homomorphism* $\psi = (\psi_0, \psi_1) : G \to G'$ *such that*

$$\kappa' \circ \psi_1 = \kappa.$$

We call it a color-preserving isomorphism if ψ_0 *and* ψ_1 *are bijective.*

10. *If* G *is simple, we call* κ *a proper edge coloring if*

$$\begin{aligned}& [\![u, v]\!], [\![u, w]\!] \in \mathrm{Edge}(G), \qquad v \neq w \\ \implies \quad & \kappa([\![u, v]\!]) \neq \kappa([\![u, w]\!]) \qquad (u, v, w \in \mathrm{Vert}(G)).\end{aligned}$$

Definition 7.4 (Bipartite graphs)**.** *Let* G *be a graph. Let* $I \subseteq \mathrm{Vert}(G)$ *and* $J \subseteq \mathrm{Vert}(G)$.

1. *We call the pair* $\mathbb{G} = (G, (I, J))$ *a bipartite graph if*

$$\mathrm{Vert}(G) = I \sqcup J, \qquad \partial_G(\mathrm{Edge}(G)) \subseteq \{[\![u, v]\!] : u \in I,\ v \in J\}.$$

2. *Let* $\mathbb{G} = (G, (I, J))$ *be a bipartite graph. We call* I *the source set and* J *the sink set of* $\mathbb{G}$. *Let* $\mathrm{Src}_{\mathbb{G}} = I$ *and* $\mathrm{Snk}_{\mathbb{G}} = J$. *For* $v \in \mathrm{Vert}(G)$, *define*

$$\begin{aligned}\mathrm{SrcAdj}_{\mathbb{G}}(v) &= \mathrm{AdjVert}_G(v) \cap I, \\ \mathrm{SnkAdj}_{\mathbb{G}}(v) &= \mathrm{AdjVert}_G(v) \cap J.\end{aligned}$$

3. Let $\mathbb{G} = (G, (I, J))$ be a bipartite graph. For a subgraph H of G, define

$$\mathrm{SubSrc}_{\mathbb{G}}(H) = I \cap \mathrm{Vert}(H), \qquad \mathrm{SubSnk}_{\mathbb{G}}(H) = J \cap \mathrm{Vert}(H).$$

Lemma 7.5 (Rigidity of properly colored trees)**.** *Let $\mathbb{T} = (T, (I, J))$ be a bipartite graph such that T is a finite tree. Let C be a set and let $\kappa : \mathrm{Edge}(T) \to C$ be a proper edge coloring. Let $\phi = (\phi_0, \phi_1) : T \to T$ be an automorphism satisfying*

$$\kappa \circ \phi_1 = \kappa, \qquad \phi_0(I) = I, \qquad \phi_0(J) = J.$$

Then, $\phi = 1_T$.

Proof. Let us use induction on $n = \#\,\mathrm{Vert}(T)$. First, suppose $n = 1$. The only vertex is fixed and there are no edges. Thus, $\phi = 1_T$.

Second, suppose $n = 2$. Let $I = \{u\}$ and $J = \{v\}$. Since $\phi_0(I) = I$ and $\phi_0(J) = J$, we have $\phi_0(u) = u$ and $\phi_0(v) = v$. The unique edge $[\![u, v]\!]$ is also fixed.

Third, suppose $n \geq 3$. Let us remove the leaves and apply induction. Define

$$L = \mathrm{Leaf}(T), \qquad V' = \mathrm{Vert}(T) \setminus L, \qquad T' = \mathrm{Subgraph}(T, V').$$

Let $(v_0, e_1, v_1, \ldots, e_\ell, v_\ell)$ be a longest simple path in T. Then,

$$\ell \geq 2, \qquad v_0, v_\ell \in L, \qquad \{v_1, \ldots, v_{\ell-1}\} \subseteq V'.$$

Let us prove that T' is connected. For $u \in V'$ and $w \in V'$, let $P = (x_0, f_1, x_1, \ldots, f_r, x_r)$ be a simple path in T from $x_0 = u$ to $x_r = w$. For $i \in [1, r-1]_{\mathbb{Z}}$, the distinct edges f_i and f_{i+1} are incident to x_i. Hence, $\deg_T(x_i) \geq 2$ and every vertex of P belongs to V'. Thus, P lies in T'. The graph T' has no cycles. Therefore, T' is a tree and

$$1 \leq \#\,\mathrm{Vert}(T') \leq n - 2.$$

Let us apply the induction hypothesis to T'. The automorphism ϕ preserves degrees. Hence, $\phi_0(L) = L$ and $\phi_0(V') = V'$. Its restriction

$$\phi' = \left(\phi_0|_{V'}, \phi_1|_{\mathrm{Edge}(T')}\right) : T' \to T'$$

preserves the coloring $\kappa|_{\mathrm{Edge}(T')}$ and the $(I \cap V', J \cap V')$-bipartition. The induction hypothesis gives

$$\phi_0(v) = v \qquad (v \in V').$$

Let us verify that $\phi_0(u) = u$ for each $u \in L$. Its unique neighbor v lies in V' because $n \geq 3$. Let $e = [\![u, v]\!]$. Since $\phi_0(v) = v$,

$$e, \phi_1(e) \in \mathrm{IncEdge}_T(v), \qquad \kappa(\phi_1(e)) = \kappa(e).$$

Properness of κ gives $\phi_1(e) = e$. Hence, $\phi_0(u) = u$. Thus, $\phi_0 = 1_{\mathrm{Vert}(T)}$. The injectivity of ∂_T then gives $\phi_1 = 1_{\mathrm{Edge}(T)}$. □

Definition 7.6 (Edge deletion and cut components)**.** *Let G be a graph.*

1. For $e \in \mathrm{Edge}(G)$, we define a graph

$$\mathrm{DelEdge}(G, e) = \left(\mathrm{Vert}(G), \mathrm{Edge}(G) \setminus \{e\}, \partial_G|_{\mathrm{Edge}(G)\setminus\{e\}}\right).$$

2. For $v \in \mathrm{Vert}(G)$, let $\mathrm{CtdComp}(G; v)$ be the connected component of G containing v.

3. *Let $\mathbb{S} = (S, (I_S, J_S))$ be a bipartite graph such that S is a tree. Let $e = [\![x, b]\!] \in \mathrm{Edge}(S)$ for $x \in I_S$ and $b \in J_S$. We define the sink-side cut component by*

$$\mathrm{SnkCtdComp}(\mathbb{S}; e) = \mathrm{CtdComp}(\mathrm{DelEdge}(S, e); b).$$

Definition 7.7 (Quivers)**.** *A quiver is a triple*

$$Q = (\mathrm{Vert}(Q), \mathrm{Edge}(Q), (s_Q, t_Q)).$$

The sets $\mathrm{Vert}(Q)$ and $\mathrm{Edge}(Q)$ are its vertex and arrow sets. The tail and head maps are

$$s_Q, t_Q : \mathrm{Edge}(Q) \longrightarrow \mathrm{Vert}(Q).$$

We call Q finite if $\mathrm{Vert}(Q)$ and $\mathrm{Edge}(Q)$ are finite. Let $\alpha : u \to v$ mean that $s_Q(\alpha) = u$ and $t_Q(\alpha) = v$. If the map $\alpha \mapsto (s_Q(\alpha), t_Q(\alpha))$ is injective, we identify each arrow α with $(s_Q(\alpha), t_Q(\alpha))$.

Definition 7.8 (Directed paths and acyclicity)**.** *Let Q be a quiver.*

1. *A directed path of length $n \in \mathbb{Z}_{\geq 0}$ is a tuple $(v_0, \alpha_1, v_1, \ldots, \alpha_n, v_n)$ such that*

$$s_Q(\alpha_i) = v_{i-1}, \qquad t_Q(\alpha_i) = v_i \qquad (i \in [n]_{\mathbb{Z}}).$$

   Here $v_i \in \mathrm{Vert}(Q)$ for $i \in [0, n]_{\mathbb{Z}}$ and $\alpha_i \in \mathrm{Edge}(Q)$ for $i \in [n]_{\mathbb{Z}}$.

2. *A directed path $(v_0, \alpha_1, v_1, \ldots, \alpha_n, v_n)$ is a directed cycle if $n \geq 1$, $v_n = v_0$, and $v_0, \ldots, v_{n-1}$ are pairwise distinct.*

3. *We call Q acyclic if it has no directed cycle.*

Definition 7.9 (Quiver notions)**.** *Let Q be a quiver.*

1. *The underlying graph $\mathrm{UnderG}(Q)$ of Q has*

$$\mathrm{Vert}(\mathrm{UnderG}(Q)) = \mathrm{Vert}(Q), \qquad \mathrm{Edge}(\mathrm{UnderG}(Q)) = \{\bar{\alpha} : \alpha \in \mathrm{Edge}(Q)\}.$$

   Here $\alpha \mapsto \bar{\alpha}$ is a bijection onto a copy of $\mathrm{Edge}(Q)$. The endpoint map of $\mathrm{UnderG}(Q)$ is

$$\partial_{\mathrm{UnderG}(Q)}(\bar{\alpha}) = [\![s_Q(\alpha), t_Q(\alpha)]\!] \qquad (\alpha \in \mathrm{Edge}(Q)).$$

   Let $\deg_Q(v) = \deg_{\mathrm{UnderG}(Q)}(v)$. For $v \in \mathrm{Vert}(Q)$, we define the set of arrows incident to v by

$$\begin{aligned}\mathrm{IncEdgeQuiv}_Q(v) &:= \{\alpha \in \mathrm{Edge}(Q) : \bar{\alpha} \in \mathrm{IncEdge}_{\mathrm{UnderG}(Q)}(v)\} \\ &= \{\alpha \in \mathrm{Edge}(Q) : s_Q(\alpha) = v \text{ or } t_Q(\alpha) = v\}.\end{aligned}$$

   We define $\mathrm{AdjVert}_Q(v) = \mathrm{AdjVert}_{\mathrm{UnderG}(Q)}(v)$. We use $\mathrm{UnderG}(Q)$ to discuss trees and connected components of Q.

2. *Let Q and Q' be quivers. A homomorphism $\phi : Q \to Q'$ is a pair of maps*

$$\phi_0 : \mathrm{Vert}(Q) \longrightarrow \mathrm{Vert}(Q'), \qquad \phi_1 : \mathrm{Edge}(Q) \longrightarrow \mathrm{Edge}(Q')$$

   satisfying

$$s_{Q'} \circ \phi_1 = \phi_0 \circ s_Q, \qquad t_{Q'} \circ \phi_1 = \phi_0 \circ t_Q.$$

   We call ϕ an isomorphism if ϕ_0 and ϕ_1 are bijective. We compose homomorphisms componentwise and let $1_Q = (1_{\mathrm{Vert}(Q)}, 1_{\mathrm{Edge}(Q)})$. The underlying graph homomorphism is

$$\mathrm{UnderG}(\phi) = (\phi_0, \bar{\phi_1}) : \mathrm{UnderG}(Q) \longrightarrow \mathrm{UnderG}(Q'), \qquad \bar{\phi_1}(\bar{\alpha}) = \overline{\phi_1(\alpha)}.$$

3. *A quiver Q' is a subquiver of Q if*

$$\mathrm{Vert}(Q') \subseteq \mathrm{Vert}(Q), \qquad \mathrm{Edge}(Q') \subseteq \mathrm{Edge}(Q)$$

and

$$s_{Q'}(\alpha) = s_Q(\alpha), \qquad t_{Q'}(\alpha) = t_Q(\alpha) \qquad (\alpha \in \mathrm{Edge}(Q')).$$

For $U \subseteq \mathrm{Vert}(Q)$, the full subquiver $\mathrm{SubQuiv}(Q, U)$ has vertex set U and arrow set

$$\mathrm{Edge}(\mathrm{SubQuiv}(Q, U)) = \{\alpha \in \mathrm{Edge}(Q) : s_Q(\alpha), t_Q(\alpha) \in U\}.$$

Definition 7.10 (Bipartite quivers)**.** *Let Q be a quiver. Let $I \subseteq \mathrm{Vert}(Q)$ and $J \subseteq \mathrm{Vert}(Q)$.*

1. *We call the pair $\mathbb{Q} = (Q, (I, J))$ a bipartite quiver if*

$$\mathrm{Vert}(Q) = I \sqcup J, \qquad (s_Q, t_Q)(\mathrm{Edge}(Q)) \subseteq I \times J.$$

The underlying bipartite graph of $\mathbb{Q}$ is $(\mathrm{UnderG}(Q), (I, J))$.

2. *Let $\mathbb{Q} = (Q, (I, J))$ and $\mathbb{Q}' = (Q', (I', J'))$ be bipartite quivers. A quiver isomorphism $(\phi_0, \phi_1) : Q \to Q'$ preserves the bipartitions if*

$$\phi_0(I) = I', \qquad \phi_0(J) = J'.$$

3. *Let $\mathbb{Q} = (Q, (I, J))$ be a bipartite quiver. We call $\mathbb{Q}$ simple if the map*

$$(s_Q, t_Q) : \mathrm{Edge}(Q) \longrightarrow I \times J$$

is injective.

See Remark 4.7(1) of [19] for arrow colors of bipartite quivers.

Definition 7.11 (Colored quivers)**.** *Let Q be a quiver. Let C be a set.*

1. *A colored quiver with color set C is a pair*

$$\widetilde{Q} = (Q, c), \qquad c : \mathrm{Edge}(Q) \longrightarrow C.$$

We define $\mathrm{QCol}(\widetilde{Q}) = c$ and use the notation

$$\mathrm{Vert}(\widetilde{Q}) = \mathrm{Vert}(Q), \qquad \mathrm{Edge}(\widetilde{Q}) = \mathrm{Edge}(Q),$$
$$s_{\widetilde{Q}} = s_Q, \qquad t_{\widetilde{Q}} = t_Q, \qquad \mathrm{UnderG}(\widetilde{Q}) = \mathrm{UnderG}(Q).$$

Let

$$\deg_{\widetilde{Q}}(v) = \deg_Q(v), \qquad \mathrm{IncEdgeQuiv}_{\widetilde{Q}}(v) = \mathrm{IncEdgeQuiv}_Q(v) \qquad (v \in \mathrm{Vert}(Q)).$$

For $v \in \mathrm{Vert}(Q)$, define

$$\mathrm{AdjVert}_{\widetilde{Q}}(v) = \mathrm{AdjVert}_Q(v).$$

2. *Let $\widetilde{Q} = (Q, c)$ be a colored quiver. Let $v \in \mathrm{Vert}(Q)$ and $d \in C$. Define the set of (v, d)-colored arrows by*

$$\mathrm{ColArrow}_{\widetilde{Q}}(v, d) := \{\alpha \in \mathrm{Edge}(Q) : t_Q(\alpha) = v,\ c(\alpha) = d\}.$$

3. *Let $\widetilde{Q} = (Q, c)$ be a colored quiver.*

(a) *For* $\alpha \in \mathrm{Edge}(Q)$*, we define* $\mathrm{DelArrow}(\widetilde{Q}, \alpha)$ *by keeping* $\mathrm{Vert}(Q)$*, replacing the arrow set by* $\mathrm{Edge}(Q) \setminus \{\alpha\}$*, and restricting* s_Q*,* t_Q*, and* c *to this set.*

(b) *For* $x \in \mathrm{Vert}(Q)$*,* $v \in \mathrm{Vert}(Q)$*, and* $d \in C$*, we define* $\mathrm{AddArrow}(\widetilde{Q}; x, v, d)$ *by keeping* $\mathrm{Vert}(Q)$ *and adjoining a new arrow* α_* *to* $\mathrm{Edge}(Q)$*. The source, target, and coloring maps extend those of* $\widetilde{Q}$ *by*

$$s(\alpha_*) = x, \qquad t(\alpha_*) = v, \qquad c(\alpha_*) = d.$$

4. *Let* $\widetilde{Q} = (Q, c)$ *and* $\widetilde{Q}' = (Q', c')$ *be colored quivers with color set* C*. A color-preserving homomorphism* $\widetilde{Q} \to \widetilde{Q}'$ *is a quiver homomorphism* $(\phi_0, \phi_1) : Q \to Q'$ *such that*

$$\mathrm{QCol}(\widetilde{Q}') \circ \phi_1 = \mathrm{QCol}(\widetilde{Q}).$$

We call this homomorphism a color-preserving isomorphism if ϕ_0 *and* ϕ_1 *are bijective.*

5. *Let* $\widetilde{Q} = (Q, c)$ *and* $\widetilde{Q}' = (Q', c')$ *be colored quivers with color set* C*. We call* $\widetilde{Q}'$ *a colored subquiver of* $\widetilde{Q}$ *if* Q' *is a subquiver of* Q *and*

$$c' = c|_{\mathrm{Edge}(Q')}.$$

For $U \subseteq \mathrm{Vert}(Q)$*, we define the full colored subquiver on* U *by*

$$\mathrm{SubColQuiv}(\widetilde{Q}, U) = \left(\mathrm{SubQuiv}(Q, U), c|_{\mathrm{Edge}(\mathrm{SubQuiv}(Q,U))}\right).$$

6. *Let* $\widetilde{Q} = (Q, c)$ *be a colored quiver. Let* $I \subseteq \mathrm{Vert}(Q)$ *and* $J \subseteq \mathrm{Vert}(Q)$*. We call the pair* $\widetilde{\mathbb{Q}} = (\widetilde{Q}, (I, J))$ *a colored bipartite quiver if* $\mathbb{Q} = (Q, (I, J))$ *is a bipartite quiver. We use*

$$\mathrm{Vert}(\widetilde{\mathbb{Q}}) = \mathrm{Vert}(Q), \qquad \mathrm{Edge}(\widetilde{\mathbb{Q}}) = \mathrm{Edge}(Q),$$
$$s_{\widetilde{\mathbb{Q}}} = s_Q, \qquad t_{\widetilde{\mathbb{Q}}} = t_Q, \qquad \mathrm{QCol}(\widetilde{\mathbb{Q}}) = c,$$
$$\mathrm{UnderG}(\widetilde{\mathbb{Q}}) = \mathrm{UnderG}(Q), \qquad \deg_{\widetilde{\mathbb{Q}}}(v) = \deg_Q(v) \quad (v \in \mathrm{Vert}(Q)).$$

7. *Let* $\widetilde{\mathbb{Q}} = (\widetilde{Q}, (I, J))$ *be a colored bipartite quiver. We call* $\widetilde{\mathbb{Q}}$ *properly colored if the restriction*

$$c|_{\mathrm{IncEdgeQuiv}_Q(v)} : \mathrm{IncEdgeQuiv}_Q(v) \longrightarrow C$$

is injective for every $v \in \mathrm{Vert}(Q)$*.*

8. *Let* $\widetilde{\mathbb{Q}} = (\widetilde{Q}, (I, J))$ *be a colored bipartite quiver. We define its set of colored-vacant slots by*

$$\mathrm{ColVacSlot}(\widetilde{\mathbb{Q}}) := \{(v, d) \in J \times C : \mathrm{ColArrow}_{\widetilde{Q}}(v, d) = \varnothing\}.$$

We call d *the color of the colored-vacant slot* (v, d)*.*

7.2 Representations

Definition 7.12 (Representations of quivers)**.** *Let* Q *be a quiver.*

1. *A representation of* Q *over* $\mathbb{C}$ *is a family*

$$R = \left((R_v)_{v \in \mathrm{Vert}(Q)}, (R_\alpha)_{\alpha \in \mathrm{Edge}(Q)}\right)$$

such that each R_v *is a finite-dimensional complex vector space and each map*

$$R_\alpha : R_{s_Q(\alpha)} \longrightarrow R_{t_Q(\alpha)} \qquad (\alpha \in \mathrm{Edge}(Q))$$

is linear.

2. *Let Q' be a quiver. Let R and R' be representations of Q and Q', respectively. For a quiver homomorphism $\phi = (\phi_0, \phi_1) : Q \to Q'$, a homomorphism $f : R \to R'$ over ϕ is a tuple $f = (f_v)_{v \in \mathrm{Vert}(Q)}$ of complex linear maps $f_v : R_v \to R'_{\phi_0(v)}$ such that*

$$f_{t_Q(\alpha)} \circ R_\alpha = R'_{\phi_1(\alpha)} \circ f_{s_Q(\alpha)} \qquad (\alpha \in \mathrm{Edge}(Q)).$$

3. *For a representation R'' of a quiver Q'' and a homomorphism $g : R' \to R''$ over $\psi : Q' \to Q''$, we define $g \circ f$ over $\psi \circ \phi$ by*

$$(g \circ f)_v = g_{\phi_0(v)} \circ f_v \qquad (v \in \mathrm{Vert}(Q)).$$

4. *Let $\mathrm{Rep}_{\mathbb{C}}(Q)$ be the category of representations of Q over $\mathbb{C}$ whose morphisms are homomorphisms over 1_Q.*

5. *For $R \in \mathrm{Rep}_{\mathbb{C}}(Q)$, we define its dimension vector and support by*

$$\dim_{\mathbb{C}}(R) = \big(\dim_{\mathbb{C}}(R_i)\big)_{i \in \mathrm{Vert}(Q)},$$
$$\mathrm{RepSupp}(R) = \{i \in \mathrm{Vert}(Q) : R_i \neq 0\}.$$

   The support quiver is $\mathrm{SubQuiv}(Q, \mathrm{RepSupp}(R))$.

6. *Let $R \in \mathrm{Rep}_{\mathbb{C}}(Q)$ and $R' \in \mathrm{Rep}_{\mathbb{C}}(Q)$. An isomorphism $\phi : R \to R'$ in $\mathrm{Rep}_{\mathbb{C}}(Q)$ is a tuple*

$$\phi = (\phi_v)_{v \in \mathrm{Vert}(Q)}$$

   such that each $\phi_v : R_v \to R'_v$ is a complex linear isomorphism and

$$\phi_{t_Q(\alpha)} \circ R_\alpha = R'_\alpha \circ \phi_{s_Q(\alpha)} \qquad (\alpha \in \mathrm{Edge}(Q)).$$

7. *Let $R \in \mathrm{Rep}_{\mathbb{C}}(Q)$. A subrepresentation W of R consists of complex linear subspaces $W_v \subseteq R_v$ for $v \in \mathrm{Vert}(Q)$ such that*

$$R_\alpha(W_{s_Q(\alpha)}) \subseteq W_{t_Q(\alpha)} \qquad (\alpha \in \mathrm{Edge}(Q)).$$

   Its arrow maps are

$$W_\alpha = R_\alpha\big|_{W_{s_Q(\alpha)}} : W_{s_Q(\alpha)} \longrightarrow W_{t_Q(\alpha)}.$$

   We denote the set of these subrepresentations by $\mathrm{SubRep}(R)$. *The zero subrepresentation has $W_v = 0$ for every $v \in \mathrm{Vert}(Q)$. We call W proper if $W \neq R$.*

Definition 7.13 (Restriction and extension by zero)**.** *Let Q be a quiver, let $U \subseteq \mathrm{Vert}(Q)$. Let $Q_U = \mathrm{SubQuiv}(Q, U)$.*

1. *Define $\mathrm{Res}_{Q,U} : \mathrm{Rep}_{\mathbb{C}}(Q) \to \mathrm{Rep}_{\mathbb{C}}(Q_U)$ by keeping the spaces R_v for $v \in U$, the maps R_α for $\alpha \in \mathrm{Edge}(Q_U)$, and the components f_v for $v \in U$ of each homomorphism f.*

2. *Define $\mathrm{ExtZero}_{Q,U} : \mathrm{Rep}_{\mathbb{C}}(Q_U) \to \mathrm{Rep}_{\mathbb{C}}(Q)$ by*

$$(\mathrm{ExtZero}_{Q,U}\, R)_v = \begin{cases} R_v & v \in U, \\ 0 & v \notin U, \end{cases} \qquad (\mathrm{ExtZero}_{Q,U}\, R)_\alpha = \begin{cases} R_\alpha & s_Q(\alpha), t_Q(\alpha) \in U, \\ 0 & \text{otherwise.} \end{cases}$$

   For a homomorphism f, its component is f_v on U and the zero map outside U.

Lemma 7.14. *Let Q be a quiver, let $U \subseteq \mathrm{Vert}(Q)$. Let $Q_U = \mathrm{SubQuiv}(Q, U)$. Then, we have the following.*

1. *Let* $R \in \mathrm{Rep}_{\mathbb{C}}(Q_U)$. *The maps*

$$(\varepsilon_R)_v = 1_{R_v} \qquad (v \in U)$$

define an isomorphism

$$\varepsilon_R : \mathrm{Res}_{Q,U}(\mathrm{ExtZero}_{Q,U} R) \longrightarrow R.$$

2. *Let* $S \in \mathrm{Rep}_{\mathbb{C}}(Q)$ *satisfy* $\mathrm{RepSupp}(S) \subseteq U$. *The maps*

$$(\eta_S)_v = \begin{cases} 1_{S_v} & v \in U, \\ 0 & v \notin U \end{cases}$$

define an isomorphism

$$\eta_S : \mathrm{ExtZero}_{Q,U}(\mathrm{Res}_{Q,U} S) \longrightarrow S.$$

3. *Let* S *satisfy the hypotheses of Claim 2. The map*

$$\rho_S : \mathrm{SubRep}(S) \longrightarrow \mathrm{SubRep}(\mathrm{Res}_{Q,U} S), \qquad W \longmapsto \mathrm{Res}_{Q,U} W$$

is a bijection. Its inverse is

$$\sigma_S : \mathrm{SubRep}(\mathrm{Res}_{Q,U} S) \longrightarrow \mathrm{SubRep}(S), \qquad Z \longmapsto \widehat{Z}.$$

For $Z \in \mathrm{SubRep}(\mathrm{Res}_{Q,U} S)$, *define*

$$\widehat{Z}_v = \begin{cases} Z_v & v \in U, \\ 0 & v \notin U \end{cases} \qquad (v \in \mathrm{Vert}(Q)).$$

For $W \in \mathrm{SubRep}(S)$ *and* $v \in U$,

$$\dim_{\mathbb{C}}(\rho_S(W)_v) = \dim_{\mathbb{C}}(W_v).$$

Proof. Proof of Claim 1. Let us compare the vertex spaces and arrow maps after restriction. By Definition 7.13,

$$\begin{aligned} (\mathrm{Res}_{Q,U}(\mathrm{ExtZero}_{Q,U} R))_v &= R_v \quad (v \in U), \\ (\mathrm{Res}_{Q,U}(\mathrm{ExtZero}_{Q,U} R))_\alpha &= R_\alpha \quad (\alpha \in \mathrm{Edge}(Q_U)). \end{aligned}$$

Thus, the identity maps define ε_R and its inverse.

Proof of Claim 2. Let us check the subrepresentation at the vertices outside U. For $v \notin U$, we have $S_v = 0$. The map $(\eta_S)_v$ is therefore the unique isomorphism between zero vector spaces. For $v \in U$, it is the identity on S_v. Let $\widehat{S} = \mathrm{ExtZero}_{Q,U}(\mathrm{Res}_{Q,U} S)$. For each $\alpha \in \mathrm{Edge}(Q)$, the equation

$$(\eta_S)_{t_Q(\alpha)} \circ \widehat{S}_\alpha = S_\alpha \circ (\eta_S)_{s_Q(\alpha)}$$

holds as an equality of maps $\widehat{S}_{s_Q(\alpha)} \to S_{t_Q(\alpha)}$. If $s_Q(\alpha) \in U$ and $t_Q(\alpha) \in U$, both sides equal S_α. Otherwise, both sides are zero. Hence, η_S is an isomorphism of representations.

Proof of Claim 3. Let us verify $\rho_S(\widehat{Z}) = Z$ and $\widehat{\rho_S(W)} = W$. Let $Z \in \mathrm{SubRep}(\mathrm{Res}_{Q,U} S)$ and $\alpha \in \mathrm{Edge}(Q)$. If both endpoints of α lie in U, the inclusion $S_\alpha(\widehat{Z}_{s_Q(\alpha)}) \subseteq \widehat{Z}_{t_Q(\alpha)}$ is the subrepresentation condition for Z. Otherwise, $S_{s_Q(\alpha)} = 0$ or $S_{t_Q(\alpha)} = 0$. Then, $S_\alpha(\widehat{Z}_{s_Q(\alpha)}) = 0 \subseteq \widehat{Z}_{t_Q(\alpha)}$. For every $W \in \mathrm{SubRep}(S)$, we have $W_v = 0$ for $v \notin U$. Consequently,

$$\sigma_S \circ \rho_S = 1_{\mathrm{SubRep}(S)}, \qquad \rho_S \circ \sigma_S = 1_{\mathrm{SubRep}(\mathrm{Res}_{Q,U} S)}.$$

The dimension equality follows from $\rho_S(W)_v = W_v$ for $v \in U$. □

Definition 7.15 (Euler and Tits forms)**.** *Let H be a finite acyclic quiver. For*

$$d = (d_v)_{v \in \mathrm{Vert}(H)} \in \mathbb{Z}^{\mathrm{Vert}(H)}, \qquad \delta = (\delta_v)_{v \in \mathrm{Vert}(H)} \in \mathbb{Z}^{\mathrm{Vert}(H)},$$

we define the Euler form and Tits quadratic form

$$\langle d, \delta \rangle_H = \sum_{v \in \mathrm{Vert}(H)} d_v \delta_v - \sum_{\alpha \in \mathrm{Edge}(H)} d_{s_H(\alpha)} \delta_{t_H(\alpha)}, \qquad Q_H(d) = \langle d, d \rangle_H.$$

Definition 7.16 (Representations of colored quivers)**.** *Let C be a set, let Q be a quiver, and let $\widetilde{Q} = (Q, c)$ be a colored quiver with color set C.*

1. *We define a representation R of $\widetilde{Q}$ over $\mathbb{C}$ as a representation of Q over $\mathbb{C}$.*
2. *Let R be a representation of $\widetilde{Q}$ over $\mathbb{C}$. We define its colored support quiver by*

$$\mathrm{SuppColQuiv}(R) := \mathrm{SubColQuiv}(\widetilde{Q}, \mathrm{RepSupp}(R)).$$

3. *(a) Let $\widetilde{Q}' = (Q', c')$ be a colored quiver with color set C. Let R and R' be representations of $\widetilde{Q}$ and $\widetilde{Q}'$, respectively. A color-preserving homomorphism $\Phi : R \to R'$ is a tuple*

$$\Phi = (\phi_0, \phi_1, (f_v)_{v \in \mathrm{Vert}(Q)})$$

such that $(\phi_0, \phi_1) : \widetilde{Q} \to \widetilde{Q}'$ is a color-preserving quiver homomorphism and the tuple $(f_v)_{v \in \mathrm{Vert}(Q)}$ is a homomorphism of representations over (ϕ_0, ϕ_1).

(b) For a composable color-preserving homomorphism $\Psi = (\psi_0, \psi_1, (g_w)_{w \in \mathrm{Vert}(Q')})$, we define

$$\Psi \circ \Phi = (\psi_0 \circ \phi_0, \psi_1 \circ \phi_1, (g_{\phi_0(v)} \circ f_v)_{v \in \mathrm{Vert}(Q)}).$$

The identity of R is $(1_{\mathrm{Vert}(Q)}, 1_{\mathrm{Edge}(Q)}, (1_{R_v})_{v \in \mathrm{Vert}(Q)})$.

(c) Let $\mathrm{ColRep}_{\mathbb{C}}(\widetilde{Q})$ denote the category of representations of $\widetilde{Q}$ and color-preserving homomorphisms. For $R \in \mathrm{ColRep}_{\mathbb{C}}(\widetilde{Q})$ and $R' \in \mathrm{ColRep}_{\mathbb{C}}(\widetilde{Q})$, let

$$\mathrm{Hom}_{\mathrm{ColRep}_{\mathbb{C}}(\widetilde{Q})}(R, R')$$

denote the set of color-preserving homomorphisms $R \to R'$.

(d) We call Φ a color-preserving isomorphism if ϕ_0 and ϕ_1 are bijective and every f_v is a complex linear isomorphism.

4. *Let $R \in \mathrm{ColRep}_{\mathbb{C}}(\widetilde{Q})$. A colored subrepresentation W of R is a representation of $\widetilde{Q}$ such that $W \in \mathrm{SubRep}(R)$. We denote the set of these colored subrepresentations by $\mathrm{SubColRep}(R)$.*
5. *Let $(Q, (I, J))$ be a bipartite quiver. Let $R \in \mathrm{Rep}_{\mathbb{C}}(Q)$ have finite support $\mathrm{RepSupp}(R)$. We define the total source dimension and total sink dimension of R by*

$$\mathrm{SrcDim}_{\mathbb{C},(I,J)}(R) = \sum_{v \in I} \dim_{\mathbb{C}} R_v, \qquad \mathrm{SnkDim}_{\mathbb{C},(I,J)}(R) = \sum_{v \in J} \dim_{\mathbb{C}} R_v.$$

We define its total dimension pair by

$$\mathrm{TotDim}_{\mathbb{C},(I,J)}(R) := (\mathrm{SrcDim}_{\mathbb{C},(I,J)}(R), \mathrm{SnkDim}_{\mathbb{C},(I,J)}(R)).$$

Definition 7.17 (Bipartite stability)**.** *Let $\mathbb{Q} = (Q, (I, J))$ be a bipartite quiver. Let $R \in \mathrm{Rep}_{\mathbb{C}}(Q)$ be nonzero and have finite support. Let $(a, b) = \mathrm{TotDim}_{\mathbb{C},(I,J)}(R)$. We call R stable if every nonzero proper $W \in \mathrm{SubRep}(R)$ satisfies*

$$a \,\mathrm{SnkDim}_{\mathbb{C},(I,J)}(W) - b \,\mathrm{SrcDim}_{\mathbb{C},(I,J)}(W) > 0.$$

Definition 7.18 (Arrow operators on representations)**.** *Let $\widetilde{Q} = (Q, c)$ be a colored quiver with color set C. Let $R \in \mathrm{Rep}_{\mathbb{C}}(Q)$.*

1. *For $\alpha \in \mathrm{Edge}(Q)$, we define* $\mathrm{DelArrow}(R, \alpha)$ *on* $\mathrm{DelArrow}(\widetilde{Q}, \alpha)$ *by keeping all vertex spaces and all arrow maps except R_α.*

2. *For $x \in \mathrm{Vert}(Q)$, $v \in \mathrm{Vert}(Q)$, $d \in C$, and a linear map $f : R_x \to R_v$, we define*

$$\mathrm{AddArrow}(R; x, v, d, f)$$

on $\mathrm{AddArrow}(\widetilde{Q}; x, v, d)$ *by keeping the vertex spaces and existing arrow maps and setting*

$$(\mathrm{AddArrow}(R; x, v, d, f))_{\alpha *} = f.$$

Definition 7.19 (Isomorphisms and markings of colored representations)**.** *Let C be a set. Let $\widetilde{Q} = (Q, c)$ and $\widetilde{Q}' = (Q', c')$ be colored quivers with color set C. Let $\widetilde{\mathbb{Q}} = (\widetilde{Q}, (I, J))$ and $\widetilde{\mathbb{Q}}' = (\widetilde{Q}', (I', J'))$ be colored bipartite quivers such that Q and Q' are finite. Let $R \in \mathrm{ColRep}_{\mathbb{C}}(\widetilde{Q})$ and $R' \in \mathrm{ColRep}_{\mathbb{C}}(\widetilde{Q}')$.*

1. *An isomorphism of $(\widetilde{\mathbb{Q}}, R)$ and $(\widetilde{\mathbb{Q}}', R')$ is a color-preserving isomorphism $R \to R'$ that takes (I, J) to (I', J'). We denote the isomorphism class of $(\widetilde{\mathbb{Q}}, R)$ by* $\mathrm{CIsomClass}_{\mathbb{C}}(\widetilde{\mathbb{Q}}, R)$.

2. *Let $k \in \mathbb{Z}_{\geq 0}$. An ordered marking of colored-vacant slots of R is a tuple*

$$\mathbf{s} = ((v_i, d_i))_{i \in [k]_{\mathbb{Z}}}, \qquad (v_i, d_i) \in \mathrm{ColVacSlot}(\widetilde{\mathbb{Q}}).$$

The marked representation is the pair $(R; \mathbf{s})$. Distinct marked colored-vacant slots means $(v_i, d_i) \neq (v_j, d_j)$ for $i \neq j$.

3. *Fix $\widetilde{\mathbb{Q}}$ and $\mathbf{s}$. We define the category* $\mathrm{MarkedColRep}_{\mathbb{C}}(\widetilde{\mathbb{Q}}; \mathbf{s})$ *as follows. Its objects are the pairs $(R; \mathbf{s})$ with $R \in \mathrm{ColRep}_{\mathbb{C}}(\widetilde{Q})$. Its homomorphisms $(R; \mathbf{s}) \to (R'; \mathbf{s})$ are the homomorphisms $\Phi = (\phi_0, \phi_1, (f_v)_v) \in \mathrm{Hom}_{\mathrm{ColRep}_{\mathbb{C}}(\widetilde{Q})}(R, R')$ such that*

$$\begin{aligned} \phi_0(I) \subseteq I, \qquad & \phi_0(J) \subseteq J, \\ \phi_0(v_i) = v_i \qquad & (i \in [k]_{\mathbb{Z}}). \end{aligned}$$

We use the identities and composition of $\mathrm{ColRep}_{\mathbb{C}}(\widetilde{Q})$.

4. *Let $\mathbf{s}' = ((v_i', d_i'))_{i \in [k]_{\mathbb{Z}}}$ be an ordered marking of R'. An isomorphism $(\widetilde{\mathbb{Q}}, R; \mathbf{s}) \to (\widetilde{\mathbb{Q}}', R'; \mathbf{s}')$ is an isomorphism $\Phi = (\phi_0, \phi_1, (f_v)_{v \in \mathrm{Vert}(Q)})$ of $(\widetilde{\mathbb{Q}}, R)$ and $(\widetilde{\mathbb{Q}}', R')$ satisfying*

$$(\phi_0(v_i), d_i) = (v_i', d_i') \qquad (i \in [k]_{\mathbb{Z}}).$$

We denote its marked color-preserving isomorphism class by $\mathrm{MarkedCIsom}_{\mathbb{C}}(\widetilde{\mathbb{Q}}, R; \mathbf{s})$.

7.3 Stable zero-sum representations and gluing

From here to the end of this section, all colored quivers have color set $[0, 2]_{\mathbb{Z}}$.

Definition 7.20 (Zero-sum representations)**.** *Let $\mathbb{U} = (Q, (I_Q, J_Q))$ be a bipartite quiver such that Q is finite. We define the zero-sum representation* $\mathrm{ZS}(\mathbb{U})$ *of Q as follows.*

1. *For $u \in J_Q$, let $\mathrm{ZS}(\mathbb{U})_u = \mathbb{C}$. For $v \in I_Q$, put $A_v = \mathrm{IncEdgeQuiv}_Q(v)$. On the space $\mathbb{C}^{A_v}$ of tuples $z = (z_\alpha)_{\alpha \in A_v}$, define*

$$\Sigma_v \colon \mathbb{C}^{A_v} \longrightarrow \mathbb{C}, \qquad \Sigma_v(z) = \sum_{\alpha \in A_v} z_\alpha, \tag{7.1}$$

and let $\mathrm{ZS}(\mathbb{U})_v = \ker \Sigma_v$.

2. *For each arrow $\alpha : v \to u$ of Q, define*

$$\mathrm{ZS}(\mathbb{U})_\alpha \colon \mathrm{ZS}(\mathbb{U})_v \longrightarrow \mathrm{ZS}(\mathbb{U})_u, \qquad z \longmapsto z_\alpha. \tag{7.2}$$

3. *Let $\widetilde{Q} = (Q, c)$ be a colored quiver. Let $\widetilde{\mathbb{Q}} = (\widetilde{Q}, (I_Q, J_Q))$ be a colored bipartite quiver. Let $\mathrm{ZS}(\widetilde{\mathbb{Q}}) = \mathrm{ZS}(\mathbb{U})$.*

Definition 7.21 (Colored source-trivalent trees and their quivers)**.** *Let $\mathbb{T} = (T, (I_T, J_T))$ be a bipartite graph such that T is a finite tree.*

1. *We call $\mathbb{T}$ source-trivalent if*

$$\deg_T(u) = 3 \qquad (u \in I_T).$$

2. *A colored source-trivalent tree is a pair $(\mathbb{T}, \kappa)$ such that $\mathbb{T}$ is source-trivalent and*

$$\kappa \colon \mathrm{Edge}(T) \longrightarrow [0, 2]_{\mathbb{Z}}$$

is a proper edge coloring.

3. *For a colored source-trivalent tree $(\mathbb{T}, \kappa)$, define the colored bipartite quiver $Q(\mathbb{T}, \kappa)$ by*

$$\begin{aligned} I_{Q(\mathbb{T},\kappa)} &= \{v \in J_T : \deg_T(v) \geq 2\}, \\ J_{Q(\mathbb{T},\kappa)} &= I_T. \end{aligned}$$

Its vertex set is $I_{Q(\mathbb{T},\kappa)} \sqcup J_{Q(\mathbb{T},\kappa)}$. We define its arrows, tail and head maps, and colors by

$$\begin{aligned} \mathrm{Edge}(Q(\mathbb{T}, \kappa)) &= \{(v, u) \in I_{Q(\mathbb{T},\kappa)} \times J_{Q(\mathbb{T},\kappa)} : [\![u, v]\!] \in \mathrm{Edge}(T)\}, \\ s_{Q(\mathbb{T},\kappa)}(v, u) &= v, \qquad t_{Q(\mathbb{T},\kappa)}(v, u) = u, \\ \mathrm{QCol}(Q(\mathbb{T}, \kappa))(v, u) &= \kappa([\![u, v]\!]). \end{aligned}$$

Lemma 7.22 (Source dimension of a subrepresentation)**.** *Let $(\mathbb{T}, \kappa)$ be a colored source-trivalent tree. Let $\mathbb{T} = (T, (I_T, J_T))$. Let $W \in \mathrm{SubColRep}(\mathrm{ZS}(Q(\mathbb{T}, \kappa)))$ satisfy $W \neq 0$. Let*

$$\begin{aligned} S &= \{u \in I_T : \dim W_u = 1\}, \qquad n = \#S, \\ u_W &= \mathrm{SrcDim}_{\mathbb{C},(I_{Q(\mathbb{T},\kappa)},J_{Q(\mathbb{T},\kappa)})}(W). \end{aligned}$$

Then, $n \geq 1$ and $u_W \leq n - 1$.

Proof. Let us bound $\dim_{\mathbb{C}} W_v$ for $v \in I_{Q(\mathbb{T},\kappa)}$. Let

$$N = \{v \in J_T : [\![u, v]\!] \in \mathrm{Edge}(T) \text{ for some } u \in S\}.$$

For $v \in J_T$, let

$$q_v = \#(S \cap \mathrm{SrcAdj}_{\mathbb{T}}(v)).$$

Fix $v \in I_{Q(\mathbb{T},\kappa)}$. The arrows starting at v are (v, u) for $u \in \mathrm{SrcAdj}_{\mathbb{T}}(v)$. For $z \in W_v$, equations (7.1) and (7.2) and the subrepresentation condition give

$$\sum_{u \in \mathrm{SrcAdj}_{\mathbb{T}}(v)} z_{(v,u)} = 0, \qquad z_{(v,u)} \in W_u \quad (u \in \mathrm{SrcAdj}_{\mathbb{T}}(v)). \tag{7.3}$$

If $u \notin S$, then $W_u = 0$ and $z_{(v,u)} = 0$. Thus, only the q_v coordinates indexed by $u \in S$ can be nonzero. Their sum is zero by (7.3). If $q_v > 0$, then $\dim_{\mathbb{C}} W_v \leq q_v - 1$. If $q_v = 0$, then $W_v = 0$.

If $n = 0$, then $W_u = 0$ for each $u \in J_{Q(\mathbb{T},\kappa)}$ by the definition of S. The bound on $\dim_{\mathbb{C}} W_v$ gives $W_v = 0$ for each $v \in I_{Q(\mathbb{T},\kappa)}$. Then, $W = 0$, contrary to the hypothesis. Hence, $n \geq 1$.

Let us count the edges of $F = \mathrm{Subgraph}(T, S \sqcup N)$. By Definition 7.3(6),

$$\mathrm{Edge}(F) = \{[\![u, v]\!] \in \mathrm{Edge}(T) : u \in S,\ v \in N\}.$$

The graph F is a forest because T has no cycles. Let $C = \#\,\mathrm{CtdComp}(F) \geq 1$. Each $u \in S$ has degree 3 in F. Each $v \in N \cap \mathrm{Leaf}(T)$ satisfies $q_v = 1$. For $v \in I_{Q(\mathbb{T},\kappa)} \setminus N$, we have $W_v = 0$. Therefore,

$$3n = \#\,\mathrm{Edge}(F) = n + \#N - C, \qquad u_W \leq \sum_{v \in N} (q_v - 1) = 3n - \#N = n - C \leq n - 1.$$

□

Lemma 7.23. *Let $(\mathbb{T}, \kappa)$ be a colored source-trivalent tree. Let $\mathbb{T} = (T, (I_T, J_T))$. Assume $\#I_T = \nu \geq 2$. Let H be the quiver underlying $Q(\mathbb{T}, \kappa)$ and let $d = \dim_{\mathbb{C}}(\mathrm{ZS}(Q(\mathbb{T}, \kappa)))$. Then, we have the following.*

1. *The representation $\mathrm{ZS}(Q(\mathbb{T}, \kappa))$ satisfies the following.*
   (a) *We have*
   $$\mathrm{TotDim}_{\mathbb{C},(I_{Q(\mathbb{T},\kappa)},J_{Q(\mathbb{T},\kappa)})}(\mathrm{ZS}(Q(\mathbb{T}, \kappa))) = (\nu - 1, \nu).$$
   (b) *We have $Q_H(d) = 1$.*
2. *Each $u \in J_{Q(\mathbb{T},\kappa)}$ satisfies $\dim_{\mathbb{C}}(\mathrm{ZS}(Q(\mathbb{T}, \kappa))_u) = 1$.*
3. *Each $v \in I_{Q(\mathbb{T},\kappa)}$ satisfies*
   $$(\dim_{\mathbb{C}}(\mathrm{ZS}(Q(\mathbb{T}, \kappa))_v), \deg_{Q(\mathbb{T},\kappa)}(v)) \in \{(1, 2), (2, 3)\}.$$
4. *The representation $\mathrm{ZS}(Q(\mathbb{T}, \kappa))$ is stable.*
5. *For each $u \in J_{Q(\mathbb{T},\kappa)}$, we have*
   $$\sum_{\substack{\alpha \in \mathrm{Edge}(Q(\mathbb{T},\kappa)) \\ t_{Q(\mathbb{T},\kappa)}(\alpha) = u}} \mathrm{im}(\mathrm{ZS}(Q(\mathbb{T}, \kappa))_\alpha) = \mathrm{ZS}(Q(\mathbb{T}, \kappa))_u.$$

Proof. Proof of Claim 1(a). Let us compute the total dimension from the tree degrees. Let $s = \#J_T$. The identity $\#\,\mathrm{Edge}(T) = \#\,\mathrm{Vert}(T) - 1$ gives

$$3\nu = \nu + s - 1, \qquad s = 2\nu + 1.$$

For $v \in J_T$, let $n_v = \deg_T(v)$. By Definition 7.20(1), we obtain

$$\mathrm{SrcDim}_{\mathbb{C},(I_{Q(\mathbb{T},\kappa)},J_{Q(\mathbb{T},\kappa)})}(\mathrm{ZS}(Q(\mathbb{T}, \kappa))) = \sum_{v \in J_T} (n_v - 1) = 3\nu - (2\nu + 1) = \nu - 1,$$
$$\mathrm{SnkDim}_{\mathbb{C},(I_{Q(\mathbb{T},\kappa)},J_{Q(\mathbb{T},\kappa)})}(\mathrm{ZS}(Q(\mathbb{T}, \kappa))) = \#I_T = \nu.$$

Proof of Claim 1(b). Claim 1(a) gives

$$Q_H(d) = \nu + \sum_{v \in J_T} ((n_v - 1)^2 - n_v(n_v - 1)) = \nu - \sum_{v \in J_T} (n_v - 1) = 1.$$

Proof of Claim 2. Let us count the sinks after deleting the leaves. The sinks of $Q(\mathbb{T},\kappa)$ belong to I_T. By Definition 7.20(1), $\mathrm{ZS}(Q(\mathbb{T},\kappa))_u = \mathbb{C}$ for every such vertex u.

Proof of Claim 3. Let us count the sources after deleting the leaves. By Definition 7.21(2), the incident edges at each vertex have distinct colors in $[0,2]_{\mathbb{Z}}$. Hence, $n_v \le 3$ for $v \in J_T$. The source vertices of $Q(\mathbb{T},\kappa)$ are the vertices $v \in J_T$ satisfying $n_v \in [2,3]_{\mathbb{Z}}$. For such a vertex, $\dim_{\mathbb{C}}(\mathrm{ZS}(Q(\mathbb{T},\kappa))_v) = n_v - 1$ and $\deg_{Q(\mathbb{T},\kappa)}(v) = n_v$.

Proof of Claim 4. Let us prove stability by bounding each subrepresentation weight. Let $W \in \mathrm{SubColRep}(\mathrm{ZS}(Q(\mathbb{T},\kappa)))$ satisfy $0 \neq W \neq \mathrm{ZS}(Q(\mathbb{T},\kappa))$. Claim 1(a) gives the dimension pair $(\nu-1,\nu)$. Define

$$S = \{u \in I_T : \dim W_u = 1\}, \qquad n = \#S,$$

and let $u_W = \mathrm{SrcDim}_{\mathbb{C},(I_{Q(\mathbb{T},\kappa)},J_{Q(\mathbb{T},\kappa)})}(W)$. Lemma 7.22 gives $n \ge 1$ and $u_W \le n-1$. For $n < \nu$, we obtain

$$(\nu-1)n - \nu u_W \ge \nu - n > 0.$$

If $n = \nu$, properness gives $u_W \le \nu - 2$. Hence,

$$(\nu-1)n - \nu u_W = \nu(\nu - 1 - u_W) \ge \nu > 0.$$

Proof of Claim 5. Let us verify surjectivity of the arrow maps in Definition 7.20(2). Since T is connected and $\nu \ge 2$, every $u \in I_T$ has a vertex $v \in \mathrm{SnkAdj}_{\mathbb{T}}(u)$ of degree $d \ge 2$. Let $e_1,\dots,e_d$ be the standard basis of $\mathbb{C}^d$. If i and j are distinct coordinate indices, then

$$e_i - e_j \in \ker\left(\mathbb{C}^d \xrightarrow{\sum} \mathbb{C}\right), \qquad (e_i - e_j)_i = 1.$$

Thus, each arrow map from this zero-sum space to $\mathbb{C}$ is surjective. □

Lemma 7.24. *Let $(\mathbb{T},\kappa)$ be a colored source-trivalent tree. Let $\mathbb{T} = (T,(I_T,J_T))$. The map sending $(u,d) \in \mathrm{ColVacSlot}(Q(\mathbb{T},\kappa))$ to the unique $v \in J_T$ satisfying*

$$\deg_T(v) = 1, \qquad [\![u,v]\!] \in \mathrm{Edge}(T), \qquad \kappa([\![u,v]\!]) = d$$

is a bijection onto $J_T \cap \mathrm{Leaf}(T)$.

Proof. Let us construct the inverse map from sink leaves to colored-vacant slots. For $u \in I_T$, the three edges in $\mathrm{IncEdge}_T(u)$ have distinct colors in $[0,2]_{\mathbb{Z}}$. The quiver $Q(\mathbb{T},\kappa)$ keeps exactly the edges from u to vertices of degree at least two. Thus, $(u,d) \in \mathrm{ColVacSlot}(Q(\mathbb{T},\kappa))$ determines the unique leaf $v \in J_T$ such that $[\![u,v]\!] \in \mathrm{Edge}(T)$ and $\kappa([\![u,v]\!]) = d$. Conversely, let $v \in J_T$ be a leaf and let $u \in I_T$ satisfy $\mathrm{SrcAdj}_{\mathbb{T}}(v) = \{u\}$. The inverse map is

$$J_T \cap \mathrm{Leaf}(T) \longrightarrow \mathrm{ColVacSlot}(Q(\mathbb{T},\kappa)), \qquad v \longmapsto (u, \kappa([\![u,v]\!])).$$

Uniqueness of the neighbor of v and of the edge of color d at u shows that the two maps are inverse. □

Definition 7.25 (Gluing zero-sum representations)**.** *Let $\mathbb{S} = (S,(I_S,J_S))$ be a source-trivalent tree. For each $b \in J_S$, choose a colored source-trivalent tree $(\mathbb{T}_b,\kappa_b)$. Let $\mathbb{T}_b = (T_b,(I_{T_b},J_{T_b}))$ for $b \in J_S$. For each $e = [\![x,b]\!] \in \mathrm{Edge}(S)$ with $x \in I_S$ and $b \in J_S$, choose*

$$(v_e,d_e) \in \mathrm{ColVacSlot}(Q(\mathbb{T}_b,\kappa_b)), \qquad 0 \neq f_e : \mathbb{C} \longrightarrow \mathrm{ZS}(Q(\mathbb{T}_b,\kappa_b))_{v_e}.$$

Let

$$\mathbf{T} = ((\mathbb{T}_b,\kappa_b))_{b\in J_S}, \qquad \mathbf{p} = ((v_e,d_e))_{e\in\mathrm{Edge}(S)}, \qquad \mathbf{f} = (f_e)_{e\in\mathrm{Edge}(S)}.$$

We define the colored representation

$$R = \mathrm{Glue}_{\mathbb{C}}(\mathbb{S};\mathbf{T},\mathbf{p},\mathbf{f})$$

by the following conditions.

1. *Its quiver consists of disjoint copies of the $Q(\mathbb{T}_b, \kappa_b)$, a new vertex x for each $x \in I_S$, and one new arrow*

$$\alpha_e : x \longrightarrow v_e, \qquad c(\alpha_e) = d_e \qquad (e = [\![x, b]\!] \in \mathrm{Edge}(S)).$$

Each v_e belongs to the copy of $Q(\mathbb{T}_b, \kappa_b)$ indexed by b. The sources are the new vertices $x \in I_S$ and the copied vertices in $I_{Q(\mathbb{T}_b,\kappa_b)}$; the sinks are the copied vertices in $J_{Q(\mathbb{T}_b,\kappa_b)}$.

2. *Its restriction to each $Q(\mathbb{T}_b, \kappa_b)$ is $\mathrm{ZS}(Q(\mathbb{T}_b, \kappa_b))$.*

3. *For each new vertex x, let $R_x = \mathbb{C}$. For each new arrow α_e, let $R_{\alpha_e} = f_e$.*

7.4 Flow sum equations

Definition 7.26 (Flow sum equations)**.** *Let $\mathbb{S} = (S, (I_S, J_S))$ be a bipartite graph with a finite graph S. Let*

$$\alpha = (\alpha_v \in \mathbb{R})_{v \in I_S}, \qquad \xi = (\xi_i \in \mathbb{R})_{i \in J_S}.$$

1. *We define $\mathrm{Flow}(S) = \mathbb{R}^{\mathrm{Edge}(S)}$. For $f \in \mathrm{Flow}(S)$, let $f_w = f(w)$. We call*

$$\sum_{w \in \mathrm{IncEdge}_S(v)} f_w = \alpha_v \quad (v \in I_S),$$
$$\sum_{w \in \mathrm{IncEdge}_S(i)} f_w = \xi_i \quad (i \in J_S)$$

the (α, ξ)-flow sum equations on S.

2. *Define the set of (α, ξ)-flows on S by*

$$\mathrm{Flow}(S, (\alpha, \xi)) = \{f \in \mathrm{Flow}(S) : f \text{ satisfies the } (\alpha, \xi)\text{-flow sum equations}\}.$$

3. *We call an (α, ξ)-flow f positive if $f_w > 0$ for every $w \in \mathrm{Edge}(S)$.*

Lemma 7.27 (Flows on a bipartite tree)**.** *Let $\mathbb{S} = (S, (I_S, J_S))$ be a bipartite graph such that S is a finite tree. Let $\alpha = (\alpha_v)_{v \in I_S} \in \mathbb{R}^{I_S}$ and $\xi = (\xi_i)_{i \in J_S} \in \mathbb{R}^{J_S}$. Assume*

$$\sum_{v \in I_S} \alpha_v = \sum_{i \in J_S} \xi_i. \tag{7.4}$$

Then, we have the following.

1. *The set $\mathrm{Flow}(S, (\alpha, \xi))$ has exactly one element f.*

2. *Let $w = [\![v, i]\!] \in \mathrm{Edge}(S)$ with $i \in J_S$. Let $S_w = \mathrm{SnkCtdComp}(\mathbb{S}; w)$. Then,*

$$f_w = \sum_{j \in \mathrm{SubSnk}_{\mathbb{S}}(S_w)} \xi_j - \sum_{u \in \mathrm{SubSrc}_{\mathbb{S}}(S_w)} \alpha_u.$$

3. *Assume $f_w > 0$ for every $w \in \mathrm{Edge}(S)$. Let $I \subseteq J_S$ and $J \subseteq I_S$. Assume $\varnothing \neq I \sqcup J \subsetneq \mathrm{Vert}(S)$. Assume $\mathrm{SnkAdj}_{\mathbb{S}}(v) \subseteq I$ for every $v \in J$. Then,*

$$\sum_{i \in I} \xi_i - \sum_{v \in J} \alpha_v > 0.$$

Proof. Proof of Claim 1. Let us prove existence and uniqueness by induction on the number of vertices.

First, suppose S has one vertex. If the vertex is a source v, equation (7.4) gives $\alpha_v = 0$. If the vertex is a sink i, that equation gives $\xi_i = 0$. The empty map is the unique (α, ξ)-flow.

Second, suppose $\#\operatorname{Vert}(S) \geq 2$. Choose $z \in \operatorname{Leaf}(S)$ and let

$$S' = \operatorname{Subgraph}(S, \operatorname{Vert}(S) \setminus \{z\}).$$

The graph S' is a tree with one fewer vertex.

First, let us remove a source leaf $z = v \in I_S$. Let $\operatorname{SnkAdj}_{\mathbb{S}}(v) = \{i\}$ and $w = [\![v, i]\!]$. The flow equation at v forces $f_w = \alpha_v$. On S', define

$$\alpha'_u = \alpha_u \quad (u \in I_S \setminus \{v\}), \qquad \xi'_j = \begin{cases} \xi_i - \alpha_v, & j = i, \\ \xi_j, & j \in J_S \setminus \{i\}. \end{cases}$$

Both sums in (7.4) decrease by α_v.

Second, let us remove a sink leaf $z = i \in J_S$. Let $\operatorname{SrcAdj}_{\mathbb{S}}(i) = \{v\}$ and $w = [\![v, i]\!]$. The flow equation at i forces $f_w = \xi_i$. On S', define

$$\xi'_j = \xi_j \quad (j \in J_S \setminus \{i\}), \qquad \alpha'_u = \begin{cases} \alpha_v - \xi_i, & u = v, \\ \alpha_u, & u \in I_S \setminus \{v\}. \end{cases}$$

Both sums in (7.4) decrease by ξ_i.

Let us extend the flow from S' to S. In either case, (α', ξ') satisfies (7.4) on S'. By induction, there is a unique (α', ξ')-flow on S'. The flow extends uniquely to S by the forced value of f_w.

Proof of Claim 2. Let us compute the weight of the cut edge. Sum the sink equations over S_w and subtract the source equations over S_w. For every $e \in \operatorname{Edge}(S_w)$, the terms f_e cancel. The edge w contributes f_w.

Proof of Claim 3. Let us cancel the terms indexed by edges $[\![v, i]\!]$ with $v \in J$ and $i \in I$. The assumption $\operatorname{SnkAdj}_{\mathbb{S}}(v) \subseteq I$ for every $v \in J$ gives

$$\sum_{i \in I} \xi_i - \sum_{v \in J} \alpha_v = \sum_{\substack{[\![v,i]\!] \in \operatorname{Edge}(S) \\ i \in I,\ v \notin J}} f_{[\![v,i]\!]}.$$

Since S is connected and $\varnothing \neq I \sqcup J \subsetneq \operatorname{Vert}(S)$, there exist $i \in I$ and $v \notin J$ such that $[\![v, i]\!] \in \operatorname{Edge}(S)$. Every summand is positive. □

Lemma 7.28 (Weights of subrepresentations of a zero-sum representation)**.** *Let $(\mathbb{T}, \kappa)$ be a colored source-trivalent tree. Let $\mathbb{T} = (T, (I_T, J_T))$, $Q = Q(\mathbb{T}, \kappa)$, and $Z = \operatorname{ZS}(Q)$. Let $a_0 \in \mathbb{Z}_{\geq 0}$ satisfy*

$$\operatorname{TotDim}_{\mathbb{C},(I_Q,J_Q)}(Z) = (a_0, a_0 + 1).$$

Let $0 \neq W \in \operatorname{SubColRep}(Z)$. Let

$$(u, t) = \operatorname{TotDim}_{\mathbb{C},(I_Q,J_Q)}(W).$$

Let $y \in \mathbb{R}_{\geq 0}$. Then, the following statements hold.

1. *We have*
$$y(t - u) - u \geq y - a_0.$$
2. *We have*
$$y(t - u) - u = y - a_0 \iff W = Z.$$

Proof. Proof of Claim 1. Let us prove $t-u \geq 1$ and $u \leq a_0$. Lemma 7.22 gives $t-u \geq 1$. For every $z \in I_Q$, the inclusion $W_z \subseteq Z_z$ gives $\dim_{\mathbb{C}} W_z \leq \dim_{\mathbb{C}} Z_z$. Summing over I_Q gives $u \leq a_0$. Therefore,

$$y(t-u)-u-(y-a_0) = y(t-u-1)+(a_0-u) \geq 0.$$

Proof of Claim 2. Let us prove that a nonzero proper subrepresentation satisfies $u < a_0$. Every $z \in J_Q = I_T$ has $Z_z = \mathbb{C}$. Hence, $\#I_T = \#J_Q = a_0+1$. If $a_0 = 0$, then $J_Q = \{z_0\}$ for one vertex z_0, $Z_{z_0} = \mathbb{C}$. Every source space is zero. The condition $W \neq 0$ gives $W_{z_0} = \mathbb{C}$. Thus, $W = Z$.

Let us treat the case in which the subrepresentation has full source dimension. Suppose that $a_0 \geq 1$ and $u = a_0$. Then,

$$0 = a_0 - u = \sum_{z \in I_Q} (\dim_{\mathbb{C}} Z_z - \dim_{\mathbb{C}} W_z).$$

Each summand is nonnegative. Thus, $W_z = Z_z$ for every $z \in I_Q$. Since $\#I_T = a_0+1 \geq 2$, Claim 5 of Lemma 7.23 applies. For every $z \in J_Q$, the subrepresentation condition gives

$$Z_z = \sum_{\substack{\beta \in \mathrm{Edge}(Q) \\ t_Q(\beta)=z}} Z_\beta(Z_{s_Q(\beta)}) = \sum_{\substack{\beta \in \mathrm{Edge}(Q) \\ t_Q(\beta)=z}} Z_\beta(W_{s_Q(\beta)}) \subseteq W_z \subseteq Z_z.$$

Consequently, $W = Z$. For $W \neq Z$, we therefore have $a_0 - u > 0$, and

$$y(t-u)-u-(y-a_0) = y(t-u-1)+(a_0-u) > 0.$$

For $W = Z$, substitution of $(u,t) = (a_0, a_0+1)$ gives equality. □

Lemma 7.29 (Weights of subrepresentations of a gluing)**.** *Let $\mathbb{S} = (S, (I_S, J_S))$ be a source-trivalent tree. For $i \in J_S$, let $(\mathbb{T}_i, \kappa_i)$ be a colored source-trivalent tree, and let*

$$Q_i = Q(\mathbb{T}_i, \kappa_i), \qquad Z_i = \mathrm{ZS}(Q_i).$$

Let $a_i \in \mathbb{Z}_{\geq 0}$ satisfy

$$\mathrm{TotDim}_{\mathbb{C},(I_{Q_i},J_{Q_i})}(Z_i) = (a_i, a_i+1) \qquad (i \in J_S).$$

For every $e = [\![v,i]\!] \in \mathrm{Edge}(S)$ with $v \in I_S$ and $i \in J_S$, fix a colored-vacant slot and a nonzero complex linear map

$$p_e = (z_e, d_e) \in \mathrm{ColVacSlot}(Q_i), \qquad 0 \neq \phi_e : \mathbb{C} \longrightarrow (Z_i)_{z_e}.$$

Let

$$\mathbf{T} = ((\mathbb{T}_i, \kappa_i))_{i \in J_S}, \qquad \mathbf{p} = (p_e)_{e \in \mathrm{Edge}(S)}, \qquad \mathbf{f} = (\phi_e)_{e \in \mathrm{Edge}(S)},$$

and let $R = \mathrm{Glue}_{\mathbb{C}}(\mathbb{S}; \mathbf{T}, \mathbf{p}, \mathbf{f})$. Use the vertex copies in Definition 7.25. Let

$$I_R = I_S \sqcup \bigsqcup_{i \in J_S} I_{Q_i}, \qquad J_R = \bigsqcup_{i \in J_S} J_{Q_i}.$$

Let

$$c = \#I_S, \qquad k = c+1, \qquad a = \sum_{i \in J_S} a_i + c, \qquad x = \frac{a}{k}.$$

Let $\alpha_v = x+1$ for $v \in I_S$ and $\xi_i = x - a_i$ for $i \in J_S$. Fix $f \in \mathrm{Flow}(S, (\alpha, \xi))$. Then, the following statements hold.

1. *Let $W \in \mathrm{SubColRep}(R)$. For $i \in J_S$, let W_i be the subrepresentation of Z_i defined by*

$$(W_i)_z = W_z \quad (z \in \mathrm{Vert}(Q_i)), \qquad (W_i)_\beta = R_\beta|_{W_{s_{Q_i}(\beta)}} \quad (\beta \in \mathrm{Edge}(Q_i)).$$

Define

$$I = \{i \in J_S : W_i \neq 0\}, \qquad J = \{v \in I_S : W_v \neq 0\}.$$

Let

$$(u_i, t_i) = \mathrm{TotDim}_{\mathbb{C},(I_{Q_i},J_{Q_i})}(W_i) \quad (i \in I), \qquad (u,t) = \mathrm{TotDim}_{\mathbb{C},(I_R,J_R)}(W).$$

Then, $\mathrm{SnkAdj}_{\mathbb{S}}(v) \subseteq I$ *for every* $v \in J$*. We have*

$$\sum_{i\in I}(x - a_i) - \#J(x+1) = \sum_{\substack{e=[\![v,i]\!]\in \mathrm{Edge}(S)\\ v\in I_S\setminus J,\ i\in I}} f_e. \tag{7.5}$$

Moreover,

$$\begin{aligned}\frac{at-(a+k)u}{k} = &\sum_{\substack{e=[\![v,i]\!]\in \mathrm{Edge}(S)\\ v\in I_S\setminus J,\ i\in I}} f_e \\ &+ \sum_{i\in I}(x(t_i - u_i - 1) + (a_i - u_i)).\end{aligned} \tag{7.6}$$

2. *Let* $w = [\![v,i]\!] \in \mathrm{Edge}(S)$ *with* $v \in I_S$ *and* $i \in J_S$*. Let*

$$S_w = \mathrm{SnkCtdComp}(\mathbb{S}; w), \qquad A_w = \mathrm{SubSnk}_{\mathbb{S}}(S_w), \qquad B_w = \mathrm{SubSrc}_{\mathbb{S}}(S_w).$$

For each vertex z *of the quiver of* R*, define the subspace*

$$W^w_z = \begin{cases} R_z, & z \in B_w \sqcup \bigsqcup_{j\in A_w} \mathrm{Vert}(Q_j), \\ 0, & \textit{otherwise.}\end{cases}$$

These subspaces and the restrictions of the arrow maps of R *define a nonzero proper subrepresentation* W^w*. For* $(u_w, t_w) = \mathrm{TotDim}_{\mathbb{C},(I_R,J_R)}(W^w)$*, we have*

$$u_w = \sum_{j\in A_w} a_j + \#B_w, \qquad t_w = \sum_{j\in A_w}(a_j+1), \qquad at_w - (a+k)u_w = kf_w.$$

Proof. Proof of Claim 1. Let us determine which restrictions of W are nonzero. For $v \in J$, the inclusion $0 \neq W_v \subseteq R_v = \mathbb{C}$ gives $W_v = \mathbb{C}$. For $e = [\![v,i]\!] \in \mathrm{Edge}(S)$ with $v \in J$, we have

$$0 \neq \phi_e(\mathbb{C}) = \phi_e(W_v) \subseteq W_{z_e}.$$

Thus, $W_i \neq 0$ and $i \in I$. This proves $\mathrm{SnkAdj}_{\mathbb{S}}(v) \subseteq I$ for $v \in J$.

Let us sum the flow equations over I and J. They give

$$\sum_{i\in I}(x-a_i) - \#J(x+1) = \sum_{i\in I}\sum_{e\in \mathrm{IncEdge}_S(i)} f_e - \sum_{v\in J}\sum_{e\in \mathrm{IncEdge}_S(v)} f_e.$$

An edge between J and I occurs in both sums. There is no edge from J to $J_S \setminus I$. Subtracting the two finite sums proves (7.5).

Let us compute the stability weight from the vertex dimensions. The disjoint vertex sets in the gluing give

$$u = \sum_{i\in I} u_i + \#J, \qquad t = \sum_{i\in I} t_i.$$

Since $a = kx$ and $k > 0$, we obtain

$$\begin{aligned}\frac{at-(a+k)u}{k} &= x(t-u) - u \\ &= \sum_{i\in I}(x(t_i - u_i) - u_i) - \#J(x+1) \\ &= \sum_{i\in I}(x - a_i) - \#J(x+1) + \sum_{i\in I}(x(t_i - u_i - 1) + (a_i - u_i)).\end{aligned}$$

Substitution of (7.5) proves (7.6).

Proof of Claim 2. Let us verify the subrepresentation condition at each arrow. For $\beta \in \mathrm{Edge}(Q_j)$, both endpoint spaces of W^w are full if $j \in A_w$ and zero if $j \notin A_w$. For an added arrow $\alpha_e : y \to z_e$ indexed by $e = [\![y, j]\!]$, suppose $W^w_y \neq 0$. Then, $y \in B_w$. The source endpoint v of w does not belong to B_w. Thus, $e \neq w$. The edge e therefore belongs to the component S_w whenever $y \in B_w$. Hence, $j \in A_w$ and

$$R_{\alpha_e}(W^w_y) = \phi_e(R_y) \subseteq R_{z_e} = W^w_{z_e}.$$

If $W^w_y = 0$, the subrepresentation condition holds because $\phi_e(0) = 0$.

Let us check that the cut subrepresentation is nonzero and proper. The index i belongs to A_w. The representation Z_i has sink dimension $a_i + 1 > 0$. Thus, $W^w \neq 0$. Also, $v \notin B_w$ gives $W^w_v = 0 \neq R_v = \mathbb{C}$. Thus, $W^w \neq R$.

Let us compute the dimension pair and its weight. The full constituent spaces over A_w and the full new source spaces over B_w give

$$u_w = \sum_{j \in A_w} a_j + \#B_w, \qquad t_w = \sum_{j \in A_w} (a_j + 1).$$

Therefore,

$$\begin{aligned} \frac{at_w - (a+k)u_w}{k} &= x(t_w - u_w) - u_w \\ &= \sum_{j \in A_w} (x - a_j) - \sum_{y \in B_w} (x+1) = f_w. \end{aligned}$$

Summing the source and sink equations for f gives (7.4). The equality $\sum_{j \in A_w}(x - a_j) - \sum_{y \in B_w}(x+1) = f_w$ follows from Claim 2 of Lemma 7.27. Multiplying by k gives $at_w - (a+k)u_w = kf_w$. □

Lemma 7.30 (Stability under gluing)**.** *Let $\mathbb{S} = (S, (I_S, J_S))$ be a source-trivalent tree. Let $c = \#I_S$. For $i \in J_S$, let $(\mathbb{T}_i, \kappa_i)$ be a colored source-trivalent tree. Let $\mathbf{T} = ((\mathbb{T}_i, \kappa_i))_{i \in J_S}$. Let $R = \mathrm{Glue}_{\mathbb{C}}(\mathbb{S}; \mathbf{T}, \mathbf{p}, \mathbf{f})$ be a gluing of their zero-sum representations using colored-vacant slots $\mathbf{p}$ and nonzero complex linear maps $\mathbf{f}$. For $i \in J_S$, let*

$$\mathrm{TotDim}_{\mathbb{C},(I_{Q(\mathbb{T}_i,\kappa_i)},J_{Q(\mathbb{T}_i,\kappa_i)})}(\mathrm{ZS}(Q(\mathbb{T}_i,\kappa_i))) = (a_i, a_i + 1).$$

Let

$$\widetilde{Q}_R = \mathrm{SuppColQuiv}(R), \qquad G_R = \mathrm{UnderG}(\widetilde{Q}_R).$$

Let $\widetilde{\mathbb{Q}}_R = (\widetilde{Q}_R, (I_R, J_R))$ for

$$\begin{aligned} I_R &= \mathrm{RepSupp}(R) \cap \left(I_S \sqcup \bigsqcup_{i \in J_S} I_{Q(\mathbb{T}_i,\kappa_i)} \right), \\ J_R &= \mathrm{RepSupp}(R) \cap \bigsqcup_{i \in J_S} J_{Q(\mathbb{T}_i,\kappa_i)}. \end{aligned}$$

The disjoint unions use the vertex copies in R. Assume the following conditions on the added arrows.

(a) Each edge $[\![x, i]\!]$ of S with $x \in I_S$ and $i \in J_S$ specifies one added (v, d)-colored arrow $x \to v$ for some $(v, d) \in \mathrm{ColVacSlot}(Q(\mathbb{T}_i, \kappa_i))$.

(b) Each $(v, d) \in \mathrm{ColVacSlot}(Q(\mathbb{T}_i, \kappa_i))$ receives at most one added (v, d)-colored arrow. The three arrows starting at each $x \in I_S$ have distinct colors.

Let

$$k = c + 1, \qquad a = \sum_{i \in J_S} a_i + c, \qquad x = \frac{a}{k}.$$

For $\alpha_v = x + 1$ and $\xi_i = x - a_i$, the (α, ξ)-flow sum equations on S are

$$\sum_{w \in \mathrm{IncEdge}_S(v)} f_w = x + 1 \quad (v \in I_S), \qquad \sum_{w \in \mathrm{IncEdge}_S(i)} f_w = x - a_i \quad (i \in J_S). \tag{7.7}$$

Then, we have the following.

1. *The following statements hold.*
   (a) *The graph G_R is a tree.*
   (b) *The coloring $\mathrm{QCol}(\widetilde{Q}_R)$ induces a proper edge coloring of G_R.*
   (c) *$\#J_S = 2c + 1$.*
   (d) *We have $\mathrm{TotDim}_{\mathbb{C},(I_R,J_R)}(R) = (a, a + k)$.*
2. *The (α, ξ)-flow sum equations (7.7) have a unique solution.*
3. *Let f be the unique element of $\mathrm{Flow}(S, (\alpha, \xi))$ from Claim 2. The representation R is stable if and only if $f_w > 0$ for every $w \in \mathrm{Edge}(S)$.*

Proof. Proof of Claim 1(a). Let us verify that G_R is a tree. Let $B = J_S$ and $b = \#B$. Counting the edges of S gives $3c = c + b - 1$. Thus, $b = 2c + 1$. For $i \in B$, let

$$G_i = \mathrm{UnderG}(\mathrm{SuppColQuiv}(\mathrm{ZS}(Q(\mathbb{T}_i, \kappa_i)))), \qquad v_i = \#\,\mathrm{Vert}(G_i), \qquad e_i = \#\,\mathrm{Edge}(G_i).$$

Definition 7.20(1) gives $\#I_{T_i} = a_i + 1 \geq 1$. By Definitions 7.20 and 7.21, G_i is obtained from T_i by deleting its sink leaves. Every path between retained vertices avoids the deleted leaves. Thus, G_i is a nonempty tree and $e_i = v_i - 1$. Replacing each $i \in B$ by G_i and adding the arrows in assumption (a) gives the connected graph G_R. Then,

$$\begin{aligned} \#\,\mathrm{Vert}(G_R) &= c + \sum_{i \in B} v_i, \\ \#\,\mathrm{Edge}(G_R) &= 3c + \sum_{i \in B} e_i = 3c + \sum_{i \in B} (v_i - 1) = \#\,\mathrm{Vert}(G_R) - 1. \end{aligned}$$

Hence, G_R is a tree. Two colored-vacant slots sharing a vertex contribute two distinct added arrows to $3c$.

Proof of Claim 1(b). Assumption (b) gives distinct colors at each new source. At each old vertex, the old incident arrows have distinct colors because κ_i is proper. Each added arrow uses a colored-vacant slot by assumption (a). Its color is therefore unused by the old arrows at its target. Assumption (b) gives distinct colors among the added arrows at each old vertex.

Proof of Claim 1(c). Let us count the sinks of the gluing tree. Let $b = \#J_S$. The identity $3c = c + b - 1$ in the proof of Claim 1(a) gives $b = 2c + 1$.

Proof of Claim 1(d). By Claim 1(c), we obtain

$$\begin{aligned} \mathrm{SrcDim}_{\mathbb{C},(I_R,J_R)}(R) &= \sum_{i \in J_S} a_i + c = a, \\ \mathrm{SnkDim}_{\mathbb{C},(I_R,J_R)}(R) &= \sum_{i \in J_S} (a_i + 1) = a - c + 2c + 1 = a + k. \end{aligned}$$

Proof of Claim 2. Let us verify equality of the two total vertex weights. Claim 1(c) gives

$$\sum_{i\in J_S}\xi_i = \sum_{i\in J_S}(x-a_i)$$
$$= (2c+1)x-(a-c) = c(x+1) = \sum_{v\in I_S}\alpha_v.$$

Claim 1 of Lemma 7.27 gives the unique (α,ξ)-flow.

Proof of Claim 3. Let us obtain stability from positive flows. Assume $f_e>0$ for every $e\in\mathrm{Edge}(S)$. Let $0\neq W\neq R$ be a subrepresentation of R. For $i\in J_S$, let

$$Q_i = Q(\mathbb{T}_i,\kappa_i), \qquad Z_i = \mathrm{ZS}(Q_i),$$

and let W_i be the restriction of W to the copy of Q_i. Define

$$I=\{i\in J_S : W_i\neq 0\}, \qquad J=\{v\in I_S : W_v\neq 0\}.$$

Let

$$(u_i,t_i)=\mathrm{TotDim}_{\mathbb{C},(I_{Q_i},J_{Q_i})}(W_i)\quad (i\in I), \qquad (u,t)=\mathrm{TotDim}_{\mathbb{C},(I_R,J_R)}(W).$$

We have $a_i\geq 0$, $a\geq 0$, $k>0$, and $x\geq 0$. Claim 1 of Lemma 7.28 gives

$$x(t_i-u_i-1)+(a_i-u_i)\geq 0 \qquad (i\in I).$$

Claim 1 of Lemma 7.29 gives $\mathrm{SnkAdj}_{\mathbb{S}}(v)\subseteq I$ for $v\in J$ and the weight formula (7.6). The dimension pair of R is $(a,a+k)$ by Claim 1(d).

First, suppose $I\sqcup J\subsetneq \mathrm{Vert}(S)$. Since $W\neq 0$, we have $I\sqcup J\neq\varnothing$. Equation (7.5) and Claim 3 of Lemma 7.27 give

$$\sum_{\substack{e=[\![v,i]\!]\in\mathrm{Edge}(S)\\ v\in I_S\setminus J,\ i\in I}} f_e = \sum_{i\in I}(x-a_i)-\#J(x+1)>0.$$

Hence, (7.6) gives $at-(a+k)u>0$.

Second, suppose $I\sqcup J=\mathrm{Vert}(S)$. Then, $I=J_S$, $J=I_S$, and $W_v=R_v=\mathbb{C}$ for every $v\in I_S$. Since $W\neq R$, there is $i_0\in J_S$ such that $0\neq W_{i_0}\neq Z_{i_0}$. Claim 2 of Lemma 7.28 and Claim 1 of Lemma 7.28 give

$$x(t_{i_0}-u_{i_0}-1)+(a_{i_0}-u_{i_0})>0.$$

For $i\in I\setminus\{i_0\}$, Claim 1 of Lemma 7.28 gives $x(t_i-u_i-1)+(a_i-u_i)\geq 0$. The boundary-edge sum in (7.6) is empty because $J=I_S$. Thus, $at-(a+k)u>0$ in this case as well. By Definition 7.17, R is stable.

Let us prove the converse. Suppose R is stable. Fix $w=[\![v,i]\!]\in\mathrm{Edge}(S)$ with $v\in I_S$ and $i\in J_S$. Claim 2 of Lemma 7.29 gives a nonzero proper subrepresentation W^w of R. For

$$(u_w,t_w)=\mathrm{TotDim}_{\mathbb{C},(I_R,J_R)}(W^w),$$

Claim 2 of Lemma 7.29 gives

$$at_w-(a+k)u_w=kf_w.$$

Stability gives $kf_w>0$. Since $k>0$, we obtain $f_w>0$. □

7.5 Exact counts for fixed colored-vacant-slot colors

Definition 7.31. *Let $j \in \mathbb{Z}_{\geq 1}$. Let $d \in [0,2]_{\mathbb{Z}}$ and $\epsilon \in [0,2]_{\mathbb{Z}} \setminus \{d\}$. We consider all colored source-trivalent trees $(\mathbb{T}, \kappa)$ such that*

$$\mathrm{TotDim}_{\mathbb{C},(I_{Q(\mathbb{T},\kappa)},J_{Q(\mathbb{T},\kappa)})}(\mathrm{ZS}(Q(\mathbb{T},\kappa))) = (j, j+1).$$

1. *We define P_j to be the number of classes*

$$\mathrm{MarkedCIsom}_{\mathbb{C}}(Q(\mathbb{T},\kappa), \mathrm{ZS}(Q(\mathbb{T},\kappa)); ((v,d))), \qquad (v,d) \in \mathrm{ColVacSlot}(Q(\mathbb{T},\kappa)).$$

2. *We define $Q_j^{=}$ to be the number of classes*

$$\mathrm{MarkedCIsom}_{\mathbb{C}}(Q(\mathbb{T},\kappa), \mathrm{ZS}(Q(\mathbb{T},\kappa)); ((v_1,d),(v_2,d)))$$

   for $(v_i, d) \in \mathrm{ColVacSlot}(Q(\mathbb{T},\kappa))$ with $i \in [2]_{\mathbb{Z}}$ and $v_1 \neq v_2$.

3. *We define $Q_j^{\neq}$ to be the number of classes*

$$\mathrm{MarkedCIsom}_{\mathbb{C}}(Q(\mathbb{T},\kappa), \mathrm{ZS}(Q(\mathbb{T},\kappa)); ((v_1,d),(v_2,\epsilon)))$$

   for $(v_1, d) \in \mathrm{ColVacSlot}(Q(\mathbb{T},\kappa))$ and $(v_2, \epsilon) \in \mathrm{ColVacSlot}(Q(\mathbb{T},\kappa))$.

Definition 7.32 (Colored branches and their series)**.** *Let $d \in [0,2]_{\mathbb{Z}}$. Let t, y_0, y_1, and y_2 be commuting indeterminates.*

1. *Let $\mathbb{B} = (B, (I_B, J_B))$ be a bipartite graph such that B is a finite tree. We define $\mathrm{Branch}(\mathbb{B}; d)$ to be the set of pairs (κ, r) satisfying the following conditions.*

   (a) *$\kappa : \mathrm{Edge}(B) \to [0,2]_{\mathbb{Z}}$ is a proper edge coloring.*

   (b) *$r \in \mathrm{Vert}(B)$ and $\kappa(e) \neq d$ for every $e \in \mathrm{IncEdge}_B(r)$.*

   (c) *For every $v \in I_B$,*

$$\deg_B(v) = \begin{cases} 2, & v = r, \\ 3, & v \neq r. \end{cases}$$

   A branch with parent color d is a tuple $\mathfrak{b} = (\mathbb{B}, \kappa, r)$ such that $(\kappa, r) \in \mathrm{Branch}(\mathbb{B}; d)$. We call r its root.

2. *We call $\mathfrak{b}$ a source branch if $r \in I_B$ and a sink branch if $r \in J_B$.*

3. *For $\mathfrak{b} = (\mathbb{B}, \kappa, r)$, form B^+ by adjoining a new vertex p in the part opposite to r and the edge $[\![p, r]\!]$. We call p the parent vertex and $[\![p, r]\!]$ the parent edge. Extend κ to $\kappa^+ : \mathrm{Edge}(B^+) \to [0,2]_{\mathbb{Z}}$ by setting*

$$\kappa^+(e) = \kappa(e) \quad (e \in \mathrm{Edge}(B)), \qquad \kappa^+([\![p,r]\!]) = d.$$

4. *Let $\mathfrak{b} = (\mathbb{B}, \kappa, r)$ and $\mathfrak{b}' = (\mathbb{B}', \kappa', r')$ be branches with parent color d. Let $\mathbb{B} = (B, (I_B, J_B))$ and $\mathbb{B}' = (B', (I_{B'}, J_{B'}))$. An isomorphism $\mathfrak{b} \to \mathfrak{b}'$ is a graph isomorphism $\psi = (\psi_0, \psi_1) : B \to B'$ such that $\psi_0(I_B) = I_{B'}$, $\psi_0(J_B) = J_{B'}$, and*

$$\psi_0(r) = r', \qquad \kappa'(\psi_1(e)) = \kappa(e) \quad (e \in \mathrm{Edge}(B)).$$

   Let $\mathcal{B}_d^{\mathrm{src}}$ and $\mathcal{B}_d^{\mathrm{snk}}$ be the sets of isomorphism classes of source and sink branches with parent color d.

5. *For $\mathfrak{b} = (\mathbb{B}, \kappa, r)$, define*
$$n(\mathfrak{b}) = \#I_B.$$
For $e \in [0,2]_{\mathbb{Z}}$, let $\ell_e(\mathfrak{b})$ be the number of vertices $v \in J_B$ such that $\deg_{B^+}(v) = 1$ and the unique edge incident to v in B^+ has color e under κ^+.

6. *We define the weight of $\mathfrak{b}$ by*
$$w(\mathfrak{b}) = t^{n(\mathfrak{b})} \prod_{e=0}^{2} y_e^{\ell_e(\mathfrak{b})}.$$

7. *Let $\mathcal{T}$ be the set of color-preserving, bipartition-preserving isomorphism classes of colored source-trivalent trees $(\mathbb{T}, \kappa)$. For $\mathbb{T} = (T, (I_T, J_T))$, define $n(\mathbb{T}) = \#I_T$. For $e \in [0,2]_{\mathbb{Z}}$, let $\ell_e(\mathbb{T}, \kappa)$ count the sink leaves of T whose unique incident edge has color e. Define*
$$w(\mathbb{T}, \kappa) = t^{n(\mathbb{T})} \prod_{e=0}^{2} y_e^{\ell_e(\mathbb{T}, \kappa)}.$$

8. *We define*
$$T_d = \sum_{[\mathfrak{b}] \in \mathcal{B}_d^{\mathrm{src}}} w(\mathfrak{b}), \qquad D_d = \sum_{[\mathfrak{b}] \in \mathcal{B}_d^{\mathrm{snk}}} w(\mathfrak{b}), \qquad U = \sum_{[(\mathbb{T}, \kappa)] \in \mathcal{T}} w(\mathbb{T}, \kappa).$$

Lemma 7.33 (Branch-series identities)**.** *Let $\{d, \epsilon, f\} = [0,2]_{\mathbb{Z}}$. Then, we have the following.*

1. *For every $d \in [0,2]_{\mathbb{Z}}$ and $n \in \mathbb{Z}_{\geq 0}$, there are finitely many classes in each of $\mathcal{B}_d^{\mathrm{src}}$, $\mathcal{B}_d^{\mathrm{snk}}$, and $\mathcal{T}$ with source count n. The weights $w(\mathfrak{b})$ and $w(\mathbb{T}, \kappa)$ are invariant under branch isomorphisms and color-preserving, bipartition-preserving tree isomorphisms, respectively. Thus, T_d, D_d, and U belong to $\mathbb{Z}[y_0, y_1, y_2][[t]]$.*

2. *We have*
$$D_d = y_d + T_\epsilon + T_f + T_\epsilon T_f, \qquad T_d = t D_\epsilon D_f. \tag{7.8}$$

3. *We have*
$$\frac{\partial U}{\partial y_d} = T_d.$$

Proof. Proof of Claim 1. Let us count the vertices for each source count n. For $\mathfrak{b} = (\mathbb{B}, \kappa, r)$ with $\mathbb{B} = (B, (I_B, J_B))$ and $\#I_B = n$, we have
$$\#\operatorname{Edge}(B) = \sum_{v \in I_B} \deg_B(v) = \begin{cases} 3n-1, & r \in I_B, \\ 3n, & r \in J_B. \end{cases}$$
Since B is a tree,
$$\#\operatorname{Vert}(B) = \#\operatorname{Edge}(B) + 1 = \begin{cases} 3n, & r \in I_B, \\ 3n+1, & r \in J_B. \end{cases}$$
For a source-trivalent tree $\mathbb{T} = (T, (I_T, J_T))$ with $\#I_T = n$, we obtain
$$\#\operatorname{Edge}(T) = \sum_{v \in I_T} 3 = 3n, \qquad \#\operatorname{Vert}(T) = 3n+1.$$

For each of these vertex counts N, relabel the vertices by $[N]_{\mathbb{Z}}$. There are finitely many choices of

$$I \subseteq [N]_{\mathbb{Z}}, \qquad J = [N]_{\mathbb{Z}} \setminus I, \qquad E \subseteq \{[\![u, v]\!] : u, v \in [N]_{\mathbb{Z}},\ u \neq v\}, \qquad \kappa : E \to [0, 2]_{\mathbb{Z}}.$$

For a branch, there are also finitely many choices of $r \in [N]_{\mathbb{Z}}$. Every class at source count n has such a representative. Thus, each set of classes at source count n is finite.

Let us verify that isomorphisms preserve the weights. Let $\psi = (\psi_0, \psi_1) : \mathfrak{b} \to \mathfrak{b}'$ be an isomorphism of branches with parent color d. Let $\mathfrak{b}' = (\mathbb{B}', \kappa', r')$ and $\mathbb{B}' = (B', (I_{B'}, J_{B'}))$. Let p and p' be their parent vertices. Extend ψ to $\psi^+ : B^+ \to B'^+$ by

$$\begin{aligned} \psi_0^+(v) &= \psi_0(v) \quad (v \in \mathrm{Vert}(B)), & \psi_0^+(p) &= p', \\ \psi_1^+(g) &= \psi_1(g) \quad (g \in \mathrm{Edge}(B)), & \psi_1^+([\![p, r]\!]) &= [\![p', r']\!]. \end{aligned}$$

Since $\psi_0(r) = r'$ and both parent edges have color d, we have

$$\begin{aligned} \deg_{B'^+}(\psi_0^+(v)) &= \deg_{B^+}(v) \quad (v \in \mathrm{Vert}(B^+)), \\ \kappa'^+(\psi_1^+(g)) &= \kappa^+(g) \qquad\quad (g \in \mathrm{Edge}(B^+)). \end{aligned}$$

Also, $\psi_0(I_B) = I_{B'}$ and $\psi_0(J_B) = J_{B'}$. Consequently,

$$n(\mathfrak{b}) = n(\mathfrak{b}'), \qquad \ell_e(\mathfrak{b}) = \ell_e(\mathfrak{b}') \quad (e \in [0, 2]_{\mathbb{Z}}),$$

and

$$w(\mathfrak{b}) = t^{n(\mathfrak{b})} \prod_{e=0}^{2} y_e^{\ell_e(\mathfrak{b})} = t^{n(\mathfrak{b}')} \prod_{e=0}^{2} y_e^{\ell_e(\mathfrak{b}')} = w(\mathfrak{b}').$$

Let us verify that isomorphisms of colored bipartite trees preserve their weights. For a color-preserving, bipartition-preserving tree isomorphism $(\mathbb{T}, \kappa) \to (\mathbb{T}', \kappa')$, preservation of degrees and colors gives

$$n(\mathbb{T}) = n(\mathbb{T}'), \qquad \ell_e(\mathbb{T}, \kappa) = \ell_e(\mathbb{T}', \kappa') \quad (e \in [0, 2]_{\mathbb{Z}}).$$

Hence, $w(\mathbb{T}, \kappa) = w(\mathbb{T}', \kappa')$. For each n, the coefficient of t^n in T_d, D_d, and U is therefore a finite sum of well-defined monomials in $\mathbb{Z}[y_0, y_1, y_2]$.

Proof of Claim 2. Let us specify the branches obtained by deleting the root. Let $\mathfrak{b} = (\mathbb{B}, \kappa, r)$ have parent color d and let $\mathbb{B} = (B, (I_B, J_B))$. Let

$$A = \{\kappa(g) : g \in \mathrm{IncEdge}_B(r)\} \subseteq \{\epsilon, f\}.$$

For each $e \in A$, proper coloring gives a unique vertex u_e satisfying

$$[\![r, u_e]\!] \in \mathrm{Edge}(B), \qquad \kappa([\![r, u_e]\!]) = e.$$

Let B_e be the connected component of $\mathrm{Subgraph}(B, \mathrm{Vert}(B) \setminus \{r\})$ containing u_e. Define

$$\begin{aligned} I_e &= I_B \cap \mathrm{Vert}(B_e), & J_e &= J_B \cap \mathrm{Vert}(B_e), \\ \kappa_e &= \kappa|_{\mathrm{Edge}(B_e)}, & \mathfrak{b}_e &= ((B_e, (I_e, J_e)), \kappa_e, u_e). \end{aligned}$$

Since B is a tree, its components after deleting r are exactly the graphs B_e for $e \in A$. Each $\mathfrak{b}_e$ has parent color e.

First, let us treat a sink root. Let $r \in J_B$. Then, $u_e \in I_e$ and

$$\deg_{B_e}(v) = \begin{cases} 2, & v = u_e, \\ 3, & v \in I_e \setminus \{u_e\}. \end{cases}$$

Thus, $\mathfrak{b}_e$ is a source branch. The map

$$[\mathfrak{b}] \longmapsto (A, ([\mathfrak{b}_e])_{e\in A})$$

gives a bijection

$$\mathcal{B}_d^{\mathrm{snk}} \longrightarrow \bigsqcup_{A\subseteq\{\epsilon,f\}} \prod_{e\in A} \mathcal{B}_e^{\mathrm{src}}.$$

To construct its inverse, choose disjoint representatives $\mathfrak{b}_e = ((B_e, (I_e, J_e)), \kappa_e, u_e)$ for $e \in A$ and a new sink r. Let

$$I_B = \bigsqcup_{e\in A} I_e, \qquad J_B = \{r\} \sqcup \bigsqcup_{e\in A} J_e,$$
$$\mathrm{Edge}(B) = \bigsqcup_{e\in A} \mathrm{Edge}(B_e) \sqcup \{[\![r, u_e]\!] : e \in A\}.$$

Let $\kappa|_{\mathrm{Edge}(B_e)} = \kappa_e$ and $\kappa([\![r, u_e]\!]) = e$ for $e \in A$. These constructions are inverse on isomorphism classes. For $A = \varnothing$, the branch consists of the sink r and has weight y_d. For $A \neq \varnothing$, the root r is not a sink leaf in B^+. Thus,

$$n(\mathfrak{b}) = \sum_{e\in A} n(\mathfrak{b}_e), \qquad \ell_h(\mathfrak{b}) = \sum_{e\in A} \ell_h(\mathfrak{b}_e) \quad (h \in [0,2]_{\mathbb{Z}}).$$

Hence,

$$\begin{aligned} D_d &= y_d + \sum_{\varnothing\neq A\subseteq\{\epsilon,f\}} \prod_{e\in A} T_e \\ &= y_d + T_\epsilon + T_f + T_\epsilon T_f. \end{aligned}$$

Second, let us treat a source root. Let $r \in I_B$. Then, $A = \{\epsilon, f\}$, $u_e \in J_e$, and every vertex of I_e has degree three in B_e. Thus, $\mathfrak{b}_\epsilon$ and $\mathfrak{b}_f$ are sink branches. The map

$$[\mathfrak{b}] \longmapsto ([\mathfrak{b}_\epsilon], [\mathfrak{b}_f])$$

gives a bijection

$$\mathcal{B}_d^{\mathrm{src}} \longrightarrow \mathcal{B}_\epsilon^{\mathrm{snk}} \times \mathcal{B}_f^{\mathrm{snk}}.$$

Its inverse adjoins a new source r to disjoint representatives of the two sink branches and adds the edges $[\![r, u_\epsilon]\!]$ and $[\![r, u_f]\!]$ of colors ϵ and f. The vertex parts are

$$I_B = \{r\} \sqcup I_\epsilon \sqcup I_f, \qquad J_B = J_\epsilon \sqcup J_f.$$

We have

$$\begin{aligned} n(\mathfrak{b}) &= 1 + n(\mathfrak{b}_\epsilon) + n(\mathfrak{b}_f), \\ \ell_h(\mathfrak{b}) &= \ell_h(\mathfrak{b}_\epsilon) + \ell_h(\mathfrak{b}_f) \quad (h \in [0,2]_{\mathbb{Z}}). \end{aligned}$$

Therefore,

$$T_d = \sum_{[\mathfrak{b}_\epsilon]\in\mathcal{B}_\epsilon^{\mathrm{snk}}} \sum_{[\mathfrak{b}_f]\in\mathcal{B}_f^{\mathrm{snk}}} t\, w(\mathfrak{b}_\epsilon) w(\mathfrak{b}_f) = t D_\epsilon D_f.$$

Proof of Claim 3. Let us specify the deletion of a marked sink leaf. Let $\mathbb{T} = (T, (I_T, J_T))$ and let $v \in J_T$ be a sink leaf. Let $u \in I_T$ be its unique neighbor and assume

$$\kappa([\![u, v]\!]) = d.$$

Let

$$\begin{aligned} B &= \mathrm{Subgraph}(T, \mathrm{Vert}(T) \setminus \{v\}), \\ \mathbb{B} &= (B, (I_T, J_T \setminus \{v\})), \qquad \kappa_B = \kappa|_{\mathrm{Edge}(B)}. \end{aligned}$$

Then, $\mathfrak{b} = (\mathbb{B}, \kappa_B, u)$ is a source branch with parent color d, since

$$\deg_B(u) = 2, \qquad \deg_B(x) = 3 \quad (x \in I_T \setminus \{u\}), \qquad \kappa_B(g) \neq d \quad (g \in \mathrm{IncEdge}_B(u)).$$

Thus, deletion defines the map on marked isomorphism classes

$$[((\mathbb{T}, \kappa); v)] \longmapsto [(\mathbb{B}, \kappa_B, u)].$$

Let us construct the inverse map. Let $(\mathbb{B}, \kappa_B, u)$ be a source branch with $\mathbb{B} = (B, (I_B, J_B))$. Adjoin a new sink v and let

$$\begin{aligned} I_T &= I_B, & J_T &= J_B \sqcup \{v\}, \\ \mathrm{Edge}(T) &= \mathrm{Edge}(B) \sqcup \{[\![u, v]\!]\}, & & \\ \kappa|_{\mathrm{Edge}(B)} &= \kappa_B, & \kappa([\![u, v]\!]) &= d. \end{aligned}$$

The marked leaf is v. Both operations preserve isomorphisms and their compositions return the original marked tree or branch up to isomorphism.

Let us compute the weight after deleting the marked leaf. Let $n = n(\mathbb{T})$ and $\ell_e = \ell_e(\mathbb{T}, \kappa)$ for $e \in [0, 2]_{\mathbb{Z}}$. Deleting v gives

$$n(\mathfrak{b}) = n, \qquad \ell_e(\mathfrak{b}) = \begin{cases} \ell_d - 1, & e = d, \\ \ell_e, & e \neq d. \end{cases}$$

By Lemma 7.5, the ℓ_d choices of v give ℓ_d distinct marked isomorphism classes. For $\ell_d \geq 1$, their total branch weight is

$$\ell_d t^n y_d^{\ell_d - 1} \prod_{e \in [0,2]_{\mathbb{Z}} \setminus \{d\}} y_e^{\ell_e} = \frac{\partial}{\partial y_d} \left(t^n \prod_{e=0}^{2} y_e^{\ell_e} \right).$$

For $\ell_d = 0$, both contributions are zero. Claim 1 makes each coefficient a finite sum. Thus,

$$\frac{\partial U}{\partial y_d} = \sum_{[\mathfrak{b}] \in \mathcal{B}_d^{\mathrm{src}}} w(\mathfrak{b}) = T_d. \qquad \square$$

Lemma 7.34. *Let $j \in \mathbb{Z}_{>0}$. Then, the following hold.*

1. *We have*
$$P_j = \frac{1}{3j+4} \binom{4j+4}{j+1}. \tag{7.9a}$$
2. *We have*
$$Q_j^{=} = 2 \binom{4j+2}{j-1}. \tag{7.9b}$$
3. *We have*
$$Q_j^{\neq} = \frac{2j+3}{2j} Q_j^{=} \geq Q_j^{=}. \tag{7.9c}$$

Proof. Proof of Claim 1. Let us recover a colored tree from its quiver. Let $(\mathbb{T}, \kappa)$ be a colored source-trivalent tree. Let $\mathbb{T} = (T, (I_T, J_T))$ and assume $\#I_T = j + 1$. Put $Q = Q(\mathbb{T}, \kappa)$. For

each $(u,e) \in \mathrm{ColVacSlot}(Q)$, choose a new vertex $v_{u,e}$. Choose these vertices distinct and outside $\mathrm{Vert}(Q)$. Up to a color-preserving isomorphism, the tree is recovered by

$$\begin{aligned} I_T &= J_Q, \\ J_T &= I_Q \sqcup \{v_{u,e} : (u,e) \in \mathrm{ColVacSlot}(Q)\}, \\ \mathrm{Edge}(T) &= \{[\![t_Q(\alpha), s_Q(\alpha)]\!] : \alpha \in \mathrm{Edge}(Q)\} \\ &\quad \sqcup \{[\![u, v_{u,e}]\!] : (u,e) \in \mathrm{ColVacSlot}(Q)\}, \\ \kappa([\![t_Q(\alpha), s_Q(\alpha)]\!]) &= \mathrm{QCol}(Q)(\alpha), \qquad \kappa([\![u, v_{u,e}]\!]) = e. \end{aligned}$$

Lemma 7.24 identifies the marked slot (u,e) with the marked leaf $v_{u,e}$.

Let us construct the induced isomorphism of zero-sum representations. Let $(\mathbb{T}', \kappa')$ be another colored source-trivalent tree. Let $\mathbb{T}' = (T', (I_{T'}, J_{T'}))$ and assume $\#I_{T'} = j+1$. Put $Q' = Q(\mathbb{T}', \kappa')$ and let $\phi = (\phi_0, \phi_1) : Q \to Q'$ be a color-preserving isomorphism preserving the bipartitions. For $v \in I_Q$, define

$$F_v : \mathrm{ZS}(Q)_v \longrightarrow \mathrm{ZS}(Q')_{\phi_0(v)}, \qquad (F_v z)_{\phi_1(\alpha)} = z_\alpha \quad (\alpha \in \mathrm{IncEdgeQuiv}_Q(v)).$$

The map ϕ_1 is a bijection on the incident arrow sets. Hence,

$$\sum_{\alpha' \in \mathrm{IncEdgeQuiv}_{Q'}(\phi_0(v))} (F_v z)_{\alpha'} = \sum_{\alpha \in \mathrm{IncEdgeQuiv}_Q(v)} z_\alpha = 0.$$

For $u \in J_Q$, set $F_u = 1_{\mathbb{C}}$. For every arrow $\alpha : v \to u$, the identity $(F_v z)_{\phi_1(\alpha)} = z_\alpha$ gives

$$\mathrm{ZS}(Q')_{\phi_1(\alpha)} \circ F_v = F_u \circ \mathrm{ZS}(Q)_\alpha.$$

The inverse maps are obtained from ϕ^{-1}. Thus, $(\phi_0, \phi_1, (F_v)_{v \in \mathrm{Vert}(Q)})$ is an isomorphism of the zero-sum representations.

Let us identify the marked isomorphism classes. The extension of ϕ_0 to the deleted leaves is

$$v_{u,e} \longmapsto v'_{\phi_0(u),e}.$$

The map $v_{u,e} \mapsto v'_{\phi_0(u),e}$ preserves the marked leaves exactly when ϕ_0 preserves the marked slots. Conversely, an isomorphism of marked representations includes an isomorphism of the marked colored quivers by Definition 7.19(4). The reconstruction therefore gives a bijection between the marked tree classes and the marked representation classes.

Let us express P_j as a coefficient. Let $d \in [0,2]_{\mathbb{Z}}$ and let $T = T_0|_{y_0=y_1=y_2=1}$. Permuting the colors gives $T_d|_{y_0=y_1=y_2=1} = T$ for every $d \in [0,2]_{\mathbb{Z}}$. Claims 2 and 3 of Lemma 7.33 give

$$T = t(1+T)^4, \qquad T(0) = 0, \qquad P_j = [t^{j+1}]T.$$

Let us extract the coefficient of t^ν for an integer $\nu \geq 1$. Let u be a formal indeterminate. Lagrange inversion gives

$$[t^\nu]T = \frac{1}{\nu}[u^{\nu-1}](1+u)^{4\nu} = \frac{1}{\nu}\binom{4\nu}{\nu-1} = \frac{1}{3\nu+1}\binom{4\nu}{\nu}.$$

Taking $\nu = j+1$ proves Claim 1.

Proof of Claim 2. Let us compute $Q_j^{=}$. Fix $d \in [0,2]_{\mathbb{Z}}$ and $\epsilon \in [0,2]_{\mathbb{Z}} \setminus \{d\}$. By Claim 3 of Lemma 7.33, differentiating T_d with respect to y_d counts an ordered pair of distinct sink leaves of color d. Lemma 7.24 identifies these pairs with the marked colored-vacant slots counted by $Q_j^{=}$.

Let us compute the derivatives in (7.8). Write

$$A = \frac{\partial T_d}{\partial y_d}\Big|_{y_0=y_1=y_2=1}, \qquad B = \frac{\partial T_\epsilon}{\partial y_d}\Big|_{y_0=y_1=y_2=1} \quad (\epsilon \neq d).$$

The two possible colors $\epsilon \neq d$ give the same B by color symmetry. Differentiating both recursions gives

$$A = \frac{2T}{1+T}(A+B), \qquad B = \frac{T}{(1+T)^2} + \frac{T}{1+T}(A+3B).$$

The equation $A = \frac{2T}{1+T}(A+B)$ gives $(1-T)A = 2TB$. Substituting $(1-T)A = 2TB$ into $B = \frac{T}{(1+T)^2} + \frac{T}{1+T}(A+3B)$ gives

$$B = \frac{T(1-T)}{(1+T)(1-3T)}, \qquad A = \frac{2T^2}{(1+T)(1-3T)}.$$

All denominators have constant term one and are invertible as formal power series.

Let us extract the coefficient that counts the two markings. Differentiating $T = t(1+T)^4$ with respect to t gives

$$tT' = \frac{T(1+T)}{1-3T}, \qquad A = tT'\frac{2T}{(1+T)^2}.$$

The derivative form of Lagrange inversion gives, for $\nu \geq 2$,

$$[t^\nu]A = [u^{\nu-1}]\frac{2u}{(1+u)^2}(1+u)^{4\nu} = 2\binom{4\nu-2}{\nu-2}.$$

Taking $\nu = j+1$ gives $Q_j^{=} = 2\binom{4j+2}{j-1}$.

Proof of Claim 3. Let us compute $Q_j^{\neq}$. Fix $d \in [0,2]_{\mathbb{Z}}$ and $\epsilon \in [0,2]_{\mathbb{Z}} \setminus \{d\}$. By Claim 3 of Lemma 7.33, differentiating T_d with respect to y_ϵ counts an ordered pair of sink leaves of colors d and ϵ. Lemma 7.24 identifies these pairs with the marked colored-vacant slots counted by $Q_j^{\neq}$. The derivative calculation in Claim 2 and color symmetry give

$$\frac{\partial T_d}{\partial y_\epsilon}\Big|_{y_0=y_1=y_2=1} = \frac{T(1-T)}{(1+T)(1-3T)} = tT'\frac{1-T}{(1+T)^2}.$$

Let us extract the coefficient for an integer $\nu \geq 2$. Lagrange inversion gives

$$[t^\nu]\left(tT'\frac{1-T}{(1+T)^2}\right) = [u^{\nu-1}](1-u)(1+u)^{4\nu-2} = \binom{4\nu-2}{\nu-1} - \binom{4\nu-2}{\nu-2}.$$

Taking $\nu = j+1$ and using Claim 2, we obtain

$$\frac{Q_j^{\neq}}{Q_j^{=}} = \frac{1}{2}\left(\frac{\binom{4j+2}{j}}{\binom{4j+2}{j-1}} - 1\right) = \frac{1}{2}\left(\frac{3j+3}{j} - 1\right) = \frac{2j+3}{2j}.$$

□

Definition 7.35 (Admissible pairs)**.** *Let $(p_j)_{j\in\mathbb{Z}_{>0}}$ and $(q_j)_{j\in\mathbb{Z}_{>0}}$ be real sequences and let $N \in \mathbb{Z}_{>0}$. We call a pair $(A, B) \in \mathbb{R}^2_{>0}$ (p_j, q_j, N)-admissible if*

$$\begin{aligned} p_j &\geq A\beta^j j^{-\frac{3}{2}}, \\ q_j &\geq B\beta^j j^{-\frac{1}{2}} \end{aligned} \qquad (j \in \mathbb{Z}_{\geq N}).$$

Lemma 7.36. *There exists a $(P_j, Q^{=}_j, 256)$-admissible pair (A, B).*

Proof. Let us obtain positive lower bounds for the normalized counts. Claims 1 and 2 of Lemma 7.34 and Stirling's formula give

$$P_j\beta^{-j}j^{\frac{3}{2}} \longrightarrow \frac{\beta}{3}\sqrt{\frac{2}{3\pi}}, \qquad Q^{=}_j\beta^{-j}j^{\frac{1}{2}} \longrightarrow \frac{32}{27}\sqrt{\frac{2}{3\pi}}.$$

Let us include the finitely many smaller indices in these bounds. Put

$$L_P = \frac{\beta}{3}\sqrt{\frac{2}{3\pi}}, \qquad L_Q = \frac{32}{27}\sqrt{\frac{2}{3\pi}}.$$

Choose $J > 256$ such that, for every integer $j \geq J$,

$$P_j\beta^{-j}j^{\frac{3}{2}} \geq \frac{L_P}{2}, \qquad Q^{=}_j\beta^{-j}j^{\frac{1}{2}} \geq \frac{L_Q}{2}.$$

The explicit formulas in Lemma 7.34 show that both counts are positive for $j \geq 256$. Hence, the finite minima

$$\begin{aligned} A &= \min\left(\left\{\frac{L_P}{2}\right\} \cup \{P_j\beta^{-j}j^{\frac{3}{2}} : 256 \leq j < J\}\right) > 0, \\ B &= \min\left(\left\{\frac{L_Q}{2}\right\} \cup \{Q^{=}_j\beta^{-j}j^{\frac{1}{2}} : 256 \leq j < J\}\right) > 0 \end{aligned}$$

give a $(P_j, Q^{=}_j, 256)$-admissible pair. □

7.6 A family of positive flows

Definition 7.37 (Labeled source-trivalent chains)**.** *Let $c \in \mathbb{Z}_{>0}$. We define the labeled source-trivalent chain* $\mathrm{LSTrivCh}(c)$ *as follows. The vertices x_t for $t \in [c]_{\mathbb{Z}}$, z_t for $t \in [0, c]_{\mathbb{Z}}$, and y_t for $t \in [c]_{\mathbb{Z}}$ are pairwise distinct.*

1. *Its bipartition is*

$$\begin{aligned} I_{\mathrm{LSTrivCh}(c)} &= \{x_t : t \in [c]_{\mathbb{Z}}\}, \\ J_{\mathrm{LSTrivCh}(c)} &= \{z_t : t \in [0, c]_{\mathbb{Z}}\} \sqcup \{y_t : t \in [c]_{\mathbb{Z}}\}. \end{aligned}$$

2. *For $t \in [c]_{\mathbb{Z}}$, define*

$$\begin{aligned} \mathrm{LeftEdge}_c(t) &= [\![x_t, z_{t-1}]\!], \\ \mathrm{MidEdge}_c(t) &= [\![x_t, y_t]\!], \\ \mathrm{RightEdge}_c(t) &= [\![x_t, z_t]\!]. \end{aligned}$$

The edge set of $\mathrm{LSTrivCh}(c)$ *consists of* $\mathrm{LeftEdge}_c(t)$, $\mathrm{MidEdge}_c(t)$, *and* $\mathrm{RightEdge}_c(t)$ *for $t \in [c]_{\mathbb{Z}}$. We call them the left, middle, and right edges at x_t, respectively.*

Lemma 7.38. *For $c \in \mathbb{Z}_{>0}$, let $S = \mathrm{LSTrivCh}(c)$ and $\mathbb{S} = (S, (I_S, J_S))$. The bipartite graph $\mathbb{S}$ is a source-trivalent tree. Its sink sets satisfy*

$$\begin{aligned} J_S \cap \mathrm{Leaf}(S) &= \{z_0, z_c\} \sqcup \{y_t : t \in [c]_{\mathbb{Z}}\}, \\ \{v \in J_S : \deg_S(v) = 2\} &= \{z_t : t \in [c-1]_{\mathbb{Z}}\}. \end{aligned}$$

Proof. First, suppose $c = 1$. The graph is a tree with center x_1 and three leaves z_0, y_1, z_1. Its only source has degree three, and it has no degree-two sinks.

Second, suppose $c \geq 2$. Let us delete $z_0, z_c, y_1, \ldots, y_c$. The remaining vertices and edges form the path

$$x_1, z_1, x_2, \ldots, z_{c-1}, x_c.$$

Each deleted vertex is joined to exactly one source by one edge. Adding these vertices to the path preserves connectedness and creates no cycle. Thus, S is a tree. Its degrees are

$$\begin{aligned} \deg_S(x_t) &= 3 \quad (1 \leq t \leq c), \\ \deg_S(z_t) &= 2 \quad (1 \leq t < c), \\ \deg_S(z_0) = \deg_S(z_c) = \deg_S(y_t) &= 1 \quad (1 \leq t \leq c). \end{aligned}$$

These formulas give the stated source degrees, sink leaves, and degree-two sinks. □

Definition 7.39 (Chain flow family and source dimensions)**.** *Let $a \in \mathbb{Z}_{>0}$ and $k \in \mathbb{Z}_{>0}$ satisfy $k \geq 2$. Let $c = k - 1$, $x = \frac{a}{k}$, and $S = \mathrm{LSTrivCh}(c)$. For $f \in \mathrm{Flow}(S)$ and $t \in [c]_{\mathbb{Z}}$, define*

$$l_t(f) = f_{\mathrm{LeftEdge}_c(t)}, \qquad m_t(f) = f_{\mathrm{MidEdge}_c(t)}, \qquad r_t(f) = f_{\mathrm{RightEdge}_c(t)}.$$

Define $\mathcal{F}_{a,k}$ to be the set of $f \in \mathrm{Flow}(S)$ such that, for every $t \in [c]_{\mathbb{Z}}$,

$$\begin{aligned} l_t(f) &\in [\frac{x}{256}, \frac{2x}{256}) \cap (tx + \mathbb{Z}), \\ r_t(f) &\in (0, \frac{x}{256}) \cap (-tx + \mathbb{Z}), \\ m_t(f) &= x + 1 - l_t(f) - r_t(f). \end{aligned}$$

For $f \in \mathrm{Flow}(S)$ and $b \in J_S$, define the source dimension

$$a_b(f) = x - \sum_{w \in \mathrm{IncEdge}_S(b)} f_w.$$

Lemma 7.40 (Flow choices)**.** *Let $a \in \mathbb{Z}_{>0}$ and $k \in \mathbb{Z}_{>0}$ satisfy $k \geq 2$ and $a \geq 100000k$. Let $c = k - 1$, $x = \frac{a}{k}$, and $S = \mathrm{LSTrivCh}(c)$. Let*

$$N = 256, \qquad \gamma = 1 - \frac{N}{x}, \qquad \lambda = \frac{1}{100}.$$

Let $f \in \mathcal{F}_{a,k}$. Let $l_t = l_t(f)$, $m_t = m_t(f)$, and $r_t = r_t(f)$ for $t \in [c]_{\mathbb{Z}}$. Then, we have the following.

1. *The flow f and the source dimensions $(a_b(f))_{b \in J_S}$ satisfy the following.*
   - *(a) We have $l_t > 0$, $m_t > 0$, and $r_t > 0$ for $t \in [c]_{\mathbb{Z}}$.*
   - *(b) For $b \in J_S$, we have $a_b(f) \in \mathbb{Z}$.*
   - *(c) We have $\sum_{b \in J_S} a_b(f) = a - c$.*
   - *(d) For every $b \in J_S$, we have*
     $$256 < a_b(f) < x.$$
2. *For $t \in [c]_{\mathbb{Z}}$, we have*
   $$\begin{aligned} \#\left([\frac{x}{N}, \frac{2x}{N}) \cap (tx + \mathbb{Z})\right) &\geq \frac{\gamma x}{N}, \\ \#\left((0, \frac{x}{N}) \cap (-tx + \mathbb{Z})\right) &\geq \frac{\gamma x}{N}. \end{aligned}$$

3. *For $f, f' \in \mathcal{F}_{a,k}$, we have*

$$(a_b(f) = a_b(f') \text{ for every } b \in J_S) \quad \Longrightarrow \quad f = f'.$$

4. *Let $\mathbf{T} = ((\mathbb{T}_b, \kappa_b))_{b \in J_S}$ be a family of colored source-trivalent trees satisfying*

$$\mathrm{TotDim}_{\mathbb{C},(I_{Q(\mathbb{T}_b,\kappa_b)},J_{Q(\mathbb{T}_b,\kappa_b)})}(\mathrm{ZS}(Q(\mathbb{T}_b,\kappa_b))) = (a_b(f), a_b(f)+1) \qquad (b \in J_S).$$

Let $\mathbf{p} = ((v_e, d_e))_{e \in \mathrm{Edge}(S)}$ be a family of colored-vacant slots and $\mathbf{g} = (g_e)_{e \in \mathrm{Edge}(S)}$ a family of nonzero complex linear maps such that

$$(v_e, d_e) \in \mathrm{ColVacSlot}(Q(\mathbb{T}_b, \kappa_b)), \qquad g_e : \mathbb{C} \longrightarrow \mathrm{ZS}(Q(\mathbb{T}_b, \kappa_b))_{v_e}$$

for $e = [\![x, b]\!] \in \mathrm{Edge}(S)$, $x \in I_S$, and $b \in J_S$. Let

$$R = \mathrm{Glue}_{\mathbb{C}}((S, (I_S, J_S)); \mathbf{T}, \mathbf{p}, \mathbf{g})$$

satisfy hypotheses (a) and (b) of Lemma 7.30. Then, R is stable.

5. *For $t \in [c-1]_{\mathbb{Z}}$, we have*

$$(l_t + r_t - 1)^{-\frac{3}{2}} (x - r_t - l_{t+1})^{-\frac{1}{2}} \geq \lambda N^2 x^{-2}. \tag{7.10}$$

The dimensions $a_{z_0}(f)$, $a_{z_c}(f)$, and $a_{y_c}(f)$ satisfy

$$(x - l_1)^{-\frac{3}{2}} (x - r_c)^{-\frac{3}{2}} (l_c + r_c - 1)^{-\frac{3}{2}} \geq x^{-\frac{9}{2}}.$$

Proof. Proof of Claim 1(a). Let us prove positivity of all edge weights. Since $f \in \mathcal{F}_{a,k}$, $l_t > 0$, $r_t > 0$, and $l_t + r_t < \frac{3x}{N}$. Hence,

$$m_t = x + 1 - l_t - r_t > x\left(1 - \frac{3}{N}\right) + 1 > 0.$$

Proof of Claim 1(b). Let us prove $a_b(f) \in \mathbb{Z}$ for $b \in J_S$. For $S = \mathrm{LSTrivCh}(c)$, the dimensions $a_b(f)$ satisfy

$$\begin{aligned} a_{z_0}(f) &= x - l_1, & a_{z_c}(f) &= x - r_c, \\ a_{y_t}(f) &= l_t + r_t - 1 & & (t \in [c]_{\mathbb{Z}}), \\ a_{z_t}(f) &= x - r_t - l_{t+1} & & (t \in [c-1]_{\mathbb{Z}}). \end{aligned}$$

The conditions $l_t - tx \in \mathbb{Z}$, $r_t + tx \in \mathbb{Z}$, and $kx = a \in \mathbb{Z}$ give

$$\begin{aligned} x - l_1 &= -(l_1 - x) \in \mathbb{Z}, \\ x - r_c &= a - (r_c + cx) \in \mathbb{Z}, \\ l_t + r_t - 1 &= (l_t - tx) + (r_t + tx) - 1 \in \mathbb{Z}, \\ x - r_t - l_{t+1} &= -(r_t + tx) - (l_{t+1} - (t+1)x) \in \mathbb{Z}. \end{aligned}$$

Proof of Claim 1(c). The dimensions $a_b(f)$ in Claim 1(b) satisfy

$$\begin{aligned} &(x - l_1) + (x - r_c) + \sum_{t=1}^{c} (l_t + r_t - 1) + \sum_{t=1}^{c-1} (x - r_t - l_{t+1}) \\ &\quad = (c+1)x - c = a - c. \end{aligned}$$

Proof of Claim 1(d). Let us bound each source dimension between 256 and x. The bound $a \geq 100000k$ gives $x \geq 100000$. First, for $t \in [c]_{\mathbb{Z}}$, the bounds $l_t \in [\frac{x}{N}, \frac{2x}{N})$ and $r_t \in (0, \frac{x}{N})$ give

$$256 < \frac{x}{N} - 1 < a_{y_t}(f) = l_t + r_t - 1 < \frac{3x}{N} - 1 < x.$$

Second, for $t \in [0, c]_{\mathbb{Z}}$, the formulas in Claim 1(b) give

$$256 < x\left(1 - \frac{3}{N}\right) < a_{z_t}(f) < x.$$

For $t = 0$, use $a_{z_0}(f) = x - l_1$ and $0 < l_1 < \frac{2x}{N}$. For $t = c$, use $a_{z_c}(f) = x - r_c$ and $0 < r_c < \frac{x}{N}$. For $t \in [c-1]_{\mathbb{Z}}$, use $a_{z_t}(f) = x - r_t - l_{t+1}$ and $0 < r_t + l_{t+1} < \frac{3x}{N}$.

Proof of Claim 2. Let us count the integers after translating each interval. For $t \in [c]_{\mathbb{Z}}$, we have

$$\#\left([\frac{x}{N}, \frac{2x}{N}) \cap (tx + \mathbb{Z})\right) = \left\lceil \frac{2x}{N} - tx \right\rceil - \left\lceil \frac{x}{N} - tx \right\rceil \geq \frac{x}{N} - 1,$$
$$\#\left((0, \frac{x}{N}) \cap (-tx + \mathbb{Z})\right) = \left\lceil \frac{x}{N} + tx \right\rceil - \lfloor tx \rfloor - 1 \geq \frac{x}{N} - 1.$$

Since $\frac{x}{N} - 1 = \frac{\gamma x}{N}$, both inequalities in Claim 2 follow.

Proof of Claim 3. Let us recover the flow from its source dimensions. Let $f \in \mathcal{F}_{a,k}$ and $\widetilde{f} \in \mathcal{F}_{a,k}$ satisfy $a_b(f) = a_b(\widetilde{f})$ for every $b \in J_S$. For $h \in \{f, \widetilde{f}\}$, we have

$$\sum_{w \in \mathrm{IncEdge}_S(v)} h_w = x + 1 \quad (v \in I_S), \qquad \sum_{w \in \mathrm{IncEdge}_S(b)} h_w = x - a_b(f) \quad (b \in J_S).$$

Claim 1 of Lemma 7.27 gives $f = \widetilde{f}$.

Proof of Claim 4. Let us prove stability. Claim 1(c) gives $\sum_{b \in J_S} a_b(f) + c = a$. Definition 7.39 gives equations (7.7) for f and the source dimensions $a_b(f)$. Claim 2 of Lemma 7.30 therefore identifies f as the unique flow used in Claim 3 of Lemma 7.30. Claim 1(a) gives $f_w > 0$ for every $w \in \mathrm{Edge}(S)$. Claim 3 of Lemma 7.30 gives stability of R.

Proof of Claim 5. Let us bound the products of inverse powers of the source dimensions. For $t \in [c-1]_{\mathbb{Z}}$, we have $a_{y_t}(f) < \frac{3x}{N}$ and $a_{z_t}(f) < \frac{(N-1)x}{N}$. Hence,

$$(l_t + r_t - 1)^{-\frac{3}{2}}(x - r_t - l_{t+1})^{-\frac{1}{2}} \geq \frac{N^2 x^{-2}}{\sqrt{3^3(N-1)}} \geq \lambda N^2 x^{-2}.$$

The inequality $\frac{N^2x^{-2}}{\sqrt{3^3(N-1)}} \geq \lambda N^2 x^{-2}$ uses $\lambda = \frac{1}{100}$ and $3^3(N-1) = 6885 < 100^2$. We also have $a_{z_0}(f)^{-\frac{3}{2}} a_{z_c}(f)^{-\frac{3}{2}} a_{y_c}(f)^{-\frac{3}{2}} \geq x^{-\frac{9}{2}}$. □

7.7 Covering quivers and dimension vectors

Definition 7.41 (Coverings, embeddings, and deck transformations)**.** *Let Q and $\widehat{Q}$ be quivers.*

1. *An embedding $\iota = (\iota_0, \iota_1) : Q \to \widehat{Q}$ consists of injective maps*

$$\iota_0 : \mathrm{Vert}(Q) \hookrightarrow \mathrm{Vert}(\widehat{Q}), \qquad \iota_1 : \mathrm{Edge}(Q) \hookrightarrow \mathrm{Edge}(\widehat{Q})$$

such that, for every $\alpha \in \mathrm{Edge}(Q)$,

$$s_{\widehat{Q}}(\iota_1(\alpha)) = \iota_0(s_Q(\alpha)), \qquad t_{\widehat{Q}}(\iota_1(\alpha)) = \iota_0(t_Q(\alpha)).$$

2. *A quiver homomorphism $p = (p_0, p_1) : \widehat{Q} \to Q$ is a covering if p_0 is surjective and, for every $v \in \mathrm{Vert}(\widehat{Q})$, the restrictions*

$$p_1 : s_{\widehat{Q}}^{-1}(\{v\}) \longrightarrow s_Q^{-1}(\{p_0(v)\}),$$
$$p_1 : t_{\widehat{Q}}^{-1}(\{v\}) \longrightarrow t_Q^{-1}(\{p_0(v)\})$$

are bijective. We call p a universal covering if $\mathrm{UnderG}(\widehat{Q})$ is a tree.

3. *A deck transformation of p is a quiver isomorphism $\delta : \widehat{Q} \to \widehat{Q}$ such that*

$$p_0 \circ \delta_0 = p_0, \qquad p_1 \circ \delta_1 = p_1.$$

We denote the group of these transformations under composition by $\mathrm{Deck}(p)$.

Definition 7.42 (Arrow traversals and lifts)**.** *Let Q be a quiver and $n \in \mathbb{Z}_{\geq 0}$. A traversal of length n is a tuple $\gamma = (v_0, \alpha_1, \varepsilon_1, v_1, \ldots, \alpha_n, \varepsilon_n, v_n)$ such that $v_i \in \mathrm{Vert}(Q)$ for $i \in [0, n]_{\mathbb{Z}}$ and*

$$\alpha_i \in \mathrm{Edge}(Q), \qquad \varepsilon_i \in \{1, -1\}, \qquad (s_Q(\alpha_i), t_Q(\alpha_i)) = \begin{cases} (v_{i-1}, v_i) & \varepsilon_i = 1, \\ (v_i, v_{i-1}) & \varepsilon_i = -1 \end{cases} \quad (i \in [n]_{\mathbb{Z}}).$$

For a covering $p : \widehat{Q} \to Q$, a lift of γ is a traversal $\widehat{\gamma} = (\widehat{v}_0, \widehat{\alpha}_1, \varepsilon_1, \widehat{v}_1, \ldots, \widehat{\alpha}_n, \varepsilon_n, \widehat{v}_n)$ satisfying $p_0(\widehat{v}_i) = v_i$ for $i \in [0, n]_{\mathbb{Z}}$ and $p_1(\widehat{\alpha}_i) = \alpha_i$ for $i \in [n]_{\mathbb{Z}}$.

Lemma 7.43 (Unique traversal lifting)**.** *Let $p : \widehat{Q} \to Q$ be a covering. Let $n \in \mathbb{Z}_{\geq 0}$ and let $\gamma = (v_0, \alpha_1, \varepsilon_1, v_1, \ldots, \alpha_n, \varepsilon_n, v_n)$ be a traversal in Q. Let $\widehat{v}_0 \in \mathrm{Vert}(\widehat{Q})$ satisfy $p_0(\widehat{v}_0) = v_0$. Then, there is exactly one traversal $\widehat{\gamma} = (\widehat{v}_0, \widehat{\alpha}_1, \varepsilon_1, \widehat{v}_1, \ldots, \widehat{\alpha}_n, \varepsilon_n, \widehat{v}_n)$ in $\widehat{Q}$ satisfying*

$$p_0(\widehat{v}_i) = v_i \quad (i \in [0, n]_{\mathbb{Z}}), \qquad p_1(\widehat{\alpha}_i) = \alpha_i \quad (i \in [n]_{\mathbb{Z}}).$$

Proof. Let us construct the lift one arrow at a time. For a traversal of length zero, the lift consists of the specified vertex $\widehat{v}_0$. For $i \geq 1$, suppose that the lifted traversal through $\widehat{v}_{i-1}$ has been determined and $p_0(\widehat{v}_{i-1}) = v_{i-1}$.

First, suppose $\varepsilon_i = 1$. The covering gives a bijection

$$p_1 : s_{\widehat{Q}}^{-1}(\{\widehat{v}_{i-1}\}) \longrightarrow s_Q^{-1}(\{v_{i-1}\}).$$

Let $\widehat{\alpha}_i$ be the unique preimage of α_i and set

$$\widehat{v}_i = t_{\widehat{Q}}(\widehat{\alpha}_i).$$

Then,

$$p_0(\widehat{v}_i) = t_Q(p_1(\widehat{\alpha}_i)) = t_Q(\alpha_i) = v_i.$$

Second, suppose $\varepsilon_i = -1$. Use the bijection

$$p_1 : t_{\widehat{Q}}^{-1}(\{\widehat{v}_{i-1}\}) \longrightarrow t_Q^{-1}(\{v_{i-1}\}).$$

Let $\widehat{\alpha}_i$ be the unique preimage of α_i and set

$$\widehat{v}_i = s_{\widehat{Q}}(\widehat{\alpha}_i).$$

Then,

$$p_0(\widehat{v}_i) = s_Q(p_1(\widehat{\alpha}_i)) = s_Q(\alpha_i) = v_i.$$

Let us prove uniqueness of the whole lift. At each step, the appropriate covering bijection determines $\widehat{\alpha}_i$, and its other endpoint determines $\widehat{v}_i$. Induction on i proves existence and uniqueness through the last vertex. □

Definition 7.44 (Lift of a traversal)**.** *Let $p : \widehat{Q} \to Q$ be a covering and let γ be a traversal in Q beginning at v_0. For $\widehat{v}_0 \in p_0^{-1}(\{v_0\})$, let*

$$\mathrm{Lift}_p(\gamma; \widehat{v}_0)$$

denote the unique lift of γ beginning at $\widehat{v}_0$ from Lemma 7.43.

Definition 7.45 (Supported dimension vectors)**.** *Let Q be a quiver.*

1. *A dimension vector is a family $d = (d_v)_{v \in \mathrm{Vert}(Q)}$ of nonnegative integers. We define*

$$\mathrm{VecSupp}(d) = \{v \in \mathrm{Vert}(Q) : d_v > 0\}.$$

We call d finitely supported if $\mathrm{VecSupp}(d)$ is finite and let

$$\mathrm{DimVec}_{\mathrm{fin}}(Q) = \{d \in \mathbb{Z}_{\geq 0}^{\mathrm{Vert}(Q)} : \#\,\mathrm{VecSupp}(d) < \infty\}.$$

Its positive support quiver is $\mathrm{SubQuiv}(Q, \mathrm{VecSupp}(d))$.

2. *Let $p : \widehat{Q} \to Q$ be a covering and let $\delta \in \mathrm{Deck}(p)$. For $d \in \mathrm{DimVec}_{\mathrm{fin}}(\widehat{Q})$, define*

$$(\delta \cdot d)_v = d_{\delta_0^{-1}(v)} \qquad (v \in \mathrm{Vert}(\widehat{Q})).$$

We call d and d' deck-equivalent if $d' = \delta \cdot d$ for some $\delta \in \mathrm{Deck}(p)$. We denote the resulting orbit by $[d]_p$.

Definition 7.46 (Covering arrow coloring)**.** *Fix a bijection $c_3 : \mathrm{Edge}(K(3)) \to [0,2]_{\mathbb{Z}}$ and a universal covering $\varpi = (\varpi_0, \varpi_1) : \widetilde{K}_3 \to K(3)$ for the rest of this section.*

1. *Let $I_{K(3)}$ consist of the first vertex and $J_{K(3)}$ of the second vertex of $K(3)$. We use the bipartition*

$$I_{\widetilde{K}_3} = \varpi_0^{-1}(I_{K(3)}), \qquad J_{\widetilde{K}_3} = \varpi_0^{-1}(J_{K(3)}).$$

2. *We define the covering arrow coloring by*

$$\mathrm{CovCol}_{\varpi} : \mathrm{Edge}(\widetilde{K}_3) \longrightarrow [0,2]_{\mathbb{Z}}, \qquad \mathrm{CovCol}_{\varpi}(\alpha) = c_3(\varpi_1(\alpha)).$$

Definition 7.47 (Cover dimension classes and stability weights)**.** *Let $a \in \mathbb{Z}_{>0}$ and $b \in \mathbb{Z}_{>0}$. Let $I = I_{\widetilde{K}_3}$ and $J = J_{\widetilde{K}_3}$.*

1. *We define*

$$\mathrm{CovDim}_{\varpi}(a,b) = \left\{ d \in \mathrm{DimVec}_{\mathrm{fin}}(\widetilde{K}_3) : \sum_{i \in I} d_i = a,\ \sum_{j \in J} d_j = b \right\}.$$

2. *For $e \in \mathrm{DimVec}_{\mathrm{fin}}(\widetilde{K}_3)$, define*

$$\theta^{\varpi}_{a,b}(e) = a \sum_{j \in J} e_j - b \sum_{i \in I} e_i.$$

3. *For $k \in \mathbb{Z}_{>0}$, we define the sets of cover dimension classes and stable cover dimension classes by*

$$\mathcal{D}_{a,k} = \{[d]_{\varpi} : d \in \mathrm{CovDim}_{\varpi}(a, a+k)\},$$
$$\mathcal{D}^{\mathrm{st}}_{a,k} = \{[d]_{\varpi} \in \mathcal{D}_{a,k} : \mathcal{M}^{\mathrm{s}}_d(\widetilde{K}_3) \neq \varnothing\}.$$

The moduli spaces here use the weight $\theta^{\varpi}_{a,a+k}$.

7.8 Counting cover representations

Definition 7.48. *Let $a \in \mathbb{Z}_{>0}$ and $k \in \mathbb{Z}_{>0}$ satisfy $k \geq 2$ and $a \geq 100000k$. Let $c = k-1$ and $S = \mathrm{LSTrivCh}(c)$. Let $\mathbb{S} = (S, (I_S, J_S))$. Define*

$$L = J_S \cap \mathrm{Leaf}(S), \qquad I = \{i \in J_S : \deg_S(i) = 2\}.$$

We call a tuple

$$M = (S, f, \tau, (B_i)_{i \in L \sqcup I})$$

marked chain data if the following conditions hold.

(a) Let $f \in \mathcal{F}_{a,k}$ and let $a_i = a_i(f)$ for $i \in J_S$.

(b) For each $t \in [c]_{\mathbb{Z}}$, the map $\tau_t \colon [3]_{\mathbb{Z}} \longrightarrow [0,2]_{\mathbb{Z}}$ is a bijection. The indices 1, 2, and 3 denote $\mathrm{LeftEdge}_c(t)$, $\mathrm{MidEdge}_c(t)$, and $\mathrm{RightEdge}_c(t)$, respectively.

(c) For each $i \in L \sqcup I$, let $(\mathbb{T}_i, \kappa_i)$ be a colored source-trivalent tree. Let $\mathbb{T}_i = (T_i, (I_{T_i}, J_{T_i}))$ for $i \in L \sqcup I$. Require the following conditions.

(i) For $i \in L$, let $\mathbf{s}_i = (s_{i,1})$ satisfy

$$s_{i,1} \in \mathrm{ColVacSlot}(Q(\mathbb{T}_i, \kappa_i)).$$

For $i \in I$, let $\mathbf{s}_i = (s_{i,1}, s_{i,2})$ satisfy

$$s_{i,j} \in \mathrm{ColVacSlot}(Q(\mathbb{T}_i, \kappa_i)) \quad (j \in [2]_{\mathbb{Z}}), \qquad s_{i,1} \neq s_{i,2}.$$

(ii) For $i \in L \sqcup I$, let

$$B_i = \mathrm{MarkedCIsom}_{\mathbb{C}}(Q(\mathbb{T}_i, \kappa_i), \mathrm{ZS}(Q(\mathbb{T}_i, \kappa_i)); \mathbf{s}_i)$$

and require

$$\mathrm{TotDim}_{\mathbb{C},(I_{Q(\mathbb{T}_i,\kappa_i)},J_{Q(\mathbb{T}_i,\kappa_i)})}(\mathrm{ZS}(Q(\mathbb{T}_i, \kappa_i))) = (a_i, a_i + 1).$$

(iii) Define the colored-vacant slots p_e for $e \in \mathrm{Edge}(S)$ by

$$p_e = s_{i,1} \quad (i \in L,\ e \in \mathrm{IncEdge}_S(i)),$$

and, for $t \in [c-1]_{\mathbb{Z}}$, let

$$p_{\mathrm{RightEdge}_c(t)} = s_{z_t,1}, \qquad p_{\mathrm{LeftEdge}_c(t+1)} = s_{z_t,2}.$$

(iv) For $e \in \mathrm{Edge}(S)$, let $p_e = (v_e, d_e)$. For every $t \in [c]_{\mathbb{Z}}$, require

$$d_{\mathrm{LeftEdge}_c(t)} = \tau_t(1), \qquad d_{\mathrm{MidEdge}_c(t)} = \tau_t(2), \qquad d_{\mathrm{RightEdge}_c(t)} = \tau_t(3).$$

1. *Let $\mathcal{H}_{a,k}$ be the set of tuples M satisfying assumptions (a)–(c). For a family $\tau = (\tau_t)_{t \in [c]_{\mathbb{Z}}}$ of bijections $\tau_t : [3]_{\mathbb{Z}} \to [0,2]_{\mathbb{Z}}$, define*

$$I_{=}(\tau) = \{z_t : t \in [c-1]_{\mathbb{Z}},\ \tau_t(3) = \tau_{t+1}(1)\}, \qquad I_{\neq}(\tau) = I \setminus I_{=}(\tau).$$

2. *For each $M \in \mathcal{H}_{a,k}$, fix a family*

$$\rho_M = ((\mathbb{T}_i, \kappa_i; \mathbf{s}_i))_{i \in J_S}$$

satisfying assumption (c) and defining the classes B_i in M. Let $\rho = (\rho_M)_{M\in\mathcal{H}_{a,k}}$. For $M \in \mathcal{H}_{a,k}$, let $\mathbf{T}^\rho_M = ((\mathbb{T}_i, \kappa_i))_{i\in J_S}$, $\mathbf{p}^\rho_M = (p_e)_{e\in\mathrm{Edge}(S)}$, and $\mathbf{1} = (1_{\mathbb{C}})_{e\in\mathrm{Edge}(S)}$. Define

$$R^\rho_M = \mathrm{Glue}_{\mathbb{C}}(\mathbb{S}; \mathbf{T}^\rho_M, \mathbf{p}^\rho_M, \mathbf{1}), \qquad \widetilde{Q}^\rho_M = \mathrm{SuppColQuiv}(R^\rho_M), \qquad d^\rho_M(v) = \dim_{\mathbb{C}}(R^\rho_M)_v.$$

Use the tagged vertices (i, v) for the copy of $Q(\mathbb{T}_i, \kappa_i)$ and distinct new vertices x_t for $t \in [c]_{\mathbb{Z}}$. Define

$$\begin{aligned} I^\rho_M &= \mathrm{Vert}(\widetilde{Q}^\rho_M) \cap \left(\{x_t : t \in [c]_{\mathbb{Z}}\} \sqcup \bigsqcup_{i\in J_S} \{(i, v) : v \in I_{Q(\mathbb{T}_i,\kappa_i)}\} \right), \\ J^\rho_M &= \mathrm{Vert}(\widetilde{Q}^\rho_M) \cap \bigsqcup_{i\in J_S} \{(i, v) : v \in J_{Q(\mathbb{T}_i,\kappa_i)}\}, \\ \widetilde{\mathbb{Q}}^\rho_M &= (\widetilde{Q}^\rho_M, (I^\rho_M, J^\rho_M)). \end{aligned}$$

Let $\iota^\rho_i : Q(\mathbb{T}_i, \kappa_i) \hookrightarrow \widetilde{\mathbb{Q}}^\rho_M$ denote the inclusion of the copy indexed by i.

3. *An isomorphism between colored bipartite quivers $\widetilde{\mathbb{G}} = (G, (I_G, J_G))$ and $\widetilde{\mathbb{G}}' = (G', (I_{G'}, J_{G'}))$ equipped with positive vertex dimensions d_G and $d_{G'}$ is a color-preserving quiver isomorphism $\phi = (\phi_0, \phi_1)$ that preserves the bipartitions and satisfies*

$$d_{G'}(\phi_0(v)) = d_G(v) \qquad (v \in \mathrm{Vert}(G)).$$

Define

$$\mathcal{S}^\rho_{a,k} = \{[(\widetilde{\mathbb{Q}}^\rho_M, d^\rho_M)] : M \in \mathcal{H}_{a,k}\}, \qquad \pi_\rho : \mathcal{H}_{a,k} \longrightarrow \mathcal{S}^\rho_{a,k}, \quad \pi_\rho(M) = [(\widetilde{\mathbb{Q}}^\rho_M, d^\rho_M)].$$

4. *Let $X = (\widetilde{\mathbb{G}}, d_G)$ represent a class in $\mathcal{S}^\rho_{a,k}$. Let $\widetilde{\mathbb{G}} = (G, (I_G, J_G))$. Define*

$$C(X) = \{v \in I_G : d_G(v) = 1,\ \deg_G(v) = 3\}.$$

5. *Define*

$$\mathcal{K}(X) = \mathrm{CtdComp}(\mathrm{Subgraph}(\mathrm{UnderG}(G), \mathrm{Vert}(G) \setminus C(X))).$$

6. *Define the bipartite graph $\mathbb{S}_X = (S_X, (I_{S_X}, J_{S_X}))$ with vertex set $C(X) \sqcup \mathcal{K}(X)$, identifying each part with its canonical image, by*

$$I_{S_X} = C(X), \qquad J_{S_X} = \mathcal{K}(X).$$

For $v \in C(X)$ and $K \in \mathcal{K}(X)$, define its edges by

$$[\![v, K]\!] \in \mathrm{Edge}(S_X) \iff \exists u \in \mathrm{Vert}(K), \quad [\![v, u]\!] \in \mathrm{Edge}(\mathrm{UnderG}(G)).$$

Lemma 7.49 (Recovery of a colored tree and its markings)**.** *Let $(\mathbb{T}, \kappa)$ and $(\mathbb{T}', \kappa')$ be colored source-trivalent trees with $\#I_T \geq 1$ and $\#I_{T'} \geq 1$. Let $Q = Q(\mathbb{T}, \kappa)$ and $Q' = Q(\mathbb{T}', \kappa')$. Let $q = (q_0, q_1) : Q \to Q'$ be a color-preserving isomorphism preserving the bipartitions. Then, we have the following.*

1. *There is a unique color-preserving isomorphism*

$$\Psi = (\Psi_0, \Psi_1) : (\mathbb{T}, \kappa) \longrightarrow (\mathbb{T}', \kappa')$$

preserving the bipartitions and satisfying $\Psi_0(v) = q_0(v)$ for $v \in \mathrm{Vert}(Q)$.

2. *Let $\ell \in \mathbb{Z}_{\geq 0}$ and let $\mathbf{s} = ((u_j, d_j))_{j \in [\ell]_{\mathbb{Z}}}$ be an ordered marking of colored-vacant slots of Q. Let $\mathbf{s}' = ((q_0(u_j), d_j))_{j \in [\ell]_{\mathbb{Z}}}$. Then, $\mathbf{s}'$ is an ordered marking of Q' and*

$$\mathrm{MarkedCIsom}_{\mathbb{C}}(Q, \mathrm{ZS}(Q); \mathbf{s}) = \mathrm{MarkedCIsom}_{\mathbb{C}}(Q', \mathrm{ZS}(Q'); \mathbf{s}').$$

Proof. Proof of Claim 1. Let us define the images of the deleted leaves. Let

$$L = \{v \in J_T : \deg_T(v) = 1\}, \qquad L' = \{v' \in J_{T'} : \deg_{T'}(v') = 1\}.$$

Connectedness, $\#I_T \geq 1$, and $\#I_{T'} \geq 1$ imply

$$\mathrm{Vert}(T) = \mathrm{Vert}(Q) \sqcup L, \qquad \mathrm{Vert}(T') = \mathrm{Vert}(Q') \sqcup L'.$$

For $v \in L$, let $u \in I_T$ be its unique neighbor and let $d = \kappa([\![u, v]\!])$. Lemma 7.24 gives $(u, d) \in \mathrm{ColVacSlot}(Q)$. Since q preserves incidence and colors,

$$(u, d) \in \mathrm{ColVacSlot}(Q) \iff (q_0(u), d) \in \mathrm{ColVacSlot}(Q').$$

Lemma 7.24 gives a unique $v' \in L'$ satisfying

$$[\![q_0(u), v']\!] \in \mathrm{Edge}(T'), \qquad \kappa'([\![q_0(u), v']\!]) = d.$$

Define $\eta(v) = v'$. Applying the construction of $\eta : L \to L'$ to $q^{-1} : Q' \to Q$ gives $\eta' : L' \to L$ satisfying

$$\eta'(\eta(v)) = v \quad (v \in L), \qquad \eta(\eta'(v')) = v' \quad (v' \in L').$$

Define

$$\Psi_0(v) = \begin{cases} q_0(v) & v \in \mathrm{Vert}(Q), \\ \eta(v) & v \in L, \end{cases} \qquad \Psi_1([\![u, v]\!]) = [\![\Psi_0(u), \Psi_0(v)]\!].$$

Let $e = [\![u, v]\!] \in \mathrm{Edge}(T)$ with $u \in I_T$ and $v \in J_T \setminus L$. The arrow $v \to u$ belongs to Q. Since q preserves incidence and colors, we have

$$\Psi_1(e) = [\![q_0(u), q_0(v)]\!] \in \mathrm{Edge}(T'), \qquad \kappa'(\Psi_1(e)) = \kappa(e).$$

For $v \in L$, the construction of $\eta(v)$ gives $\Psi_1(e) \in \mathrm{Edge}(T')$ and $\kappa'(\Psi_1(e)) = \kappa(e)$. The maps defined from q^{-1} and η' are inverse to Ψ_0 and Ψ_1. Thus, Ψ is a color-preserving, bipartition-preserving isomorphism extending q. Any such isomorphism must send a leaf v with neighbor u and color d to the unique leaf of color d adjacent to $q_0(u)$. Such an isomorphism therefore agrees with Ψ on all vertices and edges.

Proof of Claim 2. Let us give the linear maps on the vertex spaces. The colored-vacant-slot equivalence in Claim 1 shows that $\mathbf{s}'$ is a marking of Q'. For $v \in I_Q$, put

$$A_v = \mathrm{IncEdgeQuiv}_Q(v), \qquad A'_{q_0(v)} = \mathrm{IncEdgeQuiv}_{Q'}(q_0(v)).$$

The quiver isomorphism q restricts to a bijection $q_1 : A_v \to A'_{q_0(v)}$. Define $h_u = \mathrm{id}_{\mathbb{C}}$ for $u \in J_Q$. For $v \in I_Q$ and $z \in \mathrm{ZS}(Q)_v$, define

$$(h_v(z))_\beta = z_{q_1^{-1}(\beta)} \qquad (\beta \in A'_{q_0(v)}). \tag{7.11}$$

Let us prove that h_v is a linear isomorphism of the zero-sum spaces. By (7.11) and (7.1), we have

$$\sum_{\beta \in A'_{q_0(v)}} (h_v(z))_\beta = \sum_{\alpha \in A_v} z_\alpha = 0.$$

Thus, $h_v(z) \in \mathrm{ZS}(Q')_{q_0(v)}$. The map h_v is linear. Its inverse satisfies

$$(h_v^{-1}(w))_\alpha = w_{q_1(\alpha)} \qquad (\alpha \in A_v,\ w \in \mathrm{ZS}(Q')_{q_0(v)}).$$

Let us check the arrow maps and the markings. For $\alpha : v \to u$ and $z \in \mathrm{ZS}(Q)_v$, equations (7.2) and (7.11) give

$$\mathrm{ZS}(Q')_{q_1(\alpha)}(h_v(z)) = (h_v(z))_{q_1(\alpha)} = z_\alpha = h_u(\mathrm{ZS}(Q)_\alpha(z)).$$

Hence, $(q_0, q_1, (h_v)_{v\in\mathrm{Vert}(Q)})$ is a representation isomorphism. For each $j \in [\ell]_{\mathbb{Z}}$, its image of the jth marking is $(q_0(u_j), d_j)$. Hence, $(q_0, q_1, (h_v)_{v\in\mathrm{Vert}(Q)})$ is an isomorphism of the marked representations. □

Lemma 7.50 (Recovery from a labeled support)**.** *Let $a \in \mathbb{Z}_{>0}$ and $k \in \mathbb{Z}_{>0}$ satisfy $k \geq 2$ and $a \geq 100000k$. Fix a family of representatives $\rho = (\rho_M)_{M\in\mathcal{H}_{a,k}}$. Let $c = k-1$ and $S = \mathrm{LSTrivCh}(c)$. Let $\mathbb{S} = (S, (I_S, J_S))$. For $M = (S, f, \tau, (B_i)_i) \in \mathcal{H}_{a,k}$, let $\rho_M = ((\mathbb{T}_i, \kappa_i; \mathbf{s}_i))_{i\in J_S}$. Let $\iota_i = \iota_i^\rho$ for $i \in J_S$.*

Then, we have the following.

1. *For any other family of representatives ρ', we have*

$$\mathcal{S}_{a,k}^\rho = \mathcal{S}_{a,k}^{\rho'}, \qquad \pi_\rho = \pi_{\rho'}.$$

2. *Assume the following.*

   (a) *$H \in \mathcal{S}_{a,k}^\rho$ and $X = (\widetilde{\mathbb{G}}, d_G)$ represents H. Let $\widetilde{\mathbb{G}} = (G, (I_G, J_G))$.*

   (b) *$\lambda = (\lambda_0, \lambda_1) : S \to S_X$ is an isomorphism preserving the bipartition.*

   Then, there is at most one $M \in \pi_\rho^{-1}(\{H\})$ admitting an isomorphism $\phi : (\widetilde{\mathbb{Q}}_M^\rho, d_M^\rho) \to X$ such that

$$\begin{aligned} \phi_0(x_t) &= \lambda_0(x_t) && (t \in [c]_{\mathbb{Z}}), \\ \phi_0(\iota_{i,0}(\mathrm{Vert}(Q(\mathbb{T}_i, \kappa_i)))) &= \mathrm{Vert}(\lambda_0(i)) && (i \in J_S). \end{aligned} \tag{7.12}$$

3. *Let $H \in \mathcal{S}_{a,k}^\rho$ and let $X = (\widetilde{\mathbb{G}}, d_G)$ represent H. For $M \in \pi_\rho^{-1}(\{H\})$, every isomorphism $\phi : (\widetilde{\mathbb{Q}}_M^\rho, d_M^\rho) \to X$ induces an isomorphism $\lambda : S \to S_X$ preserving the bipartition and satisfying (7.12).*

Proof. Proof of Claim 1. Let us compare the two representatives of each B_i. Fix $M = (S, f, \tau, (B_i)_i) \in \mathcal{H}_{a,k}$. The equality of the marked classes gives isomorphisms

$$\Phi_i = (\phi_{i,0}, \phi_{i,1}, (h_{i,v})_v) \quad (i \in J_S)$$

between the marked representations in ρ_M and ρ'_M. They preserve the bipartitions, the arrow colors, and the ordered marked colored-vacant slots. Define maps on the glued vertices and arrows by

$$\begin{aligned} \psi_0(x_t) &= x_t && (t \in [c]_{\mathbb{Z}}), \\ \psi_0(i, v) &= (i, \phi_{i,0}(v)) && (i \in J_S,\ v \in \mathrm{Vert}(Q(\mathbb{T}_i, \kappa_i))), \\ \psi_1(\alpha_e) &= \alpha'_e && (e \in \mathrm{Edge}(S)), \\ \psi_1(i, \alpha) &= (i, \phi_{i,1}(\alpha)) && (i \in J_S,\ \alpha \in \mathrm{Edge}(Q(\mathbb{T}_i, \kappa_i))). \end{aligned}$$

For an added arrow $\alpha_e : x_t \to (i, v_e)$, preservation of its marked colored-vacant slot gives $\alpha'_e : x_t \to (i, \phi_{i,0}(v_e))$ with the same color d_e. Thus, $\psi = (\psi_0, \psi_1)$ is a color-preserving quiver

isomorphism that preserves the bipartitions. The maps $h_{i,v}$ give $d_M^{\rho'}(\psi_0(i,v)) = d_M^{\rho}(i,v)$. At each x_t, both dimensions are one. Restricting ψ to the positive supports gives

$$(\widetilde{\mathbb{Q}}_M^{\rho}, d_M^{\rho}) \cong (\widetilde{\mathbb{Q}}_M^{\rho'}, d_M^{\rho'}).$$

Hence, $\pi_\rho(M) = \pi_{\rho'}(M)$ for every $M \in \mathcal{H}_{a,k}$. Their image sets are therefore equal.

Proof of Claim 2. Let us identify the components determined by λ. Let $M \in \pi_\rho^{-1}(\{H\})$ and let $\phi : (\widetilde{\mathbb{Q}}_M^{\rho}, d_M^{\rho}) \to X$ be an isomorphism satisfying (7.12). For $i \in J_S$, let

$$K_i = \lambda_0(i), \qquad \widetilde{Q}_i = \mathrm{SubColQuiv}(G, \mathrm{Vert}(K_i)),$$
$$I_i = I_G \cap \mathrm{Vert}(K_i), \qquad J_i = J_G \cap \mathrm{Vert}(K_i),$$
$$\widetilde{\mathbb{Q}}_i = (\widetilde{Q}_i, (I_i, J_i)).$$

Assumption (b) and (7.12) give $\widetilde{\mathbb{Q}}_i \cong Q(\mathbb{T}_i, \kappa_i)$ by a color-preserving isomorphism taking (I_i, J_i) to $(I_{Q(\mathbb{T}_i,\kappa_i)}, J_{Q(\mathbb{T}_i,\kappa_i)})$.

Let us recover the colored-vacant slots of the added arrows. For $e = [\![x_t, i]\!] \in \mathrm{Edge}(S)$, there is a unique arrow

$$\alpha_e : \lambda_0(x_t) \longrightarrow u_e, \qquad u_e \in \mathrm{Vert}(K_i).$$

Existence follows from the definition of S_X and assumption (b). If there were two such arrows, their edges together with the path inside K_i would form a cycle in $\mathrm{UnderG}(G)$. This contradicts Claim 1(a) of Lemma 7.30. Define

$$d_e = \mathrm{QCol}(G)(\alpha_e), \qquad p_e = (u_e, d_e).$$

Proper coloring of G gives $p_e \in \mathrm{ColVacSlot}(\widetilde{\mathbb{Q}}_i)$.

Let us recover the color permutations and the ordered markings. For $t \in [c]_{\mathbb{Z}}$, the color permutation is

$$(\tau_t(1), \tau_t(2), \tau_t(3)) = (d_{\mathrm{LeftEdge}_c(t)}, d_{\mathrm{MidEdge}_c(t)}, d_{\mathrm{RightEdge}_c(t)}).$$

For $i \in L$, let e_i be its unique incident edge. Its marking is (p_{e_i}). For $t \in [c-1]_{\mathbb{Z}}$, its ordered marking is

$$\mathbf{s}_{z_t} = (p_{\mathrm{RightEdge}_c(t)}, p_{\mathrm{LeftEdge}_c(t+1)}).$$

These are exactly the orders required by assumption (c) of Definition 7.48.

Let us recover the marked zero-sum representations. Suppose $M' = (S, f', \tau', (B_i')_i) \in \pi_\rho^{-1}(\{H\})$ admits an isomorphism ϕ' satisfying (7.12). Let $((\mathbb{T}_i', \kappa_i'; \mathbf{s}_i'))_i = \rho_{M'}$ and let ι_i' be its inclusion maps. For $i \in J_S$, let

$$Q_i = Q(\mathbb{T}_i, \kappa_i), \qquad Q_i' = Q(\mathbb{T}_i', \kappa_i'), \qquad q_i = (\phi' \circ \iota_i')^{-1} \circ (\phi \circ \iota_i) : Q_i \longrightarrow Q_i'.$$

Both maps in this composition are isomorphisms onto $\widetilde{\mathbb{Q}}_i$. For each edge e incident to i, let $(v_{i,e}, d_e)$ and $(v_{i,e}', d_e)$ be the colored-vacant slots assigned to e in the two representatives. Uniqueness of α_e gives

$$(q_{i,0}(v_{i,e}), d_e) = (v_{i,e}', d_e).$$

The ordered markings agree under q_i because their orders are determined by S.

Let us recover each class B_i for $i \in J_S$. Assumption (c) of Definition 7.48 gives

$$\#I_{T_i} = a_i(f) + 1 \geq 1, \qquad \#I_{T_i'} = a_i(f') + 1 \geq 1.$$

Claim 1 of Lemma 7.49 recovers an isomorphism of these colored trees. Claim 2 of Lemma 7.49 gives

$$B_i = B_i' \qquad (i \in J_S).$$

Thus, G, d_G, and λ determine every B_i.

Let us recover the flow coordinates. For each $i \in J_S$, its source dimension is

$$a_i = \sum_{v \in I_i} d_G(v).$$

For $e \in \mathrm{Edge}(S)$, put $S_e = \mathrm{SnkCtdComp}(\mathbb{S}; e)$. Claim 2 of Lemma 7.27 gives

$$f_e = \sum_{i \in \mathrm{SubSnk}_{\mathbb{S}}(S_e)} \left(\frac{a}{k} - a_i\right) - \left(\frac{a}{k} + 1\right) \# \mathrm{SubSrc}_{\mathbb{S}}(S_e).$$

In particular,

$$l_t = f_{\mathrm{LeftEdge}_c(t)}, \qquad r_t = f_{\mathrm{RightEdge}_c(t)} \qquad (t \in [c]_{\mathbb{Z}}).$$

Therefore, G, d_G, and λ determine every entry of $M = (S, f, \tau, (B_i)_{i \in L \sqcup I})$.

Proof of Claim 3. Let us identify the sources in $C(X)$. Claim 1(d) of Lemma 7.40 and assumption (c) of Definition 7.48 give $\# I_{T_i} = a_i(f) + 1 \geq 2$ for $i \in J_S$. For each $t \in [c]_{\mathbb{Z}}$, the new source satisfies

$$d^{\rho}_M(x_t) = 1, \qquad \deg_{\widetilde{\mathbb{Q}}^{\rho}_M}(x_t) = 3.$$

For $i \in J_S$ and $v \in I_{Q(\mathbb{T}_i, \kappa_i)}$, the added arrows do not meet the copied source (i, v). Thus,

$$\deg_{\widetilde{\mathbb{Q}}^{\rho}_M}(i, v) = \deg_{Q(\mathbb{T}_i, \kappa_i)}(v).$$

Claim 3 of Lemma 7.23 gives

$$d^{\rho}_M(i, v) = 1 \quad \Longrightarrow \quad \deg_{\widetilde{\mathbb{Q}}^{\rho}_M}(i, v) = 2.$$

Consequently,

$$C(X) = \{\phi_0(x_t) : t \in [c]_{\mathbb{Z}}\}.$$

Let us identify the components after deleting $C(X)$. Each $Q(\mathbb{T}_i, \kappa_i)$ is connected because it is obtained from T_i by deleting its sink leaves. By Definition 7.25, deleting the new sources x_t removes every added arrow and leaves exactly the disjoint copied quivers. Thus, for each $i \in J_S$, there is a unique $K_i \in \mathcal{K}(X)$ satisfying

$$\mathrm{Vert}(K_i) = \phi_0\big(\iota_{i,0}(\mathrm{Vert}(Q(\mathbb{T}_i, \kappa_i)))\big).$$

Define $\lambda_0(x_t) = \phi_0(x_t)$ for $t \in [c]_{\mathbb{Z}}$ and $\lambda_0(i) = K_i$ for $i \in J_S$. These maps are bijections onto $C(X)$ and $\mathcal{K}(X)$, respectively.

Let us verify incidence. The added arrows in Definition 7.25 and Definition 7.48(6) give

$$[\![x_t, i]\!] \in \mathrm{Edge}(S) \quad \Longleftrightarrow \quad [\![\phi_0(x_t), K_i]\!] \in \mathrm{Edge}(S_X).$$

Hence, λ_0 induces the required isomorphism $\lambda : S \to S_X$. □

Lemma 7.51 (Marked chains and their supports)**.** *Let $a \in \mathbb{Z}_{>0}$ and $k \in \mathbb{Z}_{>0}$ satisfy $k \geq 2$ and $a \geq 100000k$. Fix a family of representatives $\rho = (\rho_M)_{M \in \mathcal{H}_{a,k}}$. Let $c = k - 1$ and $S = \mathrm{LSTrivCh}(c)$. Consider the map $\pi_\rho \colon \mathcal{H}_{a,k} \to \mathcal{S}^{\rho}_{a,k}$. Let*

$$L = J_S \cap \mathrm{Leaf}(S), \qquad I = \{i \in J_S : \deg_S(i) = 2\}.$$

Then, we have the following.

1. *For $H \in \mathcal{S}^{\rho}_{a,k}$, we have $\#\pi_\rho^{-1}(\{H\}) \leq 8$.*
2. *If $c = 1$, then $\#\pi_\rho^{-1}(\{H\}) \leq 6$ for $H \in \mathcal{S}^{\rho}_{a,k}$.*

3. *There are 6^c families $\tau = (\tau_t)_{t\in[c]_\mathbb{Z}}$ such that each $\tau_t : [3]_\mathbb{Z} \to [0,2]_\mathbb{Z}$ is a bijection.*

4. *Let $f \in \mathcal{F}_{a,k}$. Let $\tau = (\tau_t)_{t\in[c]_\mathbb{Z}}$ be a family of bijections $\tau_t : [3]_\mathbb{Z} \to [0,2]_\mathbb{Z}$. Let $a_i = a_i(f)$ for $i \in J_S$. The number of choices of $(B_i)_{i\in L\sqcup I}$ is*

$$\prod_{i\in L} P_{a_i} \prod_{i\in I_=(\tau)} Q^=_{a_i} \prod_{i\in I_\neq(\tau)} Q^\neq_{a_i}.$$

For every $(P_j, Q^=_j, 256)$-admissible pair (A,B), this number is at least

$$\beta^{a-c} A^{c+2} B^{c-1} \prod_{i\in L} a_i^{-\frac{3}{2}} \prod_{i\in I} a_i^{-\frac{1}{2}}.$$

Proof. Proof of Claim 1. Let us bound the number of marked chains over one support. Choose a representative $X = (\widetilde{\mathbb{G}}, d_G)$ of H. Let us count the bipartition-preserving isomorphisms

$$\mathcal{I}_X = \{\lambda : S \xrightarrow{\sim} S_X : \lambda_0(I_S) = I_{S_X},\ \lambda_0(J_S) = J_{S_X}\}.$$

Let us associate an element of $\mathcal{I}_X$ to each $M \in \pi_\rho^{-1}(\{H\})$. Choose an isomorphism

$$\phi_M : (\widetilde{\mathbb{Q}}^\rho_M, d^\rho_M) \longrightarrow X.$$

Claim 3 of Lemma 7.50 gives a bipartition-preserving isomorphism $\lambda_M : S \to S_X$ satisfying

$$\begin{aligned} (\lambda_M)_0(x_t) &= (\phi_M)_0(x_t) && (t \in [c]_\mathbb{Z}),\\ \mathrm{Vert}((\lambda_M)_0(i)) &= (\phi_M)_0(\iota^\rho_{i,0}(\mathrm{Vert}(Q(\mathbb{T}_i, \kappa_i)))) && (i \in J_S). \end{aligned}$$

Thus, $\lambda_M \in \mathcal{I}_X$. By Claim 2 of Lemma 7.50, each $\lambda \in \mathcal{I}_X$ determines at most one such M. Hence,

$$\#\pi_\rho^{-1}(\{H\}) \le \#\mathcal{I}_X.$$

First, suppose $c \ge 2$. Deleting the sink leaves of S gives a path with endpoints x_1, x_c and vertex set

$$\{x_t : t \in [c]_\mathbb{Z}\} \sqcup \{z_t : t \in [c-1]_\mathbb{Z}\}.$$

An isomorphism sends x_1 to one of the two endpoints of the corresponding path in S_X. For $\lambda \in \mathcal{I}_X$, choosing $\lambda_0(x_1)$ determines $\lambda_0(x_t)$ for $t \in [c]_\mathbb{Z}$ and $\lambda_0(z_t)$ for $t \in [c-1]_\mathbb{Z}$. The two sink leaves at x_1 are z_0, y_1, and those at x_c are y_c, z_c. Each pair has 2! possible permutations. For $1 < t < c$, the only sink leaf at x_t is y_t, whose image is then determined. Thus,

$$\#\pi_\rho^{-1}(\{H\}) \le \#\mathcal{I}_X \le 2 \cdot 2! \cdot 2! = 8.$$

Second, suppose $c = 1$. The only source is x_1, and its three sink neighbors are z_0, y_1, z_1. An isomorphism is determined by a permutation of these sinks. Therefore,

$$\#\pi_\rho^{-1}(\{H\}) \le \#\mathcal{I}_X \le 3! = 6 \le 8.$$

Proof of Claim 2. The case $c = 1$ in the proof of Claim 1 gives

$$\#\pi_\rho^{-1}(\{H\}) \le 6.$$

Proof of Claim 3. Let us count the color bijections independently at each source. Each τ_t is a bijection between sets of three elements. The choices are independent. Thus, their number is $(3!)^c = 6^c$.

Proof of Claim 4. Let us count the choices of zero-sum representations. Fix f and τ. Definition 7.31(1) gives P_{a_i} choices for $i \in L$. Parts (2) and (3) of that definition give $Q^=_{a_i}$

choices for $i \in I_{=}(\tau)$ and $Q^{\neq}_{a_i}$ choices for $i \in I_{\neq}(\tau)$. The choices of zero-sum representations are independent. Thus, their counts multiply.

Let us check that the counts depend only on equality of the marked colors. For colors $d, d' \in [0,2]_{\mathbb{Z}}$, choose a permutation σ of $[0,2]_{\mathbb{Z}}$ such that $\sigma(d) = d'$. For ordered pairs (d,e) and (d',e') satisfying

$$d = e \quad \Longleftrightarrow \quad d' = e',$$

choose σ such that $\sigma(d) = d'$ and $\sigma(e) = e'$. Replacing κ by $\sigma \circ \kappa$ gives a bijection of the corresponding marked classes. Its inverse uses σ^{-1}. Thus, the counts apply to every family $\tau = (\tau_t)_{t \in [c]_{\mathbb{Z}}}$.

Let us obtain a uniform lower bound for the product of marked counts. Let (A,B) be a $(P_j, Q^{=}_j, 256)$-admissible pair. Claim 3 of Lemma 7.34 gives $Q^{\neq}_j \geq Q^{=}_j$. Claims 1(c) and 1(d) of Lemma 7.40 give $\sum_i a_i = a - c$ and $a_i > 256$. The $(P_j, Q^{=}_j, 256)$-admissibility of (A,B) gives the lower bound since $\#L = c+2$ and $\#I = c-1$. □

Proposition 7.52 (Counting stable dimensioned supports)**.** *Let $k \geq 2$ and $a > 0$ be integers. Let $c = k-1$, $x = \frac{a}{k}$, $N = 256$, $\gamma = 1 - \frac{N}{x}$, and $\lambda = \frac{1}{100}$. Let $S = \mathrm{LSTrivCh}(c)$. Assume the following.*

(a) $a \geq 100000k$.

(b) Let (A,B) be a $(P_j, Q^{=}_j, 256)$-admissible pair.

Fix a family of representatives $\rho = (\rho_M)_{M \in \mathcal{H}_{a,k}}$. Then,

$$\#\mathcal{S}^{\rho}_{a,k} \geq \beta^a x^{-\frac{5}{2}} \frac{A^2}{8B\lambda N^2} \left(\frac{6AB\lambda\gamma^2}{\beta} \right)^c. \tag{7.13}$$

Proof. Let us count the marked chains for a fixed color family. Assumption (a) gives $x \geq 100000$. Fix a family $\tau = (\tau_t)_{t \in [c]_{\mathbb{Z}}}$ of bijections $\tau_t : [3]_{\mathbb{Z}} \to [0,2]_{\mathbb{Z}}$. Claim 2 of Lemma 7.40 gives

$$\#\mathcal{F}_{a,k} \geq \left(\frac{\gamma x}{N} \right)^{2c}.$$

Claim 3 of Lemma 7.40 gives distinct labeled size lists for distinct $f \in \mathcal{F}_{a,k}$. Apply Claim 4 of Lemma 7.51 and Claim 5 of Lemma 7.40. The $c-1$ pairs in (7.10) and the leaves of sizes $x - l_1$, $x - r_c$, and $l_c + r_c - 1$ give

$$\begin{aligned}
&\#\{(S, f, \tau, (B_i)_i) \in \mathcal{H}_{a,k}\} \\
&\quad \geq \left(\frac{\gamma x}{N} \right)^{2c} \beta^{a-c} A^{c+2} B^{c-1} (\lambda N^2 x^{-2})^{c-1} x^{-\frac{9}{2}} \\
&\quad = \beta^{a-c} A^{c+2} B^{c-1} \gamma^{2c} x^{-\frac{5}{2}} N^{-2} \lambda^{c-1}.
\end{aligned}$$

For $c = 1$, the product over pairs is empty and $B^{c-1}\lambda^{c-1} = 1$.

Let us sum over color families and pass to isomorphism classes. Claim 3 of Lemma 7.51 gives 6^c choices of τ. Claim 1 of Lemma 7.51 gives

$$\begin{aligned}
\#\mathcal{S}^{\rho}_{a,k} &\geq \frac{1}{8} \#\mathcal{H}_{a,k} \\
&\geq \frac{6^c}{8} \beta^{a-c} A^{c+2} B^{c-1} \gamma^{2c} x^{-\frac{5}{2}} N^{-2} \lambda^{c-1}.
\end{aligned}$$

Substitute $\beta^{a-c} = \beta^a \beta^{-c}$ to obtain (7.13). □

Lemma 7.53 (Traversals without immediate reversal in a tree)**.** *Let G be a tree and let $n \in \mathbb{Z}_{\geq 0}$. Let $v_i \in \mathrm{Vert}(G)$ for $i \in [0,n]_{\mathbb{Z}}$ and suppose*

$$e_i = [\![v_{i-1}, v_i]\!] \in \mathrm{Edge}(G) \quad (i \in [n]_{\mathbb{Z}}), \qquad e_i \neq e_{i+1} \quad (i \in [n-1]_{\mathbb{Z}}).$$

Then, $v_p \neq v_q$ for $0 \leq p < q \leq n$.

Proof. Let us suppose $v_p = v_q$ for some $0 \leq p < q \leq n$. For $i \in [p,q]_{\mathbb{Z}}$, let h_i be the length of the unique simple path from v_p to v_i. Then,

$$h_p = h_q = 0, \qquad h_{p+1} = 1.$$

In particular, $q \geq p+2$. Choose $j \in [p+1, q-1]_{\mathbb{Z}}$ satisfying

$$h_j = \max_{i \in [p,q]_{\mathbb{Z}}} h_i.$$

The unique simple paths from v_p to adjacent vertices differ by one edge. Thus, $|h_{i+1} - h_i| = 1$ for $p \leq i < q$. Maximality of h_j gives

$$h_{j-1} = h_j - 1 = h_{j+1}.$$

Both v_{j-1} and v_{j+1} are the predecessor of v_j on the unique simple path from v_p to v_j. Consequently,

$$v_{j-1} = v_{j+1}, \qquad e_j = [\![v_{j-1}, v_j]\!] = [\![v_j, v_{j+1}]\!] = e_{j+1}.$$

This contradicts $e_j \neq e_{j+1}$. □

Lemma 7.54 (Embedding a colored tree)**.** *Consider the universal covering*

$$\varpi : \widetilde{K}_3 \longrightarrow K(3)$$

with arrow coloring $\mathrm{CovCol}_{\varpi} = c_3 \circ \varpi_1$. Assume the following.

(a) $(T, (I_T, J_T))$ is a bipartite quiver such that T is finite and $\mathrm{UnderG}(T)$ is a tree. The map $\kappa : \mathrm{Edge}(T) \to [0,2]_{\mathbb{Z}}$ is a proper coloring.

(b) $v_0 \in \mathrm{Vert}(T)$ and $\widehat{v}_0 \in \mathrm{Vert}(\widetilde{K}_3)$ satisfy

$$v_0 \in I_T \quad \Longleftrightarrow \quad \widehat{v}_0 \in I_{\widetilde{K}_3}.$$

Then, there is a unique bipartition-preserving embedding $\iota : T \to \widetilde{K}_3$ such that

$$\iota_0(v_0) = \widehat{v}_0, \qquad \mathrm{CovCol}_{\varpi}(\iota_1(\alpha)) = \kappa(\alpha) \quad (\alpha \in \mathrm{Edge}(T)).$$

Proof. Let us define the vertex and arrow maps. Define $q : T \to K(3)$ by $q_0(v) = 1$ for $v \in I_T$ and $q_0(v) = 2$ for $v \in J_T$, where 1 and 2 denote the source and sink vertices of $K(3)$, respectively. Let

$$q_1(\alpha) = c_3^{-1}(\kappa(\alpha)).$$

For $v \in \mathrm{Vert}(T)$, let γ_v be the unique simple path from v_0 to v, viewed as an arrow traversal. Its projection $q\gamma_v$ starts at $\varpi_0(\widehat{v}_0)$ by assumption (b). Define $\iota_0(v)$ to be the final vertex of $\mathrm{Lift}_{\varpi}(q\gamma_v; \widehat{v}_0)$. For $\alpha : u \to v$, define $\iota_1(\alpha)$ to be the unique lift of $q_1(\alpha)$ starting at $\iota_0(u)$. The simple paths to adjacent vertices differ by one edge. Uniqueness of their lifts gives

$$s_{\widetilde{K}_3}(\iota_1(\alpha)) = \iota_0(u), \qquad t_{\widetilde{K}_3}(\iota_1(\alpha)) = \iota_0(v).$$

The equality $\varpi_1(\iota_1(\alpha)) = q_1(\alpha)$ gives $\mathrm{CovCol}_{\varpi}(\iota_1(\alpha)) = \kappa(\alpha)$.

Let us prove injectivity. Let $u \in \mathrm{Vert}(T)$ and $v \in \mathrm{Vert}(T)$ satisfy $u \neq v$. Let

$$(z_0, \bar{\alpha}_1, z_1, \ldots, \bar{\alpha}_n, z_n), \qquad z_0 = u, \quad z_n = v, \quad n \geq 1$$

be the unique simple path from u to v. For $i \in [0, n]_{\mathbb{Z}}$, let $w_i = \iota_0(z_i)$. For $i \in [n]_{\mathbb{Z}}$, let $\widehat{\alpha}_i = \iota_1(\alpha_i)$. Assumption (a) and the color identity give

$$\mathrm{CovCol}_{\varpi}(\widehat{\alpha}_i) \neq \mathrm{CovCol}_{\varpi}(\widehat{\alpha}_{i+1}) \qquad (i \in [n-1]_{\mathbb{Z}}).$$

Let $H = \mathrm{UnderG}(\widetilde{K}_3)$ and

$$e_i = [\![w_{i-1}, w_i]\!] \in \mathrm{Edge}(H) \qquad (i \in [n]_{\mathbb{Z}}).$$

The distinct colors give $\widehat{\alpha}_i \neq \widehat{\alpha}_{i+1}$ for $i \in [n-1]_{\mathbb{Z}}$. Since H has no parallel edges, $e_i \neq e_{i+1}$ for $i \in [n-1]_{\mathbb{Z}}$. Lemma 7.53 gives

$$\iota_0(u) = w_0 \neq w_n = \iota_0(v).$$

If $\iota_1(\alpha) = \iota_1(\beta)$, their two endpoints agree by injectivity of ι_0. The tree $\mathrm{UnderG}(T)$ has no parallel edges by assumption (a). Hence, $\alpha = \beta$.

Let us prove uniqueness. For any embedding $\jmath : T \to \widetilde{K}_3$ satisfying $\jmath_0(v_0) = \widehat{v}_0$ and $\mathrm{CovCol}_{\varpi} \circ \jmath_1 = \kappa$, the traversal $\jmath(\gamma_v)$ is the lift of $q\gamma_v$ from $\widehat{v}_0$. Lemma 7.43 determines this lift and hence every vertex and arrow image. □

Lemma 7.55 (Deck transformations with a fixed vertex image)**.** *Consider the universal covering $\varpi : \widetilde{K}_3 \to K(3)$. Assume the following.*

(a) $u \in \mathrm{Vert}(\widetilde{K}_3)$ *and* $v \in \mathrm{Vert}(\widetilde{K}_3)$.

(b) $\varpi_0(u) = \varpi_0(v)$.

Then, there is a unique $\delta \in \mathrm{Deck}(\varpi)$ such that $\delta_0(u) = v$.

Proof. Let us construct δ by lifting paths. For $w \in \mathrm{Vert}(\widetilde{K}_3)$, let γ_w be the unique simple path from u to w. By assumption (b), the vertex v lies over the initial vertex of $\varpi\gamma_w$. Lemma 7.43 gives its unique lift starting at v. Define $\delta_0(w)$ to be its final vertex. For an arrow $\alpha : w \to w'$, define $\delta_1(\alpha)$ to be the unique lift of $\varpi_1(\alpha)$ starting at $\delta_0(w)$. Uniqueness in Lemma 7.43 gives its endpoint $\delta_0(w')$ and

$$\varpi_0 \circ \delta_0 = \varpi_0, \qquad \varpi_1 \circ \delta_1 = \varpi_1.$$

Interchanging u and v gives a quiver homomorphism ϵ over $K(3)$. The composites $\epsilon \circ \delta$ and $\delta \circ \epsilon$ fix u and v, respectively. Applied to the paths from these vertices, uniqueness of lifting gives

$$\epsilon \circ \delta = 1_{\widetilde{K}_3}, \qquad \delta \circ \epsilon = 1_{\widetilde{K}_3}.$$

Hence, $\delta \in \mathrm{Deck}(\varpi)$. If $\delta' \in \mathrm{Deck}(\varpi)$ satisfies $\delta'_0(u) = v$, then $\delta'\gamma_w$ and $\delta\gamma_w$ both lift $\varpi\gamma_w$ from v for every $w \in \mathrm{Vert}(\widetilde{K}_3)$. Lemma 7.43 gives $\delta'_0(w) = \delta_0(w)$ for every w and $\delta'_1(\alpha) = \delta_1(\alpha)$ for every arrow α. Thus, $\delta' = \delta$. □

Lemma 7.56 (Dimension vectors from colored trees)**.** *Consider the universal covering*

$$\varpi : \widetilde{K}_3 \longrightarrow K(3)$$

with arrow coloring $\mathrm{CovCol}_{\varpi} = c_3 \circ \varpi_1$. Assume the following.

(a) $((T, \kappa), (I_T, J_T))$ *and* $((T', \kappa'), (I_{T'}, J_{T'}))$ *are properly colored bipartite quivers such that T and T' are finite and* $\mathrm{UnderG}(T)$ *and* $\mathrm{UnderG}(T')$ *are trees.*

(b) $\iota : T \hookrightarrow \widetilde{K_3}$ *and* $\iota' : T' \hookrightarrow \widetilde{K_3}$ *preserve the bipartitions and satisfy*

$$\mathrm{CovCol}_\varpi \circ \iota_1 = \kappa, \qquad \mathrm{CovCol}_\varpi \circ \iota'_1 = \kappa'.$$

(c) $\rho : \mathrm{Vert}(T) \to \mathbb{Z}_{>0}$ *and* $\rho' : \mathrm{Vert}(T') \to \mathbb{Z}_{>0}$.

Define finitely supported dimension vectors d *and* d' *by*

$$\begin{aligned} d_{\iota_0(v)} &= \rho(v) \quad (v \in \mathrm{Vert}(T)), \quad d_w = 0 \quad (w \notin \iota_0(\mathrm{Vert}(T))), \\ d'_{\iota'_0(v')} &= \rho'(v') \quad (v' \in \mathrm{Vert}(T')), \quad d'_w = 0 \quad (w \notin \iota'_0(\mathrm{Vert}(T'))). \end{aligned}$$

Then, $[d]_\varpi = [d']_\varpi$ *if and only if there is a bipartition-preserving quiver isomorphism* $\psi : T \to T'$ *satisfying*

$$\kappa' \circ \psi_1 = \kappa, \qquad \rho' \circ \psi_0 = \rho.$$

Proof. Let us restrict a deck transformation to the positive supports. Suppose $d' = \delta \cdot d$ for $\delta \in \mathrm{Deck}(\varpi)$. Assumption (c) gives

$$\mathrm{VecSupp}(d) = \iota_0(\mathrm{Vert}(T)), \qquad \mathrm{VecSupp}(d') = \iota'_0(\mathrm{Vert}(T')).$$

The images of ι and ι' are full subquivers of $\widetilde{K_3}$: an additional edge between image vertices would form a cycle together with their path in the image tree. Thus, δ restricts to an isomorphism from the image of ι to the image of ι'. The maps ι and ι' in assumption (b) give an isomorphism $\psi : T \to T'$ such that

$$\iota' \circ \psi = \delta \circ \iota.$$

Deck transformations preserve CovCol_ϖ, $I_{\widetilde{K_3}}$, and $J_{\widetilde{K_3}}$. The equality $d' = \delta \cdot d$ gives $\rho'(\psi_0(v)) = \rho(v)$ for every v.

Let us prove the converse by extending the colored tree isomorphism to the covering. Conversely, let $\psi : T \to T'$ be a bipartition-preserving quiver isomorphism satisfying $\kappa' \circ \psi_1 = \kappa$ and $\rho' \circ \psi_0 = \rho$. By assumption (a), choose $v_0 \in \mathrm{Vert}(T)$. Bipartition preservation gives

$$\varpi_0(\iota_0(v_0)) = \varpi_0(\iota'_0(\psi_0(v_0))).$$

Lemma 7.55 gives $\delta \in \mathrm{Deck}(\varpi)$ satisfying $\delta_0(\iota_0(v_0)) = \iota'_0(\psi_0(v_0))$. Their arrow colors satisfy

$$\mathrm{CovCol}_\varpi \circ \delta_1 \circ \iota_1 = \kappa = \kappa' \circ \psi_1 = \mathrm{CovCol}_\varpi \circ \iota'_1 \circ \psi_1.$$

Uniqueness in Lemma 7.54 gives $\delta \circ \iota = \iota' \circ \psi$. The identity $\rho' \circ \psi_0 = \rho$ then gives $d' = \delta \cdot d$. □

Lemma 7.57 (Finiteness of connected support classes)**.** *Consider the universal covering* $\varpi : \widetilde{K_3} \to K(3)$. *Let* $a \in \mathbb{Z}_{>0}$ *and* $b \in \mathbb{Z}_{>0}$. *There are finitely many classes in*

$$\left\{ [d]_\varpi : d \in \mathrm{CovDim}_\varpi(a, b),\ \mathrm{SubQuiv}(\widetilde{K_3}, \mathrm{VecSupp}(d)) \text{ is connected} \right\}.$$

Proof. Let us bound each connected support after a deck transformation. Let $d \in \mathrm{CovDim}_\varpi(a, b)$ and assume its positive support $D = \mathrm{VecSupp}(d)$ is connected. Since every d_v for $v \in D$ is a positive integer,

$$\#D \leq \sum_{v \in D} d_v = a + b.$$

The inequality $a > 0$ gives a source in D. Fix a source o of $\widetilde{K_3}$. Lemma 7.55 gives a representative of $[d]_\varpi$ with $o \in D$. For a vertex v, define $\mathrm{dist}(o, v)$ as the number of edges in

the unique path from o to v. Since D is connected, its path from o to each $v \in D$ has at most $\#D - 1$ edges. Hence,

$$D \subseteq \{v : \operatorname{dist}(o, v) \le a + b - 1\}, \qquad d_v \in [a + b]_{\mathbb{Z}} \quad (v \in D).$$

Every vertex of $\widetilde{K}_3$ has degree three. Thus, $\{v \in \operatorname{Vert}(\widetilde{K}_3) : \operatorname{dist}(o, v) \le a + b - 1\}$ is finite. There are finitely many sets $D \subseteq \{v : \operatorname{dist}(o, v) \le a + b - 1\}$ and finitely many functions $D \to [a + b]_{\mathbb{Z}}$. These finite choices contain a representative of every class in the statement. □

Lemma 7.58 (Cover stability and localization)**.** *Let a and k be integers. Assume the following.*

(a) $k \ge 2$ *and* $a \ge 100000k$.

(b) $\gcd(a, k) = 1$.

Fix a family of representatives $\rho = (\rho_M)_{M \in \mathcal{H}_{a,k}}$. Let $I = I_{\widetilde{K}_3}$ and $J = J_{\widetilde{K}_3}$. Then, the following hold.

1. *Let R be a representation of dimension $\widetilde{d} \in \operatorname{CovDim}_{\varpi}(a, a + k)$. For $W \in \operatorname{SubRep}(R)$, let*
$$u = \sum_{i \in I} \dim W_i, \qquad v = \sum_{j \in J} \dim W_j.$$
Then,
$$\theta^{\varpi}_{a,a+k}(\dim R) = 0, \qquad \theta^{\varpi}_{a,a+k}(\dim W) = av - (a + k)u.$$
Thus, the stability condition (1.1) with $b = a + k$ is equivalent to $\theta^{\varpi}_{a,a+k}(\dim W) > 0$ for every proper nonzero subrepresentation W.

2. *Let R be a representation of dimension $\widetilde{d} \in \operatorname{CovDim}_{\varpi}(a, a+k)$. If R satisfies the stability condition in Claim 1, then the quiver $\operatorname{SubQuiv}(\widetilde{K}_3, \operatorname{VecSupp}(\widetilde{d}))$ is connected.*

3. *We have*
$$E_k(a) \ge \#\mathcal{S}^{\rho}_{a,k}.$$

Proof. Proof of Claim 1. Let us check the dimension-vector and stability conditions for the support. Definition 7.47(1)–(2) gives

$$\theta^{\varpi}_{a,a+k}(\dim R) = a(a + k) - (a + k)a = 0, \qquad \theta^{\varpi}_{a,a+k}(\dim W) = av - (a + k)u.$$

Substituting $b = a + k$ in (1.1) proves the equivalence.

Proof of Claim 2. Let us prove connectedness of the support by contradiction. Suppose that $\operatorname{VecSupp}(\widetilde{d})$ is disconnected. Then, $R = R_1 \oplus R_2$ for two nonzero subrepresentations. Claim 1 gives

$$0 < \theta^{\varpi}_{a,a+k}(\dim R_1) + \theta^{\varpi}_{a,a+k}(\dim R_2) = \theta^{\varpi}_{a,a+k}(\dim R) = 0,$$

a contradiction.

Proof of Claim 3. Let us obtain a sum formula for $E_k(a)$ over cover dimension vectors.

Let us transfer stability to each chosen representative. We use the stability condition in Claim 1 for $\mathcal{M}^s_{\widetilde{d}}(\widetilde{K}_3)$. On $\widetilde{K}_3$, the equivalence classes of dimension vectors in [19, Corollary 3.16] are the $\operatorname{Deck}(\varpi)$-orbits. By [19, Corollary 3.16], we have

$$E_k(a) = \chi(\mathcal{M}^s_{(a,a+k)}(K(3))) = \sum_{[\widetilde{d}]_{\varpi} \in \mathcal{D}_{a,k}} \chi(\mathcal{M}^s_{\widetilde{d}}(\widetilde{K}_3)). \tag{7.14}$$

Let us restrict the sum in (7.14) to finitely many nonempty moduli spaces. Claim 2 gives connected positive support for each nonempty summand. Lemma 7.57 with $b = a + k$ gives only finitely many such classes. For each such class $[\widetilde{d}]_{\varpi}$, let $S_{\widetilde{d}} = \operatorname{VecSupp}(\widetilde{d})$.

Let us restrict stable representations to $S_{\widetilde{d}}$. Let $\mathrm{SubQuiv}(\widetilde{K}_3, S_{\widetilde{d}})$ be the full subquiver on $S_{\widetilde{d}}$. Claims 1–2 of Lemma 7.14 give inverse correspondences

$$[R] \longmapsto [\mathrm{Res}_{\widetilde{K}_3, S_{\widetilde{d}}} R], \qquad [N] \longmapsto [\mathrm{ExtZero}_{\widetilde{K}_3, S_{\widetilde{d}}} N].$$

Claim 3 of Lemma 7.14 preserves subrepresentations and their dimensions at every vertex of $S_{\widetilde{d}}$, and hence their $\theta^{\varpi}_{a,a+k}$-weights. The restriction and zero-extension formulas in Definition 7.13 apply to families of representations and induce

$$\mathcal{M}^{s}_{\widetilde{d}}(\widetilde{K}_3) \cong \mathcal{M}^{s}_{\widetilde{d}|_{S_{\widetilde{d}}}}(\mathrm{SubQuiv}(\widetilde{K}_3, S_{\widetilde{d}})). \tag{7.15}$$

The quiver $\mathrm{SubQuiv}(\widetilde{K}_3, S_{\widetilde{d}})$ is finite and acyclic.

Let us identify the classes in $\mathcal{S}^{\rho}_{a,k}$ with distinct cover dimension classes $[d]_{\varpi}$. Let $M \in \mathcal{H}_{a,k}$. Claim 4 of Lemma 7.40 gives stability of R^{ρ}_M. Claims 1(a), 1(b), and 1(d) of Lemma 7.30 give a properly colored tree support and total dimension $(a, a+k)$. Lemma 7.54 gives a color-preserving embedding

$$\iota : \widetilde{Q}^{\rho}_M \longrightarrow \widetilde{K}_3.$$

Let us extend the embedded representation by zero. Let $U_M = \iota_0(\mathrm{Vert}(\widetilde{Q}^{\rho}_M))$. An additional arrow of $\widetilde{K}_3$ between vertices of U_M would form a cycle with the path between its endpoints in the embedded tree. Thus, the image is the full subquiver $\mathrm{SubQuiv}(\widetilde{K}_3, U_M)$. Transport R^{ρ}_M to a representation N_M of this subquiver and let

$$\widehat{R}^{\rho}_M = \mathrm{ExtZero}_{\widetilde{K}_3, U_M} N_M, \qquad \widehat{d}^{\rho}_M(v) = \dim_{\mathbb{C}}(\widehat{R}^{\rho}_M)_v \quad (v \in \mathrm{Vert}(\widetilde{K}_3)).$$

Claim 3 of Lemma 7.14 preserves subrepresentations and their dimension vectors. Hence, $\widehat{R}^{\rho}_M$ is stable and $[\widehat{d}^{\rho}_M]_{\varpi} \in \mathcal{D}^{\mathrm{st}}_{a,k}$.

Let us pass to dimension classes. Lemma 7.56 gives a well-defined injective map

$$\mathcal{S}^{\rho}_{a,k} \longrightarrow \mathcal{D}^{\mathrm{st}}_{a,k}, \qquad [(\widetilde{\mathbb{Q}}^{\rho}_M, d^{\rho}_M)] \longmapsto [\widehat{d}^{\rho}_M]_{\varpi}.$$

Let us prove that a subdimension vector of weight zero is zero or the full dimension vector. For $\widetilde{d} \in \mathrm{CovDim}_{\varpi}(a, a+k)$, let $0 \le e \le \widetilde{d}$ be a subdimension vector. Let

$$u = \sum_{i \in I} e_i, \qquad v = \sum_{j \in J} e_j, \qquad u \in [0,a]_{\mathbb{Z}}, \quad v \in [0, a+k]_{\mathbb{Z}}.$$

By assumption (b), we have $\gcd(a, a+k) = \gcd(a,k) = 1$. Definition 7.47(2) gives

$$\theta^{\varpi}_{a,a+k}(e) = 0 \implies av = (a+k)u \implies (u,v) \in \{(0,0), (a, a+k)\}.$$

If $(u,v) = (0,0)$, nonnegativity gives $e = 0$. If $(u,v) = (a, a+k)$, then

$$\sum_x (\widetilde{d}_x - e_x) = 0, \qquad \widetilde{d}_x - e_x \ge 0 \quad (x \in \mathrm{Vert}(\widetilde{K}_3)).$$

Hence, $e = \widetilde{d}$. Thus, no proper nonzero subdimension vector has weight zero.

Let us bound the Euler characteristic of each nonempty summand from below. Let $X = \mathcal{M}^{s}_{\widetilde{d}}(\widetilde{K}_3)$. Suppose that $X \neq \varnothing$. Every common divisor of the entries of $\widetilde{d}|_{S_{\widetilde{d}}}$ divides a and $a+k$. Thus, $\gcd\{\widetilde{d}_x : x \in S_{\widetilde{d}}\} = 1$. Since $\theta^{\varpi}_{a,a+k}(e) \neq 0$ for every subdimension vector e satisfying $0 \le e \le \widetilde{d}$ and $e \notin \{0, \widetilde{d}\}$, $\theta^{\varpi}_{a,a+k}$-semistability and $\theta^{\varpi}_{a,a+k}$-stability coincide for dimension vector $\widetilde{d}$. By (7.15) and [8, Propositions 4.3 and 5.3, Remark 5.4], X is a smooth projective fine moduli space for the finite acyclic support quiver.

Let us compute $\chi(X)$. King–Walter's theorem [9, Theorem 3(c)] gives

$$H^{2j+1}(X;\mathbb{Q}) = 0 \qquad (j \in \mathbb{Z}_{\geq 0}).$$

Hence,

$$\chi(X) = \sum_{j=0}^{\dim_{\mathbb{C}} X} \dim_{\mathbb{Q}} H^{2j}(X;\mathbb{Q}) \geq \dim_{\mathbb{Q}} H^0(X;\mathbb{Q}) \geq 1.$$

This proves

$$\mathcal{M}_{\widetilde{d}}^{s}(\widetilde{K}_3) \neq \varnothing \quad \Longrightarrow \quad \chi(\mathcal{M}_{\widetilde{d}}^{s}(\widetilde{K}_3)) \geq 1. \tag{7.16}$$

Therefore, equations (7.14) and (7.16) give

$$E_k(a) \geq \#\mathcal{D}_{a,k}^{\mathrm{st}},$$

and $\#\mathcal{D}_{a,k}^{\mathrm{st}} \geq \#\mathcal{S}_{a,k}^{\rho}$. Proposition 7.52 bounds $\#\mathcal{S}_{a,k}^{\rho}$ from below. □

Theorem 7.59. *We have the following.*

1. *There are constants $c_* \in (0,1]$ and $q_* \in (0,1]$ with the following property. For each $a \in \mathbb{Z}$ and $k \in \mathbb{Z}$ satisfying*

   (a) $k \geq 2$ and $a \geq 100000k$,

   (b) $\gcd(a,k) = 1$,

   we have

   $$E_k(a) \geq c_* \beta^a \left(\frac{k}{a}\right)^{\frac{5}{2}} q_*^{k-1}. \tag{7.17}$$

2. *Let $((a_n,k_n))_{n\geq 1}$ be a sequence in $\mathbb{Z}_{>0}^2$ satisfying*

   $$k_n \geq 2, \qquad \gcd(a_n,k_n) = 1 \qquad (n \geq 1),$$
   $$\lim_{n\to\infty} a_n = \infty, \qquad \lim_{n\to\infty} \frac{k_n}{a_n} = 0.$$

   Then, $E_{k_n}(a_n) > 0$ for all sufficiently large n, and

   $$\liminf_{n\to\infty} \frac{\log E_{k_n}(a_n)}{a_n} \geq \log \beta. \tag{7.18}$$

Proof. Proof of Claim 1. Let us obtain the finite lower bound with constants independent of the dimensions. For each $a \in \mathbb{Z}_{>0}$ and $k \in \mathbb{Z}_{>0}$ satisfying $k \geq 2$ and $a \geq 100000k$, fix a family of representatives $\rho = (\rho_M)_{M\in\mathcal{H}_{a,k}}$.

Let us apply the counting and localization bounds. Choose a $(P_j, Q_j^{\mp}, 256)$-admissible pair (A,B) by Lemma 7.36. The inequality $a \geq 100000k$ verifies the dimension hypothesis of Proposition 7.52. The additional equality $\gcd(a,k) = 1$ verifies the coprimality hypothesis of Claim 3 of Lemma 7.58. Set $N = 256$, $\lambda = \frac{1}{100}$, $x = \frac{a}{k}$, and $\gamma = 1 - \frac{N}{x}$. These results give

$$E_k(a) \geq \#\mathcal{S}_{a,k}^{\rho} \geq \beta^a x^{-\frac{5}{2}} \frac{A^2}{8B\lambda N^2} \left(\frac{6AB\lambda\gamma^2}{\beta}\right)^{k-1}.$$

Let

$$\gamma_0 = 1 - \frac{N}{100000}, \qquad c_* = \min\left\{1, \frac{A^2}{8B\lambda N^2}\right\}, \qquad q_* = \min\left\{1, \frac{6AB\lambda\gamma_0^2}{\beta}\right\}.$$

These constants are positive. Assumption (a) gives $x = \frac{a}{k} \geq 100000$. Hence, $\gamma \geq \gamma_0$. Substituting $x = \frac{a}{k}$ and $c = k - 1$ in (7.13) proves (7.17).

Proof of Claim 2. Let us apply the finite bound to (a_n, k_n). Since $\lim_{n\to\infty} \frac{k_n}{a_n} = 0$, choose $n_0 \in \mathbb{Z}_{>0}$ such that

$$a_n \geq 100000 k_n \qquad (n \geq n_0).$$

Together with $k_n \geq 2$ and $\gcd(a_n, k_n) = 1$, this verifies the hypotheses of Claim 1. Taking logarithms in (7.17) and dividing by a_n gives

$$\frac{\log E_{k_n}(a_n)}{a_n} \geq \log \beta + \frac{\log c_* - \frac{5}{2} \log\left(\frac{a_n}{k_n}\right) + (k_n - 1) \log q_*}{a_n} \qquad (n \geq n_0). \tag{7.19}$$

Let us prove that each error term tends to zero as $n \to \infty$. The constants c_* and q_* do not depend on n. For $n \geq n_0$, we have

$$\begin{aligned} \lim_{n\to\infty} \frac{\log c_*}{a_n} &= 0, \\ 0 \leq \frac{\log(a_n/k_n)}{a_n} &\leq \frac{\log a_n}{a_n} \longrightarrow 0, \\ \left|\frac{(k_n - 1)\log q_*}{a_n}\right| &\leq \frac{k_n}{a_n} |\log q_*| \longrightarrow 0. \end{aligned}$$

Taking the lower limit as $n \to \infty$ in (7.19) proves (7.18). □

8 A central upper estimate by deletion

We obtain an upper bound for $\frac{1}{a} \log E_r(a)$ whose error tends to zero when $r = o(a)$.

Definition 8.1. *Let W be a run. We call W binary if each of its simple runs has vertical length one or two.*

Lemma 8.2. *Let U be a nonempty binary run with ν simple runs. Let D be the number of simple runs of vertical length two. For $m = 3$, we have*

$$C_0(U) \leq \binom{\nu + D}{D} 64^D C_0(R_1^{\nu+2+8D}). \tag{8.1}$$

Proof. Let us delete one vertical edge from every simple run of vertical length two. We apply Claim 2 of Proposition 4.10 with $m = 3$, $M = 2$, $K = A = 0$, and $P = 4D$. Choose $u = v = 1$ and $W_{\mathrm{L}} = W_{\mathrm{R}} = R_1^{4D}$. Let $r_1 < \cdots < r_D$ enumerate $\{j \in [\nu]_{\mathbb{Z}} : |\operatorname{RunFactor}_{j,1}(U)|_N = 2\}$. Let $U^{(0)} = U$. For $d \in [D]_{\mathbb{Z}}$, let A_d be the r_dth simple run of $U^{(d-1)}$. Let $v_d = \operatorname{End}_{\mathrm{NE}}(A_d)$ and define

$$U^{(d)} = \operatorname{Del}(U^{(d-1)}, v_d).$$

For $d \in [D]_{\mathbb{Z}}$, we have $v_d \in N(U^{(d-1)})$. All intermediate simple runs have vertical length one or two. Every added outer simple run has vertical length one. The run $U^{(D)}$ is a translated copy of R_1^{ν}. The target is $R_1^{\nu+2+8D}$. Substituting $\operatorname{Fib}_3(\nu, D) = \binom{\nu+D}{D} 64^D$ into (4.29) gives (8.1). □

Proposition 8.3 (Near-central finite comparison)**.** *Let a and r be positive integers satisfying the following conditions.*

(a) $\gcd(a, r) = 1$.

(b) We have

$$2 \leq r, \qquad 2r \leq a.$$

Let $A = a + 8r - 6$. Then, we have the following.

1. *We have*
$$E_r(a) \le \binom{a+r}{r-1} 64^{r-1} E_1(A). \tag{8.2a}$$
2. *We have*
$$\binom{a+r}{r-1} 64^{r-1} E_1(A) \le \frac{4096}{729} \binom{a+r}{r-1} 64^{r-1} \beta^{a+8r-6}. \tag{8.2b}$$

Proof. Proof of Claim 1. Let us transform $\mathcal{U}(a, a+r)$ into a translated copy of R_1^{a+1} by vertical-edge deletions. By assumption (a), the pair $(a, a+r)$ is coprime. Equation (2.10) defines the run $\mathcal{U}(a, a+r)$. Equation (2.9) gives $\mu_1(a, a+r) \ge a+1$ and $\Gamma(a, a+r) = \frac{a+r}{a} - \frac{1}{a\,\mu_1(a,a+r)}$. By assumption (b), we have $1 < \Gamma(a, a+r) < 2$. Let $W = \mathrm{RunFactor}_{1,a}(\mathcal{U}(a, a+r))$. The run $\mathcal{U}(a, a+r) = W +\!\!+ R_1$ has $\nu = a+1$ simple runs and $a+r$ vertical edges. Exactly $D = r-1$ simple runs have vertical length two.

Let us obtain $C_0(R_1^{A+1}) = 3E_1(A)$. We apply Lemma 8.2 and (2.13). Definition 2.25 gives $\mu_1(A, A+1) = A+1$ and $\mu_2(A, A+1) = A+2$. Each of the first A simple runs of $\mathcal{U}(A, A+1)$ has vertical length one. Thus, $\mathcal{U}(A, A+1) = R_1^{A+1}$. Since $A+1 = \nu + 2 + 8D$, applying (2.13) to $(a, a+r)$ and $(A, A+1)$ in (8.1) and dividing by three gives Claim 1.

Proof of Claim 2. Let us bound the adjacent Euler characteristic exponentially. By [19, Theorem 6.6], we have
$$E_1(A) = \frac{3}{(A+1)(2A+3)} \binom{4A+6}{A}. \tag{8.3}$$
The binomial theorem evaluated at $\frac{1}{3}$ gives
$$\binom{4A+6}{A} 3^{-A} \le \sum_{j=0}^{4A+6} \binom{4A+6}{j} 3^{-j} = \left(\frac{4}{3}\right)^{4A+6}.$$
Since $\frac{3}{(A+1)(2A+3)} \le 1$, we obtain
$$E_1(A) \le 3^A \left(\frac{4}{3}\right)^{4A+6} = \frac{4096}{729} \left(\frac{256}{27}\right)^A = \frac{4096}{729} \beta^A.$$
Multiplication by $\binom{a+r}{r-1} 64^{r-1}$ gives Claim 2. □

Corollary 8.4 (Upper limit at slope one)**.** *Let $((a_n, r_n))_{n\ge1}$ be a sequence in $\mathbb{Z}^2_{>0}$ satisfying the following conditions.*

(a) $\gcd(a_n, r_n) = 1$ *for every* $n \ge 1$.

(b) $\lim_{n\to\infty} a_n = \infty$.

(c) $r_n \ge 2$ *for every* $n \ge 1$, *and* $\lim_{n\to\infty} \frac{r_n}{a_n} = 0$.

Then, $E_{r_n}(a_n) > 0$ for all sufficiently large n, and
$$\limsup_{n\to\infty} \frac{\log E_{r_n}(a_n)}{a_n} \le \log \beta.$$

Proof. Let us verify the hypotheses of Proposition 8.3. Assumption (c) gives an integer $n_0 \ge 1$ such that
$$2r_n \le a_n \qquad (n \ge n_0).$$

Together with assumption (a), this proves the hypotheses of that proposition for $(a, r) = (a_n, r_n)$ and $n \geq n_0$.

Let us estimate the binomial factor. For $n \geq 1$, put

$$\nu_n = a_n + 1, \qquad D_n = r_n - 1, \qquad x_n = \frac{D_n}{\nu_n} > 0.$$

For $t > 0$, define $h(t) = (1+t)\log(1+t) - t\log t$, and set $h(0) = 0$. The limits $\lim_{t\downarrow 0} t\log t = 0$ and $\lim_{t\downarrow 0}(1+t)\log(1+t) = 0$ give

$$\lim_{t\downarrow 0} h(t) = h(0) = 0.$$

The binomial theorem gives

$$\binom{\nu_n + D_n}{D_n} x_n^{D_n} \leq (1 + x_n)^{\nu_n + D_n}.$$

Taking logarithms and using $D_n = \nu_n x_n$, we obtain

$$0 \leq \frac{1}{a_n} \log \binom{\nu_n + D_n}{D_n} \leq \frac{\nu_n}{a_n} h(x_n).$$

Let us prove that $\frac{\nu_n}{a_n} h(x_n) \to 0$ as $n \to \infty$. Assumptions (b) and (c) give

$$\lim_{n\to\infty} x_n = 0, \qquad \lim_{n\to\infty} \frac{\nu_n}{a_n} = 1.$$

Continuity of h at 0 therefore gives

$$\lim_{n\to\infty} \frac{\nu_n}{a_n} h(x_n) = 0.$$

Let us take logarithms of the finite comparison. Equation (2.13) and Lemma 3.5 give $E_{r_n}(a_n) > 0$ for $n \geq n_0$. Claims 1 and 2 of Proposition 8.3 give

$$\begin{aligned} \frac{\log E_{r_n}(a_n)}{a_n} - \log\beta \leq{}& \frac{\log\left(\frac{4096}{729}\right)}{a_n} + \frac{\nu_n}{a_n} h(x_n) \\ &+ \frac{r_n - 1}{a_n} \log 64 + \frac{8r_n - 6}{a_n} \log\beta \qquad (n \geq n_0). \end{aligned}$$

Let us compute the upper limit as $n \to \infty$. The binomial estimate and assumptions (b) and (c) give

$$\lim_{n\to\infty} \frac{\log\left(\frac{4096}{729}\right)}{a_n} = 0, \qquad \lim_{n\to\infty} \frac{\nu_n}{a_n} h(x_n) = 0,$$
$$\lim_{n\to\infty} \frac{r_n - 1}{a_n} \log 64 = 0, \qquad \lim_{n\to\infty} \frac{8r_n - 6}{a_n} \log\beta = 0.$$

Thus, $\limsup_{n\to\infty} \dfrac{\log E_{r_n}(a_n)}{a_n} \leq \log\beta.$ □

9 The three-arrow case

In this section all colored gradings use $m = 3$.

9.1 The limit at slope one

Corollary 9.1. *Let $((a_n, b_n))_{n\geq 1}$ be a sequence in $\mathcal{A}_3$ satisfying*

$$\lim_{n\to\infty} a_n = \infty, \qquad \lim_{n\to\infty} \frac{b_n}{a_n} = 1.$$

Then,

$$\lim_{n\to\infty} \frac{\log E_3(a_n, b_n)}{a_n} = \log\left(\frac{256}{27}\right).$$

Proof. Let us reduce the dimension pair to $(d_n, d_n + k_n)$. For $n \geq 1$, define

$$d_n = \min(a_n, b_n), \qquad k_n = |b_n - a_n|.$$

Since $(a_n, b_n) \in \mathcal{A}_3$, we have $\gcd(a_n, b_n) = 1$. If $a_n = b_n$, then $a_n = b_n = 1$. The condition $a_n \to \infty$ therefore gives $n_0 \in \mathbb{Z}_{>0}$ such that $k_n \geq 1$ for $n \geq n_0$. For $n \geq n_0$, we have

$$\gcd(d_n, k_n) = \gcd(\min(a_n, b_n), |b_n - a_n|) = \gcd(a_n, b_n) = 1.$$

Claim 1 of Lemma 2.1 gives

$$E_3(a_n, b_n) = E_{k_n}(d_n).$$

Let us compute the limits of the new dimensions. The hypothesis $\frac{b_n}{a_n} \to 1$ gives

$$\frac{d_n}{a_n} = \min\left(1, \frac{b_n}{a_n}\right) \longrightarrow 1, \qquad \frac{k_n}{d_n} = \frac{\left|\frac{b_n}{a_n} - 1\right|}{\min\left(1, \frac{b_n}{a_n}\right)} \longrightarrow 0.$$

In particular, $d_n \to \infty$. Define the two index sets

$$J_1 = \{n \geq n_0 : k_n = 1\}, \qquad J_2 = \{n \geq n_0 : k_n \geq 2\}.$$

First, let us compute the limit along J_1 when J_1 is infinite. Equation (8.3) and Stirling's formula give

$$\log E_1(d_n) = d_n \log \beta - \frac{5}{2} \log d_n + O(1) \qquad (n \to \infty,\ n \in J_1).$$

Dividing by d_n gives

$$\lim_{\substack{n\to\infty \\ n\in J_1}} \frac{\log E_{k_n}(d_n)}{d_n} = \log \beta.$$

Second, let us compute the limit along J_2 when J_2 is infinite. Enumerate $J_2 = \{n_1 < n_2 < \cdots\}$. The sequence $((d_{n_j}, k_{n_j}))_{j\geq 1}$ satisfies

$$k_{n_j} \geq 2, \qquad \gcd(d_{n_j}, k_{n_j}) = 1 \qquad (j \geq 1),$$

$$\lim_{j\to\infty} d_{n_j} = \infty, \qquad \lim_{j\to\infty} \frac{k_{n_j}}{d_{n_j}} = 0.$$

Claim 2 of Theorem 7.59 and Corollary 8.4 therefore give

$$\lim_{j\to\infty} \frac{\log E_{k_{n_j}}(d_{n_j})}{d_{n_j}} = \log \beta.$$

Let us combine the two cases. We have $J_1 \sqcup J_2 = \{n \geq n_0\}$. For each infinite J_i, $i \in [2]_{\mathbb{Z}}$, the limit of $\frac{\log E_{k_n}(d_n)}{d_n}$ along $n \in J_i$ is $\log \beta$. Each finite J_i contributes only finitely many terms. Thus,

$$\frac{\log E_3(a_n, b_n)}{a_n} = \frac{d_n}{a_n} \frac{\log E_{k_n}(d_n)}{d_n} \longrightarrow 1 \cdot \log \beta = \log\left(\frac{256}{27}\right).$$

□

9.2 Adjacent exchanges

Definition 9.2 (Integer lists and balance operations)**.** *Let* $\mathrm{IntList} = \bigsqcup_{n\in\mathbb{Z}_{\geq 0}} \mathbb{Z}^n$.

1. *For* $L = (a_1, \dots, a_n) \in \mathrm{IntList}$*, let* $|L| = n$*. For* $M = (b_1, \dots, b_m) \in \mathrm{IntList}$*, define*

$$L + M = (a_1, \dots, a_n, b_1, \dots, b_m).$$

For $a \in \mathbb{Z}$ *and* $s \in \mathbb{Z}_{\geq 0}$*, define*

$$\mathrm{Repeat}(a, s) = \underbrace{(a, \dots, a)}_{s \text{ entries}}, \qquad \mathrm{Repeat}(a, 0) = ().$$

2. *We define the set of balance states by* $\mathrm{BalState} = \mathbb{Z} \times \mathrm{IntList}$.

3. *For* $s \in \mathbb{Z}_{\geq 0}$*, define*

$$\mathrm{HStep}_s : \mathrm{BalState} \longrightarrow \mathrm{BalState}, \qquad \mathrm{HStep}_s(b, L) = (b + 1, \mathrm{Repeat}(b + 1, s) + L).$$

4. *For* $h \in \mathbb{Z}_{\geq 0}$*, define*

$$\mathcal{V}_h = \{(b, (a) + L_0) \in \mathrm{BalState} : a \leq b - h\}$$

and

$$\mathrm{VStep}_h : \mathcal{V}_h \longrightarrow \mathrm{BalState}, \qquad \mathrm{VStep}_h(b, (a) + L_0) = (b - h, L_0).$$

We call HStep_s *and* VStep_h *the horizontal and vertical balance operations.*

5. *For* $\delta \in \mathbb{Z}_{\geq 0}$ *and integer lists* L *and* L'*, define*

$$L' \preceq_\delta L \iff |L'| = |L| \quad \text{and} \quad L'_i \leq L_i + \delta \quad (i \in [|L|]_{\mathbb{Z}}). \tag{9.1}$$

6. *Let* Y *be a tNE path. Let* $g \in \mathrm{CtdG}(Y)$*. Let* $r \in [0, |Y|]_{\mathbb{Z}}$*. List the horizontal edges of* $Y[0, r]$ *as* $e_1, \dots, e_h$ *in path order. For* $i \in [h]_{\mathbb{Z}}$*, let*

$$b_i = \mathrm{eBal}_{Y,g}(e_i), \qquad d_i = \mathrm{Debt}_{Y,g}(e_i, r).$$

Define the boundary balance list by

$$\mathrm{BalList}_{Y,g}(r) = \mathrm{Repeat}(b_h, d_h) + \cdots + \mathrm{Repeat}(b_1, d_1).$$

For $h = 0$*, this list is* $()$.

Lemma 9.3 (Comparison of balance operations)**.** *Let* $\nu \in \mathbb{Z}_{\geq 0}$ *and* $\delta \in \mathbb{Z}_{\geq 0}$*. For* $t \in [\nu]_{\mathbb{Z}}$*, let* $\varepsilon_t \in \{E, N\}$ *and* $q_t \in \mathbb{Z}_{\geq 0}$*. Assume the following.*

(a) $z_0 = (b_0, L_0) \in \mathrm{BalState}$ *and* $L'_0 \preceq_\delta L_0$*. Let* $z'_0 = (b_0 + \delta, L'_0)$.

(b) The states $z_t = (b_t, L_t)$ *for* $t \in [\nu]_{\mathbb{Z}}$ *satisfy*

$$z_t = \begin{cases} \mathrm{HStep}_{q_t}(z_{t-1}), & \varepsilon_t = E, \\ \mathrm{VStep}_{q_t}(z_{t-1}), & \varepsilon_t = N. \end{cases}$$

For every $t \in [\nu]_{\mathbb{Z}}$ *with* $\varepsilon_t = N$*, we have* $z_{t-1} \in \mathcal{V}_{q_t}$.

Starting from z'_0, the states $z'_t = (b'_t, L'_t)$ are well-defined by

$$z'_t = \begin{cases} \mathrm{HStep}_{q_t}(z'_{t-1}), & \varepsilon_t = E, \\ \mathrm{VStep}_{q_t}(z'_{t-1}), & \varepsilon_t = N \end{cases} \qquad (t \in [\nu]_{\mathbb{Z}}).$$

For every $t \in [0, \nu]_{\mathbb{Z}}$,

$$b'_t = b_t + \delta, \qquad L'_t \preceq_\delta L_t.$$

In particular, $z'_{t-1} \in \mathcal{V}_{q_t}$ whenever $\varepsilon_t = N$.

Proof. Let us prove the assertions by induction on t. The case $t = 0$ is assumption (a). Suppose $b'_{t-1} = b_{t-1} + \delta$ and $L'_{t-1} \preceq_\delta L_{t-1}$. If $\varepsilon_t = E$, the horizontal operation increases each balance by one and prepends q_t entries. The new entries are $b_{t-1} + 1$ and $b_{t-1} + 1 + \delta$, respectively. Thus, $b'_t = b_t + \delta$ and $L'_t \preceq_\delta L_t$. If $\varepsilon_t = N$, assumption (b) gives $L_{t-1} = (a) +\!\!+ M$ and $a \le b_{t-1} - q_t$. The equal list lengths give $L'_{t-1} = (a') +\!\!+ M'$. By the induction hypothesis,

$$a' \le a + \delta \le b_{t-1} - q_t + \delta = b'_{t-1} - q_t.$$

Hence, $z'_{t-1} \in \mathcal{V}_{q_t}$. Both vertical operations delete the first entry and subtract q_t from the balance. Thus, $b'_t = b_t + \delta$ and $L'_t \preceq_\delta L_t$. □

Definition 9.4 (Adjacent exchange in a shell)**.** *Define* AdjShellExch *to be the set of tuples*

$$\xi = (P, Q, o, u, v, A, B)$$

such that P and Q are NE paths, $o \in \mathbb{Z}^2$, $u \in [2]_{\mathbb{Z}}$, $v \in [2]_{\mathbb{Z}}$, and $(A, B) \in \{(E +\!\!+ N, N +\!\!+ E), (N +\!\!+ E, E +\!\!+ N)\}$. We call an element of this set an adjacent exchange in a shell. Let $\xi = (P, Q, o, u, v, A, B) \in$ AdjShellExch.

1. *Define*

$$X_\xi = \mathrm{Trans}(E +\!\!+ P +\!\!+ A +\!\!+ Q, o), \quad X'_\xi = \mathrm{Trans}(E +\!\!+ P +\!\!+ B +\!\!+ Q, o),$$
$$Y_\xi = \mathrm{Shell}(X_\xi; u, v, 0), \qquad Y'_\xi = \mathrm{Shell}(X'_\xi; u, v, 0).$$

2. *Let $Y = Y_\xi$ and $Y' = Y'_\xi$. Let $(e_i)_{i \in [|Y|_E]_{\mathbb{Z}}}$ and $(n_j)_{j \in [|Y|_N]_{\mathbb{Z}}}$ be the horizontal and vertical edge lists of Y, respectively, in path order. Let $(e'_i)_{i \in [|Y'|_E]_{\mathbb{Z}}}$ and $(n'_j)_{j \in [|Y'|_N]_{\mathbb{Z}}}$ be the horizontal and vertical edge lists of Y', respectively, in path order. For each $j \in [|Y'_\xi|_N]_{\mathbb{Z}}$, let $p_j \in [|Y'_\xi|]_{\mathbb{Z}}$ be the unique index satisfying $(Y'_\xi)_{p_j} = n'_j$.*

3. *Define the bijection*

$$\tau_\xi : \mathrm{NE}(Y_\xi) \longrightarrow \mathrm{NE}(Y'_\xi), \qquad \tau_\xi(e_i) = e'_i, \quad \tau_\xi(n_j) = n'_j.$$

4. *Define*

$$r_\xi = u + 2 + |P|, \qquad i_\xi = |P|_E + 3, \qquad j_\xi = u + |P|_N + 1, \qquad x_\xi = e_{i_\xi}, \quad w_\xi = n_{j_\xi}.$$

5. *Define*

$$\mathcal{G}_\xi = \{f \in \mathrm{CtdG}_3(X_\xi) : \mathrm{Surp}(X_\xi, f) \ge 0\}.$$

For $f \in \mathcal{G}_\xi$, define

$$g_{\xi,f} = \mathrm{GShell}_f(X_\xi; u, v, 0), \qquad \overline{g}_{\xi,f} = g_{\xi,f} \circ \tau_\xi^{-1}.$$

6. *Let $f \in \mathcal{G}_\xi$ and let $s = g_{\xi,f}(x_\xi)$. For $s > 0$, define integers q_i for $i \in [r_\xi + 2, |Y'_\xi|]_\mathbb{Z}$ by*

$$q_{r_\xi+2} = s - 1, \qquad q_i = \begin{cases} q_{i-1} + \overline{g}_{\xi,f}((Y'_\xi)_i), & (Y'_\xi)_i \in E(Y'_\xi), \\ \max(q_{i-1} - 1, 0), & (Y'_\xi)_i \in N(Y'_\xi) \end{cases} \quad (i > r_\xi + 2).$$

Define

$$\mathcal{J}_{\xi,f} = \{j \in [|Y'_\xi|_N]_\mathbb{Z} : p_j > r_\xi + 2,\ q_{p_j-1} = 0\}.$$

For $s = 0$, let $\mathcal{J}_{\xi,f} = \varnothing$.

7. *Define the vertical repair index by*

$$J_\xi(f) = \begin{cases} \min \mathcal{J}_{\xi,f}, & A = E +\!\!+ N,\ s > 0,\ \mathcal{J}_{\xi,f} \neq \varnothing, \\ j_\xi, & \text{otherwise.} \end{cases}$$

8. *Let $J = J_\xi(f)$ and define*

$$h = g_{\xi,f}(w_\xi), \qquad h_J = \overline{g}_{\xi,f}(n'_J),$$
$$\vartheta = \mathrm{bBal}_{Y_\xi, g_{\xi,f}}(r_\xi) + 1, \qquad b_J = \mathrm{bBal}_{Y'_\xi, \overline{g}_{\xi,f}}(p_J - 1).$$

Define the grade reduction by

$$D_\xi(f) = \begin{cases} h - \max(h-1, 0), & A = E +\!\!+ N,\ s = 0, \\ 0, & A = N +\!\!+ E,\ s = 0, \\ h, & A = N +\!\!+ E,\ s > 0, \\ h_J - \min\{h_J, \max(b_J - \vartheta, 0)\}, & A = E +\!\!+ N,\ s > 0. \end{cases}$$

Define the repaired grading $\mathrm{RepGr}_\xi(f)$ on Y'_ξ by

$$\mathrm{RepGr}_\xi(f)(e'_i) = g_{\xi,f}(e_i), \qquad \mathrm{RepGr}_\xi(f)(n'_j) = \begin{cases} g_{\xi,f}(n_j) - D_\xi(f), & j = J_\xi(f), \\ g_{\xi,f}(n_j), & j \neq J_\xi(f). \end{cases}$$

Lemma 9.5 (The vertical repair index)**.** *Let $\xi = (P, Q, o, u, v, A, B) \in \mathrm{AdjShellExch}$.*

1. *If $f \in \mathcal{G}_\xi$, $A = E +\!\!+ N$, and $g_{\xi,f}(x_\xi) > 0$, then $\mathcal{J}_{\xi,f} \neq \varnothing$. The edge $n'_{J_\xi(f)}$ lies in the copy of X'_ξ in Y'_ξ.*
2. *If $f \in \mathcal{G}_\xi$, $\widetilde{f} \in \mathcal{G}_\xi$, and*

$$g_{\xi,f}(e_i) = g_{\xi,\widetilde{f}}(e_i) \qquad (i \in [|Y_\xi|_E]_\mathbb{Z}),$$

then $J_\xi(f) = J_\xi(\widetilde{f})$.

Proof. Proof of Claim 1. Let us prove $\mathcal{J}_{\xi,f} \neq \varnothing$. Let $g = g_{\xi,f}$, $s = g(x_\xi) > 0$, $r = r_\xi$, and $T = u + 1 + |X_\xi|$. The shell grading $g = g_{\xi,f}$ belongs to $\mathrm{CtdG}_3(Y_\xi)$ by Claim 2 of Lemma 3.3. Since f is connected, the last vertical edge of X_ξ has h-precursor $(X_\xi)_1$. Claim 5 of Lemma 3.3 and Claim 2 of Lemma 2.13 give

$$\mathrm{Debt}_{Y_\xi,g}((Y_\xi)_{u+2}, r) \geq 1.$$

Since g is connected, Claim 2 of Lemma 2.13 gives $\mathrm{Debt}_{Y_\xi,g}((Y_\xi)_1, r) \geq 1$. Let

$$\mathrm{BalList}_{Y_\xi,g}(r) = (a) +\!\!+ L_0.$$

The entry contributed by $(Y_\xi)_1$ follows an entry contributed by $(Y_\xi)_{u+2}$. Hence, L_0 contains the entry contributed by $(Y_\xi)_1$, and $|L_0| \geq 1$. The edge w_ξ immediately follows x_ξ, and $g(x_\xi) = s > 0$. Definition 2.6(1) gives $P_{Y_\xi,g}(w_\xi) = x_\xi$. The node balance condition (2.3) gives

$$\mathrm{eBal}_{Y_\xi,g}(x_\xi) \leq \mathrm{eBal}_{Y_\xi,g}(w_\xi) = \mathrm{eBal}_{Y_\xi,g}(x_\xi) - g(w_\xi).$$

Hence, $g(w_\xi) = 0$. Definition 9.4(6) gives $q_t \geq 0$ for $t \in [r+2, |Y'_\xi|]_\mathbb{Z}$. By Definition 9.2(6),

$$|\,\mathrm{BalList}_{Y_\xi,g}(r+2)| = s + |L_0| = q_{r+2} + 1 + |L_0|.$$

Let us prove that $q_{p_j-1} = 0$ for some $p_j \in [r+3, T]_\mathbb{Z}$. Suppose, for a contradiction, that $q_{p_j-1} > 0$ for every vertical position $p_j \in [r+3, T]_\mathbb{Z}$. For $t \in [r+3, T]_\mathbb{Z}$, we have $\tau_\xi((Y_\xi)_t) = (Y'_\xi)_t$. At a horizontal edge, both q_t and $|\,\mathrm{BalList}_{Y_\xi,g}(t)|$ increase by $g((Y_\xi)_t)$. At a vertical edge at position $t = p_j$, the inequality $q_{t-1} > 0$ gives $q_t = q_{t-1} - 1$, and Definition 2.6(1) decreases $|\,\mathrm{BalList}_{Y_\xi,g}(t)|$ by 1. Induction on t gives

$$|\,\mathrm{BalList}_{Y_\xi,g}(t)| = q_t + 1 + |L_0| \qquad (t \in [r+2, T]_\mathbb{Z}).$$

Let us apply the shell boundary identity at T. Claim 6 of Lemma 3.3, the equality $\mathrm{eBal}_{Y_\xi,g}((Y_\xi)_1) = 1$, and Definition 9.2(6) give $\mathrm{BalList}_{Y_\xi,g}(T) = (1)$. Therefore,

$$1 = |\,\mathrm{BalList}_{Y_\xi,g}(T)| = q_T + 1 + |L_0| \geq 2,$$

a contradiction. Thus, some $j \in \mathcal{J}_{\xi,f}$ satisfies $p_j \leq T$. Definition 9.4(7) gives $r+2 < p_{J_\xi(f)} \leq T$. Hence, $n'_{J_\xi(f)}$ lies in the copy of X'_ξ in Y'_ξ.

Proof of Claim 2. Let us check which grades determine the repair index. The common horizontal grades determine s and every horizontal increment in the recursion for q_i. The vertical recursion $q_i = \max(q_{i-1} - 1, 0)$ does not use a vertical grade. Hence, $\mathcal{J}_{\xi,f} = \mathcal{J}_{\xi,\widetilde{f}}$ and $J_\xi(f) = J_\xi(\widetilde{f})$. □

Lemma 9.6 (The repaired grading)**.** *Let* $\xi \in \mathrm{AdjShellExch}$ *and* $f \in \mathcal{G}_\xi$. *Let*

$$g = g_{\xi,f}, \qquad g' = \mathrm{RepGr}_\xi(f), \qquad j = J_\xi(f), \qquad \delta = D_\xi(f).$$

Then,

1. $g' \in \mathrm{CtdG}_3(Y'_\xi)$ *and* $0 \leq \delta \leq g(n_j) \leq 3$.
2. *We have*
$$\mathrm{Surp}(Y'_\xi, g') = \mathrm{Surp}(X_\xi, f) + \delta.$$

Proof. Proof of Claim 1. Let us compare the h-precursors before and after exchanging the two edges. Let $\xi = (P, Q, o, u, v, A, B)$ and $(X, X', Y, Y') = (X_\xi, X'_\xi, Y_\xi, Y'_\xi)$. Put $c = \mathrm{Surp}(X, f)$. Claim 2 of Lemma 3.3 gives $g \in \mathrm{CtdG}_3(Y)$. The formulas for $D_\xi(f)$ give $0 \leq \delta \leq g(n_j) \leq 3$. Thus, $g' \in G_3(Y')$. For $r \in [0, |Y|]_\mathbb{Z}$, let

$$b = \mathrm{bBal}_{Y,g}(r), \qquad L = \mathrm{BalList}_{Y,g}(r).$$

For $r < |Y|$, Definition 2.6(1) and the node balance condition in Definition 2.9(1) give the operations $\mathrm{HStep}_{g(Y_{r+1})}$ when $Y_{r+1} \in E(Y)$ and $\mathrm{VStep}_{g(Y_{r+1})}$ when $Y_{r+1} \in N(Y)$. Claim 3(a) of Lemma 2.15 implies that every entry of L is at most b. Claims 6 and 7 of Lemma 3.3 and Definition 9.2(6) give

$$(\mathrm{bBal}_{Y,g}(u+1), \mathrm{BalList}_{Y,g}(u+1)) = (1, (1)),$$
$$(\mathrm{bBal}_{Y,g}(u+1+|X|), \mathrm{BalList}_{Y,g}(u+1+|X|)) = (c+2, (1)).$$

The entry 1 in $L = (1)$ is contributed by Y_1. The vertical edges of X remove every entry that X contributes.

Let us construct states $z'_t = (b'_t, L'_t)$ for $t \in [0, |Y'|]_{\mathbb{Z}}$. Let $z'_0 = (0, ())$ and use the recursion

$$z'_t = \begin{cases} \mathrm{HStep}_{g'(Y'_t)}(z'_{t-1}), & Y'_t \in E(Y'), \\ \mathrm{VStep}_{g'(Y'_t)}(z'_{t-1}), & Y'_t \in N(Y'). \end{cases}$$

Let us verify $z'_{t-1} \in \mathcal{V}_{g'(Y'_t)}$ before the exchanged pair. For $t \in [0, r_\xi]_{\mathbb{Z}}$, we have

$$z'_t = (\mathrm{bBal}_{Y,g}(t), \mathrm{BalList}_{Y,g}(t)).$$

Let us compare the balance states at boundaries r_ξ and $r_\xi + 2$. Let $\tau = \tau_\xi$, $x = x_\xi$, and $w = w_\xi$. Let $s = g(x)$, $h = g(w)$, and $w' = \tau(w)$. At boundary r_ξ of Y, let $(b, (a) +\!\!+ L_0)$ be the balance state. The list L_0 is nonempty: the first edge of Y contributes its entry 1. The first edge of the copy of X has positive debt before the exchanged pair. By the definition of $g' = \mathrm{RepGr}_\xi(f)$,

$$g'(\tau(e_i)) = g(e_i) \quad (i \in [|Y|_E]_{\mathbb{Z}}), \qquad g'(\tau(n_i)) = g(n_i) \quad (i \in [|Y|_N]_{\mathbb{Z}} \setminus \{J_\xi(f)\}).$$

At boundary $r_\xi + 2$, let

$$(b_{\mathrm{old}}, L_{\mathrm{old}}) = (\mathrm{bBal}_{Y,g}(r_\xi + 2), \mathrm{BalList}_{Y,g}(r_\xi + 2)),$$
$$(b_{\mathrm{new}}, L_{\mathrm{new}}) = z'_{r_\xi + 2}.$$

First, let us verify the balance condition when $A = E +\!\!+ N$ and $s = 0$. We have $J_\xi(f) = j_\xi$, $g'(w') = \max(h-1, 0)$, and $\delta = h - \max(h-1, 0)$. Validity of VStep_h at w gives $a \le b + 1 - h$. We also have $a \le b$. Hence,

$$a \le b - \max(h-1, 0).$$

Thus,

$$L_{\mathrm{new}} = L_{\mathrm{old}}, \qquad b_{\mathrm{new}} = b_{\mathrm{old}} + \delta, \qquad \delta \in [0, 1]_{\mathbb{Z}}.$$

Second, let us verify the balance condition when $A = N +\!\!+ E$ and $s = 0$. We have $\delta = 0$ and $g'(w') = h$. The inequality $a \le b - h$ implies $a \le b + 1 - h$. Thus, $(b_{\mathrm{new}}, L_{\mathrm{new}}) = (b_{\mathrm{old}}, L_{\mathrm{old}})$.

Third, let us verify the balance condition when $A = N +\!\!+ E$ and $s > 0$. We have $J_\xi(f) = j_\xi$, $g'(w') = 0$, and $\delta = h$. The states at boundary $r_\xi + 2$ satisfy

$$b_{\mathrm{old}} = b - h + 1, \qquad L_{\mathrm{old}} = \mathrm{Repeat}(b - h + 1, s) +\!\!+ L_0,$$

$$b_{\mathrm{new}} = b + 1, \qquad L_{\mathrm{new}} = \mathrm{Repeat}(b + 1, s - 1) +\!\!+ (a) +\!\!+ L_0.$$

Since $a \le b - h$, we have $L_{\mathrm{new}} \preceq_h L_{\mathrm{old}}$ by (9.1).

Fourth, let us verify the balance condition when $A = E +\!\!+ N$ and $s > 0$. Validity of VStep_h at w immediately after x gives $b + 1 \le b + 1 - h$. Thus, $h = 0$. Let $\vartheta = b + 1$. The states at boundary $r_\xi + 2$ satisfy

$$L_{\mathrm{old}} = \mathrm{Repeat}(\vartheta, s - 1) +\!\!+ (a) +\!\!+ L_0,$$

$$L_{\mathrm{new}} = \mathrm{Repeat}(\vartheta, s) +\!\!+ L_0, \qquad b_{\mathrm{new}} = b_{\mathrm{old}} = \vartheta.$$

Let us locate the entry whose grade requires a change in the fourth case. Mark entry s of L_{new}. Claim 1 of Lemma 9.5 gives

$$r_\xi + 2 < p_j \le u + 1 + |X|.$$

The edge $T = \tau^{-1}(n'_j)$ removes the marked entry. At boundary $t \in [r_\xi + 2, p_j - 1]_{\mathbb{Z}}$, the mark has position $q_t + 1$, where q_t is the counter in Definition 9.4(6).

Let us prove that the operations before T are valid. For $t \in [r_\xi + 2, p_j - 1]_{\mathbb{Z}}$, we prove by induction that there is a list H_t satisfying

$$\begin{aligned} \mathrm{BalList}_{Y,g}(t) &= H_t +\!\!+ (a) +\!\!+ L_0, \\ L'_t &= H_t +\!\!+ (\vartheta) +\!\!+ L_0, \\ |H_t| = q_t, \qquad (H_t)_i &\geq \vartheta \quad (i \in [q_t]_{\mathbb{Z}}), \\ b'_t &= \mathrm{bBal}_{Y,g}(t) \geq \vartheta. \end{aligned}$$

At $t = r_\xi + 2$, take $H_t = \mathrm{Repeat}(\vartheta, s-1)$.

First, let us prove the induction step when $Y'_{t+1} \in E(Y')$ and $t+1 < p_j$. Both balances increase by one. Every entry inserted by this horizontal edge equals $b'_t + 1 \geq \vartheta$. Set $H_{t+1} = \mathrm{Repeat}(b'_t + 1, g'(Y'_{t+1})) +\!\!+ H_t$. The identity $q_{t+1} = q_t + g'(Y'_{t+1})$ verifies $|H_{t+1}| = q_{t+1}$. Both fixed tails, $(a) +\!\!+ L_0$ and $(\vartheta) +\!\!+ L_0$, are unchanged.

Second, let us prove the induction step when $Y'_{t+1} \in N(Y')$ and $t+1 < p_j$. The definition of $j = J_\xi(f)$ gives $q_t > 0$. Let a_t be the common first list entry. The operation $\mathrm{VStep}_{g(Y_{t+1})}$ on Y at position $t+1$ is valid, so

$$b'_t - g'(Y'_{t+1}) = \mathrm{bBal}_{Y,g}(t) - g(Y_{t+1}) \geq a_t \geq \vartheta.$$

The inequality $a_t \leq b'_t - g'(Y'_{t+1})$ verifies the condition in Definition 9.2(4) for the operation at Y'_{t+1}. Since $q_t > 0$, deleting the first entry of H_t gives H_{t+1} of length $q_{t+1} = q_t - 1$. Both fixed tails are unchanged, and $b'_{t+1} = b'_t - g'(Y'_{t+1}) \geq \vartheta$.

Let us verify the vertical operation at $\tau(T)$. Let b_T be the boundary balance immediately before T and let $q = g(T)$. We have $b_T \geq \vartheta$. At $T' = \tau(T) = n'_{J_\xi(f)}$, the definition of $D_\xi(f)$ gives

$$g'(T') = k := \min\{q, b_T - \vartheta\}, \qquad \delta = q - k.$$

Then, $0 \leq k \leq q$ and $b_T - k \geq \vartheta$. Since $q_{p_j - 1} = 0$, the lists immediately before T and T' are $(a) +\!\!+ L_0$ and $(\vartheta) +\!\!+ L_0$, respectively. Deleting their first entries gives

$$L'_{p_j} = \mathrm{BalList}_{Y,g}(p_j) = L_0, \qquad b'_{p_j} = \mathrm{bBal}_{Y,g}(p_j) + \delta.$$

Let us verify connectedness. Lemma 9.3 gives boundary balance $c + 2 + \delta$ and a list of length one at $u + 1 + |X|$. The entry contributed by Y'_1 equals 1 and stays last in every nonempty list. The original lists are nonempty throughout the copy of X. Equality of list lengths therefore prevents this entry from being removed. Hence,

$$z'_{u+1+|X|} = (c + 2 + \delta, (1)).$$

The grades on the final R_v are $v - 1$ on its horizontal edge, zero on its first $v - 1$ vertical edges, and two on its last edge. After its horizontal edge the state is

$$(c + 3 + \delta, \mathrm{Repeat}(c + 3 + \delta, v - 1) +\!\!+ (1)).$$

The zero-grade vertical operations are valid. The last vertical operation is valid because $1 \leq c + 1 + \delta$. Therefore,

$$z'_{|Y'|} = (c + 1 + \delta, ()).$$

Let us define the h-precursor from the lists L'_t. Assign each list entry to the horizontal edge that inserted it. For $t \in [|Y'|]_{\mathbb{Z}}$ with $Y'_t \in N(Y')$, let Y'_{r_t} be the horizontal edge assigned to the first entry of L'_{t-1}. The list is nonempty because $z'_{t-1} \in \mathcal{V}_{g'(Y'_t)}$. The order of the list gives

$$r_t = \max\left\{ i \in [t-1]_{\mathbb{Z}} : \begin{array}{l} Y'_i \in E(Y'), \\ g'(Y'_i) > \#\{j \in [t-1]_{\mathbb{Z}} : Y'_j \in N(Y'),\ r_j = i\} \end{array} \right\}.$$

Thus, $P_{Y',g'}(Y'_t) = Y'_{r_t}$ satisfies Definition 2.6(1).

Let us verify the node balance and the node at Y'_1. The first entry of L'_{t-1} is $\mathrm{eBal}_{Y',g'}(Y'_{r_t})$. Membership in $\mathcal{V}_{g'(Y'_t)}$ gives

$$\mathrm{eBal}_{Y',g'}(Y'_{r_t}) \le b'_{t-1} - g'(Y'_t) = \mathrm{eBal}_{Y',g'}(Y'_t) \qquad (Y'_t \in N(Y')).$$

Every list entry is removed because $L'_{|Y'|} = ()$. The entry from Y'_1 is removed at $Y'_{|Y'|}$. Thus,

$$\mathrm{Node}_{Y',g'}(Y'_1) = Y', \qquad g' \in \mathrm{CtdG}_3(Y').$$

Proof of Claim 2. Let us compute the surplus change. Only the vertical grade at n'_j changes by $-\delta$. Hence,

$$\mathrm{Surp}(Y', g') = c + \delta \in [c, c+3]_{\mathbb{Z}}.$$

□

Corollary 9.7 (The adjacent-exchange repair map)**.** *Let $\xi \in \mathrm{AdjShellExch}$. Then, the formula*

$$\Theta_\xi(f) = (\mathrm{RepGr}_\xi(f), J_\xi(f), D_\xi(f))$$

defines a map

$$\Theta_\xi : \mathcal{G}_\xi \longrightarrow \mathrm{CtdG}_3(Y'_\xi) \times [|Y_\xi|_N]_{\mathbb{Z}} \times [0,3]_{\mathbb{Z}}.$$

Proof. Let us verify the target of Θ_ξ. Fix $f \in \mathcal{G}_\xi$. Claim 1 of Lemma 9.6 gives $\mathrm{RepGr}_\xi(f) \in \mathrm{CtdG}_3(Y'_\xi)$ and $D_\xi(f) \in [0,3]_{\mathbb{Z}}$. The definition of $J_\xi(f)$ gives $J_\xi(f) \in [|Y_\xi|_N]_{\mathbb{Z}}$. □

Lemma 9.8 (An adjacent edge exchange inside a shell)**.** *Let $\xi = (P, Q, o, u, v, A, B) \in \mathrm{AdjShellExch}$. Let*

$$(X, X', Y, Y') = (X_\xi, X'_\xi, Y_\xi, Y'_\xi).$$

Let $e_1, \dots, e_{|Y|_E}$ and $n_1, \dots, n_{|Y|_N}$ be the horizontal and vertical edges of Y in path order. Let $e'_1, \dots, e'_{|Y'|_E}$ and $n'_1, \dots, n'_{|Y'|_N}$ be the horizontal and vertical edges of Y' in path order. There are maps

$$\Psi : \bigcup_{c\ge 0} \mathcal{C}_c(X) \longrightarrow \bigcup_{c\ge 0} \mathcal{C}_c(Y'),$$
$$\kappa : \bigcup_{c\ge 0} \mathcal{C}_c(X) \longrightarrow \{(h,d) : h \in [0,3]_{\mathbb{Z}},\ d \in [\tbinom{3}{h}]_{\mathbb{Z}}\}$$

with the following properties. For each $(f,\gamma) \in \bigcup_{c\ge 0} \mathcal{C}_c(X)$, let $g = \mathrm{GShell}_f(X; u, v, 0)$. Let η be the shell coloring in (3.3) and let $(g', \gamma') = \Psi(f, \gamma)$. Then, we have the following.

1. *We have*
$$\mathrm{Surp}(X, f) \le \mathrm{Surp}(Y', g') \le \mathrm{Surp}(X, f) + 3.$$

2. *For $i \in [|Y|_E]_{\mathbb{Z}}$, we have*
$$(g'(e'_i), \gamma'(e'_i)) = (g(e_i), \eta(e_i)).$$

3. *We have*
$$\#\{j \in [|Y|_N]_{\mathbb{Z}} : (g'(n'_j), \gamma'(n'_j)) \ne (g(n_j), \eta(n_j))\} \le 1.$$

4. *The map*
$$(f,\gamma) \longmapsto (\Psi(f,\gamma), \kappa(f,\gamma))$$
is injective.

Proof. Proof of Claim 1. Let us define Ψ and κ. Fix $(f,\gamma) \in \mathcal{C}_c(X)$. For every $e \in \mathrm{NE}(X)$, the condition $\gamma(e) \in [\binom{3}{f(e)}]_{\mathbb{Z}}$ gives $f(e) \le 3$. Thus, $f \in \mathcal{G}_\xi$. Use the map Θ_ξ from Corollary 9.7 and let $(g'_f, j_f, \delta_f) = \Theta_\xi(f)$. Let $\tau = \tau_\xi$. Define

$$\gamma' = \mathrm{Recol}(Y, Y', \tau, g'_f; g, \eta), \qquad \Psi(f,\gamma) = (g'_f, \gamma'), \qquad \kappa(f,\gamma) = (g(n_{j_f}), \eta(n_{j_f})).$$

Claim 2 of Lemma 9.6 gives the surplus bounds.

Proof of Claim 2. Let us check the horizontal grade and color pairs. Definition 9.4(8) keeps every horizontal grade. Equation (3.2) therefore keeps every horizontal color.

Proof of Claim 3. Let us check how many vertical grade and color pairs change. The repair in Definition 9.4(8) changes at most the vertical grade at n'_{j_f}. Equation (3.2) keeps the colors at all other vertical edges.

Proof of Claim 4. Let us recover the input from the output and the one changed pair. The pair (g', γ') determines $g(e_i) = g'(e'_i)$ for every $i \in [|Y|_E]_{\mathbb{Z}}$. Claim 2 of Lemma 9.5 therefore determines j_f from those grades. For $j \in [|Y|_N]_{\mathbb{Z}} \setminus \{j_f\}$, Claim 3 and the definition of γ' give $(g'(n'_j), \gamma'(n'_j)) = (g(n_j), \eta(n_j))$. Recover $(g(n_{j_f}), \eta(n_{j_f})) = \kappa(f,\gamma)$. Reverse the exchange and remove the two outer simple runs. The result is (f,γ). Thus, $(\Psi(f,\gamma), \kappa(f,\gamma))$ determines (f,γ). □

Definition 9.9 (Exchange decompositions)**.** *Let X be a tNE path. If $X = \emptyset$, let* $\mathrm{ExcDecomp}(X) = \emptyset$. *Suppose $X \ne \emptyset$. An exchange decomposition of X is a tuple $(B^{\mathrm{L}}, S_1, S_2, B^{\mathrm{R}})$ of NE paths such that S_1 and S_2 are simple runs and*

$$X = \mathrm{Trans}(B^{\mathrm{L}} +\!\!+ S_1 +\!\!+ S_2 +\!\!+ B^{\mathrm{R}}, \mathrm{Start}_{\mathrm{pt}}(X)), \qquad |S_1|_N \ne |S_2|_N.$$

We denote the set of these tuples by $\mathrm{ExcDecomp}(X)$.

Lemma 9.10. *Let X be a binary run. Let $u_0 \in [2]_{\mathbb{Z}}$ and $v_0 \in [2]_{\mathbb{Z}}$. Let $(B^{\mathrm{L}}, S_1, S_2, B^{\mathrm{R}}) \in \mathrm{ExcDecomp}(X)$. Let $X' = \mathrm{Exc}(X; |B^{\mathrm{L}}|, S_1, S_2)$. Let $Y = R_{u_0} +\!\!+ X +\!\!+ R_{v_0}$ and $Y' = R_{u_0} +\!\!+ X' +\!\!+ R_{v_0}$. There is a map*

$$\Phi : \bigcup_{c \ge 0} \mathcal{C}_c(X) \longrightarrow \bigcup_{c \ge 0} \mathcal{C}_c(Y')$$

with the following properties. For each $(f,\gamma) \in \bigcup_{c\ge0} \mathcal{C}_c(X)$, let $g = \mathrm{GShell}_f(X; u_0, v_0, 0)$ and define η by (3.3). Let $(g', \gamma') = \Phi(f,\gamma)$. Label the horizontal and vertical edges of Y by e_i and n_j, respectively. Label the horizontal and vertical edges of Y' by e'_i and n'_j, respectively, in path order. Then, we have the following.

1. *The grading g' satisfies the following.*
   (a) $g' \in \mathrm{CtdG}(Y')$.
   (b) *We have*
   $$\mathrm{Surp}(X,f) \le \mathrm{Surp}(Y', g') \le \mathrm{Surp}(X,f) + 3.$$
2. *We have*
   $$(g'(e'_i), \gamma'(e'_i)) = (g(e_i), \eta(e_i)) \quad (i \in [|Y|_E]_{\mathbb{Z}}).$$
3. *We have*
   $$\#\{j \in [|Y|_N]_{\mathbb{Z}} : (g'(n'_j), \gamma'(n'_j)) \ne (g(n_j), \eta(n_j))\} \le 1.$$
4. *For every $(h,\delta) \in \bigcup_{c\ge0} \mathcal{C}_c(Y')$, we have*
   $$\#\Phi^{-1}(\{(h,\delta)\}) \le 8.$$

Proof. Proof of Claim 1(a). Let us specify the tuple in Definition 9.4. Define

$$o = \mathrm{Start}_{\mathrm{pt}}(X), \qquad H = B^{\mathrm{L}} + E + N,$$
$$P = \mathrm{Trans}(H[1,|H|],(-1,0)), \qquad Q = N + B^{\mathrm{R}},$$
$$(A,B) = \begin{cases} (E+N, N+E), & (S_1,S_2) = (R_1,R_2), \\ (N+E, E+N), & (S_1,S_2) = (R_2,R_1). \end{cases}$$

Since X is binary, H begins with E. Thus, P and Q are NE paths and $E + P = H$. It follows that

$$\xi = (P,Q,o,u_0,v_0,A,B) \in \mathrm{AdjShellExch}, \qquad (X_\xi, X'_\xi, Y_\xi, Y'_\xi) = (X,X',Y,Y').$$

Apply Lemma 9.8 to ξ and let $\Phi = \Psi$. The inclusion $\Psi(f,\gamma) \in \bigcup_{c\geq 0} \mathcal{C}_c(Y')$ gives $g' \in \mathrm{CtdG}(Y')$.

Proof of Claim 1(b). Claim 1 of Lemma 9.8 gives the surplus bounds.

Proof of Claim 2. Claim 2 of Lemma 9.8 preserves every horizontal grade–color pair.

Proof of Claim 3. Claim 3 of Lemma 9.8 bounds the number of changed vertical grade–color pairs by one.

Proof of Claim 4. Let us bound each fiber by the number of recovery choices. By Claim 4 of Lemma 9.8, the map κ is injective on each fiber of Φ. The set of possible values of κ has cardinality

$$\sum_{h=0}^{3} \binom{3}{h} = 8.$$

Hence, every fiber of Φ has cardinality at most eight. □

Definition 9.11 (Wrapped exchange)**.** *Let X be a binary run. Let $u_0 \in [2]_{\mathbb{Z}}$ and $v_0 \in [2]_{\mathbb{Z}}$. Let $(B^{\mathrm{L}}, S_1, S_2, B^{\mathrm{R}}) \in \mathrm{ExcDecomp}(X)$. Let $X' = \mathrm{Exc}(X; |B^{\mathrm{L}}|, S_1, S_2)$. Let $\mathrm{WrapExch}(X, X'; u_0, v_0)$ be the set of wrapped exchanges*

$$\Phi\colon \bigcup_{c\in\mathbb{Z}_{\geq 0}} \mathcal{C}_c(X) \longrightarrow \bigcup_{c\in\mathbb{Z}_{\geq 0}} \mathcal{C}_c(R_{u_0} + X' + R_{v_0})$$

satisfying Claims 1–4 of Lemma 9.10.

9.3 Mechanical comparison by exchanges

Definition 9.12. *Let W be a binary run with n simple runs. Let*

$$W = \mathrm{Trans}(R_{w_1} + \cdots + R_{w_n}, \mathrm{Start}_{\mathrm{pt}}(W)).$$

For $r \in [2]_{\mathbb{Z}}$, define

$$\mathrm{RunCount}_r(W) = \#\{i \in [n]_{\mathbb{Z}} : w_i = r\}.$$

Definition 9.13 (Binary prefix discrepancies)**.** *Let Z and Y be binary runs with n and m simple runs, respectively. Let*

$$Z = \mathrm{Trans}(R_{z_1} + \cdots + R_{z_n}, \mathrm{Start}_{\mathrm{pt}}(Z)), \qquad Y = \mathrm{Trans}(R_{y_1} + \cdots + R_{y_m}, \mathrm{Start}_{\mathrm{pt}}(Y)).$$

For $j \in [0, \min(n,m)]_{\mathbb{Z}}$, define

$$\mathrm{PrefDiff}_j(Z,Y) = \#\{i \in [j]_{\mathbb{Z}} : z_i = 2\} - \#\{i \in [j]_{\mathbb{Z}} : y_i = 2\}.$$

For $h \in [0, \min(n,m)]_{\mathbb{Z}}$, define $\mathrm{PrefSum}_h(Z,Y) = \sum_{j=1}^{h} |\mathrm{PrefDiff}_j(Z,Y)|$.

Lemma 9.14 (Exchanges between binary runs)**.** *Let $n \in \mathbb{Z}_{>0}$. Let $\mathbf{t} \in \mathbb{Z}^2$ and let*

$$Z = \mathrm{Trans}(R_{z_1} +\!\!+ \cdots +\!\!+ R_{z_n}, \mathbf{t}), \qquad Y = \mathrm{Trans}(R_{y_1} +\!\!+ \cdots +\!\!+ R_{y_n}, \mathbf{t})$$

be binary runs. Assume $\mathrm{RunCount}_2(Z) = \mathrm{RunCount}_2(Y)$*. For $j \in [n]_{\mathbb{Z}}$, let*

$$D_j = \mathrm{PrefDiff}_j(Z, Y).$$

Then, there are $\ell \in \mathbb{Z}_{\geq 0}$ and a run-edit list $\Omega = (\omega_s)_{s \in [\ell]_{\mathbb{Z}}}$ such that

$$\mathrm{RunEResult}(\Omega, Z) = Y.$$

Every ω_s exchanges two adjacent simple runs of distinct vertical lengths, and

$$\ell = \mathrm{PrefSum}_n(Z, Y). \tag{9.2}$$

Proof. Let us decrease $\mathrm{PrefSum}_n(Z, Y)$ by adjacent exchanges to prove (9.2). If $Z = Y$, the empty run-edit list suffices. Suppose $Z \neq Y$ and let

$$a = \min\{i \in [n]_{\mathbb{Z}} : z_i \neq y_i\}.$$

The entries before a agree. Since the total counts of each vertical length agree, there is $i > a$ such that $z_i = y_a$. Let

$$b = \min\{i \in [a+1, n]_{\mathbb{Z}} : z_i = y_a\}.$$

For $j \in [a, b-1]_{\mathbb{Z}}$, minimality of b gives $z_j = 3 - y_a$. Therefore,

$$D_j = \begin{cases} -\#\{i \in [a, j]_{\mathbb{Z}} : y_i = 2\} \leq -1, & y_a = 2, \\ \#\{i \in [a, j]_{\mathbb{Z}} : y_i = 1\} \geq 1, & y_a = 1. \end{cases}$$

Let us move $R_{z_b} = R_{y_a}$ to position a. Set $Z^{(b)} = Z$. For $k = b-1, b-2, \ldots, a$, define

$$Z^{(k)} = \mathrm{Exc}\left(Z^{(k+1)}; \sum_{i=1}^{k-1}(1 + z_i), R_{3-y_a}, R_{y_a}\right).$$

For $k \in [a, b-1]_{\mathbb{Z}}$, the simple runs of $Z^{(k+1)}$ in positions k and $k+1$ are R_{3-y_a} and R_{y_a}, respectively, up to translation. Their vertical lengths are distinct. Let

$$D_j^{(k)} = \mathrm{PrefDiff}_j(Z^{(k)}, Y) \qquad (j \in [n]_{\mathbb{Z}},\ k \in [a, b]_{\mathbb{Z}}).$$

Only the prefix ending at position k changes in the exchange. Thus,

$$D_j^{(k)} = \begin{cases} D_k^{(k+1)} + 1, & j = k,\ y_a = 2, \\ D_k^{(k+1)} - 1, & j = k,\ y_a = 1, \\ D_j^{(k+1)}, & j \neq k. \end{cases}$$

The exchanges at positions greater than k do not change this prefix. Thus, $D_k^{(k+1)} = D_k$. The bounds $D_k \leq -1$ when $y_a = 2$ and $D_k \geq 1$ when $y_a = 1$ give

$$|D_k^{(k)}| = |D_k^{(k+1)}| - 1, \qquad \mathrm{PrefSum}_n(Z^{(k)}, Y) = \mathrm{PrefSum}_n(Z^{(k+1)}, Y) - 1.$$

Let us prove that the sequence of exchanges terminates. After these $b - a$ exchanges, $Z^{(a)}$ agrees with Y through position a. If $Z^{(a)} \neq Y$, use $Z^{(a)}$ as the new input Z. Recompute its vertical lengths $z_1, \ldots, z_n$, choose a and b by the defining minima, and perform the exchanges for $k = b-1, \ldots, a$. Each application increases the number of initial simple runs that agree with Y. Thus, the process ends after finitely many applications. Consequently, the resulting run-edit list ends at Y and has length

$$\ell = \mathrm{PrefSum}_n(Z, Y) - \mathrm{PrefSum}_n(Y, Y) = \mathrm{PrefSum}_n(Z, Y).$$

□

Lemma 9.15 (Completion of a binary run)**.** *Let $L \in \mathbb{Z}_{>0}$ and $C \in \mathbb{Z}_{\geq 0}$. Let $\mathbf{t} \in \mathbb{Z}^2$ and let*

$$X = \mathrm{Trans}(R_{x_1} + \cdots + R_{x_L}, \mathbf{t}), \qquad Y = \mathrm{Trans}(R_{y_1} + \cdots + R_{y_{L+C}}, \mathbf{t})$$

be binary runs. For $r \in [2]_{\mathbb{Z}}$, assume

$$b_r = \mathrm{RunCount}_r(Y) - \mathrm{RunCount}_r(X) \geq 0.$$

Let $Z = X + R_1^{b_1} + R_2^{b_2}$. Then, $b_1 + b_2 = C$, and there are $\ell \in \mathbb{Z}_{\geq 0}$ and a run-edit list $\Omega = (\omega_s)_{s \in [\ell]_{\mathbb{Z}}}$ such that

$$\mathrm{RunEResult}(\Omega, Z) = Y.$$

Every ω_s exchanges two adjacent simple runs of distinct vertical lengths, and

$$\ell \leq \mathrm{PrefSum}_L(X, Y) + \frac{C(C-1)}{2}. \tag{9.3}$$

Proof. Let us check the counts after appending the missing simple runs. The two run counts sum to the number of simple runs. Thus, $b_1 + b_2 = C$. Since $Z = X + R_1^{b_1} + R_2^{b_2}$, we have $\mathrm{RunCount}_r(Z) = \mathrm{RunCount}_r(X) + b_r = \mathrm{RunCount}_r(Y)$ for $r \in [2]_{\mathbb{Z}}$. Apply Lemma 9.14. For $j \in [L + C]_{\mathbb{Z}}$, let

$$D_j = \mathrm{PrefDiff}_j(Z, Y).$$

For $j \leq L$, the first j simple runs of Z belong to X. For $j = L + r$ with $r \in [C]_{\mathbb{Z}}$, equality of the total counts gives

$$|D_{L+r}| \leq C - r, \qquad \sum_{r=1}^{C} |D_{L+r}| \leq \sum_{r=1}^{C} (C - r) = \frac{C(C-1)}{2}.$$

Adding $\sum_{j=1}^{L} |D_j|$ gives (9.3). □

Definition 9.16 (Binary run-edit lists)**.** *Let U and V be binary runs and let $K \in \mathbb{Z}_{\geq 0}$ and $k \in \mathbb{Z}_{\geq 0}$. For a run-edit list $\Omega = (\omega_s)_{s \in [K+k]_{\mathbb{Z}}}$ from U to V, let*

$$U_s = \mathrm{RunEResult}((\omega_1, \ldots, \omega_s), U) \qquad (s \in [0, K + k]_{\mathbb{Z}}).$$

We define membership in $\mathrm{BinEdit}_{K,k}(U, V)$ as follows. We have $\Omega \in \mathrm{BinEdit}_{K,k}(U, V)$ if and only if every U_s is binary for $s \in [0, K + k]_{\mathbb{Z}}$ and there is a partition

$$[K + k]_{\mathbb{Z}} = I_R \sqcup I_X, \qquad \#I_R = K, \qquad \#I_X = k,$$

with the following conditions.

1. *For $s \in I_R$, there are $j_s \in [2]_{\mathbb{Z}}$ and $i_s \in [0, |U_{s-1}|]_{\mathbb{Z}}$ such that*

$$\omega_s(U_{s-1}) = \mathrm{Ins}(U_{s-1}, i_s, R_{j_s}), \qquad i_s = |U_{s-1}| \text{ or } (U_{s-1})_{i_s+1} \in E(U_{s-1}).$$

2. *For $s \in I_X$, the run-edit ω_s exchanges adjacent simple runs A and B such that*

$$(|A|_N, |B|_N) \in \{(1, 2), (2, 1)\}.$$

Lemma 9.17 (Comparison of mechanical factors by exchanges)**.** *Let $r \in (1, 2)$ be irrational and $0 < \eta < \frac{1}{4}$. Let*

$$\tau = \frac{1}{2} \min(r - 1, 2 - r), \qquad C = \left\lceil \frac{4}{\tau} \right\rceil.$$

Then, there are $L \in \mathbb{Z}_{>0}$ and $\delta > 0$ depending only on r and η with the following properties.

1. *We have*
$$C \le \eta L, \qquad \delta < \min\left\{\tau, \frac{\eta}{4L}\right\}.$$

2. *For each* $\sigma_1 \in (r-\delta, r+\delta)$, $\sigma_2 \in (r-\delta, r+\delta)$, $u_1 \in [0,1)$, $u_2 \in [0,1)$, $p \in \mathbb{Z}_{\ge 0}$, $V = \mathrm{MechRun}(\sigma_2, u_2)$, *and translated copy* X *of* $\mathrm{RunFactor}_{1,L}(\mathrm{MechRun}(\sigma_1, u_1))$, *there are* $q \in \mathbb{Z}_{\ge 0}$, $k \in \mathbb{Z}_{\ge 0}$, $\zeta \in \mathbb{Z}^2$, *and a run-edit list* Ω *with the following properties.*

   (a) *We have* $C + q \le \eta L$.

   (b) *We have*
$$\Omega \in \mathrm{BinEdit}_{C+q,k}(X, \mathrm{Trans}(\mathrm{RunFactor}_{p+1,L+C+q}(V), \zeta)).$$

   (c) *We have* $k \le 7\eta L$.

Proof. Proof of Claim 1. Let us choose a rational approximation common to both slopes. Choose relatively prime $p_0 \in \mathbb{Z}_{>0}$ and $q_0 \in \mathbb{Z}_{>0}$ satisfying (6.2). Choose $L \in \mathbb{Z}_{>0}$ and $\delta > 0$ such that
$$q_0 + C \le \eta L, \qquad C(C+1) \le \eta L, \qquad \delta < \min\left\{\tau, q_0^{-2}, \frac{\eta}{4L}\right\}.$$
These constants depend only on r and η. They satisfy $C \le q_0 + C \le \eta L$ and $\delta < \min\{\tau, \frac{\eta}{4L}\}$.

Proof of Claim 2(a). Let us bound the count of each simple-run length. For $i \in [2]_{\mathbb{Z}}$, we have $1 + \tau < \sigma_i < 2 - \tau$ and $|\sigma_i - \frac{p_0}{q_0}| < 2q_0^{-2}$. Let us choose q and the simple runs inserted before X. Apply Claim 1 of Lemma 6.4 with offset $b = p$. Let $q = \min J$ and $v_q = \{u_2 + (p+q)\sigma_2\}$ as in Claim 1 of Lemma 6.4. Then, $0 \le q < q_0$ and $|u_1 - v_q| < \eta$. If $q > 0$, insert $\mathrm{RunFactor}_{p+1,q}(V)$ immediately before X. We have $q + C \le q_0 + C \le \eta L$.

Proof of Claim 2(b). Let us compare the completed factor with its target. Let Y be the translated copy of $\mathrm{RunFactor}_{p+q+1,L+C}(V)$ that starts at $\mathrm{Start}_{\mathrm{pt}}(X)$.

Let us specify the simple runs appended to X. A mechanical factor of a simple runs and slope σ has $a(\sigma-1)+e$ simple runs of length two and $a(2-\sigma)-e$ of length one for some $|e| < 1$. For $j \in [2]_{\mathbb{Z}}$, define
$$b_j = \mathrm{RunCount}_j(Y) - \mathrm{RunCount}_j(X).$$
The inequalities $\sigma_2 - 1 > \tau$, $2 - \sigma_2 > \tau$, and $|\sigma_2 - \sigma_1| < 2\delta$ give
$$b_j > \tau C - 2 - 2\delta L > 0, \qquad b_1 + b_2 = C.$$
Append b_1 copies of R_1 and b_2 copies of R_2 immediately after X.

Let us construct the exchanges from $X \dashv R_1^{b_1} \dashv R_2^{b_2}$ to Y. Lemma 9.15 supplies a run-edit list $\mathcal{S} = (S_1, \ldots, S_\nu)$ from $X \dashv R_1^{b_1} \dashv R_2^{b_2}$ to Y consisting of exchanges of adjacent simple runs of distinct vertical lengths. Let H be the prefix of q simple runs inserted before X. When a local run-edit acts on the current factor Z after H, replace its edge index i by $|H| + i$. Replace $\mathrm{Exc}(Z; i, R_a, R_b)$ by
$$\mathrm{Exc}(H \dashv Z; |H| + i, R_a, R_b).$$

Let us identify the path obtained by the complete run-edit list. The q prefix insertions followed by the C appended insertions and these exchanges give Ω. Let $k = \nu$ and
$$\zeta = \mathrm{Start}_{\mathrm{pt}}(\mathrm{RunEResult}(\Omega, X)) - (p, \lfloor u_2 + p\sigma_2 \rfloor).$$
The inserted and exchanged simple runs have vertical lengths in $[2]_{\mathbb{Z}}$. We have
$$\mathrm{RunEResult}(\Omega, X) = \mathrm{Trans}(\mathrm{RunFactor}_{p+1,L+C+q}(V), \zeta).$$

Each insertion is between simple runs or at an end. Thus, Definition 9.16 gives the membership assertion in Claim 2(b).

Proof of Claim 2(c). Let us bound the number of exchanges in $\mathcal{S}$. For $i \in [L]_{\mathbb{Z}}$, the difference of the length-two prefix counts is

$$D_i = \mathrm{PrefDiff}_i(X, Y) = \lfloor u_1 + i\sigma_1 \rfloor - \lfloor v_q + i\sigma_2 \rfloor .$$

Claim 2 of Lemma 6.4 gives

$$\mathrm{PrefSum}_L(X, Y) = \sum_{i=1}^{L} |D_i| \le \frac{11}{2}\eta L.$$

The bound in Lemma 9.15 gives

$$k = \nu \le \sum_{i=1}^{L} |D_i| + \frac{C(C-1)}{2} \le \frac{13}{2}\eta L \le 7\eta L.$$

□

Lemma 9.18 (Completion with one exceptional simple run)**.** *Let $t \in \mathbb{Z}_{>0}$, $C \in \mathbb{Z}_{>0}$, $p \in \mathbb{Z}_{\ge 0}$, $\sigma_1 \in (1, 2)$, $\sigma_2 \in (1, 2)$, $u_1 \in [0, 1)$, and $u_2 \in [0, 1)$. Assume*

$$C \min(\sigma_2 - 1, 2 - \sigma_2) > 3 + t|\sigma_1 - \sigma_2|.$$

Let A_i be simple runs satisfying $|A_i|_N \in [2]_{\mathbb{Z}}$ for $i \in [t]_{\mathbb{Z}}$. Let $X = A_1 + \cdots + A_t$. Assume

$$|A_i|_N = \lfloor u_1 + i\sigma_1 \rfloor - \lfloor u_1 + (i-1)\sigma_1 \rfloor \qquad (i \in [t-1]_{\mathbb{Z}}).$$

Let

$$Y = \mathrm{Trans}(\mathrm{RunFactor}_{p+1,t+C}(\mathrm{MechRun}(\sigma_2, u_2)), \mathrm{Start}_{\mathrm{pt}}(X) - (p, \lfloor u_2 + p\sigma_2 \rfloor)).$$

Then, there are $k \in \mathbb{Z}_{\ge 0}$ and a run-edit list Ω satisfying the following.

1. *We have $\Omega \in \mathrm{BinEdit}_{C,k}(X, Y)$.*
2. *We have $k \le (t + C)^2$.*

Proof. Proof of Claim 1. Let us specify the simple runs appended to X. For $j \in [2]_{\mathbb{Z}}$, put $b_j = \mathrm{RunCount}_j(Y) - \mathrm{RunCount}_j(X)$. Replacing the final mechanical simple run by A_t changes either run count by at most one. Definition 5.2(2) gives

$$\mathrm{RunCount}_2(X) = t(\sigma_1 - 1) + e_1, \qquad \mathrm{RunCount}_2(Y) = (t + C)(\sigma_2 - 1) + e_2,$$
$$|e_1| < 2, \qquad |e_2| < 1.$$

We have

$$\mathrm{RunCount}_1(X) = t - \mathrm{RunCount}_2(X), \qquad \mathrm{RunCount}_1(Y) = t + C - \mathrm{RunCount}_2(Y).$$

Hence,

$$b_j > C \min(\sigma_2 - 1, 2 - \sigma_2) - 3 - t|\sigma_1 - \sigma_2| > 0, \qquad b_1 + b_2 = C.$$

Append $R_1^{b_1} + R_2^{b_2}$ to X. Lemma 9.14 supplies exchanges from $X + R_1^{b_1} + R_2^{b_2}$ to Y. Let Ω consist of these C insertions followed by the exchanges. Let k be the number of exchanges. Every intermediate path is binary. Hence, $\Omega \in \mathrm{BinEdit}_{C,k}(X, Y)$.

Let us bound the exchanges in the completed final factor.

Proof of Claim 2. Let $Z = X + R_1^{b_1} + R_2^{b_2}$. For $i \in [t+C]_{\mathbb{Z}}$, we have $|\mathrm{PrefDiff}_i(Z, Y)| \le t + C$. Lemma 9.14 gives

$$k \le \mathrm{PrefSum}_{t+C}(Z, Y) \le (t + C)^2.$$

□

Lemma 9.19. *Let $r \in (1,2)$ be irrational and let $0 < \eta < 1$. Then, there are constants $0 < \delta < \min(r-1, 2-r)$ and $B > 0$ depending only on r and η with the following property. For each $(2, r, \eta, \delta, B)$-admissible tuple $(l, T, \sigma_1, \sigma_2, u_1, u_2, (A_1, \dots, A_l))$ and*

$$U = A_1 \mathbin{+\!\!+} \cdots \mathbin{+\!\!+} A_l, \qquad V = \mathrm{MechRun}(\sigma_2, u_2),$$

there are $K \in \mathbb{Z}_{\ge 0}$, $k \in \mathbb{Z}_{\ge 0}$, $\zeta \in \mathbb{Z}^2$, and a run-edit list Ω such that we have the following.

1. $\Omega \in \mathrm{BinEdit}_{K,k}(U, \mathrm{Trans}(\mathrm{RunFactor}_{1,T}(V), \zeta))$.
2. $K = T - l$.
3. $k \le 7\eta l + B$.

Proof. Proof of Claim 1. Let us choose the parameters for the binary factor comparison. Let $\eta_0 = \min(\eta, \frac{1}{8})$. Then, $0 < \eta_0 < \frac{1}{4}$ and $\eta_0 \le \eta$. Choose τ, C, L, and δ from Lemma 9.17 with parameter η_0. Let $B = C + (L+C)^2 + 1$. The inequalities $|\sigma_i - r| < \delta < \min(r-1, 2-r)$ give $1 < \sigma_i < 2$ for $i \in [2]_{\mathbb{Z}}$. Thus, every simple run of U and V has vertical length in $[2]_{\mathbb{Z}}$. Let

$$h = \left\lfloor \frac{l-1}{L} \right\rfloor, \qquad t = l - hL \in [L]_{\mathbb{Z}},$$
$$X_s = A_{(s-1)L+1} \mathbin{+\!\!+} \cdots \mathbin{+\!\!+} A_{sL} \quad (s \in [h]_{\mathbb{Z}}), \qquad X_* = A_{hL+1} \mathbin{+\!\!+} \cdots \mathbin{+\!\!+} A_l.$$

Let us apply Claims 2(a)–2(c) of Lemma 9.17 to $X_1, \dots, X_h$. Let $p_0 = 0$. For $s \in [h]_{\mathbb{Z}}$, use the source intercept $\{u_1 + (s-1)L\sigma_1\}$ and the target index p_{s-1}. Claim 2(b) of Lemma 9.17 gives $q_s \in \mathbb{Z}_{\ge 0}$ and a run-edit list $\Omega_s = (\omega_{s,1}, \dots, \omega_{s,n_s})$ from X_s to a translated copy of $\mathrm{RunFactor}_{p_{s-1}+1, L+C+q_s}(V)$. Define $p_s = p_{s-1} + L + C + q_s$.

Let us bound the number of run-edits in each Ω_s. For $s \in [h]_{\mathbb{Z}}$, let K_s and k_s be its numbers of simple-run insertions and adjacent exchanges, respectively. Claims 2(a) and 2(c) of Lemma 9.17 give

$$K_s \le \eta_0 L, \qquad k_s \le 7\eta_0 L \qquad (s \in [h]_{\mathbb{Z}}).$$

Let us apply Claims 1 and 2 of Lemma 9.18 to X_*. Use the source intercept $\{u_1 + hL\sigma_1\}$ and target index p_h. Since $\tau C \ge 4$, $\min(\sigma_2 - 1, 2 - \sigma_2) > \tau$, and $2\delta t < \frac{\eta_0}{2} < 1$, we have

$$C \min(\sigma_2 - 1, 2 - \sigma_2) > 3 + t|\sigma_1 - \sigma_2|.$$

Claim 1 of Lemma 9.18 gives a run-edit list $\Omega_* = (\omega_{*,1}, \dots, \omega_{*,n_*})$ from X_* to a translated copy of $\mathrm{RunFactor}_{p_h+1, t+C}(V)$. Its insertion count is C. Claim 2 of Lemma 9.18 bounds its exchange count by $(t+C)^2$.

Let us assemble the run-edits on the full path. Apply $\Omega_1, \dots, \Omega_h$ and Ω_* successively to their factors in the whole current path. Let B^{L} and B^{R} be the prefix and suffix paths of the current factor Z, so the current path is $B^{\mathrm{L}} \mathbin{+\!\!+} Z \mathbin{+\!\!+} B^{\mathrm{R}}$. Replace a local insertion $\mathrm{Ins}(Z, i, R_j)$ by

$$\mathrm{Ins}(B^{\mathrm{L}} \mathbin{+\!\!+} Z \mathbin{+\!\!+} B^{\mathrm{R}}, |B^{\mathrm{L}}| + i, R_j).$$

Replace a local exchange $\mathrm{Exc}(Z; i, R_a, R_b)$ by

$$\mathrm{Exc}(B^{\mathrm{L}} \mathbin{+\!\!+} Z \mathbin{+\!\!+} B^{\mathrm{R}}; |B^{\mathrm{L}}| + i, R_a, R_b).$$

Induction on the local run-edits gives translated copies of $\mathrm{RunFactor}_{p_{s-1}+1, L+C+q_s}(V)$ for $s = 1, \dots, h$, followed by $\mathrm{RunFactor}_{p_h+1, t+C}(V)$. Every intermediate path is binary and the insertion and exchange counts are unchanged.

Let us count the simple-run insertions and exchanges. Let Q be the resulting path and let $T_0 = p_h + t + C$. Then,

$$Q = \mathrm{Trans}(\mathrm{RunFactor}_{1,T_0}(V), \mathrm{Start}_{\mathrm{pt}}(Q)),$$
$$T_0 = l + \sum_{s=1}^{h}(C + q_s) + C \le l + h\eta_0 L + C \le (1+\eta)l + B \le T.$$

If $T_0 < T$, append $\mathrm{RunFactor}_{T_0+1,T-T_0}(V)$. Concatenating the run-edit lists on the whole path gives Ω. Let $\zeta = \mathrm{Start}_{\mathrm{pt}}(Q)$ and write $\Omega = (\omega_1, \dots, \omega_n)$. Define

$$I_R = \{s \in [n]_{\mathbb{Z}} : \omega_s \text{ is a simple-run insertion}\},$$
$$I_X = \{s \in [n]_{\mathbb{Z}} : \omega_s \text{ exchanges adjacent simple runs}\},$$
$$K = \#I_R, \qquad k = \#I_X.$$

Then, $[n]_{\mathbb{Z}} = I_R \sqcup I_X$. Every intermediate path is binary. Each exchange has the unequal-length property in Definition 9.16. The endpoint construction and Definition 9.16 give

$$\Omega \in \mathrm{BinEdit}_{K,k}(U, \mathrm{Trans}(\mathrm{RunFactor}_{1,T}(V), \zeta)).$$

Proof of Claim 2. The simple-run insertion count is

$$K = (T_0 - l) + (T - T_0) = T - l.$$

Proof of Claim 3. The exchange counts of $\Omega_1, \dots, \Omega_h$ and Ω_* give

$$k \le 7\eta_0 hL + (t + C)^2 \le 7\eta l + B.$$

□

Proposition 9.20. *Let $m = 3$. Let $K \in \mathbb{Z}_{\ge 0}$, $k \in \mathbb{Z}_{\ge 0}$, and $P \in \mathbb{Z}_{\ge 0}$ satisfy $P \ge K + 4k$. Let U and V be binary runs and fix $\Omega \in \mathrm{BinEdit}_{K,k}(U, V)$. Let $u \in [2]_{\mathbb{Z}}$ and $v \in [2]_{\mathbb{Z}}$. Let W_{L} and W_{R} be binary runs with P simple runs each. Let $V_* = W_{\mathrm{L}} + R_u + V + R_v + W_{\mathrm{R}}$. Then, the following statements hold.*

1. *There is a map*
$$\Phi : \mathcal{C}_0(U) \longrightarrow \mathcal{C}_0(V_*)$$
such that
$$\#\Phi^{-1}(\{\xi\}) \le 8^k \qquad (\xi \in \mathcal{C}_0(V_*)). \tag{9.4}$$
2. *We have*
$$C_0(V_*) \ge \frac{C_0(U)}{8^k}. \tag{9.5}$$

Proof. Proof of Claim 1. Let us compose the maps for the individual run-edits. Let $\Omega = (\omega_s)_{s \in [K+k]_{\mathbb{Z}}}$. For $s \in [0, K+k]_{\mathbb{Z}}$, let $U_s = \mathrm{RunEResult}((\omega_1, \dots, \omega_s), U)$. Choose I_R, I_X, and $(i_s, j_s)_{s \in I_R}$ satisfying the conditions for $\Omega \in \mathrm{BinEdit}_{K,k}(U, V)$ in Definition 9.16. Let us apply $\omega_1, \dots, \omega_{K+k}$ to the copy of U in $\mathrm{Shell}(U; u, v, 0)$ and bound the surplus. Define $a_j \in [2]_{\mathbb{Z}}$ and $b_j \in [2]_{\mathbb{Z}}$ for $j \in [P]_{\mathbb{Z}}$ by

$$W_{\mathrm{L}} = \mathrm{Trans}(R_{a_P} + \cdots + R_{a_1}, \mathrm{Start}_{\mathrm{pt}}(W_{\mathrm{L}})),$$
$$W_{\mathrm{R}} = \mathrm{Trans}(R_{b_1} + \cdots + R_{b_P}, \mathrm{Start}_{\mathrm{pt}}(W_{\mathrm{R}})).$$

For $(f, \gamma) \in \mathcal{C}_0(U)$, let

$$Q_0 = \mathrm{Shell}(U; u, v, 0), \qquad (h_0, \eta_0) = \Phi_{\mathrm{sh}}(f, \gamma).$$

Here Φ_{sh} is the map in Claim 4 of Lemma 3.3 with $j = 1$ and $\epsilon = 0$. Claims 2 and 3 of Lemma 3.3 give $(h_0, \eta_0) \in \mathcal{C}_0(Q_0)$. For $s \in I_R$, let $L +\!\!+ U_{s-1} +\!\!+ R$ be the current path and apply Lemma 4.7 at $|L| + i_s$. Equation (4.23) shows that this run-edit adds one to the surplus. For $s \in I_X$, let $e_s = 1 + \#(I_X \cap [s-1]_{\mathbb{Z}})$. Let $Z = L +\!\!+ U_{s-1} +\!\!+ R$ be the current path and let $Z' = L +\!\!+ U_s +\!\!+ R$. Choose

$$\Phi_s \in \mathrm{WrapExch}(Z, Z'; a_{e_s}, b_{e_s}).$$

Lemma 9.10 gives a member of this set because Z' exchanges the two simple runs specified by ω_s inside Z. The choice of Φ_s is independent of the input colored grading. Apply Φ_s to the colored grading at step $s-1$. Claim 1(b) of Lemma 9.10 bounds the increase in surplus by three. Thus, the colored grading (h, η) on the resulting path Q satisfies

$$0 \le \mathrm{Surp}(Q, h) \le K + 3k \le P - k.$$

Let us reduce the surplus to zero. Apply Lemma 4.8 with $n = P - k$ and

$$\alpha' = (a_{k+j})_{j \in [P-k]_{\mathbb{Z}}}, \qquad \mu' = (b_{k+j})_{j \in [P-k]_{\mathbb{Z}}}.$$

Lemma 2.10 translates the final colored grading to V_*. This defines a map $\Phi \colon \mathcal{C}_0(U) \to \mathcal{C}_0(V_*)$.

Let us bound the fibers of Φ. The map $\Delta_{Q,\alpha',\mu'}$ from Lemma 4.8 is injective. Restriction recovers the colored grading before each simple-run insertion by (4.19) and (4.20).

Let us count the choices at the k exchanges. Claim 4 of Lemma 9.10 gives at most eight inputs at each exchange. Claim 4 of Lemma 3.3 recovers (f, γ) by restriction from (h_0, η_0). Recovering the inputs in the reverse order of Ω gives

$$\#\Phi^{-1}(\{\xi\}) \le 8^k \qquad (\xi \in \mathcal{C}_0(V_*)).$$

This proves (9.4).

Proof of Claim 2. Let us sum the fiber bounds to compare the grading counts. Summing (9.4) over $\xi \in \mathcal{C}_0(V_*)$ gives

$$C_0(U) = \sum_{\xi \in \mathcal{C}_0(V_*)} \#\Phi^{-1}(\{\xi\}) \le 8^k C_0(V_*).$$

Since $8^k > 0$, division by 8^k proves (9.5). □

Proposition 9.21. *We have the following.*

1. *For every irrational $r \in (1, 2)$, there is $\Upsilon_3(r) \in (0, \infty)$ such that*

$$\lim_{\substack{a \to \infty,\ (a,b) \in \mathcal{A}_3 \\ \frac{b}{a} \to r}} \frac{\log E_3(a, b)}{a} = \Upsilon_3(r).$$

2. *Let $\Upsilon_3(1) = \log\left(\frac{256}{27}\right)$. Use $\Upsilon_3(r)$ from (5.1) for rational $r \in (1, \frac{3}{2}]$ and from Claim 1 for irrational $r \in (1, \frac{3}{2})$. Then, $\Upsilon_3|_{[1,\frac{3}{2}]}$ takes values in $(0, \infty)$ and satisfies*

$$\lim_{\substack{s \to r \\ s \in [1, \frac{3}{2}]}} \Upsilon_3(s) = \Upsilon_3(r) \qquad \left(r \in [1, \frac{3}{2}]\right).$$

Proof. Proof of Claim 1. Let us choose the comparison parameters for Lemma 9.19 and Claim 2 of Proposition 9.20. Consider η with $0 < \eta < \frac{1}{1024}$. Let $d = 1 - 144\eta$. Choose δ and B from Lemma 9.19 for r and η. Choose relatively prime $p \in \mathbb{Z}_{>0}$ and $q \in \mathbb{Z}_{>0}$ such that

$q \geq 2$, $\alpha = \frac{p}{q} \in (1,2)$, and $|\alpha - r| < \delta$. Let V_* have ν simple runs and let $P = \lfloor 64\eta\nu \rfloor$. Let us translate V_* so that

$$V_* = R_{a_P} + \cdots + R_{a_1} + R_u + V + R_v + R_{b_1} + \cdots + R_{b_P}, \qquad a_j, b_j, u, v \in [2]_{\mathbb{Z}}.$$

Thus, V has $T = \nu - 2P - 2$ simple runs. For $l = d\nu + O(1)$, we have

$$T - (1+\eta)l = (15\eta + 144\eta^2)\nu + O(1) \longrightarrow \infty.$$

Hence, $T \geq (1+\eta)l + B$ for all sufficiently large ν.

Let us specify the two applications of Lemma 9.19. Consider $(a,b) \in \mathcal{A}_3$ as $a \to \infty$ and $\frac{b}{a} \to r$.

(i) For the lower bound, let

$$\nu = a + 1, \quad j = \left\lfloor \frac{d\nu}{q} \right\rfloor, \quad l = jq, \quad U = W^j_{p,q}, \quad V_* = \mathcal{U}(a,b),$$
$$(\sigma_1, \sigma_2, u_1, u_2) = (\alpha, \Gamma(a,b), 0, \{(P+1)\Gamma(a,b)\}).$$

(ii) For the upper bound, let

$$l = a + 1, \quad N = \left\lceil \frac{l}{dq} \right\rceil, \quad \nu = Nq, \quad U = \mathcal{U}(a,b), \quad V_* = W^N_{p,q},$$
$$(\sigma_1, \sigma_2, u_1, u_2) = (\Gamma(a,b), \alpha, 0, \{(P+1)\alpha\}).$$

Let us verify admissibility for each of the two comparisons. In each case, let $A_1, \ldots, A_l$ be the simple runs of U in path order. Equations (2.9), (2.10), and (5.1) show that $(l, T, \sigma_1, \sigma_2, u_1, u_2, (A_1, \ldots, A_l))$ is $(2, r, \eta, \delta, B)$-admissible for all sufficiently large a. The path V is a translated copy of $\mathrm{RunFactor}_{1,T}(\mathrm{MechRun}(\sigma_2, u_2))$. In case (i), the final R_1 of $\mathcal{U}(a,b)$ belongs to W_{R}. In case (ii), we have $A_l = R_1$ and $|A_l|_N = 1 \in [2]_{\mathbb{Z}}$, as permitted by condition (a) of Definition 6.2.

Let us bound the insertions and exchanges in each comparison. Lemma 9.19 gives an insertion count $K = T - l$ and an exchange count k satisfying

$$\lim_{\nu \to \infty} \frac{K}{\nu} = 16\eta, \qquad k \leq 7\eta l + B \leq 8\eta\nu, \qquad K + 4k \leq P$$

for all sufficiently large ν. Apply Claim 2 of Proposition 9.20 and Lemma 2.10 in each case.

First, let us obtain the lower bound from case (i), keeping η, p, and q fixed. Equation (2.13) gives

$$\frac{\log E_3(a,b)}{a} \geq \frac{j}{a} \frac{\log C_0(W^j_{p,q})}{j} - \frac{k}{a} \log 8 - \frac{\log 3}{a}, \qquad \frac{j}{a} \longrightarrow \frac{d}{q}, \quad \frac{\nu}{a} \longrightarrow 1.$$

Second, let us obtain the upper bound from case (ii). Equation (2.13) gives

$$\frac{\log E_3(a,b)}{a} \leq \frac{N}{a} \frac{\log C_0(W^N_{p,q})}{N} + \frac{k}{a} \log 8 - \frac{\log 3}{a}, \qquad \frac{N}{a} \longrightarrow \frac{1}{dq}, \quad \frac{\nu}{a} \longrightarrow \frac{1}{d}.$$

Let us apply the periodic limit to both bounds. We have $j \to \infty$ in case (i) and $N \to \infty$ in case (ii). Use $k \leq 8\eta\nu$ and Claim 2 of Proposition 3.8 to obtain

$$\begin{aligned} \liminf_{\substack{a \to \infty, \ (a,b) \in \mathcal{A}_3 \\ \frac{b}{a} \to r}} \frac{\log E_3(a,b)}{a} &\geq d\Upsilon_3(\alpha) - 8\eta \log 8, \\ \limsup_{\substack{a \to \infty, \ (a,b) \in \mathcal{A}_3 \\ \frac{b}{a} \to r}} \frac{\log E_3(a,b)}{a} &\leq \frac{\Upsilon_3(\alpha) + 8\eta \log 8}{d}. \end{aligned} \tag{9.6}$$

For each fixed η, we choose $\alpha = \frac{p}{q}$, δ, and B before letting $a \to \infty$ and $\frac{b}{a} \to r$ through $(a,b) \in \mathcal{A}_3$.

Let us bound the difference between the upper and lower bounds in (9.6). Lemma 3.5 with $m = 3$ and $L = 2$, together with Claim 2 of Proposition 3.8 and (5.1), gives

$$\Upsilon_3(\alpha) \le 3 \log 8.$$

For each fixed η, equation (9.6) bounds the lower and upper limits for every sequence $((a_n, b_n))_{n \ge 1}$ in $\mathcal{A}_3$ satisfying

$$\lim_{n\to\infty} a_n = \infty, \qquad \lim_{n\to\infty} \frac{b_n}{a_n} = r.$$

Let us prove that the upper bound minus the lower bound in (9.6) tends to zero as $\eta \downarrow 0$. The difference is at most

$$3 \log 8 \left(\frac{1}{d} - d \right) + 8\eta \log 8 \left(1 + \frac{1}{d} \right) \longrightarrow 0.$$

Since $r \in (1,2) \subset I_3$, Lemma 6.6 gives a sequence $((a_n, b_n))_{n \ge 1}$ in $\mathcal{A}_3$ satisfying

$$\lim_{n\to\infty} a_n = \infty, \qquad \lim_{n\to\infty} \frac{b_n}{a_n} = r.$$

For any two such sequences, denote their lower and upper limits by L_i and U_i, $i \in [2]_{\mathbb{Z}}$. They are finite by Lemma 3.5 and (2.13). The common interval in (9.6) gives

$$U_i - L_j \le 3 \log 8 \left(\frac{1}{d} - d \right) + 8\eta \log 8 \left(1 + \frac{1}{d} \right) \qquad (i, j \in [2]_{\mathbb{Z}}).$$

Letting $\eta \downarrow 0$ gives $U_i \le L_j$ for all i, j. Since $L_i \le U_i$, both sequences converge and their limits are equal. Define $\Upsilon_3(r)$ to be this common limit.

Let us check positivity of the limit. The run $\mathcal{U}(a,b)$ has $a+1$ simple runs of vertical length at most two for all sufficiently large a. Lemma 3.5 and (2.13) give

$$\frac{\log E_3(a,b)}{a} \ge \frac{\lceil \frac{a+1}{2} \rceil - 1}{a} \log 3 \longrightarrow \frac{1}{2} \log 3.$$

Hence, $\Upsilon_3(r) \ge \frac{1}{2} \log 3 > 0$.

Proof of Claim 2. Let us combine the three kinds of slopes before applying the continuity criterion. Claim 1, Theorem 5.8, and Corollary 9.1 give, respectively, the irrational, noninteger rational $r > 1$, and $r = 1$ cases of

$$\lim_{\substack{a\to\infty,\ (a,b)\in\mathcal{A}_3 \\ \frac{b}{a}\to r}} \frac{\log E_3(a,b)}{a} = \Upsilon_3(r) \qquad \left(r \in [1, \frac{3}{2}] \right).$$

For rational $r \in (1, \frac{3}{2}]$, Claim 2 of Proposition 3.8 and (5.1) give $\Upsilon_3(r) > 0$. At $r = 1$, we have $\Upsilon_3(1) = \log \frac{256}{27} > 0$. Lemma 6.6 gives hypothesis (b) of Lemma 6.8 for $J = [1, \frac{3}{2}]$. Claim 2 of Lemma 6.8 therefore gives continuity on $[1, \frac{3}{2}]$. Claim 3(b) of Lemma 6.8 specifies the endpoint limits:

$$\lim_{s\downarrow 1} \Upsilon_3(s) = \Upsilon_3(1), \qquad \lim_{s\uparrow \frac{3}{2}} \Upsilon_3(s) = \Upsilon_3\left(\frac{3}{2}\right) \qquad \left(s \in \left[1, \frac{3}{2}\right] \right).$$

□

10 Reflection, endpoints, and the central value

10.1 Limits on the full interior interval

Lemma 10.1 (A specified reduction word and its scale factor)**.** *Let $m \in \mathbb{Z}_{\geq 3}$. Let*

$$\lambda_\pm = \frac{m \pm \sqrt{m^2-4}}{2}, \qquad I = (\lambda_-, \lambda_+), \qquad T(t) = \frac{1}{m-t}, \qquad z(t) = \frac{t-\lambda_-}{\lambda_+ - t} \quad (t \in I).$$

Define the iterates by $T^0(t) = t$ and $T^{j+1}(t) = T(T^j(t))$ for $j \in \mathbb{Z}_{\geq 0}$. Then, the following statements hold.

1. *The maps $t \mapsto \frac{1}{t}$, $t \mapsto m - t$, and T preserve I. The function z is positive and strictly increasing on I. We have*

$$z(1) = \lambda_-, \qquad z(m-1) = \lambda_+, \qquad z(T^j(t)) = \lambda_+^{-2j} z(t) \quad (t \in I,\ j \in \mathbb{Z}_{\geq 0}).$$

In particular,

$$T^j(t) = \frac{\lambda_- + \lambda_+^{1-2j} z(t)}{1 + \lambda_+^{-2j} z(t)}. \tag{10.1}$$

2. *For $r \in I$, define*

$$\nu = \begin{cases} 1, & r < 1, \\ 0, & r \geq 1, \end{cases} \qquad s = \begin{cases} \frac{1}{r}, & \nu = 1, \\ r, & \nu = 0, \end{cases}$$

$$A = \frac{\log(\frac{z(s)}{\lambda_+})}{2\log\lambda_+}, \qquad N = \max\{0, \lceil A \rceil\}, \qquad x = T^N(s),$$

$$\varepsilon = \begin{cases} 1, & x > \frac{m}{2}, \\ 0, & x \leq \frac{m}{2}, \end{cases} \qquad r_* = \begin{cases} m - x, & \varepsilon = 1, \\ x, & \varepsilon = 0. \end{cases}$$

Then, $N \in \mathbb{Z}_{\geq 0}$, $1 \leq x \leq m-1$, and $1 \leq r_* \leq \frac{m}{2}$. Moreover,

$$N = \min\{j \in \mathbb{Z}_{\geq 0} : T^j(s) \leq m-1\}.$$

3. *Use ν, N, ε, and r_* from Claim 2. Define the integer matrices*

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 0 \\ m & -1 \end{pmatrix}, \qquad G = R^\varepsilon (SR)^N S^\nu = (G_{ij})_{i,j \in [2]_\mathbb{Z}}.$$

For $t \in I$, define

$$c_G(t) = G_{11} + G_{12} t, \qquad \rho_G(t) = \frac{G_{21} + G_{22} t}{G_{11} + G_{12} t}. \tag{10.2}$$

We have $G_{11} + G_{12} t > 0$, $\rho_G(t) \in I$, and $\rho_G(r) = r_*$. Both functions are continuous on I. If $(a, b) \in \mathcal{A}_m$ and $t = \frac{b}{a} \in I$, define

$$\begin{pmatrix} \widetilde{a} \\ \widetilde{b} \end{pmatrix} = G \begin{pmatrix} a \\ b \end{pmatrix}.$$

Then,

$$(\widetilde{a}, \widetilde{b}) \in \mathcal{A}_m, \qquad E_m(\widetilde{a}, \widetilde{b}) = E_m(a, b), \qquad \frac{\widetilde{a}}{a} = c_G(t), \qquad \frac{\widetilde{b}}{\widetilde{a}} = \rho_G(t).$$

In particular, $c_m(r) := c_G(r) > 0$ depends only on m and r. If $r \in [1, \frac{m}{2}]$, then G is the identity matrix, $c_m(r) = 1$, and $r_* = r$.

Proof. Proof of Claim 1. Let us use $\lambda_\pm = \frac{m\pm\sqrt{m^2-4}}{2}$. We have

$$\lambda_- + \lambda_+ = m, \qquad \lambda_-\lambda_+ = 1, \qquad 0 < \lambda_- < 1 < m-1 < \lambda_+ < m.$$

For the strict inequality $m-1 < \lambda_+$, evaluate the polynomial $t^2 - mt + 1$ at $m-1$: its value is $2-m < 0$. Thus, $m-1$ lies between the two roots. The other inequalities follow from their positivity, product one, and sum $m > 2$. Consequently, if $t \in I$, then

$$\lambda_- < \frac{1}{t} < \lambda_+, \qquad \lambda_- < m - t < \lambda_+.$$

Their composition $T(t) = \frac{1}{m-t}$ also belongs to I. We have $m - t > 0$.

Let us check that the change of variable is strictly increasing. For $\lambda_- < u < v < \lambda_+$,

$$z(v) - z(u) = \frac{(\lambda_+ - \lambda_-)(v-u)}{(\lambda_+ - v)(\lambda_+ - u)} > 0.$$

For $t \in I$, we have $t - \lambda_- > 0$ and $\lambda_+ - t > 0$. The identities $\lambda_-\lambda_+ = 1$ and $\lambda_- + \lambda_+ = m$ give

$$z(1) = \frac{1-\lambda_-}{\lambda_+ - 1} = \lambda_-, \qquad z(m-1) = \frac{\lambda_+ - 1}{1-\lambda_-} = \lambda_+.$$

They also give

$$\begin{aligned} z(T(t)) &= \frac{1-(m-t)\lambda_-}{(m-t)\lambda_+ - 1} \\ &= \frac{\lambda_-(t-\lambda_-)}{\lambda_+(\lambda_+ - t)} = \lambda_+^{-2} z(t). \end{aligned}$$

Induction on j gives $z(T^j(t)) = \lambda_+^{-2j} z(t)$. Solving $z(u) = \frac{u-\lambda_-}{\lambda_+ - u}$ gives

$$u = \frac{\lambda_- + \lambda_+ z(u)}{1 + z(u)}.$$

Substituting $u = T^j(t)$ and $z(T^j(t)) = \lambda_+^{-2j} z(t)$ proves (10.1).

Proof of Claim 2. Let us verify the specified stopping index. Claim 1 gives $s \in I$ and $s \geq 1$. Since $z(s) > 0$ and $\lambda_+ > 1$, the real number A is defined. For every integer $j \geq 0$, Claim 1 and strict monotonicity of z give

$$\begin{aligned} T^j(s) \leq m-1 &\iff \lambda_+^{-2j} z(s) \leq \lambda_+ \\ &\iff j \geq A. \end{aligned}$$

Therefore, $N = \max\{0, \lceil A \rceil\}$ is exactly the least nonnegative integer with $T^N(s) \leq m-1$. If $N = 0$, then $x = s \geq 1$. If $N \geq 1$, minimality gives $y := T^{N-1}(s) > m-1$, while $y \in I$ gives $y < m$. Hence,

$$0 < m - y < 1, \qquad x = \frac{1}{m-y} > 1.$$

This proves $x \in [1, m-1]$. If $x \leq \frac{m}{2}$, then $r_* = x \in [1, \frac{m}{2}]$. If $x > \frac{m}{2}$, then

$$1 \leq m - x < \frac{m}{2},$$

so $r_* = m - x \in [1, \frac{m}{2}]$.

Proof of Claim 3. Let us factor $G = R^\varepsilon (SR)^N S^\nu$. Put $d = \nu + 2N + \varepsilon$. Define matrices L_i, $i \in [d]_{\mathbb{Z}}$, in application order by

$$\begin{aligned} L_1 &= S \quad \text{if } \nu = 1, \\ L_{\nu+2j-1} &= R, \quad L_{\nu+2j} = S \quad (j \in [N]_{\mathbb{Z}}), \\ L_d &= R \quad \text{if } \varepsilon = 1. \end{aligned} \tag{10.3}$$

The index ranges in (10.3) are disjoint and cover $[d]_{\mathbb{Z}}$. We have $G = L_d \cdots L_1$. For $d = 0$, this product is the identity. For $t \in I$, put $t_0 = t$ and, for $i \in [d]_{\mathbb{Z}}$, define

$$t_i = \begin{cases} \frac{1}{t_{i-1}}, & L_i = S, \\ m - t_{i-1}, & L_i = R. \end{cases}$$

Let us compute the first coordinate after applying G. Claim 1 gives $t_i \in I$ for $i \in [0, d]_{\mathbb{Z}}$. For $i \in [d]_{\mathbb{Z}}$ and $a > 0$, we have

$$S\begin{pmatrix} a \\ at_{i-1} \end{pmatrix} = at_{i-1}\begin{pmatrix} 1 \\ 1/t_{i-1} \end{pmatrix}, \qquad R\begin{pmatrix} a \\ at_{i-1} \end{pmatrix} = a\begin{pmatrix} 1 \\ m - t_{i-1} \end{pmatrix}.$$

Both first coordinates are positive since $t_{i-1} \in I \subset (0, m)$. Starting with $(1, t)^{\mathsf{T}}$ and applying $L_1, \ldots, L_d$ gives

$$c_G(t) = \prod_{\substack{i \in [d]_{\mathbb{Z}} \\ L_i = S}} t_{i-1} > 0, \qquad \rho_G(t) = t_d \in I.$$

The empty product equals one. The construction in Claim 2 gives $t_d = r_*$ when $t = r$.

Let us prove continuity of c_G and ρ_G at an arbitrary $t \in I$. For any sequence $(t_n)_{n \geq 1}$ in I satisfying $\lim_{n \to \infty} t_n = t$, we have

$$\lim_{n \to \infty} (G_{11} + G_{12} t_n) = G_{11} + G_{12} t = c_G(t) > 0.$$

Equation (10.2) and the quotient limit give

$$\lim_{n \to \infty} c_G(t_n) = c_G(t), \qquad \lim_{n \to \infty} \rho_G(t_n) = \frac{G_{21} + G_{22} t}{G_{11} + G_{12} t} = \rho_G(t).$$

Let us identify the corresponding transformations of dimension pairs. On integer dimension pairs the two factors act by

$$S\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} b \\ a \end{pmatrix}, \qquad R\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \\ ma - b \end{pmatrix}.$$

Each intermediate pair is positive because its slope belongs to I. Claims 1, 2(a), and 2(b) of Lemma 2.1 and (1.3) preserve membership in $\mathcal{A}_m$ and the Euler characteristic at each step. More explicitly,

$$\gcd(b, a) = \gcd(a, b), \qquad \gcd(a, ma - b) = \gcd(a, b),$$
$$mba - b^2 - a^2 = mab - a^2 - b^2,$$
$$ma(ma - b) - a^2 - (ma - b)^2 = mab - a^2 - b^2.$$

The Euler identities are those of Claim 1 of Lemma 2.1. Multiplication by G gives

$$\frac{\widetilde{a}}{a} = G_{11} + G_{12} t = c_G(t), \qquad \frac{\widetilde{b}}{\widetilde{a}} = \frac{G_{21} + G_{22} t}{G_{11} + G_{12} t} = \rho_G(t).$$

If $1 \leq r \leq \frac{m}{2}$, then $\nu = 0$, $r \leq m - 1$, $N = 0$, and $\varepsilon = 0$. Thus, $G = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, $c_m(r) = 1$, and $r_* = r$. □

Lemma 10.2 (Endpoint folding)**.** *Let $m \in \mathbb{Z}_{\geq 3}$, put $I = (\lambda_-, \lambda_+)$, and let $\rho \in [1, \frac{m}{2}]$. Define*

$$\delta_\rho = \begin{cases} \frac{1}{2} \min\{\rho - 1, \frac{m}{2} - \rho\}, & 1 < \rho < \frac{m}{2}, \\ \frac{m-2}{2m}, & \rho = 1, \\ \frac{m-2}{4}, & \rho = \frac{m}{2}. \end{cases}$$

The cases are disjoint because $m > 2$. We have $\delta_\rho > 0$. For $(a, b) \in \mathcal{A}_m$, $t = \frac{b}{a} \in I$, and $|t - \rho| < \delta_\rho$, define

$$\mathrm{Fold}_\rho(a, b) = (a', b') = \begin{cases} (a, b), & 1 < \rho < \frac{m}{2}, \\ (\min(a, b), \max(a, b)), & \rho = 1, \\ (a, \min(b, ma - b)), & \rho = \frac{m}{2}. \end{cases}$$

Then,

$$(a', b') \in \mathcal{A}_m, \qquad E_m(a', b') = E_m(a, b), \qquad 1 \le \frac{b'}{a'} \le \frac{m}{2}.$$

Moreover, the ratios are exactly

$$\left(\frac{a'}{a}, \frac{b'}{a'}\right) = \begin{cases} (1, t), & 1 < \rho < \frac{m}{2}, \\ (\min(1, t), \max(t, \frac{1}{t})), & \rho = 1, \\ (1, \min(t, m - t)), & \rho = \frac{m}{2}. \end{cases}$$

Consequently, let $((a_n, b_n))_{n \ge 1}$ be a sequence in $\mathcal{A}_m$ satisfying

$$\lim_{n \to \infty} \frac{b_n}{a_n} = \rho.$$

Then, there is $N \in \mathbb{Z}_{>0}$ such that $\mathrm{Fold}_\rho(a_n, b_n) = (a'_n, b'_n)$ is defined for every $n \ge N$ and satisfies

$$(a'_n, b'_n) \in \mathcal{A}_m, \qquad E_m(a'_n, b'_n) = E_m(a_n, b_n), \qquad 1 \le \frac{b'_n}{a'_n} \le \frac{m}{2}.$$

We have

$$\lim_{n \to \infty} \frac{a'_n}{a_n} = 1, \qquad \lim_{n \to \infty} \frac{b'_n}{a'_n} = \rho.$$

Proof. First, suppose $1 < \rho < \frac{m}{2}$. Then,

$$1 < \rho - \delta_\rho < t < \rho + \delta_\rho < \frac{m}{2}.$$

The pair is unchanged. Thus, all conclusions follow.

Second, suppose $\rho = 1$. The chosen radius satisfies

$$1 - \delta_1 = \frac{m + 2}{2m} > \frac{2}{m}, \qquad 1 + \delta_1 = \frac{3m - 2}{2m} < \frac{m}{2},$$

where $\frac{3m-2}{2m} < \frac{m}{2}$ follows from $(m - 1)(m - 2) > 0$. Therefore,

$$\frac{2}{m} < t < \frac{m}{2}, \qquad 1 \le \max(t, \frac{1}{t}) < \frac{m}{2}.$$

If $t \ge 1$, then $\mathrm{Fold}_1(a, b) = (a, b)$. If $t < 1$, then $\mathrm{Fold}_1(a, b) = (b, a)$. These two cases give

$$\frac{a'}{a} = \min(1, t), \qquad \frac{b'}{a'} = \max(t, \frac{1}{t}).$$

Vertex exchange preserves membership in $\mathcal{A}_m$ and the Euler characteristic by Claims 1, 2(a), and 2(b) of Lemma 2.1 and (1.3).

Third, suppose $\rho = \frac{m}{2}$. In this case

$$1 < \frac{m + 2}{4} = \rho - \delta_\rho < t < \rho + \delta_\rho = \frac{3m - 2}{4} < m - 1.$$

Therefore,

$$1 < \min(t, m-t) \le \frac{m}{2}.$$

For $t \le \frac{m}{2}$, the pair $\mathrm{Fold}_{\frac{m}{2}}(a,b)$ equals (a,b). For $t > \frac{m}{2}$, it equals $(a, ma-b)$. Thus, $\frac{a'}{a} = 1$ and $\frac{b'}{a'} = \min(t, m-t)$. Claims 1, 2(a), and 2(b) of Lemma 2.1 and (1.3) give $(a',b') \in \mathcal{A}_m$ and $E_m(a',b') = E_m(a,b)$.

Let us pass to the limit in each case. We have $[1, \frac{m}{2}] \subset I$. Convergence to ρ puts $t_n = \frac{b_n}{a_n}$ in $I \cap (\rho - \delta_\rho, \rho + \delta_\rho)$ for all sufficiently large n. The limits $\frac{a'_n}{a_n} \to 1$ and $\frac{b'_n}{a'_n} \to \rho$ follow from continuity of min, max, and inversion at a positive number. At $\rho = 1$, these limits are $\min(1,1) = 1$ and $\max(1,1) = 1$. At $\rho = \frac{m}{2}$, the slope limit is $\min(\frac{m}{2}, m - \frac{m}{2}) = \frac{m}{2}$. □

Lemma 10.3 (Reduction of sequences of dimension pairs)**.** *Let $m \in \mathbb{Z}_{\ge 3}$ and $r \in (\lambda_-, \lambda_+)$. Then, there are numbers $r_* \in [1, \frac{m}{2}]$ and $c_m(r) > 0$ depending only on m and r with the following property. For each sequence $((a_n, b_n))_{n \ge 1}$ in $\mathcal{A}_m$ such that*

$$\lim_{n\to\infty} a_n = \infty, \qquad \lim_{n\to\infty} \frac{b_n}{a_n} = r, \tag{10.4}$$

there are an integer $N \ge 1$ and pairs $(a'_n, b'_n) \in \mathcal{A}_m$ for $n \ge N$ such that

$$1 \le \frac{b'_n}{a'_n} \le \frac{m}{2}, \qquad E_m(a'_n, b'_n) = E_m(a_n, b_n) \qquad (n \ge N), \qquad \text{and} \tag{10.5}$$

$$\frac{a'_n}{a_n} \longrightarrow c_m(r), \qquad \frac{b'_n}{a'_n} \longrightarrow r_* \qquad (n \longrightarrow \infty). \tag{10.6}$$

Proof. Let us apply one fixed reduction word to the sequence. Put $I = (\lambda_-, \lambda_+)$. Let G, r_*, and $c_m(r) = G_{11} + G_{12} r > 0$ be the objects specified in Lemma 10.1. By (10.4), the slopes $\frac{b_n}{a_n}$ belong to I for all sufficiently large n. For these indices, define

$$\begin{pmatrix} \widetilde{a}_n \\ \widetilde{b}_n \end{pmatrix} = G \begin{pmatrix} a_n \\ b_n \end{pmatrix}.$$

Claim 3 of Lemma 10.1 gives

$$\begin{gathered} (\widetilde{a}_n, \widetilde{b}_n) \in \mathcal{A}_m, \qquad E_m(\widetilde{a}_n, \widetilde{b}_n) = E_m(a_n, b_n), \\ \frac{\widetilde{a}_n}{a_n} = G_{11} + G_{12} \frac{b_n}{a_n} \longrightarrow c_m(r), \\ \frac{\widetilde{b}_n}{\widetilde{a}_n} = \frac{G_{21} + G_{22}(\frac{b_n}{a_n})}{G_{11} + G_{12}(\frac{b_n}{a_n})} \longrightarrow r_*. \end{gathered}$$

Apply Lemma 10.2 with $\rho = r_*$. Define

$$(a'_n, b'_n) = \mathrm{Fold}_{r_*}(\widetilde{a}_n, \widetilde{b}_n)$$

for $n \ge N$, where $N \ge 1$ is large enough for Lemma 10.2 to apply. That lemma gives $(a'_n, b'_n) \in \mathcal{A}_m$, $E_m(a'_n, b'_n) = E_m(\widetilde{a}_n, \widetilde{b}_n)$, and $1 \le \frac{b'_n}{a'_n} \le \frac{m}{2}$. Thus (10.5) holds. Lemma 10.2 gives

$$\frac{a'_n}{a_n} = \frac{a'_n}{\widetilde{a}_n} \frac{\widetilde{a}_n}{a_n} \longrightarrow 1 \cdot c_m(r) = c_m(r), \qquad \frac{b'_n}{a'_n} \longrightarrow r_*.$$

This proves (10.6). The definitions of G, r_*, and $c_m(r)$ use only m and r. □

Proposition 10.4. *For every integer $m \ge 3$, there is a function $F_m : (\lambda_-, \lambda_+) \to (0, \infty)$ such that the following hold.*

1. *We have*

$$\lim_{\substack{a\to\infty,\ (a,b)\in\mathcal{A}_m\\ \frac{b}{a}\to r}} \frac{\log E_m(a,b)}{a} = F_m(r) \qquad (r\in(\lambda_-,\lambda_+)).$$

2. *The function F_m is continuous.*

Proof. Proof of Claim 1. Let us first obtain the limit for $r\in[1,\frac{m}{2}]$. Fix such an r and consider $(a,b)\in\mathcal{A}_m$ such that

$$a\longrightarrow\infty, \qquad \frac{b}{a}\longrightarrow r, \qquad 1\le\frac{b}{a}\le\frac{m}{2}.$$

First, let us obtain the limit when $m\ge 4$. Define $\Upsilon_m(1)=\ell(R_1)$ and $\Upsilon_m(m-2)=\ell(R_{m-2})$. For all sufficiently large a, we have $b>a\ge 2$ and $1<\Gamma(a,b)<\frac{b}{a}\le\frac{m}{2}\le m-2$ by (2.9). The floor differences in (2.10) and the appended simple run have vertical lengths in $[m-2]_{\mathbb{Z}}$. Claims 1 and 2 of Proposition 5.1, Theorem 5.8, and Theorem 6.7 give the limits on $[1,\frac{m}{2}]$. Lemma 3.5 and (2.13) give positivity.

Second, let us obtain the limit when $m=3$. Theorem 5.8, Claim 1 of Proposition 9.21, and Corollary 9.1 give the limits on $[1,\frac{3}{2}]$. Claim 2 of Proposition 9.21 gives positivity. Thus, the positive limits on $[1,\frac{m}{2}]$ equal $\Upsilon_m(r)$.

Let us obtain the limit at each $r\in(\lambda_-,\lambda_+)$. Let $((a_n,b_n))_{n\ge1}$ be any sequence in $\mathcal{A}_m$ satisfying

$$\lim_{n\to\infty} a_n=\infty, \qquad \lim_{n\to\infty}\frac{b_n}{a_n}=r.$$

Apply Lemma 10.3 to this sequence. Since $\frac{a'_n}{a_n}\to c_m(r)>0$, we have $a'_n\to\infty$. Hence,

$$\frac{\log E_m(a_n,b_n)}{a_n}=\frac{a'_n}{a_n}\frac{\log E_m(a'_n,b'_n)}{a'_n}\longrightarrow c_m(r)\Upsilon_m(r_*)>0.$$

Let us define F_m on (λ_-,λ_+). The numbers r_* and $c_m(r)$ depend only on m and r. Lemma 6.6 supplies at least one sequence approaching r. Define $F_m(r)$ to be the common limit of $\frac{1}{a_n}\log E_m(a_n,b_n)$ along all such sequences. Then,

$$F_m(r)=c_m(r)\Upsilon_m(r_*) \qquad (r\in(\lambda_-,\lambda_+)).$$

For $r\in[1,\frac{m}{2}]$, Claim 3 of Lemma 10.1 gives $r_*=r$ and $c_m(r)=1$. Thus, Claim 1 holds at every interior slope.

Let us identify $F_m(r)$ with $\Upsilon_m(r)$ on the entire interval $(1,M_{\mathrm{run}}(m))$. Fix r in this interval and choose a sequence $((a_n,b_n))_{n\ge1}$ as in (6.4).

If $m\ge4$ and r is an integer, equations (2.9) and (2.10) give $(a_n,b_n)\in\mathcal{S}_m$ for all sufficiently large n. Claim 1 of Proposition 5.1 and (5.1) then give the limit $\Upsilon_m(r)=\ell(R_r)$.

For noninteger rational r, Theorem 5.8 and (5.1) give the limit $\Upsilon_m(r)$. For irrational r, Theorem 6.7 applies when $m\ge4$, and Claim 1 of Proposition 9.21 applies when $m=3$.

Comparing each limit with Claim 1 gives

$$F_m(r)=\Upsilon_m(r) \qquad (1<r<M_{\mathrm{run}}(m)).$$

Let $r\in(\lambda_-,\lambda_+)$. First, let us prove $F_m(r)=rF_m(1/r)$. By Lemma 6.6, choose a sequence $((a_n,b_n))_{n\ge1}$ in $\mathcal{A}_m$ satisfying

$$\lim_{n\to\infty} a_n=\infty, \qquad \lim_{n\to\infty}\frac{b_n}{a_n}=r.$$

Since $r>0$, we have $b_n\to\infty$ and $\frac{a_n}{b_n}\to\frac{1}{r}$. Claim 1 of Lemma 2.1 gives

$$\frac{\log E_m(a_n,b_n)}{a_n}=\frac{b_n}{a_n}\frac{\log E_m(b_n,a_n)}{b_n}\longrightarrow rF_m\left(\frac{1}{r}\right).$$

Second, let us prove $F_m(r) = F_m(m-r)$. Since $m - r > 0$, choose $n_0 \in \mathbb{Z}_{>0}$ such that

$$m - \frac{b_n}{a_n} > \frac{m-r}{2} > 0 \qquad (n \geq n_0).$$

Thus, $ma_n - b_n > 0$ for $n \geq n_0$, and

$$\frac{ma_n - b_n}{a_n} \longrightarrow m - r, \qquad \frac{\log E_m(a_n, b_n)}{a_n} = \frac{\log E_m(a_n, ma_n - b_n)}{a_n} \longrightarrow F_m(m-r).$$

Both transformed sequences belong to $\mathcal{A}_m$ by Lemma 2.1 and (1.3). Comparing these limits with Claim 1 gives

$$F_m(r) = rF_m\left(\frac{1}{r}\right), \qquad F_m(r) = F_m(m-r). \tag{10.7}$$

Proof of Claim 2. Let us prove continuity at each interior slope. Let $r \in (\lambda_-, \lambda_+)$ and choose $\lambda_- < h_1 < r < h_2 < \lambda_+$. Claim 1 and Lemma 6.6 verify the hypotheses of Lemma 6.8 for $F_m|_{[h_1,h_2]}$. Claim 2 of Lemma 6.8 gives continuity at r. □

10.2 Proof of Theorem 1.1 and upper bound

Proof of Theorem 1.1. Let us prove the limit of $\frac{1}{a}\log E_m(a,b)$ at both endpoints. Claim 1 of Proposition 10.4 gives positivity and (1.2) on (λ_-, λ_+). Let $F_m(\lambda_\pm) = 0$. Fix $r \in \{\lambda_-, \lambda_+\}$ and any sequence $((a_n, b_n))_{n\geq 1}$ in $\mathcal{A}_m$ such that $a_n \to \infty$ and $r_n = \frac{b_n}{a_n} \to r$. Since $mr - 1 - r^2 = 0$, we have

$$\max(mr_n - 1 - r_n^2, 0) \longrightarrow 0, \qquad \frac{\log(\frac{2^m}{m})}{a_n} \longrightarrow 0 \qquad (n \to \infty).$$

The bound (3.12) gives

$$0 \leq \frac{\log E_m(a_n, b_n)}{a_n} \leq m \log 2 \sqrt{\frac{m+2}{m-2}} \sqrt{\max(mr_n - 1 - r_n^2, 0)} + \frac{\log(\frac{2^m}{m})}{a_n} \longrightarrow 0.$$

Thus, we obtain (1.2) for $r = \lambda_-$ and $r = \lambda_+$.

Let us prove continuity on the closed interval I_m. For every $r \in I_m$, Lemma 6.6 supplies a sequence satisfying (6.4). Apply Claim 2 of Lemma 6.8 with $J = I_m = [\lambda_-, \lambda_+]$. □

Corollary 10.5 (Quantitative upper bound)**.** *For each integer $m \geq 3$, the function F_m in Theorem 1.1 satisfies*

$$F_m(r) \leq m \log 2 \sqrt{\frac{m+2}{m-2}} \sqrt{(\lambda_+ - r)(r - \lambda_-)} \qquad (r \in I_m). \tag{10.8}$$

Proof. Let us obtain (10.8) from (3.12). For $r \in (\lambda_-, \lambda_+)$, choose a sequence $((a_\nu, b_\nu))_{\nu\geq 1}$ in $\mathcal{A}_m$ satisfying (6.4). Put $r_\nu = \frac{b_\nu}{a_\nu}$. Equations (1.2) and (3.12) give

$$\begin{aligned} F_m(r) &= \lim_{\nu\to\infty} \frac{\log E_m(a_\nu, b_\nu)}{a_\nu} \\ &\leq \lim_{\nu\to\infty} \left(m \log 2 \sqrt{\frac{m+2}{m-2}} \sqrt{\max(mr_\nu - 1 - r_\nu^2, 0)} + \frac{\log(\frac{2^m}{m})}{a_\nu} \right) \\ &= m \log 2 \sqrt{\frac{m+2}{m-2}} \sqrt{mr - 1 - r^2}. \end{aligned}$$

The identity $mr - 1 - r^2 = (\lambda_+ - r)(r - \lambda_-)$ proves (10.8). For $r = \lambda_\pm$, both sides are zero. □

10.3 The value at slope one and some figures for F_m

Corollary 10.6. *For every $m \geq 3$,*

$$\begin{aligned}\lim_{\substack{a\to\infty,\ (a,b)\in\mathcal{A}_m \\ \frac{b}{a}\to 1}} \frac{\log E_m(a,b)}{a} &= F_m(1) \\ &= (m-1)^2\log((m-1)^2) - m(m-2)\log(m(m-2)).\end{aligned} \tag{10.9}$$

Proof. Let us obtain the limit along dimensions $(a, a+1)$. The adjacent-dimension formula of [19, Theorem 6.6] is

$$E_m(a,a+1) = \frac{m}{(a+1)((m-1)a+m)}\binom{(m-1)^2a+(m-1)m}{a}.$$

Fix m and let $c=(m-1)^2$ and $d=m(m-1)$. Stirling's formula $\log(n!) = n\log n - n + O(\log n)$ gives, as $a\to\infty$ through positive integers,

$$\begin{aligned}\log\binom{ca+d}{a} &= (ca+d)\log(ca+d) - a\log a \\ &\quad - ((c-1)a+d)\log((c-1)a+d) + O(\log a) \\ &= a\big(c\log c - (c-1)\log(c-1)\big) + O(\log a).\end{aligned}$$

The constants in the error terms depend only on m. For the prefactor, we have

$$\log\frac{m}{(a+1)((m-1)a+m)} = -2\log a + O(1).$$

Dividing by a gives

$$\begin{aligned}\frac{1}{a}\log\binom{ca+(m-1)m}{a} &\longrightarrow c\log c - (c-1)\log(c-1), \\ \frac{1}{a}\log\frac{m}{(a+1)((m-1)a+m)} &\longrightarrow 0.\end{aligned}$$

Since $c-1 = m(m-2)$, equation (1.2) gives (10.9). □

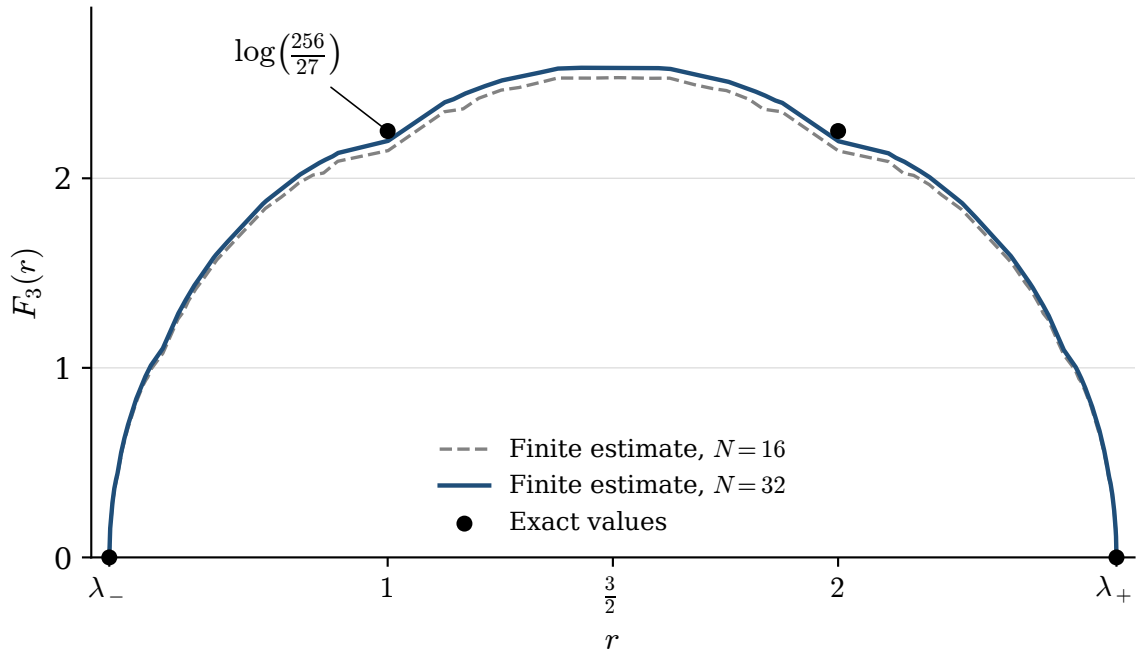


Figure 2: Numerical illustration of F_3 on I_3. Here $\lambda_\pm = \frac{3\pm\sqrt{5}}{2}$. For each reduced $\alpha = \frac{p}{q} \in [1, \frac{3}{2}]$ with $q \le 8$, let $N \in \{16, 32\}$ and $j = \left\lfloor \frac{N}{q} \right\rfloor$. We use $\widehat{F}_N(\alpha) = \frac{\log C_0(W_{p,q}^{2j}) - \log C_0(W_{p,q}^{j})}{jq}$. We interpolate these values linearly and extend them by (10.7). The dots mark the exact values $F_3(\lambda_\pm) = 0$ and $F_3(1) = F_3(2) = \log\left(\frac{256}{27}\right)$. The curves are finite approximations; we do not assert an error bound.

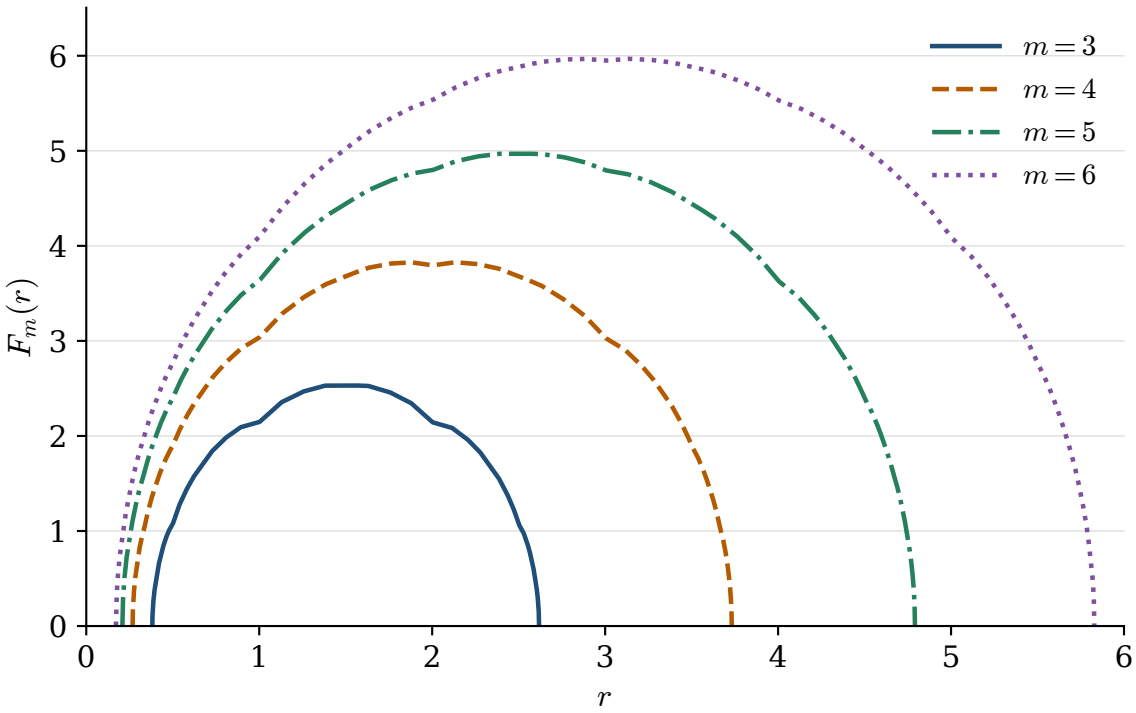


Figure 3: Finite approximations to $F_m(r)$ for $m = 3$, 4, 5, and 6, each on its interval I_m. For $\alpha = 1 + \frac{k}{8}$ with $k \in [0, 4(m-2)]_{\mathbb{Z}}$, let $\alpha = \frac{p}{q}$ in lowest terms and let $j = \frac{16}{q}$. We evaluate C_0 for the indicated m and use $\widehat{F}_m(\alpha) = \frac{\log C_0(W_{p,q}^{2j}) - \log C_0(W_{p,q}^{j})}{jq}$. We interpolate these values linearly on $[1, \frac{m}{2}]$ and extend them by (10.7). We use the exact value zero at both endpoints. The curves are finite approximations; we do not assert an error bound.

11 Conjectural log-concavity

We state the following conjecture on Euler characteristics near slope one. See [2, 13] for related studies and [12, Section 1.12] for a study of log-concavity in mathematics and statistical mechanics.

Conjecture 11.1 (Euler-characteristic ratios near slope one)**.** *Let* $a \in \mathbb{Z}_{\geq 5}$ *satisfy* $\gcd(a, 6) = 1$. *Then,*

$$\frac{E_3(a, a+4)}{E_3(a, a+3)} \leq \frac{E_3(a, a+3)}{E_3(a, a+2)}.$$

We provide the following figure on Conjecture 11.1.

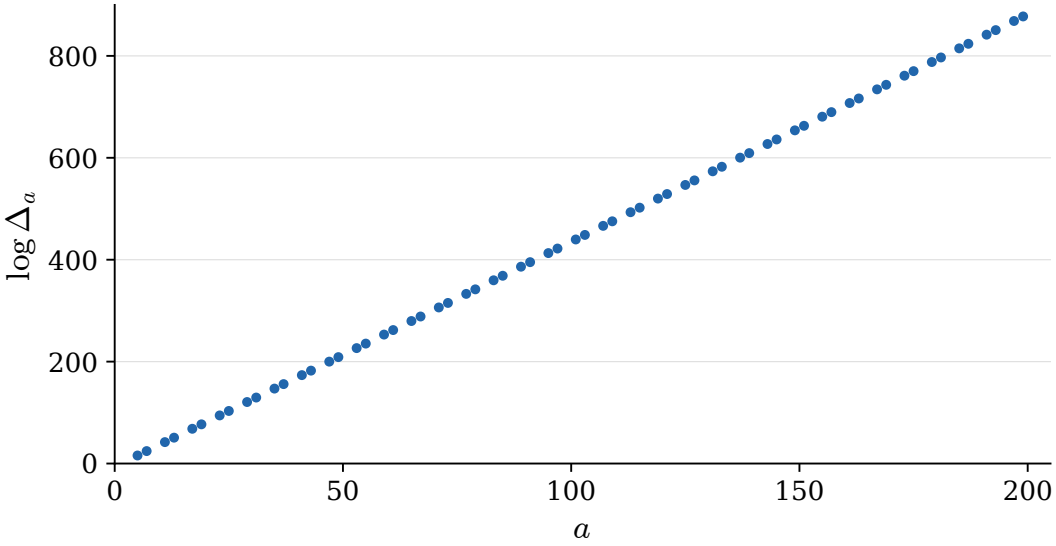


Figure 4: Let $\Delta_a = E_3(a, a+3)^2 - E_3(a, a+2)E_3(a, a+4)$. The points plot $\log \Delta_a$ by explicit computation for $a \in [5, 200]_{\mathbb{Z}}$ satisfying $\gcd(a, 6) = 1$. All 66 differences are positive.

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